



VOLUME 31 ISSUE 2

Administratio Publica



ASSADPAM

Journal of the Association of
Southern African Schools and Departments
of Public Administration and Management

ADMINISTRATIO PUBLICA

Chief Editor: Prof Danielle Nel-Sanders (University of Johannesburg)

Deputy Editor: Thokozani Nzimakwe (University of KwaZulu-Natal)

Technical Editor: Liza van Jaarsveldt (University of South Africa)

Editorial Committee

Ms Adèle Burger (University of Stellenbosch)

Prof Enaleen Draai (Nelson Mandela Metropolitan University)

Prof Liezel Lues (University of the Free State)

Prof Gerda van Dijk (University of Pretoria)

Prof Michelle Esau (University of Western Cape)

Prof Sipho Mokoena (University of Limpopo)

Prof Melvin Diedericks (North West University)

Dr Kambidima Wotela (University of the Witwatersrand)

Dr Onkgopotse Madumo (University of Johannesburg)

Editorial Council

Prof C Auriacombe (University of Johannesburg, South Africa)

Prof M J (Jide) Balogun (Former Senior Advisor, UNDESA)

Prof B Basheka (University Kabale, Uganda)

Prof G Bouckaert (EGPA, Leuven)

Prof R Cameron (University of Cape Town, South Africa)

Prof V Clapper (University of South Africa, South Africa)

Prof M S de Vries (University of Nijmegen, Netherlands)

Prof Natasja Holtzhausen (University of Pretoria)

Prof V Jarbandhan (University of Johannesburg, South Africa)

Prof G K Karyeija (Uganda Management Institute, Uganda)

Prof J Kuye (Chief Editor: African Journal of Public Affairs (AJPA), South Africa)

Prof J Lutabingwa (Appalachian State University, North Carolina)

Prof C Lynch (Southern University, Louisiana)

Prof M Macaulay (Victoria University, New Zealand)

Prof A McLennan (University of the Witwatersrand, South Africa)

Prof R K Mishra (Osmania University, India)

Prof D Olowu (ISS, The Hague, Netherlands)

Prof L A Picard (University of Pittsburgh, Pennsylvania)

Prof P S Reddy (University of KwaZulu-Natal, South Africa)

Prof C Reichard (EAPAA, University Potsdam, Germany)

Prof D Thakhathi (University of Fort Hare, South Africa)

Prof A van Nieuwkerk (University of the Witwatersrand, South Africa)

Prof M van der Steen (University of Tilburg, Netherlands)

ISSN 1015-4833

About the Journal

Administratio Publica is a double blind, peer-reviewed journal accredited with the South African Department of Higher Education and Training (DoHET) and produced four times a year and aims to promote academic scholarship in public administration and management and related fields. Analytical articles in the form of original theoretical and empirical articles, debates, research viewpoints, review articles and book reviews in English will be considered for publication. *Administratio Publica* is published under the auspices of the Association of Southern African Schools and Departments of Public Administration and Management (ASSADPAM).

Page fees

Authors who are ASSADPAM members are required to pay a fee of R400.00 per page, as well as R200.00 per graphic for any manuscript printed in the journal (excluding the costs of language editing where needed). Authors who are not ASSADPAM members are required to pay a fee of R500.00 per page, as well as R200.00 per graphic for any manuscript printed in the journal (excluding the costs of language editing where needed). Payment is due on receiving confirmation from the Editor that a contribution is to be published in a specific issue of the Journal.

Where to send articles

All correspondence concerning articles should be directed to:

Prof Danielle Nel-Sanders

School of Public Management, Governance and Public Policy

College of Business and Economics

P O Box 524

Auckland Park 2006

Tel: +27 82 463 1776

Email: daniellen@uj.ac.za

Articles should be submitted online at <https://journals.co.za/journal/adminpub>.

ASSADPAM membership enquiries

Prof M Subban

School of Management, IT & Governance

College of Law & Management Studies

Westville Campus, University of KwaZulu-Natal

Postal address: Private Bag X54001, Durban 4000

E-mail: subbanm@ukzn.ac.za

Tel: +27 31 260 7763

Contents

Editorial	iii
<i>O Madumo</i>	
New Public Governance Theory	1
A Framework for Lesotho Policymakers to Enhance Community Participation during Climate Change Policy Formulation and Implementation	
<i>R Chatanga and M Biljohn</i>	
The Quest for a New Local Government Model in South Africa	25
A Visionary Perspective	
<i>J Breed, C Jones, P Pillay and S Zondi</i>	
South Africa's Renewable Energy Auctions	47
An Exploration of Factors, Processes and Outcomes	
<i>W Kruger</i>	
Transition from Paper-based Analogue Government to Sustainable Digital Governance Outcomes in Africa	70
<i>F Cloete</i>	
Preparing for Digital Governance	91
Mapping Competency Domains for Postgraduate Programmes in Public Administration and Management	
<i>G van der Waldt</i>	
POSTGRADUATE FORUM	
Fiscal Decentralisation	113
Lesotho Subnational Government Financing	
<i>T J Mohapi, V T Sambo and G M Ferreira</i>	
Service Delivery Challenges within Rural Communities	136
The Case of the Nkomazi Local Municipal Area	

B R Shongwe and D F Meyer

**Organisational Variables Influencing Strategic
Management in Public Organisations**

160

Critical Considerations for Strategy Execution

C B Sibiya and M Subban

Data and Information Security Governance

183

A Prerequisite for Improved Protection of Public

Sector Data and Information Practices

L Masilela and D Nel-Sanders

Editorial

O Madumo

Acting Chief Editor

Article 6 of the United Nations Framework Convention on Climate Change (UNFCCC) obliges its parties to involve communities in the fight against climate change. In line with this, the African Union has prioritised the UNFCCC's call for commitment to climate governance. In **'New Public Governance Theory: A Framework for Lesotho Policymakers to Enhance Community Participation during Climate Change Policy Formulation and Implementation'**, R Chatanga and M Biljohn argue that this Southern African country, as an African Union signatory through its 2017 National Climate Change Policy, is yet to facilitate such participation in climate change policy processes.

In line with this, the article explores Lesotho policymakers' role in enhancing community participation in climate change policy formulation and implementation with the New Public Governance (NPG) theory as the theoretical framework. By following a qualitative research approach, the authors provide a narrative literature review and content analysis of peer-reviewed research and grey literature.

Using key NPG components as theoretical lens, the authors state that, "[To] facilitate sustainable policy implementation and ultimate climate change mitigation, policymakers should engage the community sector at all five stages of the policy process". The authors propose a theoretical framework for policymakers' role in enhancing community participation during climate change policy formulation and implementation.

In conclusion, the article draws attention to the extent to which the NPG theory and its components can be applied to other policy contexts where community participation is required, such as social welfare and environmental management. The authors call for further empirical research on the application of the NPG components, especially in policy areas where the voice and influence of the community are required.

In South Africa, extensive resources are poured into local institutionalism to facilitate social change, socio-economic growth, and community development. In **'The Quest for a New Local Government Model in South Africa: A Visionary Perspective'**, J Breed, C Jones, P Pillay and S Zondi state that the expected rate of return on these attempts has failed to help fast-track the country's development. The authors argue that, while local government's unlocked potential can have a game-changing impact, this institutional mechanism "produces too little, at too high a price". In fact, the authors highlight that, in many communities, local

governments have failed to address communities' growing frustration, with negative consequences and damage.

The article highlights that drastic trajectory changes are needed. According to the authors, local governments should unlock and maximise untapped human capital to create new opportunities. As local government is a country's biggest socio-economic development asset, it should be used to introduce a new enterprising institutional process, as well as maximise human, social capital and lateral solution capital. To this end, the article calls for a framework to facilitate a hybrid business model for developing "exceptional technocratic and institutional behaviour, thought and performance modification". In conclusion, "A new model should...envision and trace a completely new strategic and transformative local governmental model aimed at unlocking all possible assets available to and latent in local societal silos".

South Africa's renewable energy auction programme has been widely hailed as a regional and global success, with more than 6000 MW of renewable energy capacity secured and built in four bidding rounds at increasingly competitive prices. In his article **'South Africa's Renewable Energy Auctions: An Exploration of Factors, Processes and Outcomes'**, W Kruger argues that topic-related literature does not analyse how the implementation of a competitive procurement programme's implementation contributes to a project's price and ultimate success. By following an exploratory case study approach, this article aims to fill this knowledge gap.

A key auction implementation aspect highlighted by the South African case is the impact of developing and maintaining bidder and investor trust to drive competition and lower project prices. According to Kruger, considerable resources were invested in ensuring that the bidding process was secure, transparent and well-structured. Kruger states that the IPP office's approach to the design and implementation of REI4P was to a large degree shaped by an emphasis on building and maintaining trust, with the ultimate goal of reducing perceived risk levels in the long run. Despite this, the article cautions that South Africa's political economic forces beyond the control of the auction process can ultimately derail these investments. The author shows that "it is crucial for the success of future auction programmes that the country provides policy certainty and maintains the market's trust".

Digital government has proved to be more cost-effective, efficient and sustainable than traditional analogue governance practices. However, many lesser developed societies are not ready for digital transformation, as reflected by several failed transitions. In **'Transition from Paper-based Analogue Government to Sustainable Digital Governance Outcomes in Africa'**, F Cloete shares key conclusions on a qualitative inquiry into the latest international research findings and conclusions regarding the state of digital governance transformation in Africa, and how such approaches and practices can be bolstered.

The article highlights and clarifies the key aspects that promote countries' preparedness for, and the success of, digital governance transformation. In general, e-government in Africa is weak due to the cumulative impact of resource, technological, cultural, social, financial and political policy governance constraints. According to the author, "The global e-government paradigm shift is still future music in most African nations, and the successful implementation of 4IR in this regard in Africa faces an uphill battle". Despite this grim picture, the article highlights that South Africa was an early adopter of e-government at the beginning of the 21st century and is regarded as one of the leaders in this regard in developing countries. However, over the last two decades, little progress has been made towards more mature digital government transitions.

In **'Preparing for Digital Governance: Mapping Competency Domains for Postgraduate Programmes in Public Administration and Management'**, G Van der Walddt states that rapid technological advancement and a highly dynamic world of governance place new demands on public managers to prepare for the digitalisation of the public sector. To meet these demands, public managers must be prepared to adopt new technology and understand the impact it has on public institutions and the public. According to the author, public managers should harness technology's potential to improve governance, enhance government efficiency, engage stakeholders and better serve citizens. They must also be able to effectively use data analytics, digital communication tools, and cybersecurity measures to improve the efficiency and effectiveness of their operations. "This requires investing in skills, competencies and knowledge to successfully navigate public sector institutions towards digital governance," states Van der Walddt.

Desktop thematic analysis methodology was used to extract seven core digital competency domains public managers should master from an extensive literature survey. According to the author, "Curriculation efforts should enhance public managers' comprehension of these seven domains to support digital transformation and modernisation endeavours in the public sector". To this end, this article proposes a baseline curriculum framework for postgraduate programmes in Public Administration and Management to prepare current and prospective public managers in the field of digital governance.

POSTGRADUATE FORUM

Fiscal decentralisation is a core component of the decentralisation process. Moreover, the financing of subnational governments is central to fiscal decentralisation reforms. In **'Fiscal Decentralisation: Lesotho Subnational Government Financing'**, T J Mohapi, V T Sambo and G M Ferreira state that this country's government implemented decentralisation as a policy to promote popular

participation and better local public service to ensure socio-economic prosperity among the Basotho. Lesotho's decentralisation was operationalised in 1997 through the Local Government Act 6 of 1997 (Lesotho Local Government Act 6 of 1997). In 2014, the country's government adopted the national decentralisation policy as a comprehensive reform framework. In particular, decentralisation reforms intend to address the fiscal aspect of the decentralisation process.

In the article, the authors consider the conceptual nature of fiscal decentralisation and subnational government financing in Lesotho. The study applied mixed methods research methodology. Data was collected through document analysis, closed-ended questionnaires and interviews within local councils in the Maseru, Berea, Mophale's Hoek and Thaba-Tseka Districts to ensure representation of all the regions of the country. The purposively selected respondents included elected councillors and technical staff.

The study found that Lesotho's decentralisation process lacks the fiscal component, because fiscal decentralisation is practised as intergovernmental transfers. According to the authors, "The decentralisation process does not have fiscal decentralisation legislation to guide and manage a subnational government financial management system". To this end, the authors recommend the development of a comprehensive subnational government framework to improve resource mobilisation for subnational government financing.

Municipalities across South Africa continuously experience challenges concerning service provision under their jurisdiction. As a result, communities that receive poor service delivery often voice their frustration through service delivery protests. In **'Service Delivery Challenges within Rural Communities: The Case of the Nkomazi Local Municipal Area'**, B R Shongwe and D F Meyer state that these service delivery protests are typically triggered by municipalities' failure to provide communities with basic quality services such as electricity, water, tarred roads, sanitation and waste removal.

In line with this, this research was based on the premise that South African municipalities have a poor track record in terms of service delivery. To shed light on the issue, the article identifies and discusses key challenges relating to service delivery in the Nkomazi Local Municipality area. This article adopted a qualitative research approach for data collection. Data was collected using semi-structured interviews with municipal administrators within various departments, six ward councillors and four PR councillors. The findings of the qualitative interviews were rigorously interpreted and analysed in this article. Based on the findings, the authors argue that poor planning, financial mismanagement, corruption and lack of community participation are key challenges that hinder the municipality from delivering quality basic services. "In general, corruption, financial mismanagement and unproductive operations are among the challenges that contribute to ineffective service delivery by municipalities," state the authors.

Strategic planning is a tool to ensure the benefits of realising predetermined strategic objectives. However, literature indicates there are organisational factors that need to be effectively managed to ensure that strategic objectives are effectively achieved. **‘Strategic Management Processes in Public Organisations: Critical Considerations’**, by C B Sibiyi and M Subban focuses on the organisational variables that influence strategy implementation in public sector organisations. After comprehensive strategies have been developed, the authors state that significant challenges are often encountered during the strategy execution process. “The growing needs of communities coincide with the shrinking resources in public administration where cost-effective means and efficiency are main concerns to contribute to more effective service delivery with less resources” state the authors.

The article is based on the public service strategy execution perspective, where leaders are involved in strategy formulation and in taking ownership of the strategy execution. To shed light on the issue, the article conceptualises the public strategic management process, provides several perspectives for strategic management in the public sector and highlights the public service planning framework and other plans. The article also discusses the influence of organisational variables on the implementation of pre-determined objectives and strategic goals that impact on the beneficiaries. According to the authors, the ability to develop and implement new strategies can mean the difference between effective and poor service delivery. However, most strategies in public organisations fail in the strategy execution phase.

The government is one of the biggest collectors and distributors of Public Sector Data and Information (PSDI) assets. As such, it must be constantly aware of the risks linked to handling such large volumes of data and information. In **‘Data and Information Security Governance: A Prerequisite for Improved Protection of Public Sector Data and Information Practices’**, L Masilela and D Nel-Sanders argue that, “to counteract sensitive data and information theft in the government, the government must understand the importance of Data and Information Security Governance (DISG) and how it can be utilised as an effective approach to improve data and Information Security (IS) systems”. According to the authors, DISG helps ensure that a comprehensive approach is followed to ensure the overall protection of data and information, ensure confidentiality, as well as guard the integrity of any organisation in the public or private sector.

To reach key conclusions, the researchers investigate the role and importance of DISG practices and principles within the national sphere of government in the South African public sector, namely the Department of Energy (DoE), former Department of Science and Technology (DST) and former Department of Environmental Affairs (DEA). The researchers followed a qualitative approach using a case study strategy; primary data was collected through semi-structured

interviews. Findings are presented in an analytical framework consisting of the people, systems, processes, risks and controls.

The study results point to a need for research and development initiatives and institutionalisation of DISG management practices in the public sector. “In order for the government to improve its efforts towards the institutionalisation of DISG management practices, its public sector institutions must have clearly defined roles, responsibilities, and functions”. In conclusion, the authors highlight that the security landscape of IT on a local and international level is constantly evolving and changes daily. “The government must therefore implement proactive measures and approaches...for the adequate identification and mitigation of internal and external risks that could significantly hamper the protection of PSDI”.

ACTING CHIEF EDITOR

Dr Onkgopotse Madumo

Email: onkgopotsem@uj.ac.za

New Public Governance Theory

A Framework for Lesotho Policymakers to Enhance Community Participation during Climate Change Policy Formulation and Implementation

R Chatanga

Department of Public Administration and Management
University of the Free State

M Biljohn

Department of Public Administration and Management
University of the Free State

ABSTRACT

Article 6 of the United Nations Framework Convention on Climate Change (UNFCCC) obliges its parties to engage communities in debates to respond collectively to climate change. Likewise, the African Union has prioritised this call for commitment to climate governance by the UNFCCC. However, Lesotho – as an African Union signatory through its 2017 National Climate Change Policy – has not been successful in facilitating such participation in climate change policy processes. Accordingly, the study, on which this article is based, explored the role of Lesotho policymakers to enhance community participation during climate change policy formulation and implementation through the New Public Governance (NPG) theory as the theoretical framework. Through a qualitative research approach, a narrative literature review is provided based on content analysis of peer-reviewed research and grey literature. Using the NPG theory as the theoretical framework, the literature suggests that three aspects of governance, namely socio-political governance, administrative governance, and public policy governance are fundamental to enhancing community participation during climate change policy formulation and implementation. From this review, it was clear that, for sustainable policy implementation and ultimate climate change mitigation, policymakers should provide an opportunity for the legitimate and equal voice of the community sector at all five stages of the policy process.

INTRODUCTION

Community participation during climate change policy formulation and implementation processes is high on the global agenda of the United Nations as a sustainable development goal number 13 (United Nations 2015:23). This global agenda has highlighted the role of policymakers in developed and developing countries to facilitate community participation during the respective policy formulation and implementation stages. If this global problem is not addressed through climate governance and the adoption of appropriate policies, climate change may cause 100 million people to live in poverty by 2030 (Worker and Northrop 2018:21). Despite this global problem, there seems to be limited literature and research available about how community participation could be facilitated at every stage of the climate change policy process in practice (Burton, Malone and Huq 2004:263; Burton and Mustelin 2013:411; Runhaar 2009; Uittenbroek *et al.* 2019:2530; Wright *et al.* 2014:323). Further, what the role of the policymaker should constitute when facilitating community participation during climate change policy processes is likewise limited (Hegger *et al.* 2017; Mees *et al.* 2019:199; Uittenbroek *et al.* 2019). The policymakers' role is of particular significance in areas that are vulnerable to climate change and where the participation of communities living in such areas is critical for climate change adaptation and mitigation. However, research by Mburia (2015:19) points out that on the African continent and in certain countries, the policymakers' role in enhancing community participation in climate change policy formulation and implementation remains elusive.

On the African continent, a wide gap exists between climate change policy and its governance. This gap is attributed to a variety of factors, such as a lack of adaptation policies, weak climate change participation frameworks, poor coordination between implementation of existing legislation, and the absence of leadership foresight in national development planning concerning climate change (Mburia 2015:19). Lesotho is an example of a southern African country where environmental legislation and policy are not well implemented due to incoherence of coordination and fragmentation across a number of government ministries and departments (Machepha 2010:18). This is exacerbated by a lack of community participation during the formulation of the 2017 Lesotho National Climate Change Policy (NCCP) and its implementation processes (Ministry of Energy and Meteorology 2017:24). With a population of about 2 007 201 million citizens, 80% reside in the lowlands and 20% reside in the mountains (Bureau of Statistics 2018:19). The greater part of the Lesotho population, such as the Manonyane community, who lives in the lowlands puts pressure on its resources because less than 20% of the country's surface is lowlands and more than 80% of the country's surface is mountainous (Bureau of Statistics 2018:19). Located at the foothills of the Lesotho mountains, the Manonyane community is no exception to climate change risks and is vulnerable

to and adversely affected by climate change. A lack of participation by this community was also prevalent during the 2017 NCCP formulation and implementation (Ministry of Energy and Meteorology 2017:24; Chatanga and Biljohn 2022:nd).

Against this background, this article drew on the NPG theory components as a framework for Lesotho policymakers to enhance community participation during climate change policy formulation and implementation. The research question posed was: How could the components of the NPG theory serve as a framework for Lesotho policymakers to enhance community participation during climate change policy formulation and implementation? Through a qualitative research approach, deductive analysis of the content of documents was undertaken about relevant research concerning NPG theory, policy formulation and implementation, and climate change, and climate change in Lesotho. Through a range of electronic search engines, the concepts of NPG theory, policy formulation and implementation, and climate change were applied respectively as search terms and adapted as required to identify research that could be included in a narrative literature review. From the content analysis of documents – such as peer-reviewed journal articles, published and unpublished research and grey literature – the topic was explored through a deductive approach (Braun and Clarke 2006:8). Against this background, the first section discusses foundational values that should underpin the role of policymakers during the respective policy stages. The subsequent section discusses how the NPG theory components of administrative governance, socio-political governance and public policy governance apply to the respective policy stages. This is followed by community participation in climate change policy formulation and implementation in the Lesotho context. Using the NPG components, the article concludes by proposing a conceptual framework for the role of policymakers to enhance community participation during climate change policy formulation and implementation.

FOUNDATIONAL VALUES FOR POLICYMAKERS

This section discusses foundational values that should underpin the role of policymakers during the respective policy stages. In this article, policymakers refer to politicians who approve policies and the public administrators who formulate and implement policies. For the current study, the focus on policymakers was the role of the public administrator in climate change policy formulation and implementation. During agenda setting, the role of public administrators should be embedded in values of facilitating interaction and collaboration, legitimacy and trust in participation, and should lend integrity to the process (McNamara *et al.* 2020; Mees *et al.* 2019:200). These values are elements of the NPG theory component of public policy governance (Reed *et al.* 2018). The facilitation of the interaction process

occurs when public administrators take the lead in bringing policy networks and communities together (Wu *et al.* 2018:7). Once interaction has been achieved, collaboration should be facilitated between communities and other actors, which involves cooperation to achieve a defined goal, bargaining, coordination and collective future planning (Wanna 2008:3). Such collaboration could occur through actors co-producing, deliberating, collaborating and interacting to choose the policy alternatives that assist to build support, legitimacy and trust, not only for policy implementation, but also for the entire policy process (Easton 1957; Reed *et al.* 2018). Legitimacy in participation is premised on the procedures of the policy process and the way policymakers ensure that policy inputs or policy outcomes of actors are considered part of the climate change policy (Karlsson-Vinkhuyzen 2016:198; Quick and Bryson 2016:4). Trust as an element of public policy governance is defined as having subjective confidence and holding positive perceptions based on actual experiences, or expectations about the actions of an individual or an organisation (Bovaird 2005:224, Organisation for Economic Co-operation and Development (OECD) 2017:15). During agenda setting, the values of policymakers, such as integrity, openness and fairness, influence the extent to which communities and policy networks find the rest of the policy stages trustworthy. Lastly, integrity relates to good and ethical behaviour of the policymaker and the way this makes the policy process inclusive for communities (OECD 2017:25).

During policy formulation, the values that should guide the role of the policymaker are similar to agenda setting, namely the facilitation of interaction and collaboration, legitimacy and trust in participation (Berkes, Berkes and Fast 2007; Brink and Wamsler 2018). Interaction and collaboration can be facilitated through a focus on the interrelationships and interdependences for resources and ideas among public administrators, policy networks and the communities when formulating policy solutions (Ross, Baldwin and Carter 2016:126; Vangen and Huxham 2010:166). Once this has been achieved, the legitimacy of this policy stage is found grounded in the availability of supportive structures and processes to enhance interaction, trustworthiness and collaboration among actors when drafting the implementation plan as part of this stage (Crosby, Bryson and Stone 2010:203; Hill 2005:21). About trust, it should be sustained by policymakers during policy formulation to enhance the trust of communities in the policy process. Trust recognises responsiveness as a dimension with a major focus on how citizens receive public services vis-à-vis government listening to the communities to give them feedback (Keping 2018:6; OECD 2017:22). To build more trust, policymakers can enhance community participation by ensuring the availability of opportunities to participate, an accessible policy process, timeous responses during communication, and the ability to involve all willing people (OECD 2017).

During policy adoption, the values that should guide the role of the policymakers comprise interaction, legitimacy, trust, and management of power dynamics. The

first value of interaction, which is common among all the policy stages, is achieved during policy adoption through participatory supportive structures, techniques, and processes to ensure that a legitimate policy is adopted. By facilitating interaction between policy networks and communities, the latter participate in the co-production of a policy which increases its legitimacy when interaction is adhered to (Reed *et al.* 2018). Agreed-upon alternative courses of action about participation and suggested contribution by communities should be established in writing during agenda setting and policy formulation to ensure legitimacy during policy adoption (Few, Brown and Tompkins 2007; Reed 2008). It is also important for policymakers to maintain the trust of communities in the policy adoption process. Power dynamics – referring to the influence of party loyalty, personal values, public interest, among other factors – should be managed to ensure that community interests determine how policymakers support the policy for adoption (Anderson 2003:129).

During policy implementation, values such as interaction, collaboration, trust and legitimacy emerge as pillars of public policy governance that seek to enhance community participation (Brink and Wamsler 2018; Mees *et al.* 2019). Concerning interaction, policymakers have to manage power dynamics, by considering who does what, when and how to ensure equal participation of all groups of actors involved (Fawcett and Daugbjerg 2012:197). A plan that guides interaction would be useful not only to manage these power dynamics but also to provide supportive structures and processes for policy implementation and its monitoring (Crosby *et al.* 2010:203). The emphasis is on community participation as a process rather than only on the relevant techniques used for participation, which lead to policy implementation (Reed 2008:2417). Consequently, collaboration requires regular communication through the usual established means of communication to build trust during the policy stage. On the other hand, clear and consistently applied legislation may reassure all actors that the process is equitable and fair (Ansell and Gash 2007:557). The success or failure of policy implementation is dependent on the extent to which collaboration between actors can be achieved (Klijn and Koppenjan 2000:141). Joint initiatives between the public administrator, politicians and policy networks could be undertaken to enhance community participation during this stage. During this stage, legitimacy is increased by policymakers who ensure that policy programmes and outcomes are linked to community preferences determined during policy formulation (Boedeltje and Cornips 2004:2).

During policy evaluation, public policy governance that aims to enhance community participation should ensure public administrators adhere to values of interaction, collaboration, legitimacy and trust as already noted in other policy stages (agenda setting, formulation, adoption, and implementation). Similarly, as indicated during agenda setting, policy formulation, policy adoption, and policy implementation increased interaction, among actors, require all-inclusive supportive structures and processes. During interaction, public administrators should

manage power dynamics to ensure equal participation of all actors. The process of collaboration during policy evaluation should obtain information about how the policy process as well as its policy stages can be improved (Ansell and Gash 2007:557). The policy evaluation process can be perceived to be legitimate depending on the extent to which the policy outcomes effectively solve social problems and the openness of the policy evaluation process to exploration of considered mutual outcomes (Boedeltje and Cornips 2004:6). Nevertheless, a lack of legitimacy during policy evaluation may have severe implications for agenda setting, thereby affecting how communities would want to participate again in future policy processes (Boedeltje and Cornips 2004:15). Trust among actors is also key during policy evaluation (Peters 2010). This can be gained through:

- direct consultation with the beneficiaries of the policy to compile policy outcomes;
- public reporting of the policy outcomes; and
- the collection and use of data representative of the whole population from the beneficiaries of the policy programme (Peters 2010).

NEW PUBLIC GOVERNANCE COMPONENTS AND THE POLICY STAGES

The purpose of the NPG theory is to contextualise the policymakers' role in community participation during climate change policy formulation and implementation. Specifically, the role of public administrators as policymakers is highlighted in this article through the lens of the components that underpin NPG, as discussed in the remainder of this section and depicted in Figure 1.

Policymakers' role during agenda setting

During agenda setting, multiple processes guide the behaviour of politicians, public administrators, experts and community representatives on how to ensure community input in the climate change agenda. Administrative and socio-political governance are the main components of NPG that influence how politicians and public administrators can facilitate community participation through the policy agenda stage. Administrative governance is defined as aiming at applying public administration effectively (Osborne 2010:7) through prioritising citizens in public service, as well as changing the mindset of public administrators through their values and behaviour and increasing administrative capacity (Robinson 2015:15) for enhancing community participation. In this article, 'administrative governance' is defined as the public administrators' capacity, values and behaviour towards prioritising community participation in policy formulation and implementation.

Socio-political governance occurs when the government relies on other societal actors, for example, the communities, for its legitimacy and impact (Osborne 2010:6). Socio-political governance entails the combination and integration of the social and political systems (Kooiman 2008:173) to enhance policy formulation and implementation.

In socio-political governance, policymakers should coordinate co-production, facilitate deliberations, and manage cross-sector collaborations during climate change agenda setting to ensure that the views of community members are considered from their perspectives (Wu *et al.* 2018:7). Socio-political governance is relevant to the policy agenda because it specifies to policymakers the direction of transition from hierarchical and market forms of governance to the use of social and political values and indigenous knowledge systems as more robust ways of involving communities during the defining of policy problems (Ahearn 2019). Through administrative governance, policymakers can instil social and political values of communities during the formal processes used to define policy problems and democratise agenda setting through the active participation of communities in setting the climate change policy agenda. It is the policymakers' role to encourage communities to participate as equal decision-makers and social learners (by sharing indigenous knowledge systems), implying co-production (Glucker, Driessen, Kolhoff and Runhaar 2013:106). The socio-political governance system concurs that actors are shaped continuously by their interactions within their social and political systems, which bring together actors to relate to each other about shared policy problems and societal challenges (Kennet 2010:24). Socio-political governance highlights the relationship among public administrators (bureaucracies), society (communities and other stakeholders) and politicians (legislature and executive) during policy formulation and implementation. It is important for policymakers to know how forms of collective action (co-production, deliberations, and cross-sector collaborations) through societal actors, such as community groups, organisations or individuals, influence policy formulation and implementation during these interactions and interrelationships (Kooiman and Jentoft 2009:834).

Policymakers' role during policy formulation

Administrative and socio-political governance are the main components that enhance contribution by the community towards selecting policy alternatives during climate change policy formulation. During this stage, public administrators apply administrative governance to facilitate community participation that contributes to developing acceptable alternatives to deal with climate change (Brink and Wamsler 2018; Mees *et al.* 2019). Among plural actors (multiple actors), public administrators would lead collaboration and coordinate the co-production of policy alternatives during this policy stage (Ahearn 2019). Notably, consideration of socio-political

governance factors is important to guide public administrators in their coordination of co-production efforts among plural actors. Enhancing community participation requires consideration of social factors among which are indigenous knowledge systems of communities during formulation of climate change policy alternatives.

Policymakers' role during policy adoption

Administrative and public policy governance are the main components which enhance community participation during decision-making for policy adoption. At this stage, administrative governance is vested in how public administrators champion collective decision-making through pluralist processes that enable communities to legitimise the adopted policy. During policy adoption, public policy and administrative governance seek to guide how climate change policy decisions should be co-produced through pluralist processes during this policy stage (Reed *et al.* 2018; Sprain 2016, Uittenbroek *et al.* 2019). This means creating opportunities for communities to have an equal say in decision-making, just like any other actor (Reed 2008; Reed *et al.* 2018; Sprain 2016; Uittenbroek *et al.* 2019). Public policy governance guides different actors on how to co-produce climate change policy decisions with communities (De Coning and Cloete 2006:40). Public policy and administrative governance mainly influence public administrators, to promote the co-production of climate change policy decisions with communities through non-linear interaction (Ewert 2020; Glucker *et al.* 2013; Reed *et al.* 2018; Uittenbroek *et al.* 2019). Public administrators are expected to enhance community participation by ensuring that –

- proposed alternatives by the communities are technically sound;
- policy alternatives address contextual climate change issues;
- they estimate the budgetary costs of implementing the policy; and
- the preferred policy alternatives by the community have political support (Wu *et al.* 2018:7).

Policymakers' role during policy implementation

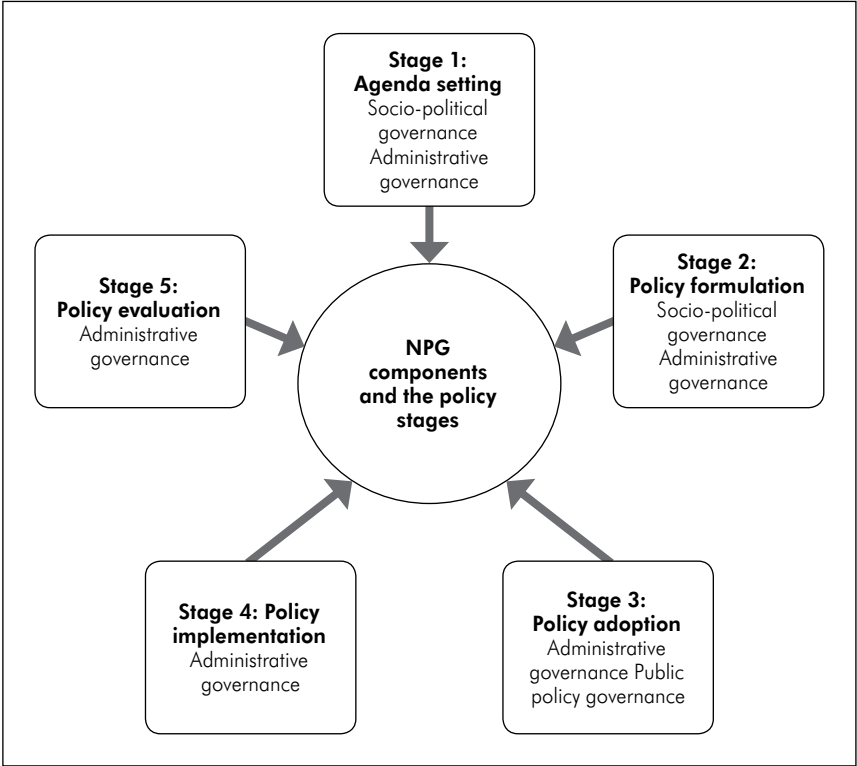
The main component of NPG that dominates climate change policy implementation is administrative governance. Through administrative governance, public administrators are guided on how to champion collective decisions among plural actors within a pluralist implementation process (Brink and Wamsler 2018; Mees *et al.* 2019). It is noteworthy that administrative governance is synonymous with community participation during policy implementation (Burton and Mustelin 2013:404). The practice of administrative governance, is characterised by broad courses of action which include plural actors and pluralist processes all aimed at one goal (Carvalho and Gupta 2012; Glucker *et al.* 2013; Reed 2008; Sprain 2016; Uittenbroek *et al.* 2019).

Public administrators bring these diverse plural actors together with communities to implement policy programmes collectively (Mees *et al.* 2019). The policymakers facilitate policy implementation because of their legal and authoritative power (Anderson 2003:197). Guiding collective decisions are also predicated on taking the lead in coordinating financial resources, knowledge exchange, skills, and legal advice for policy implementation (Carvalho and Gupta 2012).

Policymakers’ role during policy evaluation

The main component of the NPG which dominates policy evaluation is administrative governance. The relevance of administrative governance during policy evaluation relates to how communities together with other actors are part of pluralist processes to choose the evaluation criteria for the policy process but also its outcome (Fosbenner 2009; Mees *et al.* 2019). Public administrators

Figure 1: NPG components and the policy stages



Source: (Authors’ own compilation)

can either enhance or hinder such participation of communities during policy evaluation. It is for this reason that public administrators should coordinate the co-production of evaluation decisions and facilitate deliberations about climate change policy evaluation among multiple actors by way of administrative governance (Burton and Mustelin 2013:404; Leroy and Crabb 2012:153). The public administrators should facilitate how communities can participate as co-planners, co-designers, co-implementers and co-evaluators of the climate change policy evaluation decisions. Inherent to this is managing how cross-sector collaboration occurs when evaluation decisions are determined, to ensure that such collaboration benefits communities (Burton and Mustelin 2013:404; Leroy and Crabb 2012:153).

COMMUNITY PARTICIPATION IN CLIMATE CHANGE POLICY FORMULATION AND IMPLEMENTATION – LESOTHO CONTEXT

The Lesotho NCCP notes a lack of citizen participation in climate change processes, yet the country is a critical watershed due to its mountainous nature that gives rise to the most utilised headwaters that feed major river systems (Senqu or Orange River) in southern Africa (Ministry of Energy and Meteorology 2017:24). Tailored for its diverse African mountain regions, this country needs to clearly define the policymakers' roles to enhance community participation in climate change policy formulation and implementation. In 2017, Lesotho undertook the formulation of its NCCP which will be reviewed in 2027 (Ministry of Energy and Meteorology 2017:41). To date, the NCCP is the only climate change policy formulated by the Lesotho government in its response to climate governance. During the formulation of the NCCP in 2017, Lesotho's Ministry of Energy and Meteorology involved communities in a consultative format through surveys, interviews, and workshops during the agenda setting stage of the policy process (Ministry of Energy and Meteorology 2017:24). Due to limited time and financial resources, communities were not included in the four remaining policy stages of the NCCP (Chatanga and Biljohn 2022:nd). Despite the consultative nature of community participation during the NCCP, the governance processes and activities for relevant actors to enhance community participation is not explicitly outlined in this policy (Ministry of Energy and Meteorology 2017:24).

Community members from the Manonyane communities who participated in a climate change research study in 2022 said that they were not aware of climate change decision-making processes undertaken by the Ministry of Energy and Meteorology (Chatanga and Biljohn 2022:nd). They did not participate in the policy stages towards development of the 2017 NCCP (Chatanga and Biljohn

2022:nd). It is apparent that despite the role of local chiefs and the Manonyane councils as actors who can assist with coordinating community participation during policy processes, their assistance was seemingly not effectively used by the relevant Lesotho department for developing the NCCP. Instead, reasons for the lack of community participation were attributed to their limited understanding and awareness of climate change as well as lacking knowledge of climate change decision-making processes (Chatanga and Biljohn 2022:nd). The Manonyane community members who participated in this research through focus group discussions understood climate change from a lack of adherence to indigenous knowledge systems (culture, norms, indigenous knowledge and values) (Chatanga and Biljohn 2022:nd).

The significance of climate change for the Manonyane communities emanates from their location in the foothills of Lesotho. These foothills are vulnerable to climate change and other environmental problems, such as land degradation, wetland degradation, food insecurity and extreme weather events, such as droughts, floods, strong winds, and prolonged cold weather seasons (Bureau of Statistics 2010:2; Department of Environment 2014:41). These communities are affected by climate change in reduced agricultural production, frequent rain storms, floods, strong winds and sudden snowfalls (United Nations Development Programme (UNDP) 2013:8). Consequently, the long-term impact of climate change for these communities includes drought, insufficient drinking water, reduced agricultural production and damage to property and crops (UNDP 2013:8).

THE POLICYMAKERS' ROLE AND THE NEW PUBLIC GOVERNANCE COMPONENTS

From the Lesotho context which includes the Manonyane communities, it is apparent that community participation has been limited to the agenda stage at the national level and not involving communities during each policy stage (Ministry of Energy and Meteorology 2017:24). This article proposes a framework for the role of policymakers to enhance community participation during climate change policy formulation and implementation. This framework describes how community participation should be structured at all the climate change policy stages.

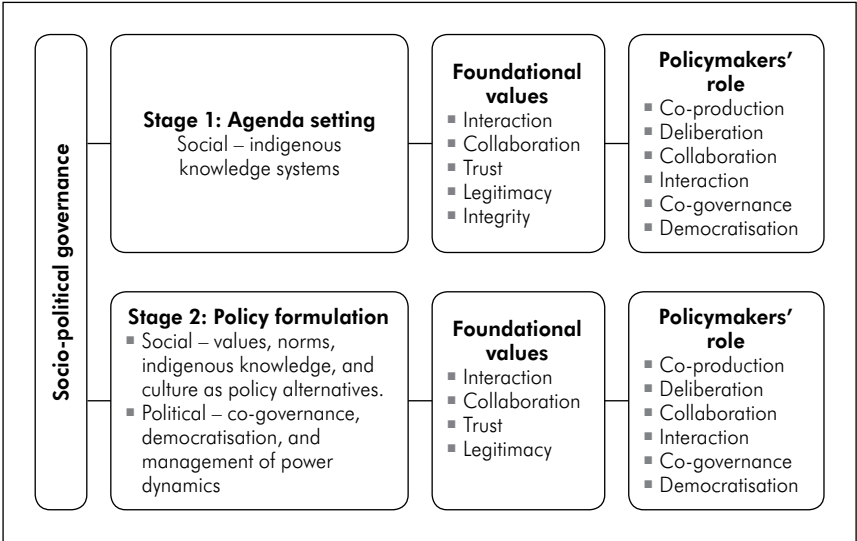
Socio-political governance

The literature shows that socio-political governance, is applicable during agenda setting and policy formulation (Ahearn 2019; Smyth 2008). During agenda setting,

policymakers are required to facilitate co-production of the climate change agenda by considering community indigenous knowledge systems in the context of socio-political governance. Co-production during agenda setting involves communities participating as co-decision-makers, co-planners, co-designers and co-evaluators of the climate change policy agenda. From this, it can be deduced that the consideration of indigenous knowledge systems as social factors during problem definition may influence how communities understand the process as they would have to relate to what they do and what they believe. Agenda setting should specify the targeted community, the community's location, attitude, norms, values, cultural beliefs and indigenous knowledge in terms of the problem to be defined. This implies that policymakers should include the communities when defining climate change policy problems, using community social factors to influence their participation during the agenda setting stage (Katsamunskia 2016). Foundational values of the policymakers' role during agenda setting, comprises facilitating interaction and collaboration, legitimacy and trust in participation, and lending integrity to the process.

During policy formulation, the application of socio-political governance requires that co-production of mitigation and adaptation of policy alternatives consider societal factors (values, norms, indigenous knowledge, and culture) and political factors (co-governance, democratisation and management of power dynamics) (Collins and Ison 2009; Kooiman 1999; Newig, Schulz and Jager 2016; Reed *et al.* 2018:1). This

Figure 2: Socio-political governance and the policymakers' role



Source: (Authors' own compilation)

means that the duties of policymakers would be to stimulate community initiatives continuously as would have been proposed during climate change agenda setting through facilitating collaboration and deliberation of the policy alternatives suggested by communities, while coordinating cross-sector collaborations (Friis-Hansen 2017). Similarities in how policymakers are expected to apply socio-political governance in both stages of agenda setting and policy formulation, relate to the policymakers' role being premised on co-production, deliberation, collaboration and democratisation. Contrary to these similarities, one difference in the policymakers' role during agenda setting is its focus on the facilitation of co-production, deliberation, collaboration, interaction, co-governance and democratisation when defining climate change problems, while policy formulation focuses on the same, but for policy alternatives. It could be inferred that socio-political governance is critical to guide policymakers when facilitating co-production, collaboration, deliberation and democratisation during the climate change agenda setting and policy formulation. In fact, interaction and co-governance are embedded in co-production, collaboration, deliberation and democratisation. To conclude, during policy formulation, foundational values for the policymakers comprise facilitating interaction and collaboration, legitimacy and trust in participation. Against this background, Figure 2 outlines the policymakers' role as it is embedded in socio-political governance and foundational values.

Administrative governance

The literature suggests that, unlike socio-political and public policy governance, administrative governance is applicable throughout the climate change policy process stages. During agenda setting, policymakers are required to facilitate co-production, deliberation, and building trust that will lead to the desired results. Further, policymakers should manage conflicts and power dynamics not only for communities, but also for other stakeholders, and promote interaction among actors to develop shared understanding and commitment at all levels (community grassroots, executive, and administrative) (Brink and Wamsler 2018; Mees *et al.* 2019; Reed *et al.* 2018:5–6). In the context of administrative governance, this means that a single group of actors does not dominate the climate change agenda but rather works with other groups of actors during agenda setting (Ahearn 2019; Chanza and De Wit 2016). Policymakers are required to ensure that every actor has a role to play to influence the climate change agenda, which is community-based (Sprain 2016:63).

During policy formulation, administrative governance requires policymakers to facilitate interrelationships, collaborations, deliberations, and interdependence for resources among actors to benefit community participation (Mees *et al.* 2019; Woodruff and Stults 2016). This means that administrative

governance during policy formulation guides policymakers on how to facilitate communities to develop acceptable alternatives to deal with climate change (Brink and Wamsler 2018; De Vries 2007:146; Etchart 2017; Mees *et al.* 2019). Administrative governance requires policymakers to enhance community participation by making the climate change policy formulation process open to the communities (Anderson 2003:474; Carvalho and Gupta 2012; Sprain 2016). This means that administrative governance during policy formulation guides policymakers on how to facilitate communities to develop acceptable alternatives to deal with climate change (Brink and Wamsler 2018; De Vries 2007:146; Etchart 2017; Mees *et al.* 2019).

In policy adoption, administrative governance requires policymakers to interact frequently with communities, and to use both bottom-up and top-down deliberation approaches and equal distribution of decision-making power, and to promote a high level of community trust in the adoption process. Policymakers ensure that communities participate as authors and subjects of the climate change policy (Samaddar *et al.* 2019). Interactions referred to here are power relations among actors (i.e. politicians, public administrators, experts, community representatives and communities) in decision-making (Leroy and Crabb 2012:13). Collaborations during climate change policy adoption are conducted by policymakers to act on preferred community policy alternatives. This is done to communicate to communities the rejected policy alternatives, and to communicate to communities the modifications being made on the preferred policy alternatives. The equal distribution of decision-making power during policy adoption refers to a scenario where the contribution by the community significantly influences the adopted climate change policy (Karlsson-Vinkhuyzen 2016:201; Karlsson-Vinkhuyzen and McGee 2013:70; Quick and Bryson 2016:5). This implies that policymakers should represent policy proposals by the community fairly (adhering to such policy proposals) and reliably (sticking to principles of co-production, public policy governance, and objectives of community participation) during the technical process (formulation of legislation) of climate change policy adoption (Reed *et al.* 2018).

In policy implementation, administrative governance requires policymakers to facilitate collaboration, interaction and co-production (Diver 2017; Teder and Kaimre 2018). Policymakers facilitate the collaboration process, manage the cross-sector collaboration process and interaction, and coordinate climate change policy implementation programmes (Mattijssen, Buijs, Elands and Arts 2017; Mees *et al.* 2019:203). Public administrators also manage relevant government ministries and departments as well as manage power dynamics during interactions when actors are interdependent in terms of sharing resources with communities (Friis-Hansen 2017; Funder, Mweemba and Nyambe 2017). Communities participate as co-planners, co-designers, co-implementers and co-evaluators of

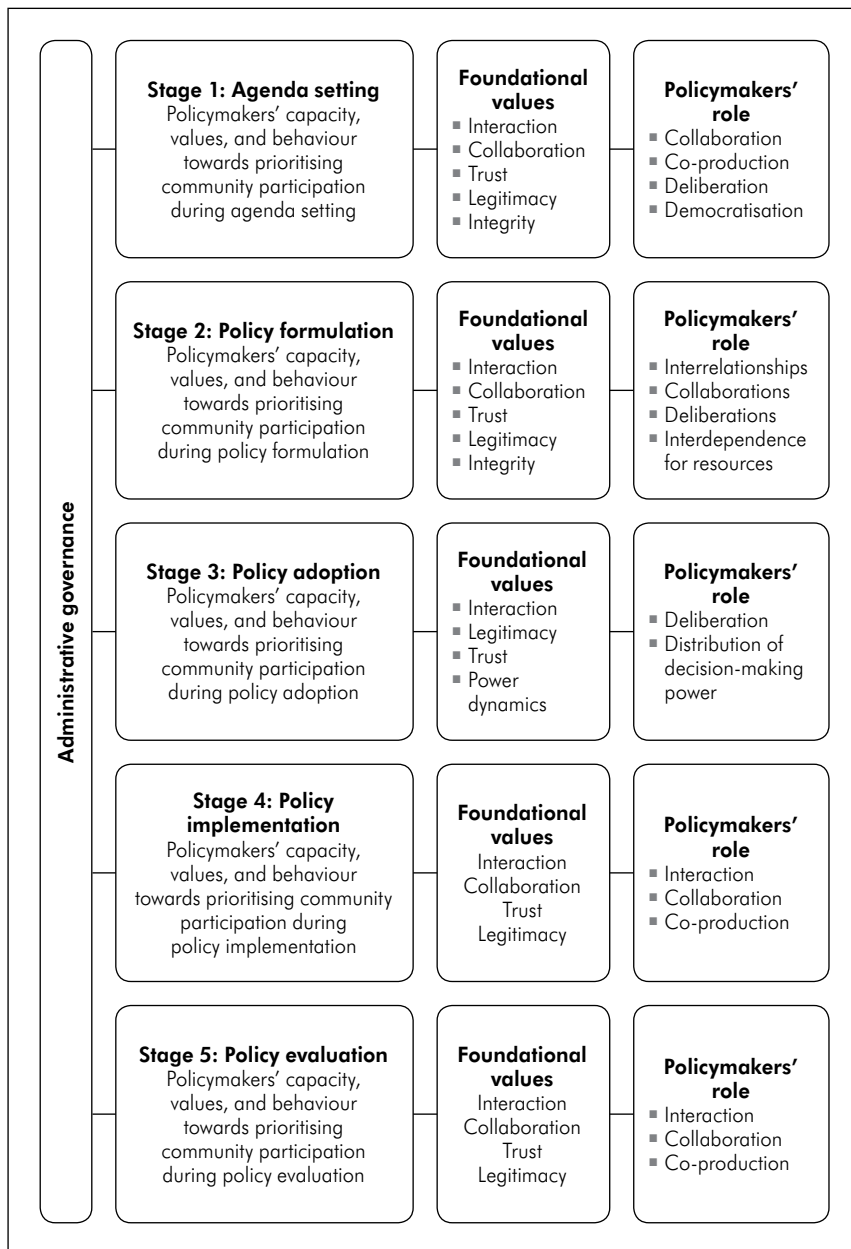
the implementation programmes. This means that administrative governance guides policymakers on how to champion collective decisions among plural actors within a pluralist policy implementation process (Brink and Wamsler 2018; Mees *et al.* 2019).

In policy evaluation, administrative governance requires policymakers to manage power dynamics and to facilitate co-production of the policy evaluation decisions that relate to interpretation and release of the evaluation results (Stufflebeam 2001:62). When choosing the evaluation criteria, issues of power dynamics should be managed to encourage communities to participate as co-producers of the climate change policy evaluation criteria (Samaddar *et al.* 2015:125). Communities participate as part of the deliberation process to co-plan, co-produce, co-implement and co-evaluate the climate change evaluation criteria (Crabb and Leroy 2008:154; King 2013; Schoenefeld and Jordan 2017). This implies that policymakers should put the interests of the communities first when defining the evaluation criteria for climate change policy (Samaddar *et al.* 2015). During policy evaluation, public administrators adhere to values of interaction, collaboration, legitimacy and trust as noted in other policy stages (agenda setting, formulation, adoption and implementation) while politicians oversee the policy evaluation process.

Against this background, the role of policymakers in applying administrative governance shares similarities during all the policy stages. All policy stages capture the role of policymakers in facilitation of collaboration, co-production, interaction and deliberation, as was evident from the literature review. It could be inferred that, whenever administrative governance is applied, policymakers are required to build support for policy implementation at every policy stage by facilitating co-production, deliberation, collaboration, cross-sector collaboration and democratisation. All the other values, such as trust, legitimacy, management of power dynamics and interaction, are embedded in co-production, deliberation, collaboration, cross-sector collaboration and democratisation.

Concerning differences, the role of policymakers in applying administrative governance is different to what the literature proposes in stages 1, 2, 3, 4 and 5. Stage 3, for instance, captures the policymakers' role as being embedded in values of facilitation of deliberation, sharing decision-making power, and building of trust. Literature proposes the policymakers' role as different in stage 1, namely facilitation of co-production, deliberation, building of trust, managing conflicts and power dynamics, and interaction. It is deduced that similarities during policy stages 1, 2, 3, 4 and 5 are grounded in four main roles, namely collaboration, co-production, deliberation and democratisation. It could be inferred that the role of policymakers in administrative governance should be embedded in the climate change policy process in stages 1, 2, 3, 4 and 5 as outlined in Figure 3.

Figure 3: Administrative governance and the policymakers' role



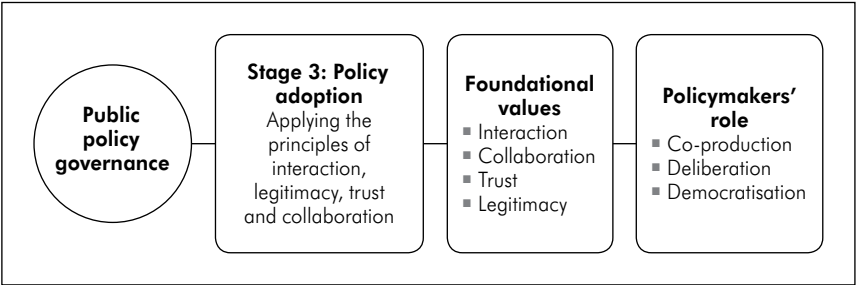
Source: (Authors' own compilation)

Public policy governance

Public policy governance is dominant during the climate change policy adoption stage. During policy adoption, policymakers are required to facilitate interaction, collaborations, equal distribution of decision-making power and a high level of trust in the adoption process. Public policy governance is relevant to policy adoption because it indicates how policymakers, as the main actors, should provide a framework (legislation and institutions) through which community participation could be facilitated in a democratic way to ultimately come up with a legitimate policy (Carvalho and Gupta 2012:146, De Vries, 2007). Policymakers appear to be the main actors, whose role in enhancing community participation serves to ensure empowerment of marginalised communities through an open system of climate change policy adoption processes (Mees and Driessen 2018:198; Sprain 2016; Uittenbroek *et al.* 2019). Public policy governance gives the formal final authority to policymakers to adopt a policy (Anderson 2003:119). This means that the contribution by the community would be through interactions, collaborations and equal distribution of power to consider the adopted policy legitimate (Brink and Wamsler 2018; Carvalho and Gupta 2012:1; Friis-Hansen 2017:7). For this, trust is required and communities should, at least, perceive policymakers’ behaviour during interactions as open, reliable and fair (Carvalho and Gupta 2012:1; Friis-Hansen 2017:7). Against this background, the role of policymakers in applying public policy governance shares similarities with agenda setting, policy formulation, policy implementation and policy evaluation. The policy adoption stage captures the role of policymakers in values of interaction, collaboration, legitimacy and trust, which underpin the policymakers’ role of deliberation, collaboration and democratisation (Figure 4).

Concerning differences, the role of policymakers in applying public policy governance is different from what the literature proposes in stages 1, 2, 4 and 5. Accordingly, stage 3 captures the policymakers’ role as mainly facilitating

Figure 4: Public policy governance and the policymakers’ role



Source: (Authors’ own compilation)

interactions and collaborations during decision-making with the main aim to legitimise the policy and share power equally with communities (collaboration, deliberation and democratisation). Likewise, the literature proposes the policy-makers' role in stages 1, 2, 4 and 5, which points to co-production, collaboration, deliberation and democratisation.

CONCLUSION

This article discussed the NPG theory components as a framework for Lesotho policymakers to enhance community participation during climate change policy formulation and implementation. The research question posed was: How could the components of the NPG theory serve as a framework for Lesotho policymakers to enhance community participation during climate change policy formulation and implementation? This aim was achieved, and the research question answered by discussing how the NPG components apply to the respective policy stages. Further foundational values that should underpin the role of policymakers during the respective policy stages were discussed. Notably, using the theoretical lens of the NPG components, a framework was proposed for the policymakers' role in enhancing community participation during climate change policy formulation and implementation.

This article argued that the NPG components of socio-political, administrative and public policy governance are critical for policymakers to enhance community participation during the climate change policy process. The policymakers' role was delineated considering these governance aspects across the policy stages. Inherent to this are values varying across the policy stages, such as interaction, collaboration, trust, legitimacy and integrity, which should guide policymakers' engagement with communities. These values provide the foundation upon which policymakers and communities should work together during each policy stage.

The practical implications of this article are threefold. First, in Lesotho, the local chiefs and councils are the actors to access the community voice and to enhance community participation in climate change policy processes. Policymakers as the main actors should ensure the empowerment of local chiefs and councils through an open system of climate change process to enhance community participation. Second, in other regions or internationally, agricultural associations, third-sector organisations and advocacy groups involved in climate change adaptation and mitigation could serve as access points to facilitate and access the community's voice in climate change policy processes. Policymakers have a foundation upon which they plan the nature of collaboration, interaction, co-production, deliberation, democratisation, legitimacy and trust to enhance community participation in climate change policy processes. Third, the framework guides policymakers

on how to obtain community input into the climate change policy process in a plural and pluralist environment. This gives communities an opportunity to plan for their community and to provide solutions to their climate change problems. The article also draws attention to the extent to which the NPG theory and its components can be applied to other policy contexts; for example, social welfare and environmental management, where the influence of community participation is required. To conclude, further empirical research about the application of the NPG components during policy processes could be of benefit, especially in policy areas where the voice and influence of the community are required.

REFERENCES

- Ahearn, A. 2019. Surviving extreme weather: Mongolian indigenous knowledge, local institutions and Governance innovations for adaptation. In Ahearn, A., Oelz, M. and Dhir, R.K. (Eds.). 2019. *Indigenous Peoples and Climate Change Emerging Research on Traditional Knowledge and Livelihoods*. Geneva: International Labour Organization. Available at: https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=Surviving+extreme+weather%3A+ongolian+indigenous+knowledge%2C+local+institutions+and+Governance+innovations+for+adaption&btnG=. (Accessed on 20 July 2020).
- Anderson, E.J. 2003. *Public Policy Making: An Introduction*. Boston: Houghton Mifflin Company.
- Ansell, C. and Gash, A. 2007. Collaborative governance in theory and practice. *Journal of Public Administration Research and Theory*.18(4):543–571.
- Berkes, F., Berkes, M.K. and Fast, H. 2007. Collaborative integrated management in Canada's north: The role of local and traditional knowledge and community-based monitoring. *Coastal management*. 35(1):143–162.
- Boedeltje, M.M. and Cornips, J.J. 2004. *Input and output legitimacy in interactive governance*. Netherlands Institute for Government. Annual Work Conference 2004 Rotterdam. Available at: <https://repub.eur.nl/pub/1750/NIG2-01.pdf>. (Accessed on 20 July 2020).
- Bovaird, T. 2005. Public governance: Balancing stakeholder power in a network society. *International Review of Administrative Sciences*. 71(2):217–228.
- Braun, V. and Clarke, V. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology*. 3(2):77–101.
- Brink, E. and Wamsler, C. 2018. Collaborative governance for climate change adaptation: Mapping citizen–municipality interactions. *Environmental Policy and Governance*. 28(2):82–97.
- Bureau of Statistics. 2010. Statistical Yearbook 2010. Maseru: The KOL Minsitry of Finance and Development Planning Bureau of Statitics. Available at: <http://www.bos.gov.ls>. (Accessed on 4 January 2017).
- Bureau of Statistics. 2018. *2016 Population and Housing Census Analytic Report: Population Dynamics, Vol. III A*. Maseru: Government of Lesotho. Available at: <http://www.bos.gov.ls>. (Accessed on 25 April 2020).

- Burton, I., Malone, E. and Huq, S. 2004. *Adaptation Policy Frameworks for Climate Change: Developing Strategies, Policies and Measures*. Cambridge: The Press Syndicate of the University of Cambridge. Available at: <https://adapt.epa.gov.tw/disPageBox/files/751.pdf>. (Accessed on 22 July 2020).
- Burton, P. and Mustelin, J. 2013. Planning for climate change: Is greater public participation the key to success? *Urban Policy and Research*. 31(4):399–415.
- Carvalho, A. and Gupta, J. 2012. State commitment to promoting public participation: The UNFCCC and citizens' roles in climate change politics. In Carvalho, A. and Peterson, T.R. (Eds.). 2012. *Climate Change Politics: Communication and Public Engagement*. Amherst: Cambria Press.
- Chanza, N. and De Wit, A. 2016. Enhancing climate governance through indigenous knowledge: Case in sustainability science. *South African Journal of Science*. 112(3/4):1–7.
- Chatanga, R. and Biljohn, M. 2022. *Public Administrators and Administrative Governance Facilitating Community Participation in Climate Change Policy Formulation and Implementation in Lesotho*. ASSADPAM Conference, Durban South Africa. 23–25 November 2022.
- Collins, K. and Ison, R. 2009. Jumping off Arnstein's ladder: Social learning as a new policy paradigm for climate change adaptation. *Environmental Policy and Governance*. 19:358–373.
- Crabb, A. and Leroy, P. 2008. *The handbook of environmental policy evaluation*. London: Earthscan.
- Crosby, C.B., Bryson, M.J. and Stone, M.M. 2010. Leading Across Frontiers: How visionary leaders integrate people, processes, structures and resources. In Osborne, P.S. (Eds.). 2010. *The New Public Governance? Emerging Perspectives on Theories and Practice of Public Governance*. London: Routledge Taylor and Francis Group.
- De Coning, C. and Cloete, F. 2006. Theories and models for analysing public policy. In Cloete F., Wissink, H. and De Coning, C. (Eds.). 2006. *Improving Public Policy from Theory to Practice*. 2nd edition. Pretoria: Van Schaik.
- Department of Environment. 2014. *Lesotho Environment Outlook Report: Environment for National Prosperity*. Maseru: Ministry of Tourism, Environment and Culture, Government of Lesotho.
- De Vries, M.S. 2007. *Public Participation in Policy Processes: Towards a Research Agenda*. *Administratie si Management Public*. 8:144–153.
- Diver, S. 2017. Negotiating indigenous knowledge at the science-policy interface: Insights from the Xáxl'i'p Community Forest. *Environmental Science & Policy*. 731–11.
- Easton, D. 1957. An approach to the analysis of political systems. *World Politics*. 9(3):383–400.
- Etchart, L. 2017. The role of indigenous peoples in combating climate change. *Palgrave Communications*. 3(1):1–4.
- Ewert, B. 2020. Moving beyond the obsession with nudging individual behaviour: Towards a broader understanding of behavioural public policy. *Public Policy and Administration*. 35(3):337–360.
- Fawcett, P. and Daugbjerg, C. 2012. Explaining governance outcomes: Epistemology, network governance and policy network analysis. *Political Studies Review*. 10(2):195–207.
- Few, R., Brown, K. and Tompkins, E.L. 2007. Public participation and climate change adaptation: Avoiding the illusion of inclusion. *Climate Policy*. 746–59.
- Fosbenner, L. 2009. Public Participation in Municipal Climate Action Planning: Exploring

- Deliberative Methods. Available at: <https://www.sustainlv.org/one/wpcontent/uploads/2019/08/Comm-Engage-for-CAP-LF.pdf>. (Accessed on 5 August 2020).
- Friis-Hansen, E. 2017. Implementing African national climate change policies. In Friis-Hansen, E. (Eds.). 2017. *Decentralized Governance of Adaptation to Climate Change in Africa*. Oxfordshire: CABI: United Kingdom. Available at: https://books.google.cz/books?hl=sv&lr=&id=TxwxDwAAQBAJ&oi=fnd&pg=PA11&dq=Implementing+African+national+climate+change+policies&ots=qCNXEukNvA&sig=puseVwhxEh7VyWsvtZUI-mwYlbM&redir_esc=y#v=onepage&q=Implementing%20African%20national%20climate%20change%20. (Accessed on 16 May 2020).
- Funder, M., Mweemba, C. and Nyambe, I. 2017. Climate change adaptation and decentralisation politics: the case of local governments in rural Zambia. In Friis-Hansen, E. (Eds.). 2017. *Decentralized Governance of Adaptation to Climate Change in Africa*. Oxfordshire: CABI: United Kingdom. Available at: https://books.google.cz/books?hl=sv&lr=&id=TxwxDwAAQBAJ&oi=fnd&pg=PA11&dq=Implementing+African+national+climate+change+policies&ots=qCNXEukNvA&sig=puseVwhxEh7VyWsvtZUI-mwYlbM&redir_esc=y#v=onepage&q=Implementing%20African%20national%20climate%20change%20policies&f=false. (Accessed on 16 May 2020).
- Glucker, N.A., Driessen, P.J.P., Kolhoff, A. and Runhaar, A.C.H. 2013. Public participation in environmental impact assessment: Why, who and how? *Environmental Impact Assessment Review*. 43:104–111.
- Hegger, D.L.T., Mees, H.L.P., Driessen, P.P.J. and Runhaar, H.A.C. 2017. The roles of residents in climate adaptation: A systematic review in the case of the Netherlands. *Environmental Policy and Governance*. Available at: <https://onlinelibrary.wiley.com/doi/abs/10.1002/eet.1766>. (Accessed on 26 August 2020).
- Hill, M. 2005. *The Public Policy Process*. England: Pearson Longman.
- Karlsson-Vinkhuyzen, S.I. 2016. Legitimacy. In Ansel, C. and Torfing, J. (Eds.). 2016. *Handbook on Theories of Governance*. Cheltenham: Edward Elgar.
- Karlsson-Vinkhuyzen, S.I. and McGee, J. 2013. Legitimacy in an era of fragmentation: The case of global climate governance. *Global Environmental Politics*. 13(3):56–78.
- Katsamunskaa, P. 2016. The concept of governance and public governance theories. *Economic Alternatives*. (2):133–141.
- Kennet, P. 2010. Governance and globalisation. In Osborne, S. P. (Eds.). 2010. *The New Public Governance? Emerging Perspectives on the Theory and Practice of Public Governance*. London: Routledge Taylor & Francis Group.
- Keping, Y. 2018. Governance and good governance: A new framework for political analysis. *Journal of the Humanities Social Science*. 11(1):1–8.
- King, J.A. 2013. Getting people involved: The origin of interactive evaluation. In Alkin, C.M. (Eds.). 2013. *Evaluation Root*. London: Sage.
- Klijn, E.H. and Koppenjan, J.F.M. 2000. Public Management and Policy Networks. *Public Management: An International Journal of Research and Theory*. 2(2):135–158.
- Kooiman, J. 1999. Social-Political Governance. *Public Management: An International Journal of Research and Theory*. 1(1):67–92.

- Kooiman, J. 2008. Exploring the concept of governability. *Journal of Comparative Policy Analysis*. 10(2):171–190.
- Kooiman, J. and Jentoft, S. 2009. Meta-Governance: Values, Norms and Principles, and the Making of Hard Choices. *Public Administration*. 87(4):818–836.
- Leroy, P. and Crabb, A. 2012. *The Handbook of Environmental Policy Evaluation*. London: Earthscan.
- Machepha, M.M. 2010. *Parliamentary Role and Its Relationship With Its Relevant Institutions in Effectively Addressing Climate Change Issues-Lesotho: A study Commissioned and Supported by the International Institute for Environment and Development (IIED) in Partnership with European Parliamentarians for Africa (AWEPA)*. Available at: <https://policycommons.net/artifacts/2018712/parliamentaryrole-and-its-relationship-with-its-relevant-institutions-in-effectively-addressing-climate-changeissues-lesotho/2771155/>. (Accessed on 15 July 2019).
- Mattijssen, T., Buijs, A., Elands, B. and Arts, B. 2017. The ‘green’and ‘self’ in green self-governance: a study of 264 green space initiatives by citizens. *Journal of Environmental Policy & Planning*. 20(1):96–113.
- Mburia, R. 2015. *Africa Climate Change Policy: An Adaptation and Development Challenge in a Dangerous World*. Nairobi: Climate Emergency Institute.
- McNamara, K.E., Clissold, R., Westoby, R., Piggott-McKellar, A.E., Kumar, R., Clarke, T., Namoumou, F., Areki, F., Joseph, E., Warrick, O. and Nunn, P.D. 2020. An assessment of community based adaptation initiatives in the Pacific Islands. *Nature Climate Change*. Available at: <https://www.nature.com/articles/s41558-020-0813-1>. (Accessed on 22 December 2020).
- Mees, H. and Driessen, P. 2018. A framework for assessing the accountability of local governance arrangements for adaptation to climate change. *Journal of Environmental Planning and Management*. 62(4):1–21.
- Mees, H.L.P., Uittenbroek, C.J., Hegger, D.L. and Driessen, P.P.J. 2019. From citizen participation to government participation: An exploration of the roles of local governments in community initiatives for climate change adaptation in the Netherlands. *Environmental Policy and Governance*. 29(3):198–208.
- Ministry of Energy and Meteorology, 2017. *The Kingdom of Lesotho National Climate Change Policy 2017–2027*. Maseru: Government of Lesotho. Available at: <https://www.bing.com/search?q=The+Kingdom+of+Lesotho+National+Climate+Change+Policy&cvid=51fe0c366a8b49148376fcbb42caf529&aqs=edge..69i57j69i60.5097j0j1&pglt=43&FORM=ANNTA1&PC=U531>. (Accessed on 19 November 2017).
- Newig, J., Schulz, D. and Jager, N.W. 2016. Disentangling puzzles of spatial scales and participation in environmental governance – The case of governance re-scaling through the European water framework directive. *Environmental Management*. 58(6):998–1014.
- Organisation for Economic Co-operation and Development. 2017. *Trust and Public Policy: How Better Governance Can Help Rebuild Public Trust, OECD Public Governance Reviews*. Paris: OECD Publishing.
- Osborne, P.S. 2010. The new public governance: A suitable case for treatment? In Osborne, S.P. (Eds.). 2010. *The New Public Governance? Emerging Perspectives on the Theory and Practice of Public Governance*. London: Routledge Taylor & Francis.

- Peters, B.G. 2010. Meta-governance and public management. In Osborne, S.P. (Eds.). 2010. *The New Public Governance? Emerging Perspectives on the Theory and Practice of Public Governance*. London: Routledge Taylor & Francis Group.
- Quick, S.K. and Bryson, M.J. 2016. Public participation. In Torbing, J. and Ansell, C. (Eds.). 2016. *Handbook in Theories of Governance*. Cheltenham: Edward Elgar.
- Reed, M.S. 2008. Stakeholder participation for environmental management: A literature review. *Biological conservation*. 141(10):2417–2431.
- Reed, M.S., Vella, S., Challies, E., De Vente, J., Frewer, L., Hohenwallner-Ries, D., Huber, T., Neumann, R.K., Oughton, E.A. and Sidoli del Ceno, J. 2018. A theory of participation: What makes stakeholder and public engagement in environmental management work? *Restoration Ecology*. 26:1–18.
- Robinson, M. 2015. *From Old Public Administration to the New Public Service: Implications for Public Sector Reform in Developing Countries*. Heng Mui Keng Terrace: UNDP Global Centre for Public Service Excellence. Available at: <https://www.undp.org/publications/old-publicadministration-new-public-service>. (Accessed on 4 April 2018).
- Ross, H., Baldwin, C. and Carter, B.R.W. 2016. Subtle implications: Public participation versus community engagement in environmental decision-making. *Australasian Journal of Environmental Management*. 23(2):123–129.
- Runhaar, H. 2009. Putting SEA in context: A discourse perspective on how SEA contributes to decision-making. *Environmental Impact Assessment Review*. 29(3):200–209.
- Samaddar, S., Ayaribilla, A.J., Oteng-Ababio, M., Dayour, F. and Yokomatsu, M. 2019. Stakeholders' perceptions on effective community participation in climate change adaptation. In Sarkar, A., Sensar, S.R. and VanLoon, G.W. (Eds.). 2019. *Sustainable Solutions for Food Security*. Cham: Springer.
- Samaddar, S., Yokomatsu, M., Dayour, F., Oteng-Ababio, M., Dzivenu, T., Adams, M. and Ishikawa, H. 2015. Evaluating effective public participation in disaster management and climate change adaptation: Insights from Northern Ghana through a user-based approach. *Risk, Hazards & Crisis in Public Policy*. 6(1):117–143.
- Schoenefeld, J. and Jordan, A. 2017. Governing policy evaluation? Towards a new typology. *Evaluation*. 23(3):274–293.
- Smyth, P. 2008. Collaborative governance: The community sector and collaborative network governance. In O'Flynn, J. and Wanna, J. (Eds.). 2008. *Collaborative Governance: A New Era of Public Policy in Australia?* Canberra: ANU.
- Sprain, L. 2016. Paradoxes of public participation in climate change governance. *The Good Society*. 25(1):62–80.
- Stufflebeam, D. 2001. Evaluation models. *New Directions for Evaluation*. 2001(89):7–98.
- Teder, M. and Kaimre, P. 2018. The participation of stakeholders in the policy processes and their satisfaction with results: A case of Estonian forestry policy. *Forest Policy and Economics*. 89:54–62.
- Uittenbroek, C.J., Mees, H.L., Hegger, D.L. and Driessen, P.P. 2019. The design of public

- participation: Who participates, when and how? Insights in climate adaptation planning from the Netherlands. *Journal of Environmental Planning and Management*. 62(14):2529–2547.
- United Nations. 2015. *Transforming our World: the 2030 Agenda Sustainable Development*. New York: United Nations Department of Economic and Social Affairs. Available at: <https://sdgs.un.org/2030agenda>. (Accessed on 4 February 2018).
- United Nations Development Programme. 2013. Reducing Vulnerability from Climate Change in the Foothills, Lowlands and the Lower Senqu River Basin Maseru, Lesotho. Available at: <https://www.thegef.org/project/reducing-vulnerabilityclimate-change-foothills-lowlands-and-lower-senqu-river-basin>. (Accessed on 2 February 2018).
- Vangen, S. and Huxham, C. 2010. Introducing the theory of collaborative advantage. In Osborne, P.S. (Eds.). 2010. *The New Public Governance? Emerging Theories and Practise of Public Governance*. London: Routledge Taylor & Francis Group.
- Wanna, J. 2008. Collaborative Government: Meanings, Dimensions, Drivers and Outcomes. In O'Flynn, J. and Wanna, J. (Eds.). 2008. *Collaborative Governance: A new Era of Public Policy in Australia?* Canberra: ANU E Press.
- Woodruff, S.C. and Stults, M. 2016. Numerous strategies but limited implementation guidance in US local adaptation plans. *Nature Climate Change*. 6(8):796–802.
- Worker, J. and Northrop, E. 2018. Assessing and Addressing Climate Governance Challenges In Low- and Middle-Income Countries: An Operational Guide. Paris: AFD. Available at: <https://www.afd.fr/sites/afd/files/2018-01-03-39-38/39-notes-techniques.pdf>. (Accessed on 4 February 2018).
- Wright, H., Vermeulen, S., Laganda, G., Olupot, M., Ampaire, E. and Jat, M.L. 2014. Farmers, food and climate change: Ensuring community-based adaptation is mainstreamed into agricultural programmes. *Climate and Development*. 6(4):318–328.
- Wu, X., Ramesh, M. and Howlett, M. 2018. Policy capacity: Conceptual framework and essential components. In Wu, X., Howlett, M. and Ramesh, M. (Eds.). 2018. *Policy Capacity and Governance: Assessing Governmental Competences and Capabilities in Theory and Practice*. Cham: Palgrave Macmillan.

AUTHORS' CONTACT DETAILS

Mrs Redemption Chatanga

Department of Public Administration and Management
Faculty of Economic and Management Sciences
University of the Free State
P O Box 339, Bloemfontein 9300
Tel: 051-401 7185
Email: redemutoozi@gmail.com

Dr Maréve Biljohn

Department of Public Administration and Management
Faculty of Economic and Management Sciences
University of the Free State
P O Box 339, Bloemfontein 9300
Tel: 051-401 7185
Email: BiljohnMIM@ufs.ac.za

The Quest for a New Local Government Model in South Africa

A Visionary Perspective

J Breed

Beyond Local Government
Mogale City

C Jones

Faculty of Theology
Stellenbosch University

P Pillay

School of Public Leadership and Faculty of Economic and
Management Sciences Stellenbosch University

S Zondi

School of Management, IT and Governance
University of KwaZulu-Natal

ABSTRACT

Enormous efforts, capital and energy are spent in local institutionalism in South Africa to impact social change, socio-economic growth, and community development towards a better life for citizens. The expected rate of return on these endless attempts does not produce the desired outcomes for the development of the country. Local government produces too little, at too high a price. This institutional mechanism's unlocked potential is capable of a game-changing impact. However, in many communities, local government does not realise the expressions of frustration that have been escalating over years, with enormous negative consequences and damage. Drastic trajectory changes have become evident. For local governments, the potential to be found in the immense vaults of human capital to create and invent opportunities, should be unlocked. The model's framework that is envisioned in this article should mediate and discuss a completely new hybrid business model for the development of exceptional technocratic and institutional behaviour, thought and performance modification.

Unfortunately, local government's credo to create, produce, innovate, and fulfil opportunities is often strangled by obsolescence, narrow mindedness, and traditional conventionalism. A new model should not focus on repair – it should rather envision and trace a completely new strategic and transformative local governmental model aimed at unlocking all possible assets available to and latent in local societal silos. Although local institutionalism's nemesis is generally regarded as service delivery, local government should be much more than this pursuit. It also needs to act as a co-creator of wealth. All these objectives, among others, should be characteristics of a new envisioned local government model.

INTRODUCTION

The focus of this introduction is mainly on the context of local governments in South Africa to which local communities have been exposed to since the attainment of democracy in 1994. The scholarly discourse on local governance in the developing countries is inundated with case studies of ineffective administrations and failing systems in local government. Hence, a new local government model should have new structures, norms and values and be guided by strong political leadership which is able to make policy decisions. Local government produces too little, at too high a cost, and at too slow a pace. Bureaucratic institutionalism has frustrated visionaries / idealists / functionaries for almost three decades of South African democracy. The time for change has long passed. Most of the volumes of advice and improvement strategies that have been introduced in local government, including, Operation Clean Audit, Turn Around Strategies and the Back to Basic Initiative have achieved little or no notable significant socio-economic improvements. The new envisioned model should provide an opportunity for the total transformation of the local government sphere in South Africa. If many of those experts' prescriptions to address institutional inadequacies and social

Table 1: Summary of findings of the State of Local Government Report, 2021

Dysfunctional (red)	Medium risk (orange)	Low risk (yellow)	Stable (green)
64	111	66	16
24.90%	43.19%	25.68%	5.46%

Source: (CoGTA State of Local Government Report 2021:6–7)

dysfunctionality were so well informed, why then is South African institutionalism still in such doldrums despite all their efforts and financial investments? The COGTA Portfolio Committee’s “State of Local Government Report” (2021:6–7), “confirms the correlation between failures in governance and political oversight as the primary causes underpinning the increase in the number of dysfunctional municipalities”.

Unbundled this “barometer” quantifies the status of all municipalities as follows:

Table 2: Status of all municipalities

Province	Number of municipalities	Dys-functional (red)	Medium risk (orange)	Low risk (yellow)	Stable (green)
Eastern Cape	39	11	14	14	0
Free State	23	11	11	1	0
Gauteng	11	2	7	1	1
KwaZulu-Natal	54	11	20	22	1
Limpopo	27	3	21	3	0
Mpumulanga	20	6	9	4	1
North West	22	10	7	5	0
Northern Cape	31	9	16	5	1
Western Cape	30	1	7	10	12
Total	257	64	111	66	16

Source: (CoGTA State of Local Government Report 2021:6–7)

This necessitates fresh thinking to collaborate with multiple partners to solve complex issues and to generate various forms of capital – intellectual, human, financial, social and physical, to the benefit of society (Hassan 2015:n.p.).

Against the backdrop of innovation in service delivery, a brief discussion on New Public Management (NPM) is important in the context of this article. NPM was coined by Christopher Hood in 1991 and emphasised decentralisation, devolution, and modernisation of public services. NPM reforms had various objectives, political as well as technical and challenged traditional public organisations to shift from the old, rule-bound Weberian form, to scale down the large public sectors and create more “business-like” public services (Ferlie 2017).

After the 1994 democratic elections, the South African Government promulgated several pieces of legislation to deepen decentralisation at the local

Figure 1: The shift to New Public Management



Source: (<https://schoolofpoliticscience.com/new-public-management/>)

government level. Some writers argue that NPM has several advantages such as the strong focus on financial controls, value for money and increasing efficiency and improving quality in service delivery. Others argue that the people are viewed as customers and there is difficulty with holding officials accountable for poor service delivery (Cameron 2009; Heywood 2013).

NPM makes a citizen-friendly administration from a rigid, hierarchical, disciplined bureaucratic administration that needs to make weak public administration strong and effective (<https://schoolofpoliticscience.com/new-public-management/>).

PROBLEM STATEMENT

Local governance in South Africa is in a crisis. It is *inter alia* characterised by poor performance, credibility, and corporate governance problems, being unable to execute its mandates to address job creation, poverty alleviation, unsatisfactory levels of service delivery, the lack of skills, and huge demands from society for better and more efficient services (Brand 2017:1).

In the Media Release of 15 June 2022, the Office of the Auditor-General (AG) calls on municipal leaders “to transform local government to improve people’s lives” (AG 2022:1). The AG reports that during the year under review (2020–2021),

78 (30%) of the 257 municipalities received “qualified audits with findings” and 25 of the 257 (10%) “disclaimed with findings”. In the AG’s Consolidated General Report on local government audit outcomes (MFMA 2020–2021), the AG remarks: “The South African Constitution envisaged that citizens will live in towns, cities and settlements where they have access to clean water, sanitation, electricity, refuse service, and good roads and infrastructure ... Sadly, this is not the lived reality of most South African citizens in the country. Local government is characterised by accountability and service delivery failures, poor governance, weak institutional capacity, and instability” (AG 2022:1). The AG further remarks: “This is especially significant if one considers that municipalities and municipal entities were responsible for an estimated expenditure budget of R509 billion in 2020–21” (AG 2022:5). Financial reporting consultants “have become permanent features in municipalities’ financial reporting processes, with the cost of these consultants amounting to R1,26 billion in 2020–21. The cost of consultants doubled over the term of the previous administration”. “When combining the money spent on finance units and consultants, it is clear that financial reporting carried a substantial price tag in 2020–2021 of just over R11,67 billion” (AG 2022:16).

AIMS AND GOALS

The main emphasis of the article is that communities should be encouraged to assist local governments to excel, to assemble and breed the necessary capacity and competence, and to be allowed to liberate caged industriousness as a mechanism that espouses innovation, to redesign and co-create, among others, a 21st century hybrid local government business model (Breed 2020:n.p.). This can never be a single or simplistic traditional effort, but rather a multi-faceted exercise that transforms service delivery and unshackles adherent suffocating bureaucratic governance.

Local authorities must also realise that they should act as the government’s development agent for Local Economic Development (LED), which in turn should exploit the development of the human and social capital of a country – to create the so-called new currency or economy (Breed 2018:2). A more appropriate approach to invention, innovation, entrepreneurial and industrial policy, should be transfixed with such a model which involves search, experimentation, monitoring, learning and adaptation, all of which need to occur in a context of “... [visionary] openness to knowledge, trade, investment, and competition” (Dutz *et al.* 2014:19–33). Furthermore, it should focus on social and economic impact and the delivery of policy advice – what some call a “science of delivery” (Dutz *et al.* 2014:19).

A further aim of this article is to indicate that South Africa needs a local government model without any substantial analogues of the past, to achieve mainly three goals, according to Breed (2020:n.p.), namely:

- To exponentially increase the intrinsic value of local institutionalism by reinventing the “engine room” (becoming a liberated, foresighted technocratic social innovation partner) and re-engineering the rusted, conventional structure of local governments to suit each individual institution’s unique circumstances and visions. The traditional model should be substituted by a non-generic model. It should be a crossover between the most proved contemporary business principles and concepts; for example, the cost-to-company business trajectory and multiplier, blended with undiscovered and lenient bureaucratic support.
- To combine and join the unlocked potential in local government’s empowerment mechanisms and the undiscovered asset base extractable from human capital’s knowledge and idealism. A mechanism to leverage the Gross Domestic Product (GDP) per person and Gross Geographic Product (GGP) of the specific geographic region in the quest towards an individual wealth creation formula (Sachs 2005:38). The necessities and conveniences of life should be in proportion to and regulated by the skill, dexterity, and judgement with which its labour is generally applied (Smith 2017:1).
- To catalytically free the human mind, its spirit, and its ability to achieve the seemingly impossible.

VALUE OF STUDY

If such a new local government model could be achieved, the fibre and texture of the South African socio-economic landscape could be changed, and the country could thrive. A new innovative, multi-disciplinary, and unconventional model for local governance in South Africa could unlock the potential and asset base of local institutionalism that could benefit many destitute, struggling, and hopeless people, among others. It could “create the structures to facilitate sustainable growth” and it could be seen “as the delivery mechanism for improvements in housing, healthcare, and infrastructural development” (Pycroft in Fitzgerald *et al.* 1997:391).

METHODOLOGY

The methodological approach of the study is a practical, analytical, and interpretative approach based on the authors’ experience and long exposure to local government in South Africa, enhanced by interdisciplinary research and expertise.

DELIMITATIONS

The article is limited to the following coherent, carefully selected and most needed building blocks of the envisioned new local government model. The authors start with a brief focus on the government's commitment to the process of local government restructuring and transformation. The authors also argue that local governments, even though many of them currently do not function well, is the country's biggest socio-economic development asset which has to be transformed, exploited maximally and harnessed carefully in the interest of the citizens of South Africa. The possible gearing of such a new envisioned model; what the keys should look like to unlock this new enterprising institutional process in order to create and apply the new currency or economy the country needs, will also be discussed.

LOCAL GOVERNMENTAL RESTRUCTURING AND TRANSFORMATION IN SOUTH AFRICA

The traditional local institutionalism's model argues that it possesses the keys to unlock the potential of local government. A new model, however, should change the locks. The value of such a model should lie in its ability to provide vertical and lateral leniency to empower local government to discover and process data and valuable information buried in human and social capital. It should exhibit what human and social capital can contribute (see later) when included in the menu of success recipes of high performing institutions.

It is important to note that local government is the support engine for the economy – it is not the engine itself. A municipal institution must aim to act as a stimulant, as the spark that combusts the engine (local community) into a forceful locomotive that gently and determinately moves on its tracks (mission) to its destination (vision). According to Breed (2020:n.p.), local institutionalism should have broadly the following interrelated and critical attributes in accordance with the constitutional objectives of local government in South Africa augmented by expressed systemic mandatory societal needs and expectations. They are:

- To create, innovate and fulfil opportunities – an efficient vehicle that endeavours to grow a more mature society and (some degree of) political maturity (Reddy 2016:n.p.). It should provide communities with the means to flourish in accordance with their human and social capital's potential. To test the boundaries of human endeavour and extract from its asset base all possible potentially beneficial thoughts and partnerships. To care for its people and their future through mechanisms, initiatives, and structures suitable for the times they are living in, that will inspire them to obtain bigger goals;

- To deliver better services sooner, faster, and more smoothly in all spectrums of its proclaimed institutional ability, including the needs, aspirations and expectations of its society, as outlined in the government's *Batho Pele* Vision (Department of Public Service and Administration (DPSA) 2014);
- To carry out its mandate in accordance with the prescripts of local government in South Africa (Constitution, RSA 1996: Section 152(1)) guided by moral compasses and ethical leadership, as discussed in the Supplement for Municipalities in the *King IV Report on Corporate Governance for South Africa* (Institute of Directors Southern Africa (IODASA) 2016); and
- To create capable leadership and an environment that will fertilise socio-economic growth (Constitution, RSA 1996: Section 152(1)). This is confirmed by Thornhill (2012:132) that "another requirement is that capable officials acting with impeccable integrity and expertise should be employed to carry out the complex policies in order to promote socio-economic development with the co-operation of the members of society".

To achieve these goals, local government needs to think and act revolutionarily. Although this study concurs with academics such as Sam Koma and Jerry Kuye (2014:96) that "a core of trained and committed Local Economic Development (LED) experts at local level is lacking", arguments throughout the article reverberate the now obsolete trend that tacitly subjects LED to an organisational submission within an integrated development plan (IDP), regulated by a bureaucratic management system. The 21st century logic shared in this article disputes the adherence to that conventionalism.

Since the attainment of South Africa's democracy in 1994, the government has shown greater commitment towards the process of local governmental restructuring and transformation. This process has been essential to alleviate the inequalities of the apartheid legacy and to accelerate service delivery for local communities (Msomi, Reddy & Zondi 2018:105). To this end, the success of many government policy initiatives in South Africa, including the National Development Plan (NDP) – Vision 2030 depends on the effective and efficient roles of a transformed and democratic local governance system. In addition, Van der Walddt (2014:88) posits that local government in South Africa is crucial for community development, community participation, poverty alleviation and social cohesiveness, given the diverse population of approximately 58 million citizens. The current structure of local government consists of 226 local, 8 metropolitan and 44 district municipalities that are established as provided by the Local Government: Municipal Structures Act 117 of 1998 (RSA 1998). These municipalities represent the socio-economic aspirations of local communities and are regarded as key instruments for the realisation of human rights through quality service delivery at the community levels (Parnell, Pieterse & Swilling 2002).

This view is further endorsed by Reddy and De Vries (2015:12) who point out that local government is the government sphere operating at the community level to give expression to the essence of participatory democracy. During the three decades leading to the democratic dispensation in 1994, local government did not play an embracing role for building a capable state. Instead, it facilitated the agenda of the apartheid government to facilitate racial inequalities, according to Reddy and De Vries (2015:12).

In light of the current local government democratisation, there are various structures and policy initiatives that have been created to boost service delivery mandates (and demands) in local government. For example, the mandate enshrined in Section 152 of the 1996 Constitution which requires municipalities to serve as enabling institutions for the promotion of a democratic and responsible governance at the community levels; municipalities are also required to coordinate service delivery planning in a sustainable manner and promote socio-economic development for local communities; create safe and healthy living conditions for communities and to allow for a conducive environment to allow public participation in local government affairs (RSA 1996). To achieve these objectives, there are various oversight mechanisms that provide hands-on support in the form of capacity development focusing on critical areas of local government such as the IDP, policy formulation, local economic development, community participation and service delivery (COGTA 2018:3). Despite the challenges that threaten the sustainability of municipalities, especially those in rural areas, South African local government has managed to cement its democratic position in a complex and dynamic environment.

LOCAL GOVERNMENT – A COUNTRY’S BIGGEST SOCIO-ECONOMIC DEVELOPMENT ASSET

Local government is a serious, complex, and remarkable depository where the care, conduct, values, needs and aspirations of all, including its officials, should be deposited, analysed, contemplated, interacted with, and prioritised. Buried in the depths of local government is definite nobility and pride, a proverbial ‘soul’ that needs rediscovery and enactment. A new model should propose a modern 21st century strategic and creative operational plan between the present and the ideal.

Despite excellent functioning pockets, local government has become an apparatus with which to obtain power, income, influence and authority, and is leveraged for corruption (see Thornhill 2012:140–141; Pillay 2018:22–39); and “self-enrichment, patronage and the favouring of family members and friends” (Tshishonga 2015:n.p.). Reddy (2016) refers in this respect to Zybrands (2012:82) and states that “such activities have been so endemic in local government that drastic measures have to be taken for this state of affairs to be remedied”. Local

government has gradually become eroded by paradoxical choices and strange coalitions of insufficient options to address the needs of its people. It has further become politicised with resultant “implications for municipal service delivery in the broader context of developmental and good local governance” (Reddy 2016:n.p.). Reddy (2016:n.p.) also “critically reviews the ‘cadre deployment’ policy of the ruling party, the African National Congress (ANC), which has generated a great deal of debate and discussion in political, academic and social circles, as such policies have provided the impetus for the election and appointment of senior municipal functionaries in local government”. Reddy (2016:n.p.) further postulates that “[p]olitical infighting and related clashes between the political and management components in local government in South Africa have ... adversely affected municipal service delivery” and that “municipal functionaries have to take into account the passion, commitment, competence, qualifications and ethical leadership of those in line to be appointed, or elected”.

Local government owns the means of production (Gaarder 1995:329) and is the pivotal key to a successful South Africa. Many municipalities buckle under the pressure in their fatigued endeavours to find some relief for so many increased and perplexing challenges appearing daily – only to inevitably find that their quests were mere distorted attempts for today and tomorrow’s demands, possibly because the future and expectations outran their efforts. Few partially succeed. The future is incessantly shadowing local government. The new model envisioned should intercept that future.

To conquer the future, local government needs to develop what Larry Page, American computer scientist, termed a “healthy disrespect for the impossible” (Sculley 2014:4) and an appetite for braveness. To reach the future early, local institutionalism needs to dislodge itself from traditional and conventionally comprised formulas. According to Dostal *et al.* (2005:n.p.), it should get away from “current futures”. In contrast to this is the creation of an “ideal future”. While the logic of “current futures” is an extension of the past (“more of the same logic”), a deliberate new creation is necessary to define the “ideal future”.

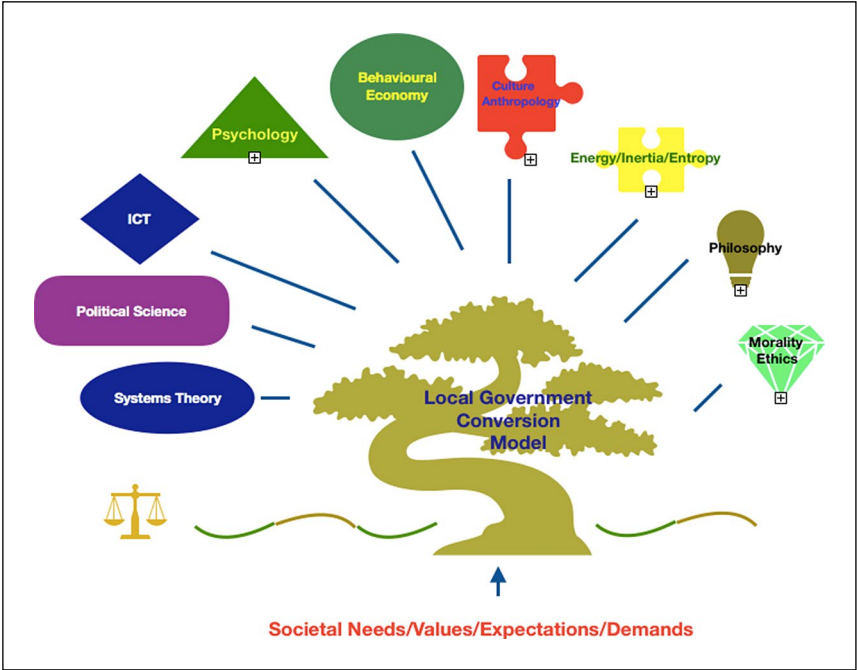
Local government must realise that it is the biggest business in town and continually needs to adopt economically induced competencies and business intelligence to at least balance its budget. Unfortunately, the padded mechanisms of control and unnecessary duplication are unable to spark the anticipated results. The dialogue should be about a totally new thinking, an innovative approach to harness interdependency, a crossover to reflect modalities of public institutional (enterprising) entrepreneurship. The thinking of the authors is about how best to accelerate the country’s social and economic muscle to present a new thinking of how to sculpture prosperity from the marble block which local institutionalism has become. The vision must not be about repair – it rather should encompass the new great local government transformation.

While there is increasing literature and empirical evidence advocating for the inclusion of an innovative business model in local government, some scholars opine that municipalities should maintain their traditional mandates for the facilitation of public goods and engaging communities in policy formulation and implementation (Madumo 2014). This view has always been defended on the basis that municipalities exist to fulfil service delivery obligations and the democratisation of local communities (Stewart & Woods 2017). In essence, according to these scholars, the process of local government restructuring and transformation will be considered incomplete if it does not prioritise service delivery and mass participation of the citizenry in policy planning.

THE GEARING OF A POSSIBLE NEW MODEL

Local government, according to Breed (2020:n.p.), should be an interaction of *inter alia* nine inescapable fundamental disciplines. The local government management system should contain inputs from: 1) Systems Theory (interdisciplinary

Figure 2: Local government conversion model



Source: (Co-author Johann Breed designed the diagram for this article, 2022)

study of systems (Von Bertalanffy 1968:1); 2) Political science; 3) Information and Communication Technology (ICT); 4) Psychology; 5) Behavioural economy (Kahneman 2011:412); 6) Culture and anthropology; 7) Energy, entropy, and inertia; 8) Philosophy (wherein there is ideological reductions); and 9) Morality and ethics (from the philosophical and spiritual *gestalt*). This is indicated/explained with the following diagram in Figure 2.

All these disciplines should be incessantly diffused in multiple energies present in business operational and local government processes. Unfortunately, all these respective inputs cannot be explained in this article, due to limited space. Although a new model should, according to Breed (2020), assimilate all these disciplines, it should more than usually pay attention to *inter alia* the communicative, scientific, entrepreneurial and systems theory components. Such a model should signal and symbolise the serious intent of a local government enterprise to attack unemployment, poverty, inequality, and social dysfunctionality uncompromisingly and unconventionally as an interactive partnership whose sole objective is the pursuit of wealth and happiness. A new model should also deal with the influence of inertia on the performance of municipal institutions. According to Newton's first law of motion, "an object at rest stays at rest and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force". To change local institutionalism according to this law, means that a greater force, another force with more energy needs to be applied to change local government's existent trajectory. The authors consider the new envisioned model as that energy (<http://www.physicsclassroom.com/class/newtlaws/Lesson-1/Newton-s-First-Law>).

Claude Shannon, an eminent scientist, postulated that there are basically three stages of communication. The information source, the conduit/transmitter, and the receiver (Shannon 1948:379–423; 623–656). Uncertainty occurs in at least two of these three stages. Uncertainty that results in entropy is *inter alia* caused in these stages by vagueness; encoding/interpretation of the contents; credibility/reliability; "political" scenarios operative in the conduit; ambiguousness; and context.

The importance of the information (communication) theory (ICT) is for the receiver (local institutionalism) to ascertain whether what it received and duly interprets, was the source's correct content and intent and that the same quality of information that was originally communicated has in fact been transmitted, received and interpreted (Breed 2020:n.p.). In a new, exciting initiative (Nothnagel 2018:5), a municipal institution could be regarded as basically a hub of communication, where entropy is ever-present because the bigger the institution, the more communication is required, the easier interpretation-confusion erupts and hence the more entropy with consequential attention and action.

THE KEYS THAT SHOULD UNLOCK THE NEW ENTERPRISING INSTITUTIONAL PROCESS

Government should not really be expected to ignite badly needed sustainable economic booms. These economic booms originate in the minds and souls of individuals and in an atmosphere and culture of great leaders, in great towns and great cities. A new local governmental model should purport to distribute the advice of how to realise the aspirations of people more inclusively and laterally. It should distil into the following surrogate value-chain links, and according to Breed (2020:n.p.), consist of among others 10 practical keys that should unlock the new enterprising institutionalism process and subsequent path to transformation. They are:

- The Game theory – the necessity for the initiation of the non-zero-sum game (Schelling 1980:85,89);
- Value-chains – the inclusion of a unique (human) algorithmic computation – contextualising the concept “the speed of the game is determined by the slowest player”;
- Business and emotional intelligence through serious technocratic transformation;
- Intrinsic value – increase the potency of the engine room performance, and monitoring and evaluating the impact of the strategic intent’s resource component;
- Multiplier-effect – the so called “silk road” (Frankopan 2015) of the modern era is needed. Over the last 40 years, 750 million people all over the world have been lifted out of poverty as a result of the increase in trade from US\$2 trillion to a staggering US\$90 trillion. Trade created a new route to prosperity and wealth (Norberg 2016). According to Alcock (in Nieselow 2019), the informal sector (or hidden economy), which is much larger than just the township economy alone, is completely unquantified and offers tremendous potential for employment and economic growth. Alcock (in Nieselow 2019) estimates that the informal sector makes up 7% of the country’s GDP and there have been reports that this could go up to 20%. Thus, this sector provides an internalised interdependent, interactive, and synergetic new prosperity coalition with unbelievable triumph over wealth stagnation, and entrepreneurial non-development, that needs to be included in such a model. It could, for example, be done by elevating “stokvels” into a major economic strategy, and contextualising the question, “what is wealth” and how it is created. About 11.5 million South Africans participate in stokvels – and ZAR45 billion is collected in any given year, according to Hogg (2015:n.p.). Minanawe (2015) postulates that the average hawker in South Africa earns ZAR3 000 p/m and that there are more than 500 000 – many of these deals not traceable through official channels. The muti (traditional herbs) market or industry is worth ZAR3 billion a year,

with 27 million consumers (Nieselow in Alcock 2019). “The spaza sector is ‘disrupting’ the formal sector and is worth ZAR250-billion, with local stores offering goods at between 5% to 10% cheaper than low-cost grocery giant Shoprite” (Nieselow in Alcock 2019). Research shows “[s]ales through South Africa’s 134 000 Traditional Trade (TT) outlets currently amount to ZARR46-billion per annum, which equates to one of every five rand spent and one third of all consumer goods packages sold in South Africa” (Nielson Holdings 2016). Globally, over 90% of enterprises are small, micro and medium enterprises (SMMEs) (Nieselow 2019);

- Systemic transformation from a local government institution to become a morally compassed local government enterprise;
- Innovation – to deepen the transformation to a business model through which local government would be advised of adaptations of ideas in accordance with a more entrepreneurial minded modular performance system; and to help innovative thinkers to be more proactive and confident in their imagination and design processes;
- Empowerment – employees are only as good as they are allowed to be;
- Local government is a complex business that requires inclusive innovative development and collaborative ways in solving them. One size does not fit all; and
- Return on investment – calculating what the result, harvest and/or contribution caused by the appointment and elected investment in human resources in local institutionalism are.

The social, economic, political, and ideological renaissance’s heartland for South Africa is dependent on and embedded in the hearts and minds, the *corpus* and *modus vivendi* achieved through harvesting the seed planted by human capital, tenderly nurtured by an enabled visionary local enterprise. Presently, institutionalism is compromised by centrist control. Although the importance of corporate governance should never be disputed, South African socio-economic development now is in desperate need of a more flexible, informal, and indigenous management mechanism. Over-regulation, over-subscriptions to overly imagined possible transgressions suffocates free, unencumbered bartering between entrepreneurial-minded have-nots and a market economy.

A NEW ECONOMY NEEDED – HUMAN (AND SOCIAL) CAPITAL

In (almost) no budget of any household, business, industry, enterprise, or organisation one will find any vote-appropriated human capital. Yet without human capital there will be no budget, institution, organisation, exploration, and spiritual

fraternity – no development. Human capital, as supported by the World Bank's Human Capital Project (World Bank 2020), is the source of wealth creation and meaning in the world and drives its economies. According to Patrinos and Angist (2018:2) "a country's human capital is crucial for its economic success" and they found that "human capital comprises the largest component of a nation's wealth, at 64%". Murray and Meyer (2020) reason that "85% of the current market value of US corporations is secreted in intangibles such as human, intellectual and social capital. On the other hand, forty years ago, 80% of the market value of US corporations was attributable to tangible assets" (Patrinos and Angist 2018:2). Unfortunately, according to The Human Capital Index 2018, South Africa is indexed at 126th out of 157 countries.

This product of the human mind is exalted by Fukuyama (1995:45) when he contrasts the quality not quantity of human capital's value-chain contribution in the production of modern wealth. Central to a new model of local institutionalism should be the following synopsis: 1) What "wealth" fundamentally is; 2) How it can be created (Beinhocker 2007:4); and 3) Whether local government can act as stewards in this process, and if so, how?

What has been discovered from young South Africans from dysfunctional families during the accredited Community Development Workers (CDW) course (Stellenbosch University) in the Northern Cape (Emthanjeni Local Municipality) in fact was that wealth is an oxymoron and generally misconstrued. One can be rich – but in fact poor; one can be poor – but in fact rich; and one can be poor but happily nurtured within the fact that all humans can never be dispossessed from the potential of deploying their own personalised "currency", namely human capital.

Local government's contribution to the GDP of individuals left alone societies and the business fraternity contracted to the conditions and levels it is in now. The burgeoning number of laws, regulations, prescripts, and bureaucracy piled onto the management of local government that were supposed to fertilise growth and development, in the process has achieved little, and in many cases, has had no effect. It seems as if local institutionalism does not grasp the fact that poverty alleviation and job creation can never be solely sustainably addressed by itself. These two serious illnesses in South African society can only be addressed by inclusive, socially innovative, moral compassing discovery and deployment of human capital, growth, and economic development via an effective organisational structure.

The new economy local institutionalism should trade in is linked to social outcomes, reputation, and social impact (Eggers & MacMillan 2013:4). A new model for local governance theorised herein should advance proven initiatives to cultivate cooperation, co-production, and co-creativity – favoured and hypothesised in the non-zero-sum game. Local government does not need to be big and strong

to conquer its challenges. Small, hungry institutions with little resources can easily outscore big, shallow institutions.

LATERAL SOLUTION CAPITAL

One of the objectives of this article is that communities should be encouraged to assist local governments to excel, to assemble and breed the necessary capacity and competence, and to be allowed to liberate caged industriousness as a mechanism that espouses innovation, to redesign and co-create, among others, a 21st century hybrid local government business model. This can be done through lateral solution capital which has the potential of transforming service delivery and unshackling adherent suffocating bureaucratic governance.

According to the GBSB Global Business School (2017), lateral thinking “is a process of idea generation and problem solving where new concepts are created by looking at things from a new angle. Logical thinking is considered vertical thinking and carries an idea forward by means of conventionality and meeting the challenges head on. Sideways thinking, or Lateral Thinking provokes fresh ideas or alters the frame of reference in order to procure a new approach to the issue. In Lateral Thinking, individuals try to bypass or topple problems by thinking up radically new approaches conformists would never have thought possible” (No author 2017:n.p.)

Maltese-born United Kingdom-based psychologist Edward de Bono (GBSB Global Business School 2017:n.p.) reasons that “Lateral Thinking is concerned not with playing with the existing pieces but with seeking to change those very pieces”. He continues by saying that there “is no doubt that creativity is the most important human resource of all. Without creativity, there would be no progress, and we would be forever repeating the same patterns” (GBSB Global Business School 2017:n.p.). The opportunity space opened up by dysfunctional local governments, can be occupied by this lateral thinking option, embedded in human capital.

Government is no longer the only role-player when it comes to societal problem solving (Eggers & MacMillan 2013:3). With the new envisioned local government model in mind, society could witness a change in how it deals with its own problems to find a new direction – a shift from a government dominated model to one in which government is just one player among many that has emerged. This proposed model acknowledges that in today's era of fiscal constraints and political gridlock communities can no longer turn to government alone to tackle towering social problems. What is required is a lateral, more collaborative economy- and prosperity-generating local government system, with enormous monetary results. For this to be realised, South Africa needs a mechanism such as a social laboratory where societal and economic problems can be addressed

by relevant stakeholders and where hope and solutions can be created. Social laboratories have three core characteristics. They are first, social (“social labs start by bringing together diverse participants from different sectors of society, such as government, civil society, and the business community, to work in a team that acts collectively. The participation of diverse stakeholders beyond consultation, as opposed to teams of experts or technocrats, represents the social nature of social labs”); second, experimental (“social labs are not one-off experiences. They are ongoing and sustained efforts. The team doing the work takes an iterative approach to the challenges it wants to address, prototyping interventions and managing a portfolio of promising solutions. This reflects the experimental nature of social labs, as opposed to the project-based nature of many social interventions”); and third, systemic (“the ideas and initiatives developing in social labs, released as prototypes, aspire to be systemic in nature. This means trying to come up with solutions that go beyond dealing with a part of the whole or symptoms and address the root cause of why things are not working in the first place”) (Schwella & Pretorius 2017:28). This should reveal just how a burgeoning new need where gifted players from across the spectrum of business, government, academia, philanthropy, and social enterprise converge to solve big problems and create public value that can change the trajectory of societal needs and aspirations. By erasing public-private sector boundaries, the solution economy could unlock billions of rand in social benefit and commercial value.

This article exhibits the description of an awakening and building of an unpolluted and silent gossamer-threaded alternative camaraderie for South African society. A creative instrument better suited to deal with societal deprivation. An enterprise wherein each member of society possesses varied forms of solutions. Solution capital that could deal with unique and commonly experienced societal circumstances. The future incessantly shadows and, in many instances, outpaces local institutionalism. This model therefore should not be about a methodology rooted in dogma alone but in substance, relevance, idealism, hope, vision, exposure, experience, skills, and expertise. According to Pretorius and Schurink (2007:23) “in leadership, people that are skilled have got expertise, are capable and confident and have vision, so that in return the people that are recruited to the organisation have the necessary skills, necessary expertise, they share in values of the organisation, they are willing to go out and do things, and that is the only way you can enhance service delivery”.

Undoubtedly this new model should be percolated by the entropic phenomenon that is ever intensely present in local government. Dysfunctional institutions are in a state of entropy, which provides the conditions wherein solutions can be sought.

However, a transformed institutionalism needs to become an enterprise. The concept of local institutionalism needs to be replaced by a new modern

description, dynamic local government enterprise that has a healthy disrespect for the impossible. Many local institutions major in minor things. This model should operationalise the ability to deepen, personalise and enrich, if you would, the contact between a cold, lifeless machine, and the people it is involved with. It should humanise a clinical bureaucratic operation and this caged catalyst with a personality and with emotional intelligence and lateral thinking. It should convince local institutionalism of the many opportunities that are buried in the minds and idealism of its inventory and should provide potential by unlocking and advocating for the discovery of human capital formation that leverages in turn societal and political capital. It should envision local government enterprises that benchmark performance standards that do not yet exist in local institutionalism.

However, Corporate Social Responsibility needs to shoulder this quest as it was so gallantly expected to do. In the words of Bobby Godsell, South African businessman, “you cannot do good business in a bad society” (Gleason *et al.* 2011:145). Corporate Social Responsibility needs to share the burden – it is part of the problem that is faced but fortunately also part of the solution. This model should incorporate a process which could provide that new millionaires can be created through lateral solution capital discussed earlier; and to give this machine a personality and to convince local institutionalism of the many opportunities that are buried in the depths of thousands of papers of legislations and smothering regulations and in the imagination of its society.

CONCLUDING REMARKS

The article started with an introduction, focusing mainly on the context of local governments in South Africa to which local communities have been exposed to since the attainment of democracy in 1994. This was followed by a focus on local governmental restructuring and transformation in South Africa; that local government is a country’s biggest socio-economic development asset; what the gearing of a possible new model should look like; the keys that should unlock the new enterprising institutional process; a new economy that is needed, namely human and social capital; as well as lateral solution capital. One must remember that the future is an inevitable territory and can never be evaded. The future, in this context, is where individual local institutions envision it to be.

REFERENCES

- Alessandro, A. 2005. Towards Quality e-Services in the Public Sector. *Managing Service Quality*. 15(1):6–23.

- Auditor General, South Africa, Media Release. 2022. Auditor-General calls on municipal leaders to transform local government to improve people's lives. Available at: <https://www.agsa.co.za/Portals/0/Reports/MFMA/2020-21/MFMA2020-21%20Media-release%2015%20June%20final%20approved.pdf?ver=2022-06-15-105855-130>. (Accessed on 22 August 2022).
- Auditor General, South Africa. 2021. Consolidated general report on local government audit outcomes, MFMA 2020–21. Available at: <https://mfma-2021.agsareports.co.za/downloads/agsa-mfma-general-report-2021>. (Accessed on 22 August 2022).
- Alcock, G.G. 2015. *Kasinomics: African informal economies and the people who inhabit*. Johannesburg: Jonathan Ball.
- Beinhocker, E.D. 2007. *The Origin of Wealth*, Harvard Business School Press. Boston, Massachusetts, USA: McKinsey & Company.
- Bevir, M. 2009. *Key Concepts in Governance*. Available at: <http://dx.doi.org/10.4135/9781446214817.n23>. (Accessed on 14 May 2020).
- Brand, D.J. 2017. Social Laboratories – An Innovative Approach to Co-production, Paper delivered at the annual conference of the International Co-production Study Group in Washington DC.
- Breed, J.A. 2020. Beyond Local Government: A New Model for Local Entrepreneurial Institutionalism. (unpublished manuscript).
- Breed, J.A. 2018. New Transformational Model for Local Government in South Africa (unpublished submission to the Honourable Minister of Corporate Governance and Traditional Affairs). 2/5/2018. P. 2.
- Business Encyclopaedia*. 2020. Available at: <https://www.shopify.com/encyclopedia/enterprise>. (Accessed on 29 May 2020).
- Cameron, R. 2009. New Public Management Reforms in the South African Public Service: 1999–2009. *Journal of Public Administration*. 44(4.1):910–942.
- Co-operative Governance and Traditional Affairs (COGTA). 2018a. *Monitoring Performance of Municipalities. Republic of South Africa (RSA)*. Pretoria: Government Printer.
- Co-operative Governance and Traditional Affairs (COGTA). 2018b. *List of Dysfunctional and Distressed Municipalities*. Available at: <http://www.cogta.gov.za/?p=4088>. (Accessed on 28 May 2020).
- Department of Cooperative Governance on State of Local Government Report – Local government support and interventions package*. 2021. Available at: <https://pmg.org.za/committee-meeting/33432/>. (Accessed on 22 August 2022).
- Department of Provincial and Local Government. 2006. *Stimulating and developing local economies, The National Framework for LED*, 2006/2011.
- Dostal, E., Cloete, A. and János, G. 2005. *Biomatrix: A systems approach to organisational and societal change*. BiomatrixWeb.
- Dutz, M.A., Kuznetsov, Y., Lasagabaster, E. and Pilat, D. 2014. *Making Innovation Policy Work*. Paris: OECD/World Bank Publishing.
- Eggers, W.D. and MacMillan, P. 2013. *The Solution Revolution*, Deloitte Global Services. Boston, Massachusetts, USA: Harvard Business Review Press.

- Ferlie, E. 2017. Exploring 30 years of UK public services management reform – the case of health care. *International Journal of Public Sector Management*. 30(6–7):615–625.
- Fitzgerald, P., McLennan, A. and Munslow, B. (Eds.). 1997. *Managing sustainable development in South Africa*. Cape Town: Oxford University Press.
- Frankopan, P. 2015. *The Silk Roads*. London, Great Britain: Bloomsbury Paperbacks.
- Fukuyama, F. 1996. *Trust*. New York: Simon & Schuster.
- Gaarder, J. 1995. *Sophie's World*, (Karl Marx). London: Phoenix House.
- GBSB Global Business School. 2017. *Lateral thinking: Goodbye to conventional solutions*. Available at: <https://www.global-business-school.org/announcements/lateral-thinking-goodbye-conventional-solutions>. (Accessed on 1 June 2020).
- Gleason, D., Nkomo, S. and De Jongh D. 2011. *Courageous Conversations*, Faculty of Economic and Management Sciences. University of Pretoria: Van Schaik Publishers.
- Heywood, A. 2013. *Politics*. Fourth Edition. London, UK: Palgrave MacMillan.
- Hogg, A. 2015. Stokvels Interview, BizNews Web, Tim Modise, 5/11/2015.
- Hood, C. 1991. A Public Management for all Seasons? *Public Administration*. 69(1):3–19.
- Human Capital Index. 2018. The World Bank. Available at: https://photius.com/rankings/human_capital_index_country_rankings_2018.html. (Accessed on 1 June 2020).
- Institute of Directors Southern Africa. 2016. *KING IV, Report on Corporate Governance for South Africa*. Johannesburg: Institute of Directors Southern Africa.
- Kahneman, D. 2011. *Thinking Fast and Slow*. London: Penguin Books.
- Koma S.B. and Kuye J.O. 2014. The Synchronisation of the Integrated Development Plan & Local Economic Development Policy in South African Municipalities. *African Journal of Public Affairs*. 7(1):1–14.
- Madumo, O.S. 2014. Fostering Effective Service Delivery Through Public Participation. *Administratio Publica*. 22(3):130–147.
- Msoni, E.S., Reddy P.S. and Zondi, S.I. 2018. Local Government Democratisation and Service Delivery in South Africa: Quo Vadis? The Case of Ndwedwe Local Municipality, South Africa. *Administratio Publica*. 26(3).
- Murray, A. and Meyer, D. 2020. *Stakeholder capitalism and technological transformation are connected*. Fortune CEO Daily, (27/01/2020). Available at: <https://fortune.com/2020/01/27/stakeholder-capitalism-technological-transformation-ceo-daily/> (Accessed on 1 June 2020).
- Nielson Holdings. 2016. "South Africa's Trade Sector Thriving with R46 Billion in Annual Sales". Available at: <https://www.nielsen.com/za/en/press-releases/2016/south-africas-trade-sector-thriving-with-r46-billion-in-annual-sales/>. (Accessed on 30 May 2020).
- Nieselow, T. 2019. "How to grow the informal sector", *Mail & Guardian*, 12/4/2019. Available at: <https://mg.co.za/article/2019-04-12-00-how-to-grow-the-informal-sector/>. (Accessed on 30 May 2020).
- Norberg, J. 2020. "The Real Adam Smith-Ideas that Changed the World", Free to Choose Media, YouTube, 28/3/2016. (Accessed on 30 May 2020).

- Nothnagel, G. 2018. *Entropy and Inertia in the Context of Organisations*. BLG's Hexagonal "Currency" 'Ideal Local Government' Leveraging Agents: Abrief. P.5.
- Patrinos, H.A. and Angrist, N. 2018. *Global Dataset on Education Quality: A Review and Update (2000–2017) (English)*. Policy Research working paper; no. WPS 8592. Washington, D.C. World Bank Group. Available at: <http://documents.worldbank.org/curated/en/390321538076747773/Global-Dataset-on-Education-Quality-A-Review-and-Update-2000-2017>. (Accessed on 1 June 2020).
- Pillay, P. 2018. Ethics, Public Trust and Corruption: Reflections on South African Realities. *Administratio Publica*. 26(4):22–39.
- Pretorius, D. and Schurink, W. 2007. 'Enhancing service delivery in local government: The case of a district municipality', *South African Journal of Human Resources Management*. 5(3):19–29. Available at: <http://dx.doi.org/10.4102/sajhrm.v5i3.148>. (Accessed on 1 June 2020).
- Reddy, P.S. and De Vries. 2015. *Quo Vadis: Local Governance and Development in South Africa Post 1994*. P.S. Belgium: Bruylant Publishers.
- Reddy, P.S. 2016. 'The politics of service delivery in South Africa: The local government sphere in context', *The Journal for Transdisciplinary Research in Southern Africa* 12(1), a337. Available at: <http://dx.doi.org/10.4102/td.v12i1.337>. (Accessed on 16 May 2020).
- Republic of South Africa. 1996. *The Constitution of the Republic of South Africa of 1996*. Pretoria: Government Printer.
- Republic of South Africa. 1998. Local Government Structures Act 117 of 1998. Pretoria: Government Printer.
- Sachs, J.D. 2005. *The End of Poverty*. London: Penguin Books.
- Schelling, T.C. 1980. *The Strategy of Conflict*. Massachusetts: The President and Fellows of Harvard College.
- Schwella, E. and Pretorius, M.A. 2017. *Innovation in Inclusive Development. Systemic and policy issues in the complex problem of extreme economic inequality: A Case Study*. Bellville, South Africa: SPL.
- Sculley, J. 2014. *Moonshot, Game-Changing Strategies to Build Billion-Dollar Businesses*. Kindle: Rosetta Books.
- Shannon, C.E. 1948. A Mathematical Theory of Communication, Reprinted with corrections from *The Bell System Technical Journal*. 27(1):379–423, 623–656.
- Smith, A. 2017. *An Inquiry into the Nature and Causes of the Wealth of Nations*, 1776. Introduction to the plan of work. (1) Jonathan Bennett.
- Stewart, G.H. and Woods. P. 2017. Good Practice in Local Governance Management. *The International Journal of Public Sector Management*. 13(2):167–191.
- The Batho Pele Vision. 2014. A better life for all South Africans by putting people first, 2014. Available at: <http://www.dpsa.gov.za/documents/Abridged%20BP%20programme%20July2014.pdf>. (Accessed on 14 May 2020).
- The Physics Classroom, Newton's Laws - Lesson 1 – "Newton's First Law of Motion, Inertia and Mass, Newton's First Law, Inertia and Mass, State of Motion". Available at: <http://www.physicsclassroom.com/class/newtlaws/Lesson-1/Newton-s-First-Law>.

- The World Bank. 2020. Human Capital Project. Available at: www.worldbank.org/en/publication/human-capital?cid=ECR_E_NewsletterWeekly_EN_EXT&deliveryName=DM42770. (Accessed on 31 May 2020).
- Thornhill, C. 2012. Improving local government transformation in South Africa, *Administratio Publica*. 20(3):128–143.
- Tshishonga, N. 2015. 'The increased politicisation of human resources recruitment'. In Reddy, P.S. and De Vries, M. (Eds.). *Quo Vadis?: Local governance and development in South Africa post 1994*: 1 Pp.29–146. Belgium: Bruylant Publishers.
- Van der Walddt., G., Khalo, T., Nealer, E., Phutiage, K., Van der Walddt., Van Niekerk, D. and Venter, D. 2014. *Municipal Management: Serving the People*. Cape Town: Juta.
- Von Bertalanffy, L. 1968. *General System Theory, Foundations, Development, Applications*. New York: Braziller.
- Zybrands, W. 2012. 'Municipal structures and finances – Predicaments and performance challenges'. In Booysen, S. (Ed.), *Local elections in South Africa: Policies, parties and politics*. 67–88. Bloemfontein: Sun Press.

AUTHORS' CONTACT DETAILS

Chris Jones

Chief Researcher: Department of Systematic
Theology and Ecclesiology
Head: Unit for Moral Leadership
Faculty of Theology
Stellenbosch University
South Africa
Tel: (021)808 9224
Email: chrisjones@sun.ac.za

Prof Pregala Pillay

School of Public Leadership
Stellenbosch University
Pregala@sun.ac.za
Cell: 084 365 7860

South Africa's Renewable Energy Auctions

An Exploration of Factors, Processes and Outcomes

W Kruger

Graduate School of Business
University of Cape Town

ABSTRACT

South Africa's renewable energy auction programme has been widely hailed as a regional and global success, with more than 6000 MW of renewable energy capacity secured and built in four bidding rounds at increasingly competitive prices. The renewable energy auction literature does not analyse nor identify how a competitive procurement programme's implementation (as opposed to its design) contributes to a project's price and realisation outcomes (success). This study aims to fill this knowledge gap by using an exploratory case study approach to show how an auction implementation approach that emphasised trust-building resulted in South Africa's programme, and also how this was undermined through a number of political-economy factors that continue to shape the country's approach to the energy transition.

INTRODUCTION

South Africa has emerged as an unlikely leader for competitively procured renewable energy-based private power investment in sub-Saharan Africa. Prior to 2011, the country had almost no utility-scale renewable energy power plants and all generation assets (>90% coal) were owned and operated by the state-owned, vertically integrated utility, Eskom. By 2015, South Africa had procured 112 renewable energy independent power projects, with a total installed capacity of more than 6300 MW. Initially, prices were high: too much capacity was auctioned, there was insufficient competition and equipment prices were much

higher. Project prices fell remarkably between 2011 and 2015 to levels below Eskom, the national utility's average selling price.

An auction is a structured procurement programme where contracts are awarded to projects with the lowest tariff. The level of competition in an auction is determined by the number of bidders (more bidders mean more competition) and also by the kinds of bidders (technically and financially stronger bidders mean more competition).

Globally, most research conducted on renewable energy auctions tend to focus on the impact of various auction design choices on outcomes (Kitzing and Weber 2015; Del Río 2017; Lucas, Del Río and Sokona 2017; Haufe and Ehrhart 2018; Kitzing *et al.* 2019; Kruger, Nygaard and Kitzing 2021) with modifications. Auction design factors deal mainly with three choices: how the procurement volume (amount of energy or power) is determined and divided (auction volume); how auction winners are selected (winner selection), and how an auction ensures the timely execution (realisation) of procured projects (auction effectiveness).

Auction theorists generally take for granted that an auction programme is a functioning institution embedded within an effective institutional and organisational context. Both theory and empirical evidence show that auction design choices are often rendered irrelevant in the face of weak and ineffective institutions that undermine the capability, commitment and integrity of the procurer and the auction programme (del Río 2017; Lucas *et al.* 2017; Milgrom 2004; Saussier *et al.* 2009).

This institutional blind spot leads to auction theory often neglecting two key areas that play an important role in determining project price and realisation outcomes: the impact of trust on potential bidders' decisions to participate in an auction, and the capability, authority and commitment of the implementing agency to help bring procured projects to commercial operation. Both of these goals – building trust, and supporting project realisation – rely on the auction being implemented by a capable, well-resourced auctioning authority in a way that is seen as transparent and fair (Colquitt 2001; Chiu, Huang and Yen 2010). The level of trust is based on the auctioneer's perceived ability, integrity and benevolence (Mayer, Davis and Schoorman 1995; Zitron 2006).

The above background rationale leads to the following research question:

How have auction implementation factors contributed to the price and project realisation outcomes of renewable energy auctions in South Africa?

There is a growing body of literature in the public policy, administration and management fields that specifically identifies factors supporting the successful implementation of the South African Renewable Energy Independent Power Project Procurement Programme (REI4P). This includes the management of governance risks and barriers in hybrid public-private partnerships (PPPs) (Nel 2013, 2014), understanding the political, policy and technical underpinnings of the programme

(Montmasson-Clair and Ryan 2014) and identifying how the programme's institutional design contributed to the development of the capacity and commitment required (Eberhard, Kolker and Leigland 2014; Kolker 2014; Leigland 2018). None of these studies explicitly trace how these factors influence the private sector's response to the programme. In particular, understanding the influence of trust on the private sector's response, as well as the factors underpinning this trust-response, is key.

This study aims to fill this knowledge gap by using an exploratory case study approach to show how an auction implementation approach that emphasised trust-building resulted in South Africa's programme, and also how this was undermined. It does so by locating this research within the broader bodies of scholarship on renewable energy auction design and Independent Power Project (IPP) investments, thereby showing how the trust-building implementation factors interact with the broader enabling environment, auction design decisions and project-level elements to determine outcomes.

The research was conducted using a primarily exploratory case study design, in line with the exploratory nature of the main research question. A case study is especially useful where phenomena, such as a government procurement programme, are not well understood and need to be studied within their context – and where the boundaries between the context and the phenomena are not clear (Babbie and Mouton 2001; Yin 2003; Kumar 2011). The rich description that results from case study research is not only useful for tracing causal paths and mechanisms, but also supports the transference of findings to other settings by allowing readers "... to make decisions regarding transferability..." to other settings that have shared characteristics (Creswell and Poth 2018:263)

Data was gathered by collecting and analysing documentation (including requests for qualification and proposals documents, contracts, project agreements, financing and credit enhancement offers, media reports, press statements, relevant policies and regulations, meeting minutes and scholarly publications) and conducting unstructured interviews (guided conversations) with officials from government agencies and departments, the private sector (including successful and unsuccessful bidders and project developers, debt and equity providers, engineering, procurement and construction and operations and maintenance contractors) and advisors intimately involved in the design and implementation of the renewable energy (RE) auction programme. The interviews were held in 2018–2021.

A purposive sampling strategy (Saunders, Lewis and Thornhill 2016; Yin 2018) was used to select interview participants, based on their involvement and familiarity with the auction programmes. 21 interviews were conducted in person, as well as telephonically. Most of the interviews were in-depth conversations taking anywhere between one and three hours, in line with the fluid "intensive interview" method typically employed as part of the case study approach (Weiss

1994; Rubin and Rubin 2012) development of an interview guide, the conduct of the interview, analysis, and preparation of data. . . . He offers specific techniques and guidelines—equally helpful to the seasoned professional or beginning graduate student—which he has found useful over many years. [“Learning from Strangers”] recounts the emotional experience of being interviewed, the likelihood of respondents telling the truth, and the ethical defensibility of interviewing. (PsycINFO Database Record (c. Interview notes were coded using NVivo, mainly using theoretical concepts from the literature in a combined analytical framework (Kruger and Eberhard 2023), but also allowing for the data to highlight important new themes. Case study drafts were shared with key informants for validation. I was additionally provided with access to the bidding documents (including draft contracts) for bidding windows 1 to 4, all relevant policies and regulations, official bidding results and relevant sector- and project-level data.

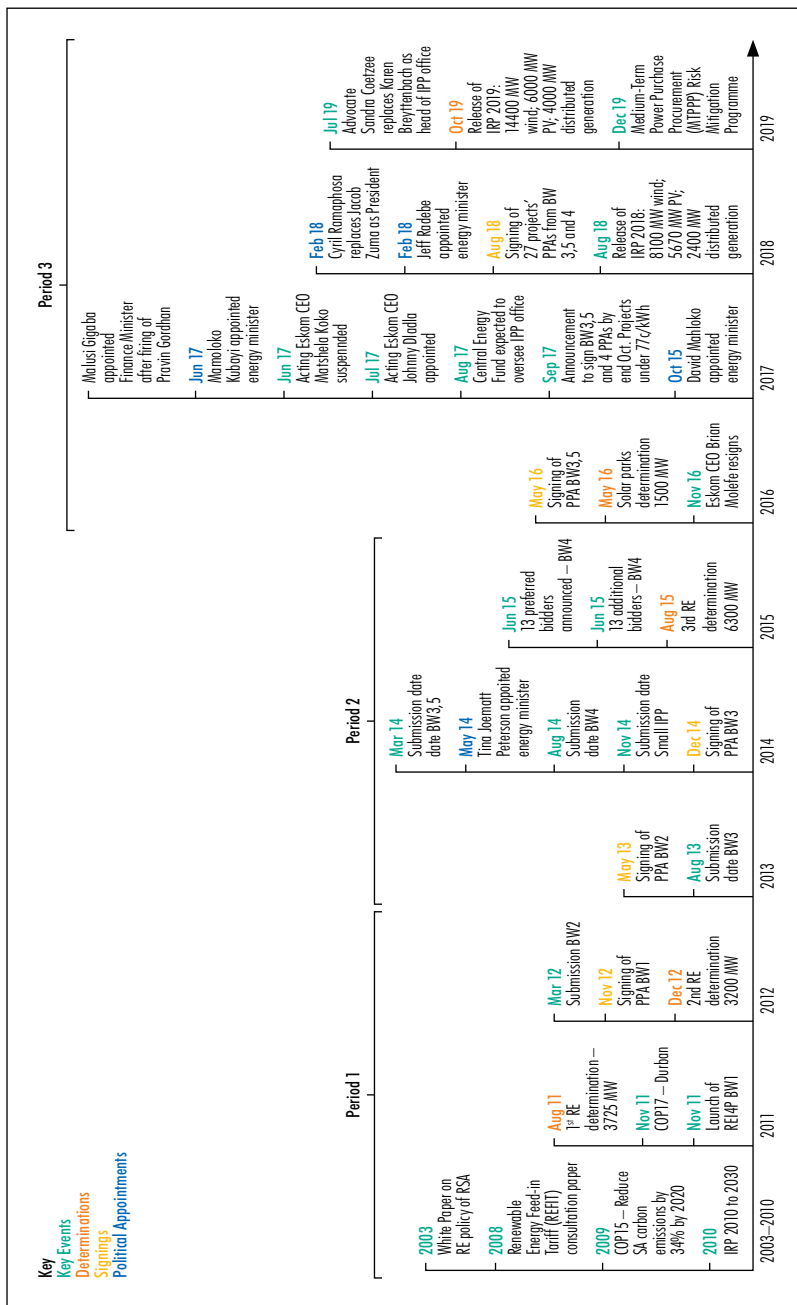
THE SOUTH AFRICAN REI4P

South Africa’s auction programme, from inception to the financial close of projects awarded in the fourth round (bid window 4), can be structured according to three relatively distinct periods (Figure 1). An electricity supply crisis initially disrupted the status quo, creating the conditions for new ideas and institutions (such as auction-based renewable energy IPP procurement programmes) to emerge while limiting the power of incumbent interests to suppress competition. This period saw South Africa committing to increasingly ambitious carbon mitigation goals on the international stage, which opened up the space for the ambition contained in the country’s energy policies to be realised through the auction programme.

The second phase involved a second disruption – this time caused by the price and investment outcomes of the auctions. By proving that renewable energy IPPs were cost-competitive and effective at rapidly deploying capital at scale, the auction programme undermined the ideas that supported the power-bases of incumbent interests – whether located around the utility or adjacent industrial or political configurations. This techno-economic disruption coincided with institutional disruptions as new entities were created or repurposed to design and implement the procurement programmes, causing a shift in institutional power dynamics that extended the threat posed by these auctions to the political-administrative spheres.

In the third phase reactions to the impacts of the auctions coalesced around trying to restrain the scope and scale of the transition. In South Africa, the auction results sparked a concerted and coordinated programme of resistance – including aggressive social media and public relations (PR) campaigns, persistent lobbying, direct court challenges and protests – encompassing interests as wide-ranging as

Figure 1: Timeline of key events and analytical periods



Source: (Author's compilation)

organised labour, the free market foundation, the coal mining industry, Eskom's board and management, various government departments and officials (most prominently the Department of Public Enterprises – Eskom's direct shareholder, but also certain sections of the Department of Energy), the upper echelons of the ruling African National Congress (ANC) and even the former president's office. As a consequence, the implementation of bid window 4 projects was delayed by three years while government failed to launch new procurement rounds, even as South Africa faced its most severe and devastating electricity supply crisis in decades. This coincided with increasing instability at a leadership level in the electricity supply industry as ministers and Eskom's chief executive officers (CEOs) were replaced at an alarming rate.

The following sub-sections unpack this phased evolution, tracing the impacts of the auction implementation choices on auction design decisions and outcomes.

Period 1: First shoots

The right conditions

A unique set of circumstances was instrumental in South Africa realising its long-standing policy goals of allowing private power investment and increasing renewable energy generation. An important initial driver was power cuts or rolling electricity blackouts (loadshedding, in Eskom parlance) that started in November 2007 and extended into 2008. The crisis engendered by these power shortages, provided the impetus to National Treasury and the Department of Energy to procure private power generation, despite opposition from Eskom and the Department of Public Enterprises.

The second contributing factor was South Africa's president's voluntary commitment at the United Nations climate change conference (COP15) in Copenhagen in 2009 to cap greenhouse gas emissions in a peak-plateau-decline trajectory (Figure 1). This commitment subsequently made its way into the Integrated Resource Plan (IRP) for electricity for 2010–2030 (Figure 1) where for the first time renewable energy, in the form of solar and wind, was included in the least-cost power generation expansion plan. The IRP is a crucial element in South Africa's electricity planning framework as it sets the parameters of what can, and cannot, be built to meet electricity demand. South Africa hosted the next COP17 in Durban in 2011 (Figure 1), which acted as a catalyst for demonstrating government's climate change mitigation actions.

Third was South Africa's national energy regulatory agency's (NERSA) initiation of a generous (but ultimately unsuccessful) renewable energy feed-in tariff (REFIT) programme from 2008 onwards (Figure 1). The REFIT programme failed to adequately address investor concerns and opposition from within certain sections of government (most notably National Treasury and the Department of Energy).

Bankable power purchase agreements were never finalised and no projects were procured. The REFIT programme was eventually replaced by the REI4P auctions at the “eleventh hour” (Figure 1), but caused developers to begin to secure suitable sites and undertake resource and environmental assessments, ensuring that these projects were well placed to participate later in the REI4P (Eberhard 2014; Eberhard, Kolker and Leigland 2014; Martin and Winkler 2014; Montmasson-Clair and Ryan 2014; Morris *et al.* 2020).

A focused, agile approach built on trust

While the conditions seemed ripe for a South African renewable energy investment programme, the REI4P ultimately owes its existence to the dedication, hard work and results-oriented approach of a handful of committed public sector officials, supported by legal and financial consultants. The group consisted of two officials from National Treasury’s PPP unit – Karen Breytenbach (Finance) and Lena Magondo (Legal) – and two officials from the Department of Minerals and Energy (DME) – Ompi Aphane (Deputy Director General: Policy, Planning and Clean Energy) and Maduna Ngobeni (Energy Regulation). These officials first started working together in early 2007 on the contracting of two open cycle gas turbine (OCGT) IPPs (1005MW) that were meant to help Eskom meet peak demand (hereafter called peaker IPPs), after which they turned their attention to the renewables sector.

This small group was able to set up an office outside the department, working with agility, finding ways around obstacles and emphasising problem-solving to achieve results. The Development Bank of Southern Africa (DBSA) was convinced to “host” and administer the accounts of the IPP office and procurement programme by being offered 25% senior debt on all procured projects, although it eventually took up only a fraction of this. Having the IPP office under the aegis of the DBSA meant that it would enjoy more independence and was able to operate outside of the normal governmental bureaucratic processes. It was also less vulnerable to political meddling.

The group was able to work in this way because it had the trust and support of key stakeholders, secured through various informal networks and relationships that had been built up over time. The programme also had the critical support of the most powerful ministry – National Treasury – and in particular that of Andrew Donaldson (deputy director general). National Treasury provided ZAR 100 million (from the ZAR 500 million Jobs Fund that had been sitting unutilised at DBSA) to the programme as seed funding based in large part on the reputation of and trust in the programme’s leadership.

Guiding design and implementation principles

Three sets of factors influenced how the programme was initially implemented. The first set of factors was influenced by the team’s difficult prior experience with

the procurement of the IPP peaker projects, which at that stage (2011), after three years, had still not resulted in a successful contract and would turn out to be a five-year long negotiation process. It was decided early on that the REI4P would not allow project contracts to be the subject of any negotiation by project developers or bidders. A great deal of time and resources was invested in ensuring that the quality of the programme documentation (especially contracts) was acceptable to the lender community (i.e. bankable contracts that appropriately allocate risks) before the procurement process started. The programme was able to secure the trust of South Africa's commercial and development banks through a series of informal Friday afternoon meetings where various funding and risk allocation arrangements were discussed and tested prior to the programme's launch. Because the programme had the buy-in of local banks, it was able to offer local currency denominated (ZAR), non-negotiable 20-year power purchase agreements (PPAs), indexed to local inflation through the official consumer price index (CPI).

The second set of factors was a recognition that these types of large government procurement programmes can, and often do, end up being targets of corruption and rent-seeking and that secure arrangements needed to be instituted around potential conflicts of interest, transparency and independent auditing. The programme was being conceived and implemented during the first years of the Zuma administration – a presidential term that would later be viewed as a period of accelerated kleptocracy in the South African state (Godinho and Hermanus 2018; Mokhele 2019; Bowman 2020). No IPP office or other government officials were thus directly involved in the bid evaluation processes, and several layers of auditing and security were set up to ensure the integrity of the programme. Bid evaluation was conducted by consultants under strict security conditions (including closed circuit television (CCTV) surveillance) and results were reviewed and audited by a second, independent team. The IPP office also employed an additional team of consultants to review and monitor the overall governance of the programme and alert the implementing team to any possible problems.

The third factor influencing the design and implementation of the programme was effectiveness – meaning the timely realisation of high-quality projects. The political pressure created by South Africa's hosting of COP17 played an important role, but there was also the need to demonstrate to the broader market that South Africa was serious about implementing a programme that would work.

As a result, auction design decisions were initially more concerned with project quality and preparedness than with price competition. The programme accordingly set several late-stage qualification criteria at levels more demanding than most international auction programmes. Bids were essentially required to be effectively ready for financial-close by the time of submission, needing to have secured all land rights, resource assessments, grid connection agreements, permits, bankable commitments, etc.

Finance providers also had to provide signed term sheets and signed letters of support that required them to make firm commitments and pledged that they had conducted due diligence of the proposal. These funder commitments were particularly powerful in proving bidders' bona fide project preparation credentials and effectively shifted much of the due diligence to banks, rather than the bid evaluators (Eberhard and Naude 2016). The seriousness and commitment of the bidders in reaching financial close, construction and commercial operation was also tested through the requirement to lodge bid and performance bonds, and by penalties for non-performance (Department of Energy 2014; Eskom Holdings 2014; Eberhard and Naude 2016).

This overriding concern with timely project execution also played a determining role in how the auction volumes were set and divided in the first auction round. The IPP office made 3626MW available in the first bid window (Figure 1) to maintain the market's confidence by easing the transition from a feed-in tariff-based approach to a competitive one by limiting competition – a decision which has since been acknowledged as a mistake. Project size limits (see Table 1) were similarly set to diversify realisation risk, through ensuring that the entire volume for particular technologies would not be awarded to a single project.

Table 1: Project size limits

Technology	Minimum Capacity (MW)	Maximum Capacity (MW)
Onshore Wind	1	140
Concentrated Solar Power (CSP)	1	75
Solar PV	1	100
Biomass	1	25
Biogas	1	10
Landfill gas	1	20
Small hydro	1	40

Source: (Eberhard & Naude 2016)

Results

The chosen approach proved to be effective, but initial bid prices were high. The programme attracted significantly more attention than initially expected, but was unable to take full advantage of this fact because overall volumes had been set so high, resulting in limited price competition between bidders. In bid window 1 (3670 MW offered), 53 projects were submitted, with a total volume of 2128MW. Of these, 28 projects met the qualification criteria and were awarded (1425 MW). Average bid tariffs were close to the announced ceiling prices (Table 2) and close

to double the global average weighted levelised cost of electricity (LCOE) for each technology at that time. Risk perceptions and resultant markups and return expectations, as well as advisory costs, were high. Projects were on average also about 50% smaller than the project size limits, which played a further role in pushing up prices as many of them were not able to take full advantage of economies of scale.

Table 2: BW1 average prices vs. price ceilings

	Average awarded price (ZARc/kWh)	Price ceiling (ZARc/kWh)	Average project size (MW)	Project size limits (MW)
Onshore Wind	114	115	85	140
Solar PV	276	285	36	75
CSP	269	285	75	150

Source: (Adapted from Eberhard & Naude 2016)

Bid window 2 was launched four months later (Figure 1) and did not present much opportunity for reflection and learning. One important lesson that was taken on board was the reduction in the offered bid volume to 1276 MW, resulting in more competition. In total, 79 projects totaling 3233 MW were submitted, out of which the 19 highest ranked projects (1040 MW) were awarded. Average prices were considerably lower – particularly for onshore wind and solar photovoltaic (PV) – than in bid window 1, despite average project sizes remaining more or less the same (Table 3).

Table 3: BW2 average prices and project sizes

	Average awarded price (ZARc/kWh)	Price ceiling (ZARc/kWh)	Average project size (MW)	Project size limits (MW)
Onshore Wind	90	115	80	140
Solar PV	165	285	46	75
CSP	251	285	50	150

Source: (Author compilation based on Eberhard & Naude 2016; IPP Office 2019).

In a marked departure – and a sign of things to come – international developers began to play a much more prominent role, with only three of the awarded projects having South African entities as lead developers. These results, in quick succession, achieved at a time of significant upheaval and uncertainty in the Organisation for Economic Cooperation and Development (OECD) renewable energy markets, set up the programme for a step-change in the next rounds.

Period 2: A step-change

The time between 2013 and 2015 was a period of increasing sophistication and a paradigm shift in terms of how the programme was being implemented and reacted to (Figure 1). This was a particularly busy time for the IPP office, with the country's first 47 utility-scale renewable energy projects moving towards financial close and commercial operation. Bid windows 3 (2013) and 4 (2014), along with bid window 3,5 (an additional round for concentrated solar power (CSP) projects), the expedited round (where previously unsuccessful bidders could rebid prices) and some small IPP rounds were also being prepared and implemented, and even though bid window 3 was launched a year after the submission date for bid window 2, the opportunities for strategic reflection remained limited. The changes introduced to the design of the programme during this time were incremental, reflecting some of the more obvious lessons learned up to that point.

The IPP office started tweaking the design of the programme to reduce transaction costs and broaden its impacts. As a result, some qualification requirements were relaxed over this period, though they did not have a marked impact on bid preparation costs or broader participation rates since they did not address the more expensive components of the bid (e.g. environmental permitting) nor did they relax the high access requirements (e.g. net asset and track record tests) for equity participation. The fact that bidders were still required to provide signed commitment letters from finance providers also meant that even though some qualification criteria might have been relaxed from the procurer's point of view, these would remain relevant as part of finance providers' due diligence processes.

The IPP office also started treating the market in a more nuanced way during this period. It, for example, ran a separate bid window (bid window 3.5) for CSP (Figure 1) and, in recognition of the value of these plants to the system, incentivised them to invest in storage by offering a 270% premium on their submitted "base price" during the afternoon peak (16:30–21:30). The office also launched the small IPP procurement programme (October 2013), with the aim of providing opportunities for smaller, South African firms that had not been able to qualify for or effectively compete in the previous bidding rounds. An expedited bidding window (1800 MW) was also launched after bid window 4 (Figure 1), which would allow previously unsuccessful bidders a second opportunity to bid (Eberhard and Naude 2016; Kruger and Eberhard 2018; Filipova *et al.* 2019) investment and price outcomes. Since 2011 a total of 6 328 Megawatts of wind, solar photovoltaic and other RE generation capacity has been procured, amounting to USD 20.5 billion in investment. Bid tariffs have fallen sharply over the course of the programme and the most recently awarded projects are amongst the lowest priced grid-connected RE projects in the world. Considering South Africa's success in achieving more investment via independent power

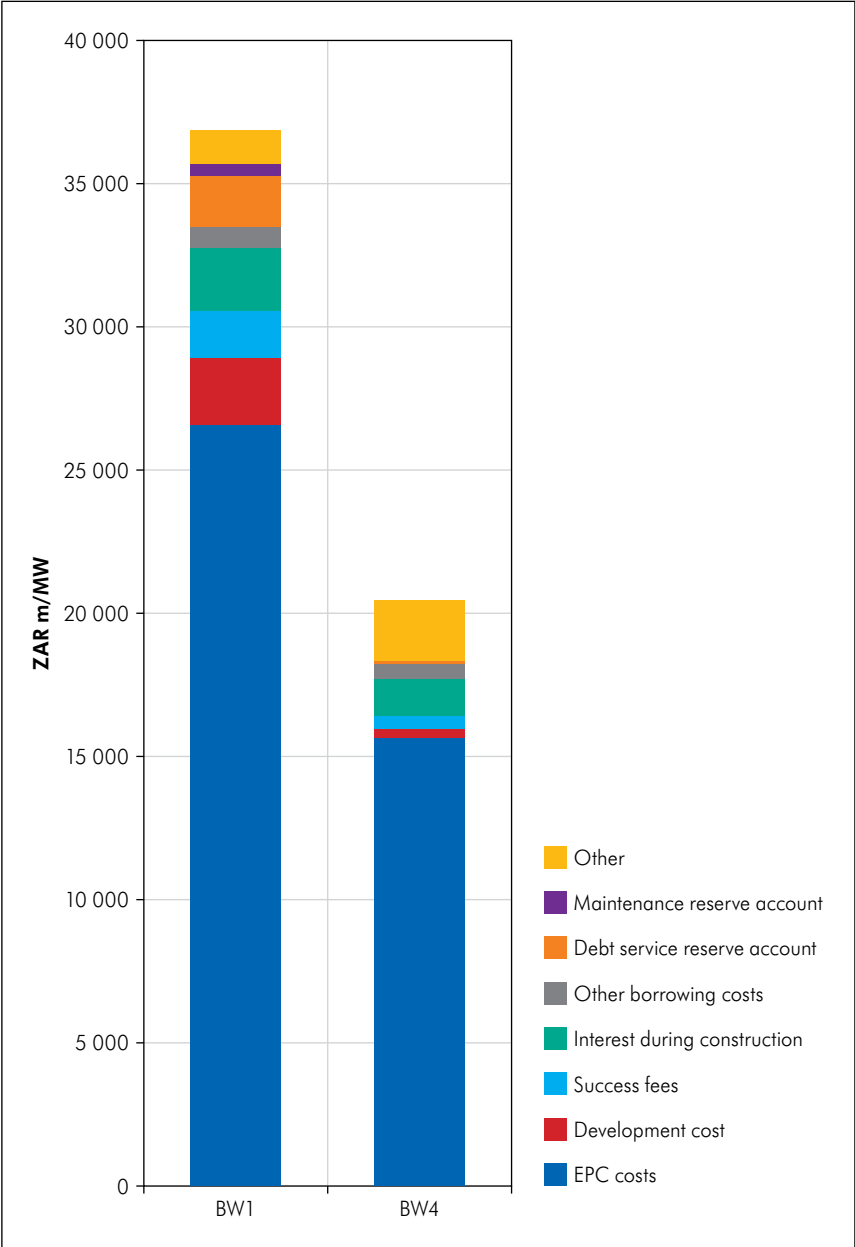
producers in four years than in the rest of Sub-Saharan Africa over the past 25, the REIPPPP's design and management is likely to be of interest to policymakers in African (and other developing) .

Whereas changes to the programme were incremental, changes in bidding behaviour and the market were more disruptive. Bid windows 3 and 4 saw large international utilities bidding much more aggressively and strategically. Larger companies started bringing more services and aspects of the value chain in-house to cut costs and margins. These companies were bidding multiple projects and using their international project portfolios to take advantage of both economies of scale and their stronger negotiating position relative to suppliers and service providers.

Large bidders were also able to use cheaper corporate finance (as opposed to more costly project finance) to lower the cost of capital (and thereby prices) on their bids (Steffen 2020; Kruger, Nygaard and Kitzing 2021). ENEL Green Power, for example, used corporate finance in its bid windows 3 and 4 bids, and after award negotiated better priced debt (on a project finance basis) with South African banks. Many developers had complained, up to this point, that the big five South African banks were simply too risk-averse and acting in a cartel-like fashion, offering almost identical loan terms and pricing. ENEL's approach broke this pattern and the relative power of the banking community in the programme – at least for their awarded projects. The result was that, while equipment costs had gone down considerably over this period, there were also notable reductions in margins across all project cost components (Figure 2).

As a result of these bidding strategies, continued reductions in equipment costs, an increase in competition and the maturation of the sector, the bid windows 3 and 4 auctions saw steep declines in pricing, especially for solar PV, but also for onshore wind. 93 projects (6023 MW) were submitted for bid window 3, 79 projects (5805 MW) for bid window 4 and 106 projects for the expedited bid window. 17 projects (1457 MW) were awarded in bid window 3, and the IPP office chose to double the number of awarded projects from 13 to 26 for bid window 4 (2205 MW). Average prices for onshore wind in bid window 3 fell to ZARc 74/kWh and ZARc 99/kWh for solar PV. In bid window 4 these prices fell to an average of ZARc 67/kWh for onshore wind and ZARc 82/kWh for solar PV. The average prices for the expedited round, although never “officially” announced, were even more dramatic at ZARc 62/kWh for both onshore wind and solar PV (Kruger and Eberhard 2018; Filipova *et al.* 2019). Inc. Sub-Saharan Africa desperately needs more electricity. Recent years have seen private investment in renewable energy projects breaking through in the region, primarily driven by well-designed and implemented auction programs. We review three renewable energy auction programs in the region to improve our understanding of the auction design and implementation elements that have enabled this important transition: the South African Renewable Energy Independent Power Producers Procurement Program (REIPPPP).

Figure 2: Project cost components, BW1 vs BW4



Source: (Author compilation based on bidding data from IPP office; Eberhard & Naude 2016)

Results from these rounds illustrate the power of competition, not merely for price discovery but also price reduction. These results also reflect the maturation of the sector, with stakeholders across the value chain (especially investors) learning and becoming increasingly comfortable with the technologies and associated risks. Loan terms, for example, increased from 15 to 17 years during this period, the internal rates of returns (IRRs) dropped to below 15% and project gearing ratios also increased over the rounds, with some projects in round 4 getting as much as 85% debt.

Period 3: Running to a standstill

The REI4P programme came to a grinding halt in 2016 when Eskom's CEO refused to sign the PPAs of projects awarded in bid windows 4 and 3.5 as well as those awarded under the small IPP programme (Figure 1). The results have been devastating for the programme, the sector and the country. The first shoots of a local renewable energy manufacturing industry have withered, as two wind tower factories, one inverter factory and several solar PV assembly plants closed down (GreenCape 2020). There were no new REI4P procurement rounds until 2021, despite South Africa experiencing its deepest and most disruptive episodes of loadshedding in 2018–2022. This situation also undermined investor confidence in South Africa as an investment destination and played some part in the country's loss of its investment-grade credit rating in 2018.

There are a number of political economy factors that have come into play to both disrupt and threaten the continued rollout of the programme. The period from 2015 onwards has seen growing opposition to IPPs and RE. Those that had benefited from other government procurement, many politically connected, felt excluded from the REI4P. Influence peddling and rent-seeking were much more difficult because of the transparency of the programme and where decisions on investment partners were made by the private sector. Those who had been benefitting from coal contracts with Eskom (the one mining sector that has seen significant entry by emerging local investors) and those who hoped to benefit from a future large nuclear programme became alarmed at the potential unbundling and break-up of Eskom and the increasing competitiveness and success of solar and wind.

Rounds 1 and 2 were more or less under the political radar, but then the volumes increased and prices were competitive. This spawned the growth of a number of lobby and activist groups and social media platforms organised against IPPs and RE, with relative success in promoting a counter-narrative that RE is expensive and is causing Eskom's financial demise. This opposition has been conflated with important, legitimate concerns about the loss of jobs in coal-mining regions, and the extent to which the RE programme has truly afforded South African

companies opportunities for participation (Filipova *et al.* 2019; Bowman 2020; Eberhard 2020; Fakir 2020; Morris *et al.* 2020).

Ending the stalemate between Eskom and the REI4P ultimately required a change of leadership at the highest level. Matters came to a head in 2017 when a coalition of labour (COSATU) and big business (BUSA and BLSA) mobilised against Jacob Zuma under the banner of rolling back state capture to carry Cyril Ramaphosa to victory in the ANC electoral conference (Bowman 2020) (Figure 1).

One of president Ramaphosa's first orders of business was to replace the entire Eskom board and senior management, as well as appoint a new minister of Finance (former reserve bank governor Tito Mboweni), a new minister of Public Enterprises (Pravin Gordhan) and a new minister of Energy (Jeff Radebe) (Figure 1) (Nicolaides and Motau 2018; Rabkin and Pather 2018). Minister Radebe ensured that the PPAs for preferred bidder projects from bid windows 3, 5 and 4 were finally signed in April 2018 (Gosling 2018), but at that stage the IPP office had already broken its non-negotiable contracting approach to increase projects' black economic empowerment (BEE) shareholding (at the expense of broader community-based shareholding) in an attempt to convince government to proceed with bid signing. Projects awarded as part of the expedited round and the small IPP programme were not included due to either being more expensive than the bid window 4 projects (in the case of the smalls) or bid validity periods having lapsed (in the case of the expedited round).

Apart from the broader impacts on South Africa's energy sector, this period also had several direct impacts on the IPP office. Eskom's refusal to sign the awarded PPAs precipitated a funding crisis for the programme. The fact that the IPP office did not form part of any official departmental budget meant that it was entirely financially dependent on development fees (1% of total project costs) paid by awarded projects on reaching financial close. But for the signing of bid window 4 PPAs in April 2018, the office – which was a couple of months away from not being able to pay salaries – would have had to close down. Another important factor was the departure two months after Malusi Gigaba's appointment of Andrew Donaldson, deputy director general with National Treasury, who had long been the chief supporter of the IPP programme (Figure 1) (Donnelly 2017). This moment signaled a shift in the IPP office's strategic relationships with government, as the Department of Energy quickly became the more prominent – albeit slightly more ambivalent – partner to the programme.

The weakening of National Treasury's direct support also had long-lasting impacts on the programme and its leadership. The first was an attempt in 2017 to move the IPP office to the tainted Central Energy Fund – a state institution originally set up as a sanctions-busting company by the apartheid regime and which has been at the centre of several controversial deals and State Owned Enterprise (SOE) management failures in the post-1994 period (Creamer 2017). The proposed move

was fiercely and ultimately successfully resisted by the IPP office leadership, but illustrated how the unofficial institutional setting that made it possible for the office to work in a flexible way was also a potential vulnerability (alongside the office's funding model). Finally, Karen Breytenbach's contract as head of the IPP office with DBSA was not renewed in June 2019 (Figure 1), a surprising move that was widely criticised by the industry (Creamer 2019; Mahlaka 2019; Watson 2020).

Despite the changes initiated by president Ramaphosa's administration, the future of the REI4P programme and South Africa's energy sector remains uncertain and highly contested. Jeff Radebe was replaced in May 2019 by former National Union of Mineworkers secretary general, and ANC chairman, Gwede Mantashe, an influential power broker in the ruling party (Figure 1) (Morris *et al.* 2020). Minister Mantashe oversaw the (re-)amalgamation of the Department of Energy with the Department of Mineral Resources, a prominent reassertion of the staying power and importance of the minerals energy complex. None of the minister or (new) department's announcements or decisions has provided much certainty to the renewables sector (Eberhard 2020).

Still, there are signs that the current hiatus may be broken.

South Africa's IRP was finally approved by cabinet and gazetted in 2019 (Figure 1), and although some coal and nuclear capacity was forced in through a "policy adjustment" process, it makes provision for large volumes of renewables, including 8.2GW solar PV and 17.7 GW of wind (Department of Mineral Resources and Energy 2019). A section 34 determination by the minister has since been approved by the national energy regulator for concurrence, after which the IPP office proceeded with round 5 of the programme (Creamer 2020).

Price results from bid window 5 have been the most competitive to date: 25 projects were awarded at average prices of ZARc 49.5/kWh for onshore wind and ZARc 42.9/kWh for solar PV. There are, however, concerns about the viability of these prices, as equipment costs have increased since bid awards and financial close deadlines have been missed. Bid window 5 also sparked further concern about market concentration, as the 25 projects were awarded to three bidder consortia. Further evidence of greater concentration among BEE shareholders (with one shareholder present in 21 of the 25 projects) and reduced margins is prompting further concerns about the ability of the programme to stimulate and maintain upstream socio-economic benefits.

Several utility-scale renewable energy tenders were launched by commercial and industrial entities (including mines) as well as municipalities in South Africa in 2019–2021, powerfully signaling industry's recognition of the inexorability of the transition (Mfobo 2019; Bellini 2020; Kew 2020).

The challenge now is to ensure that South Africa's transition from a coal-dominated power system to one built on low-cost renewables is just – meaning that the impacts on workers and communities currently dependent on the coal industry

are mitigated through a range of measures, including reskilling and building of renewable infrastructure in coal areas (Cruywagen, Swilling and Davies 2019; Bloom 2020). The IPP office's next challenge will be to identify and integrate transition measures into the programme's design, also being careful to recognise the limitations of a procurement programme to carry and drive macro-level policy agendas in isolation. The country also needs to come to terms with the realities of managing such a transition amid a power sector reform process, with important decisions around the role and home of procurement in future energy markets still not having been made.

CONCLUSION

A key auction implementation aspect highlighted by the South African case is the impact of developing and maintaining bidder and investor trust to drive competition and lower project prices. Competition levels ultimately depend on multiple "decisions to bid" by many potential bidders – some stronger or weaker than others – influenced not only by the auction's design, but also by bidders' risk perceptions of the bidding process and implementing institution, as well as the broader investment context (Zitron 2006; Saussier, Staropoli and Yvrande-Billon 2009; Chiu, Huang and Yen 2010). Investors' risk perceptions also influence the cost of capital (Vaaler, James and Aguilera 2008; Hatzilambros 2016), which – in combination with the competition levels in the programme – played an important role in determining the price levels of awarded projects.

Many of South Africa's IPP office's auction design and implementation decisions were not only motivated by their established theoretical auction impacts, but also by whether they sent the right kind of trust-building and risk mitigating signals to the market – something that most auction design literature does not take into account when discussing auction design. In a certain sense this represents a novel application of signaling theory in an auction context, which is normally concerned with how and what bidders signal to other bidders or possible investors with their bidding behaviour (Levy and Lazarovich-Porat 1995; Connelly *et al.* 2011), and not what the auctioning authority is signaling to the market. This also builds on the work done by public administration and policy scholars on South Africa's programme (Nel 2013, 2014; Eberhard, Kolker and Leigland 2014; Kolker 2014; Montmasson-Clair and Ryan 2014) by locating many of the factors highlighted in these analyses in relation to how they influence bidder trust.

The IPP office's approach to the design and implementation of REI4P was to a large degree shaped by an emphasis on building and maintaining trust, with the ultimate goal of reducing perceived risk levels in the long run. In both its conception and initial execution, REI4P emphasised stability, predictability and respect

for the “rules of the game”. Considerable resources were invested in ensuring that the bidding process was secure, transparent and well structured.

Finally, the reputation of the IPP office – with its proven track record in terms of ability, integrity and a favourable view of the private sector, all factors that support trust building (Mayer, Davis and Schoorman 1995; Zitron 2006) – played a pivotal role in securing and maintaining the private sector’s trust. The South African auction designers also understood the importance of equity and debt providers to the ultimate success of the auction programme and consequently placed their concerns at the centre of the auction process.

The willingness of major international developers to invest in the programme, the convergence of average project prices in later rounds with price levels in lower risk investment contexts and the fact that all projects reached financial close in reasonable time frames (Kitzing *et al.* 2021) speaks to the market’s growing levels of trust in the programme. These compelling results – increasing competition, lower prices, high realisation rates – support this trust-emphasising focus of the auction implementation process.

This case also shows how South Africa’s political economy impacted IPP investment sustainability. While Del Rio (Del Río *et al.* 2015; Del Río 2017) includes the “socio-political” feasibility of an auction as both an auction outcome and possible determinant of future auction success – influenced by prices, realisation rates and local impacts – this case has shown how political economy forces beyond the control of the auction process can ultimately derail these investments. Surprisingly, this case illustrates how the auction outcomes normally expected to support sociopolitical feasibility – low prices, large investments, rapid realisation rates, local (socio-) economic impacts, increased foreign direct investment – might be so disruptive that they threaten existing political and economic interests. As a consequence, policy certainty and predictability has been undermined as REI4P procurement was halted and resulted in the erosion of much of the hard-won trust that is so important to the programme’s success.

The preceding analysis has shown that it is crucial for the success of future auction programmes that the country provides policy certainty and maintains the market’s trust. This is not only important for maintaining high competition levels and strong investor interest from experienced developers, but is essential for achieving the country’s renewable energy manufacturing localisation goals. This requires a renewed commitment to the factors that underpinned the programme’s initial success, underpinned by a robust energy policy framework and strengthened institutional capacity and autonomy of the IPP office.

A country’s political economy – the structural relationships between interests, power relations, institutions and ideas that shape processes and outcomes (Godinho 2018) – is a cross-cutting issue that impacts the entire sector, including the investment climate, the policy and regulatory framework as well as the

planning-procurement nexus. The case study analysis highlighted the pervasive influence of political economy factors on the evolution and outcomes of the auction programme. Political-economy literature and analysis were explicitly excluded from the scope of this article but the preceding discussion suggests that it might be a fruitful area of research in the future to better understand the influence of these factors on both short- and long-term auction processes and outcomes.

REFERENCES

- Athias, L. and Saussier, S. 2007. Contractual flexibility or rigidity for Public Private Partnerships? Theory and evidence from infrastructure concession contracts. (Unpublished paper). Paris: University of Paris Sorbonne, France. doi: 10.2139/ssrn.828944.
- Babbie, E.R. and Mouton, J. 2001. *The practice of social research*. Cape Town: Oxford University Press.
- Bellini, E. 2020. South African chemicals producer seeks to procure 600 MW of renewables. *PV Magazine International*. 18 March. Available at: <https://www.pv-magazine.com/2020/05/18/south-african-chemicals-producer-seeks-to-procure-600-mw-of-renewables/>. (Accessed on 18 March 2020).
- Bloom, K. 2020. Just transition, redux–Cosatu’s bid to save Eskom, the climate and South Africa. *Daily Maverick*, 28 January. Available at: <https://www.dailymaverick.co.za/article/2020-01-28-just-transition-redux-cosatus-bid-to-save-eskom-the-climate-and-south-africa/>. (Accessed on 29 January 2020).
- Bowman, A. 2020. Parastatals and economic transformation in South Africa: The political economy of the Eskom crisis. *African Affairs*. 119(476):395–431. doi: 10.1093/afraf/adaa013.
- Chiu, C.M., Huang, H.Y. and Yen, C.H. 2010. Antecedents of trust in online auctions. *Electronic Commerce Research and Applications*. 9(2):148–159. doi: 10.1016/j.elerap.2009.04.003.
- Colquitt, J.A. 2001. On the dimensionality of organizational justice: A construct validation of a measure. *Journal of Applied Psychology*. 86(3):386–400. doi: 10.1037/0021-9010.86.3.386.
- Connelly, B.L., Certo, S.T., Ireland, R.D. and Reutzel, C.R. 2011. Signalling theory: A review and assessment. *Journal of Management*. 37(1):39–67. doi: 10.1177/0149206310388419.
- Creamer, T. 2017. IPP Office to be incorporated into overhauled Central Energy Fund. *Engineering News*. 19 May, pp. 1–4. Available at: <http://www.engineeringnews.co.za/article/ipp-office-to-be-incorporated-into-overhauled-central-energy-fund-2017-05-19>. (Accessed on 19 May 2017).
- Creamer, T. 2019. Respected head of IPP Office, Breytenbach, asked to leave post, *Engineering News*. pp. 1–5. Available at: <https://www.engineeringnews.co.za/article/respected-head-of-ipp-office-breytenbach-asked-to-leave-her-post-2019-07-22>. (Accessed on 22 June 2019).
- Creamer, T. 2020. Next renewables bidding round to be launched only in second quarter of 2021. *Engineering News*, 23 June. Available at: <https://www.engineeringnews.co.za/article/next-renewables-bidding-round-to-be-launched-only-in-second-quarter-of-2021-2020-06-23>. (Accessed on 23 June 2020).

- Creswell, J.W. and Poth, C.N. 2018. *Qualitative enquiry and research design: choosing among five approaches*. Fourth. London: SAGE.
- Cruywagen, M., Swilling, M. and Davies, M. 2019. R6-billion: First estimate of Just Transition in South Africa. *Daily Maverick*, 10 December. Available at: <https://www.dailymaverick.co.za/article/2019-12-10-r6-billion-first-estimate-of-just-transition-in-south-africa>. (Accessed on 10 December 2019).
- Department of Energy. 2014. *Request for qualification and proposals for new generation capacity under the IPP procurement programme. Volume 3: Technical requirements*. (Tender no: DOE/003/13/14) Updated for the Fourth Bid Submission Date: May 2014. Republic of South Africa. Available at: https://www.tlanelo.com/RFP_PART_A-GENERAL_REQUIREMENTS_RULES-BW3.pdf. (Accessed on 15 July 2015).
- Department of Mineral Resources and Energy 2019. *Integrated Resource Plan (IRP2019)*. Tshwane: Republic of South Africa. Available at: <http://www.energy.gov.za/IRP/2019/IRP-2019.pdf>. (Accessed on 14 February 2020).
- Donnelly, L. 2017. Andrew Donaldson leaves national treasury after more than 20 years. *Mail & Guardian*, 8 May, pp. 5–9. Available at: <https://mg.co.za/article/2017-05-08-andrew-donaldson-quits-treasury-after-more-than-20-years/>. (Accessed on 15 July 2020).
- Eberhard, A. 2014. Feed-in tariffs or auctions? Procuring renewable energy supply in South Africa. *Energize RE: Renewable Energy Supplement*, June 2014. Pp. 36–38.
- Eberhard, A., Gratwick, K., Morella, E. and Antmann, P. 2017. Independent Power Projects in Sub-Saharan Africa: Investment trends and policy lessons. *Energy Policy*. 108:390–424. doi: 10.1016/j.enpol.2017.05.023.
- Eberhard, A. 2020. New coal and nuclear power proposals undermine prospects of a post-Covid-19 economic recovery. *Daily Maverick*. 17 June. Available at: <https://www.dailymaverick.co.za/article/2020-06-17-new-coal-and-nuclear-power-proposals-undermine-prospects-of-a-post-covid-19-economic-recovery/>. (Accessed on 17 June 2020).
- Eberhard, A., Kolker, J. and Leigland, J. 2014. *South Africa's renewable energy IPP Procurement Program: Success factors and lessons*. Public-Private Infrastructure Advisory Facility (PPIAF), World Bank. Available at: <https://www.gsb.uct.ac.za/files/ppiafreport.pdf>. (Accessed on 23 July 2015).
- Eberhard, A. and Naude, R. 2016. The South African Renewable Energy Independent Power Producers Procurement Programme (REIPPPP): A review and lessons learned. *Journal of Energy in Southern Africa*. 27(4):14. doi: <http://dx.doi.org/10.17159/2413-3051/2016/v27i4a1483>.
- Eskom Holdings 2014. *Power Purchase Agreement–Photovoltaic Projects: The seller and Eskom holding SOC limited*. Appendix K2: Available at: <https://static.pmg.org.za/RNW-509-2017-04-03-Anex-B.pdf>. (Accessed on 14 July 2019).
- Fakir, S. 2020. The political economy of South Africa's energy transition for outsiders. *Daily Maverick*, 22 June. Available at: <https://www.dailymaverick.co.za/opinionista/2020-06-22-the-political-economy-of-south-africas-energy-transition-for-outsiders/>. (Accessed on 23 June 2020).
- Filipova, A. et al. 2019. *South Africa country report*. (Report 6: Energy and Economic Growth Research Programme). Cape Town: Power Futures Lab, Graduate School of Business, University of Cape Town. doi: 10.1007/3-540-26722-0_20.

- Godinho, C. 2018. *Political economy of power sector reform*. Cape Town: [Lecture presentation], 'Managing power sector reform and regulation in Africa'. Cape Town: Graduate School of Business, University of Cape Town.
- Godinho, C. and Hermanus, L. 2018. *(Re)conceptualising state capture: With a case study of South African power company Eskom*. Conference Paper prepared for the Public Affairs Research Institute's 'State Capture and Its Aftermath: Building Responsiveness Through State Reform', 22 -24 October, Johannesburg. Available at: <http://www.gsb.uct.ac.za/files/ReconceptualisingStateCapture.pdf>. (Accessed on 14 January 2019).
- Gosling, M. 2018. Renewable energy burst: Jeff Radebe to sign off wind and solar power projects after a two-year hiatus. *Daily Maverick*, 9 March. Available at: <https://www.dailymaverick.co.za/article/2018-03-09-renewable-energy-burst-jeff-radebe-to-sign-off-wind-and-solar-power-projects-after-a-two-year-hiatus/>. (Accessed on 9 December 2018).
- Hatzilambros, C. 2016. *Determinants of the cost of credit for project finance debt in Africa*. (MCom in Financial Management) Cape Town: University of Cape Town. Available at: <https://open.uct.ac.za/handle/11427/22839>. (Accessed on 15 July 2018).
- Haufe, M.-C. and Ehrhart, K. 2018. Auctions for renewable energy support – Suitability, design, and first lessons learned. *Energy Policy*. 121:217–224. doi: 10.1016/j.enpol.2018.06.027.
- Kew, J. 2020. AB InBev turns to solar power as South Africa suffers blackouts. *Bloomberg*, 17 January, Pp. 6–8. Available at: <https://www.bloomberg.com/news/articles/2020-01-17/ab-inbev-turns-to-solar-power-as-south-africa-suffers-blackouts>. (Accessed on 18 January 2020).
- Kitzing, L., Anatolis, V., Fitch-Roy, O., Klessmann, C., Kreiss, J., Del Rio, P., Wigand, F. and Woodman, B. 2019. Auctions for renewable energy support: Lessons learned in the AURES project. *IAEE Energy Forum*, p. Third Quarter, 11–14. Available at: <https://www.iaee.org/en/publications/newsletterdl.aspx?id=809>. (Accessed on 21 January 2020).
- Kitzing, L., Khan, B.S., Nygaard, I. and Kruger, W. 2021. Worth the wait: Why South Africa's renewable energy auctions are slower but more effective than Europe's. *Energy Policy (Submitted)*.
- Kitzing, L. and Weber, C. 2015. Support mechanisms for renewables: How risk exposure influences investment incentives. *International Journal of Sustainable Energy Planning and Management*. 7:117–134. doi: 10.5278/ijsepml.2015.7.9. (Accessed on 13 January 2018).
- Kolker, J. 2014. Developing Renewable Energy through an Independent Power Producer Procurement Program. In Alam, A., Mokate, R., Plangemann, K.A. (Eds.) *Making it happen: selected case studies of institutional reforms in South Africa*. World Bank Publications, Washington D.C. Pp. 179–194.
- Kruger, W. and Eberhard, A. 2018. Renewable energy auctions in sub-Saharan Africa: Comparing the South African, Ugandan, and Zambian programs. *WIRES: Energy and Environment*. Advanced online, e295. Pp 1–13. doi: 10.1002/wene.295. (Accessed on 19 January 2019).
- Kruger, W. and Eberhard, A. 2023. The impact of competition , trust and capital on Renewable Energy Auction Outcomes in sub-Saharan Africa : Analysing auctions in South Africa, Zambia and Namibia. *Energy Policy (Submitted)*.
- Kruger, W., Nygaard, I. and Kitzing, L. 2021. Counteracting market concentration in renewable energy auctions: Lessons learned from South Africa. *Energy Policy*. 148(PB): 111995. doi: 10.1016/j.enpol.2020.111995.

- Kumar, R. 2011. *Research methodology : a step-by-step guide for beginners*. Thousand Oaks. SAGE.
- Leigland, J. 2018. Public-private partnerships in developing countries: The emerging evidence-based critique. *World Bank Research Observer*. 33(1):103–134. doi: 10.1093/wbro/lkx008. (Accessed on 19 June 2019).
- Levy, H. and Lazarovich-Porat, E. 1995. Signaling theory and risk perception: An experimental study. *Journal of Economics and Business*. 47(1):39–56. doi: 10.1016/0148-6195(94)00022-6. (Accessed on 14 June 2019).
- Lucas, H., Del Río, P. and Sokona, M.Y. 2017. Design and assessment of renewable electricity auctions in sub-Saharan Africa. *IDS Bulletin*. 48(5):79–100. Available at: <https://www.jstor.org/stable/4006908>. (Accessed on 13 February 2018).
- Mahlaka, R. 2019. Ousting of renewable energy head shrouded in mystery. *Daily Maverick*, 25 July. Pp. 1–7. Available at: <https://www.dailymaverick.co.za/article/2019-07-25-ousting-of-renewable-energy-head-shrouded-in-mystery/#gsc.tab=0> 1/7. (Accessed on 26 July 2019).
- Martin, B. and Winkler, H. 2014. *Procurement models applied to Independent Power Producer Programmes in South Africa*. (ERC Research Paper). Cape Town: Energy Research Centre, University of Cape Town. doi: 10.13140/2.1.1633.4721. (Accessed on 15 July 2015).
- Mayer, R.C., Davis, J.H. and Schoorman, F.D. 1995. An integrative model of organisational trust. *Academy of Management Review*. 20(3):709–734. Available at: <http://www.jstor.org/stable/258792>. (Accessed on 19 July 2018).
- Mfobo, V. 2019. PPC turns to solar as power cuts bite. *Concrete Trends*, September. Available at: <https://www.concretetrends.co.za/news/ppc-turns-to-solar-as-power-cuts-bite>. (Accessed on 20 October 2019).
- Milgrom, P.R. 2004. *Putting auction theory to work*. Cambridge: Cambridge University Press.
- Mokhele, K. 2019. *Political risk analysis of the Renewable Energy Sector in South Africa and the effect on Foreign Direct Investment*. Cape Town: University of Cape Town. Available at: https://open.uct.ac.za/bitstream/item/8977/thesis_sci_2014_kaaya_lt.pdf?sequence=1. (Accessed on 18 December 2020).
- Montmasson-Clair, G. and Ryan, G. 2014. Lessons from South Africa's renewable energy regulatory and procurement experience. *Journal of Economic and Financial Sciences*. 7(4):507–526. doi: 10.4102/jef.v7i4.382.
- Morris, M., Robbins, G., Hansen, U.E. and Nygaard, I. 2020. *Energy and industrial policy failure in the South African wind renewable energy global value chain: The political economy dynamics driving a stuttering localisation process*. Cape Town: (Working Paper 2020-3), Policy Research on International Services and Manufacturing. Cape Town: University of Cape Town.
- Nel, D. 2013. Risk governance in public private partnerships. *Administratio Publica*. 21(4):8–28.
- Nel, D. 2014. International Best Practice in Public Private Partnerships and Risk Management. *Administratio Publica*. 22(2):46–67.
- Nicolaides, G. and Motau, K. 2018. Ramaphosa makes major changes at Eskom. *Eyewitness News*, 21 January. Available at: <https://ewn.co.za/2018/01/21/ramaphosa-makes-major-changes-at-eskom>. (Accessed on 21 July 2019).

- Rabkin, F. and Pather, R. 2018. New deputy president, finance minister announced in major Cabinet reshuffle. *Mail & Guardian*, 26 February.
- Del Río, P., Kitzing, L., Haufe, M.C., Wigand, F., Wellisch, M., Steinhilber, S., and Fitch-Roy, O. 2015. *A methodological note on the links between components for the assessment of design elements in auctions for RES*. (Report D2.2(c)) Aures Project.
- Del Río, P. 2017. Designing auctions for renewable electricity support. Best practices from around the world. *Energy for Sustainable Development*. International Energy Initiative. 41:1–13. doi: 10.1016/j.esd.2017.05.006.
- Rubin, H.J. and Rubin, I.S. 2012. *Qualitative interviewing: The art of hearing data*. (3rd ed.). Thousand Oaks: SAGE.
- Saunders, M., Lewis, P. and Thornhill, A. 2016. *Research methods for business students*. Essex: Pearson Education Limited.
- Saussier, S., Staropoli, C. and Yvrande-Billon, A. 2009. Public-private agreements, institutions, and competition: When economic theory meets facts. *Review of Industrial Organization*. 35(1–2):1–18. doi: 10.1007/s11151-009-9226-z.
- Steffen, B. 2020. Estimating the cost of capital for renewable energy projects. *Energy Economics*. 88:104783. doi: 10.1016/j.eneco.2020.104783.
- Vaalor, P.M., James, B.E. and Aguilera, R.V. 2008. Risk and capital structure in Asian project finance. *Asia Pacific Journal of Management*. 25(1):25–50. doi: 10.1007/s10490-007-9045-4.
- Watson, A. 2020. “Ousting” of IPP Office head Karen Breytenbach raises eyebrows. *The Citizen*, 25 July.
- Weiss, R.S. 1994. *Learning from strangers: The art and method of qualitative interview studies*. New York, NY, US: Free Press.
- Yin, R.K. 2003. *Case study research : design and methods*. Los Angeles: SAGE.
- Yin, R.K. 2018. *Case study research and applications: Design and methods*. Sixth. Los Angeles: SAGE.
- Zitron, J. 2006. Public-private partnership projects: Towards a model of contractor bidding decision-making, *Journal of Purchasing and Supply Management*. 12(2):53–62. doi: 10.1016/j.pursup.2006.04.002.

AUTHOR'S CONTACT DETAILS

Dr Wikus Kruger

Power Futures Lab, West Turret, Graduate School of Business
Breakwater Campus
University of Cape Town
South Africa
Email: Wikus.kruger@uct.ac.za; jlkrgr@gmail.com
Tel: +27744668834

Transition from Paper-based Analogue Government to Sustainable Digital Governance Outcomes in Africa

F Cloete

Department of Public Administration and Management
University of the Free State

ABSTRACT

Digital government has proven to be more cost-effective, efficient and sustainable than traditional analogue governance practices. However, many lesser developed societies are not ready for digital transformation, as many failed transitions have so far illustrated. This article comprises a summary of a qualitative inquiry into the latest international research findings and conclusions about the current state of digital governance transformation in Africa, and how such approaches and practices on the continent can be improved. It identifies and clarifies the main issues that promote readiness for and the success of digital governance transformation. The research found that specific strategic transformative initiatives and priorities are crucial to create the minimum readiness conditions for digital transformation of society and government. Appropriate digitisation strategies can improve the potential for successful migration of public sector management and governance systems from unsustainable analogue systems to more sustainable digital platforms. This can also facilitate organisational and societal cultures that are more in congruence with the requirements of complex new 4IR conditions.

INTRODUCTION

Over the first two decades of the new millennium the increasing transition from paper-based, analogue management and governance systems to digital management and governance systems has proven to be more cost-effective, efficient and sustainable than traditional analogue practices (e.g. MacLean & Titah 2021; Pratt

2021). However, many lesser developed societies are not yet ready for digital transformation, as many failed transitions have also proven (e.g. Davenport & Westerman 2018; Lang *et al.* 2022; Lues 2021:179). Appropriate digital management and governance project interventions and priorities are increasingly regarded as imperative to initiate, lead, monitor and manage successful transitions of this nature. How this might be achieved in the best way, is, however, not always fully understood.

This article attempts to identify, clarify, assess and overcome the main constraints on fundamental digital transformation within an African context. The nature of societal and digital transformations is summarised, followed by an assessment of the latest research findings and conclusions about the current state of digital government in Africa. This is followed by a summary and assessment of the main strategic requirements for successful digital transformation of public services. The article concludes with evidence-informed recommendations by different digital governance specialists about how specific strategic digital transformation approaches, strategies and practices, can optimise the potential for successful digital transformation programmes from unsustainable analogue to more sustainable digital governance systems in Africa. These proposals include the role of dedicated and authoritative visionary and democratic leadership.

NATURE AND MANAGEMENT OF GENERAL FUNDAMENTAL SOCIETAL TRANSFORMATION

Societal and organisational transformation does not refer to small changes, refinements or improvements to the way things have been done in the past. It implies deep fundamental changes to the nature, structure, functioning and values of a society or organisation (Feola 2015; Kvint 2023; Linnér *et al.* 2020; Rabie 2013). The replacement of colonial settler rule in Africa with fully independent indigenous governments, brought about significant changes in the institutions, processes and values in those states, although some deep-seated colonial trappings still persist (Cloete 2018a).

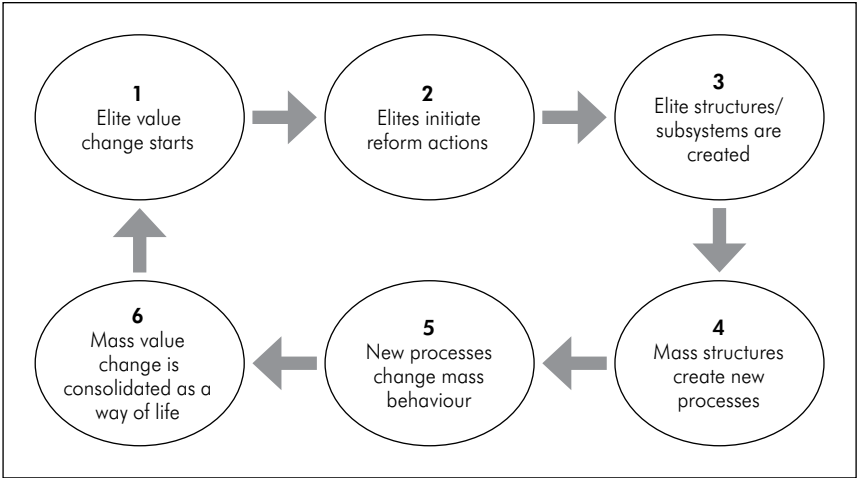
In South Africa, the restrictive, closed apartheid government, dominated by the descendants of successive European colonial settler regimes since 1952, was replaced in 1994 with a new inclusive, indigenous democratic government. This post-apartheid government was based on new governing institutions (policymaking bodies, administrative departments, courts of law). It was staffed by new indigenous elites who were formerly excluded from governing but took power in 1994 and started to govern the country via new political governance processes to achieve totally different goals, including promoting full and equal human rights for all citizens according to a liberal written constitution and prohibiting former

racially discriminatory practices (Cloete 2018b:305–308). Similar fundamental transformations occurred in virtually all colonial states when they became independent from their colonial occupying powers.

Democratic organisational or societal transformation is a complex process which must be initiated by government and societal thought leaders and managed well in order to succeed. Figure 1 visually explains and illustrates this general dynamic in society. It indicates how individual thought and opinion leaders play an important role in initiating changes, starting out with normative changes in the minds and views of specific individuals, followed by behavioural changes and some structural changes in elite/leadership institutions. These influences seep through to the rest of their followers in the organisation or society at large through their strategic motivation, intervention initiatives, guidance and monitoring activities, gradually transforming organisational institutions, behaviour and (hopefully), eventually, culture and values at general mass levels, into more appropriate reflections of the new status quo. Once cultural/value transformation has occurred at most mass (community or employee) levels, and is widely accepted, transformational change has been consolidated. If full and durable value internalisation does not take place both at leadership and follower levels, consolidation of transformational change will fail. The consolidation of transformational change therefore leads to the creation of new organisational or societal cultures that are more in congruence with new realities that the organisation or society faces (Cloete 2018b:304).

Organisational culture refers to the way in which collectives of people generally think and act (WEF 2021:9). It is the relatively consistent and coherent prevailing

Figure 1: Transformational change in society



Source: (Cloete 2018b:304)

mental and action frameworks that are widely accepted by most of the members of an organisation, community or society. These transformation processes are difficult to achieve in current volatile, uncertain, complex and ambiguous (VUCA) systems (Baran & Woznyj 2021:3), and normally require dedicated, systematic top-down interventions from leaders and managers promoting or driving such transformations.

Several transformation constraints must be overcome for this purpose. Baran and Woznyj (2021:3) identify the main managerial constraints on fundamental transformation that must be overcome in VUCA contexts as "...inertia of the status quo; time and organisational design barriers; unresponsiveness to customer needs; growing pains; slow communication and bureaucracy, and outdated processes". Traditional, evidence-informed change management approaches and strategies (e.g. Errida & Lotfi 2021; Philips & Klein 2023; Rousseau & Ten Have 2022) need to be considered and applied to overcome fundamental transformation constraints and resistance. Baran and Woznyj (2021:3) recommend the following agile management approaches and practices to overcome transformation obstacles: "communication and transparency; knowledge sharing and teamwork; fostering agile behaviour, iterative improvement and learning; deep customer focus; leadership and role modelling, and strategic alignment and managing talent" (see also Delioğlu & Uysal 2022). Braams, Wesseling, Meijer and Hekkert (2023), Phillips and Klein (2023) and Rahman, *et al.* (2023) took many of these ideas further and developed detailed transition agendas for transformation-oriented public officials to overcome and manage different forms of resistance against organisational change. These transition agendas are also relevant for digital transformation of society (e.g. EU 2019a; Korachi & Bounabat 2020).

NATURE AND MANAGEMENT OF DIGITAL TRANSFORMATION

Digital transformation is a specific type of societal transformation. It refers in general to a move from a paper-based analogue system of operation and management by an individual or group to a largely paperless electronic/digital way of operating. Digital transformation of the public sector implies that traditional, paper-based approaches to the provision of public services and other governmental functions are substantially replaced by digital strategies, interactions and processes. UNDESA (2022:xxiv) concluded that: "Digital government has reached a critical point. It is no longer a stand-alone or auxiliary tool, nor does it represent a panacea for government deficiencies or inefficiencies; it should be seen as an integral and thoroughly integrated aspect of the physical functioning of public institutions and services delivery. Digital development is inexorable, and inaction or the wrong action can be costly (in terms of missed economic and social

development opportunities) and deepen risks (in particular those linked to cyber-security and privacy issues)".

Digital transformation must be distinguished from digitisation and digitalisation (EU 2019a:10). **Digital transformation** of society comprises "fundamental changes in the organizational structure of...(an) ecosystem...as a whole by optimal integration of traditional processes and digital technologies with their gradual implementation at all levels" (Morze & Strutynska 2021:5). This fundamental restructuring of society takes place through **digitisation** (the technical conversion of "analog data streams into digital format"), and **digitalisation** (the "process of transformation and/or improvement of enterprise activities, business models, business functions, communications, use of online platforms, training and retraining of staff to work in new conditions", through the widespread application of digital technologies and data) (Morze & Strutynska 2021:5).

Transforming a traditional 20th century paper-based organisational and work culture into a new 21st century digital organisational culture in the public service and in its society in general, also needs dedicated government intervention (Misuraca *et al.* 2020; UNDESA 2022:xxv). This is not an easy task because it requires ensuring that government itself is ready and has the capacity to guide or steer this transition process. Digital transformation in other organisations, communities and societies implies similar fundamental changes to the way those agencies must function in future, and bringing especially communities and individuals who are not digitally literate, both within the public service and in society in general, successfully into the information era.

The fundamental changes in banking and other commercial services since the development of automated teller machines (ATMs) and internet access to such services, illustrate the far-reaching digital transformation of those services. Instead of physically having to visit a bank or a shop to select goods for purchase, one can now get information, interact with suppliers, order goods for delivery and pay electronically in different ways to conclude transactions. Public sector examples of current and envisaged fundamental future changes in government functions and services include the choice to apply and pay electronically for certain approvals or licences instead of visiting a government agency personally for this purpose, and having the required products delivered to you in many cases. There is also an exponential global increase in the replacement of traditional printed ID and passport documents, paper-based census surveys and elections, with digital documents, more digital interactions with various government agencies, digital census surveys and digital voting via electronic devices such as computers, mobile phones and other digital access points to various public sector services.

Other illustrations include fundamental changes to the security industry where physical patrols are increasingly being replaced with security cameras and the automatic recording and availability of cutting-edge data and information about

societal activities. The latest development in this regard is the exponential escalation in popularity of artificial general intelligence (AGI) systems. They have the potential to significantly improve the efficiency, effectiveness and productivity of individual and policy performances to improve societal issues and problems through evidence-informed lessons learnt, if these technologies are appropriately utilised and regulated (Fields & Huesig 2019).

STATE OF DIGITAL GOVERNMENT GLOBALLY AND IN AFRICA

Ducass (2017) reviewed and assessed the development of digital government in Africa, while the latest 2022 UNDESA E-Government Survey that assessed countries' digitisation strengths, challenges, opportunities and how their progress has also informed their policies and strategies, found that progress has been made across all regions, even in the least developed countries. The survey assessed the nature of digital services, status of electronic infrastructure and staff. The findings concluded that a "growing number of countries are moving towards seamless, invisible government in which fully automated and personalized services are made accessible to anyone anytime from anywhere...(w)ith the increased focus on cognitive government, agile and adaptive government, and the development of predictive capabilities, governments are setting themselves up to better anticipate and respond to the needs of all members of society" (UNDESA 2022:xxx).

However, "in Africa, 50 out of 54 countries (home to 95 per cent of the region's population) have EGDI values below the global average" (UNDESA 2022:xxix). African countries are generally among the lowest scores. E-government in Africa is regarded as still generally very weak, mainly because of different cumulative resource, technological, cultural, social, financial and political policy governance constraints. Most African countries crowd the bottom of the 2022 UN e-government index.

An earlier comprehensive EU (2019b) report classified African countries in **three** main groups on the basis of their respective levels of e-governance maturity:

- 12 countries were the most advanced, with many digital platforms, services, and regulatory frameworks. They can be regarded as digital front-runners. They include Botswana, Cape Verde, Egypt, Ghana, Kenya, Mauritius, Morocco, Namibia, Rwanda, Seychelles, South Africa, Tunisia.
- Another 26 countries have made some progress towards digital governance maturity, but have so far not reached the same level as the first group. The best of these includes Algeria, Lesotho and Nigeria (EU 2019b:37).
- The last 16 countries all exhibit high degrees of political conflict and poverty. They lag far behind the leading countries in Africa. They include Burundi,

Central African Republic, Chad, Congo, Democratic Republic of Congo, Djibouti, Eritrea, Equatorial Guinea, Guinea, Guinea Bissau, Malawi, Niger, Sierra Leone, Somalia, South Sudan (EU 2019b:8).

The 2019 EU report concluded that most African states have already started with digitisation of their public records and services but that progress is difficult and slow (EU 2019b:7). Fast internet access is reasonable in North and southern Africa but weak in Central Africa. Mobile technologies are, however, widespread throughout the continent. Different forms of resistance to the transition to digitisation is a major obstacle and complicates digital transformation in all sectors (EU 2019b:8). Progress is very uneven among and within African countries, where small, isolated pockets of high technology are frequently surrounded by large areas where no digital facilities and services are available.

South Africa was an early adopter of e-government at the beginning of the 21st century and regarded as one of the leaders in this regard in developing countries (Cloete 2012). Since then, it has built up one of the best ICT backbones and local content on the road to e-government in Africa. However, the coordinated integration of digital technologies into existing public governance processes in the country has so far been weak.

The most important constraints on South Africa's e-government programme include a serious lack of e-governance leadership, coordination and integration in the absence of a coherent e-governance policy and sectoral strategies for about two decades until 2017. This situation resulted in inadequate prioritisation of resources and consequently in inadequate ICT education and training of specialists as well as of the general public, and also inadequate ICT infrastructure to escalate South Africa's e-government maturity level from its current enhanced presence to the next, transactional level. It boils down to half-hearted and bad management and governance of the programme at national governmental level.

The African Union (AU) (2020:6) adopted a comprehensive digital transformation strategy for 2020–2030, based on the principles of general solidarity and cooperation, comprehensive integration to avoid silos and promote sustainability, digitally inclusive and transformative, homegrown solutions that require new mindsets and take full cognisance of the opportunities and risks involved. The strategy is structured as follows:

The 2020 AU digital transformation strategy comprises a comprehensive framework of action that envisages full digital governance by 2030. Its main objective is: "to harness digital technologies and innovation to transform African societies and economies to promote Africa's integration, generate inclusive economic growth, stimulate job creation, break the digital divide, and eradicate poverty for the continent's socio-economic development and ensure Africa's ownership of modern tools of digital management" (AU 2020:2).

Table 1: The digital transformation strategy for Africa (2020–2030)

foundation pillars	Enabling environment, policy and regulation, digital infrastructure, digital skills and human capacity, digital innovation and entrepreneurship
critical sectors	Digital industry, digital trade and financial services, digital government, digital education, digital health, digital agriculture
cross cutting themes	Digital content and applications, digital ID, emerging technologies, cybersecurity, privacy and personal data protection, research and development

Source: (Adopted from AU 2020:6)

This African digital transformation agenda is ambitious, because of many political, economic and social constraints that have so far delayed significant progress towards sustainable development in Africa in the information era. The main reasons for this situation are summarised in Cloete and Mmakola (2018). Ndlovu, Ochara and Martin (2022) emphasised the influence of digital government innovation on transformational government in resource-constrained contexts, while Mutula (2013) also assessed the implications of digital transformation in sub-Saharan Africa.

DIGITAL GOVERNANCE TRANSFORMATION READINESS REQUIREMENTS

Successful digital transformation of government activities entails full digitisation of existing manual and paper-based structures, processes and cultures in an organisation or in society. Such fundamental transformation does not happen by itself, because it implies destroying people’s comfort zones and creating new comfort zones that promise to be more beneficial, but which have no guarantees of success (Cloete 2018b:304).

The main strategies to do this have been summarised by the WEF (2021:7) as purposeful and sustainable integration of environmental, social and governance (ESG) decisions and actions throughout an organisation as pillars of a new digital culture, including through taking calculated risks; collaborating with others to co-produce creative solutions to problems; using evidence for public value addition and continuous improvement of customer satisfaction.

The EU (2019b) proposed a systematic digital government deployment matrix consisting of criteria to measure the key digital and analogue indicators for four maturity levels: minimum, basic, useful and sustainable levels. It also identifies different roadmaps to implement this deployment matrix for each of these countries, containing the most strategic business plan outlines for them to fast-track improving their respective e-government development outcomes in the shortest feasible manner with the assistance and support of the EU (2019b:39–43).

Dunleavy and Margetts also developed an Essentially Digital Governance (EDGE) model as a new paradigm to guide the transition from analogue to a new, fundamentally different digital public sector process. They conclude that “governments need a reform agenda that directly confronts the continuing predominance of the Weberian model, discards the tattered remains of NPM and provides a route map for the digital future” (2015:2–5).

The EDGE model summarises the most appropriate organisational cultures for this purpose, as well as the core strategic issues that need to be dealt with to achieve and consolidate such value changes. The authors also propose several principles for the development and application of digital transformation processes for this purpose. These principles are:

■ **Action and design principles**

- Free digital public services.
- More transparent, innovative and effective use of data.
- No duplication but integrated, decentralised services delivery systems.
- Competitive economies of scale-based services.
- DIY government: reduce citizen dependence on government.

■ **Choice principles**

- Prioritise outcomes and not processes: e.g. responsive to the public.
- Protect and promote digital rights: e.g. privacy and security.
- Duty to collect, digitise, store and use data interactively in the public interest.
- Optimal, experiential, agile, nudging, sustainable governance strategies.

Dunleavy and Margetts (2015:26) concluded that their proposals could transform the current government paradigm into a digital network with government in the centre, providing services on demand to different autonomous nodes in the network for co-production, implementation and management. This has the potential to create a “post-bureaucratic, Information State”.

These useful insights into the required principles of and approaches to digital transformation have been taken further by Bwalya (2018a:84), who identified the following general factors that will influence the success of digital transformation of government: innovative, entrepreneurial leadership-driven digital transformation with active citizen participation in especially implementation, will increase citizen acceptance and trust. To achieve these goals, digital transformation strategies would also have to include an appropriate government business model, and effective open communication among all stakeholders (Bwalya 2018a:85). Trust for these purposes has two dimensions: 1) trust in the government based on past experiences (ethical, professional, impartial, fair, transparent, responsive, reliable), and 2) trust in the technologies that are used (useful, easy to use, improvement on analogue systems, protecting privacy and secure transactions and communications).

Digital transformation success necessitates adequate political support for this difficult transformation, as well as adequate resources, technologies and skills that exist on both the supply and demand sides of their societies (Bwalya 2018a:90–91). The main policy strategies to achieve successful digital transformation programmes include “brand building and management, compliance monitoring, regulation and central reporting, community engagement and securing funding” (Bwalya 2018a:97).

Kirvan (2021) also provides a concrete roadmap for successful digital transformation that is compatible with the above suggestions. The 14 elements of his roadmap are the following:

- Start with doing systematic research into digital transformation.
- Identify and assess optimal ways in which digital transformation can improve the *status quo*.
- Brief senior management on findings and conclusions.
- Get senior management support for strategic transition and budgetary frameworks.
- Develop with key internal and external stakeholders a digital transformation vision and strategy.
- Create project plans and budgets adaptable to changing conditions and time frames.
- Choose technologies that align with operational priorities.
- Continuously solicit and maintain the collaboration and support of employees and beneficiaries.
- Systematically monitor, track, measure and improve performance and progress towards vision achievement.
- Train, train, train.
- Test, test, test.
- Be ready for unplanned events and resilient in setbacks.
- Implement user feedback and revise strategies and processes for better results.
- Regularly review system performance against vision and strategic goals.

The OECD (2020:16) further developed a Digital Government Policy Framework based on the best digital transformation performers among OECD countries (South Korea, the UK, Colombia, Denmark, Japan, Canada, Spain, Israel, Portugal, France, New Zealand and Norway). The framework identified the attributes of a mature digital government as: digital by design, data-driven, acts as platform for services delivery, open by default, user-driven and proactive. Following these findings, the OECD (2021a & 2021b) developed a digital transformation toolkit and handbook for digital transformation leaders around 12 principles.

Other recent proposals for systematic criteria to measure and assess the digital transformation status of a country, include those of Bumann and Peter (2019), Bwalya (2018c & 2018d), Carrijo *et al.* (2023), EU (2020), Kaisara and Pather

(2012), Lang *et al.* (2022), Melhem *et al.* (2020), Peixoto *et al.* (2022), PWC (2022), Reyes (2021) and Stanimirovic and Vintar (2013).

The need for better governance of 4IR technologies in Africa to enable better sustainable development outcomes on the continent, is clear. However, this is an ambitious goal, although there is evidence (EU 2019b) that it is at least feasible to start with a dedicated digital government initiative if a critical mass of governmental support for such an initiative can be mobilised. The EU (2019b) report on the potential of successful digital transformation in Africa, is, for example, upbeat about this prospect and states that “Africa, with over 400 million mobile internet users and numerous sectorial leap-frogging innovations, demonstrates unprecedented growth of digital technologies...There are unique opportunities to use ICT for economic growth, productivity increase and better service delivery. It is timelier than ever for governments to create strong, enabling e-governance systems. Governments should ensure the best possible use of digital technologies for the benefit of the people, act as facilitators, enablers and regulators, involving all stakeholders through transparent cooperation” (EU 2019b:6).

The EU report (2019b:13) identified the following strategic issues to be improved in Africa to maximise the potential for digital government:

- **Key analogue issues:** International frameworks; legal framework (legislation and regulations enabling digital transactions and protecting privacy); coordinating institutions; political will and change management; access to services, and awareness raising.
- **Key digital issues:** Government portal; digital databases and digitisation of records; secure exchange of data; secure digital identity and digital signature and infrastructure issues.

The first systematic monitoring and evaluation system to determine progress with e-government programmes in South Africa, was the e-barometer initiative (RSA-DoC 2011). The e-barometer was developed to measure the changes in digitisation of government services between 2000 and 2010. The instrument ambitiously set out to measure the progress with e-government in South Africa in terms of digitisation progress in South Africa with access, uptake and usage, in nine societal sectors (individuals, households, communities, business, government, health, education, digital local content and ICT sectors), compared to the government’s own objectives, as well as those of its peers: Brazil, Russia, India and China (the so-called BRICS alliance) and the objectives of the rest of the globe.

The 2011 e-barometer report concluded that digitisation was imperative and not only a nice-to-have for South Africa. However, the report found that the country was not ready for this transformation and over the assessment period regressed because of a lack of commitment and resources (RSA-DoC 2011:44–46). The country’s ranking over the last few years before 2010 was “stagnant or slipping,

while its BRICS peers have generally improved their respective rankings” (RSA-DoC 2011:17). Since the 2011 assessment was done, little has changed.

During 2016–17 the South African government published a revised digital government White Paper, strategy and roadmap (RSA-DTPS 2016 and RSA-DTPS 2017). It identified a range of digital transformation risks in South Africa at that time, as well as concrete proposals to overcome or minimise those risks (RSA-DTPS 2017:39).

There is no record of any repeat barometer measurement undertaken after 2011. This is a further indication of the lack of commitment and dedication of successive governments in South Africa towards making e-government work in this country. This sad situation is the result of a lack of strategic vision and commitment of government towards consolidating the migration of paper-based governance systems in the country to an e-governance system (Cloete 2012). Despite these constraints, the current e-government system in South Africa is regarded by the UN as internationally competitive with many more developed nations in Europe and Asia. It is still the best digital government performer in Africa (UNDESA 2022:63). Research ICT Africa (2020) and Jegede and Ncube (2021) have recently proposed practical improvements for South Africa to implement digital transformation in the country better.

SIGNIFICANCE OF AUTHORITATIVE, DEMOCRATIC LEADERSHIP FOR DIGITAL TRANSFORMATION SUCCESS

There is clearly a need for fundamentally changing the culture of the public sector in the medium and long-term, to consolidate a new digital way of looking at and doing things in future, as the latest digital government report of UNDESA (2022) has pointed out. This process has to be driven, coordinated and managed by transformation-orientated leaders in the public interest (Bwalya 2018a:77; Savoldelli & Codagnone 2013). Digital transformation leaders in the public sector in Africa should especially have clear digital visions, strategies and schedules to facilitate this fundamental shift from analogue to digital public services and eventually achieve and consolidate new digital organisational cultures that are more appropriate for eventual sustainable development results on the continent.

Professor Tshilidzi Marwala, former Deputy Chairperson of the Presidential Council on 4IR, is very outspoken about the need and requirements for strategic leadership to achieve digital transformation in Africa and states that: “The 21st century is proving to offer technological advancements at an exponential rate...due to the complexity of problems that face humanity today, those who do not know should not lead...those who are interested in human and social sciences should also be required to study science and technology. ...we need to understand issues of automation, technology and human capital to improve the

quality of life of our people. Any collective knowledge gap in our understanding of technology or global economics, or our knowledge of human capital competency, will derail our ambition of creating a developed society in the 21st century” (Marwala 2021).

Leadership is a critical guiding dimension of management. A leader is accepted by others as a person who can and should take charge to guide and/or direct them, to do what the leader thinks are the right things to do. The leader is accepted as someone who influences and commands, and who is to be obeyed and followed (Lues 2021). Leadership is the process of influencing others to achieve goals that the leader convinces them to pursue and try to achieve. This ‘convincing’ can occur through peaceful rational, ideological or emotional ‘persuasion’ (Cloete 2018b), or through more violent unilateral intimidation and/or enforcement tactics. Different types of leadership approaches and styles can be identified. They are all potentially useful in specific conditions. The main types include (e.g. Alblooshi *et al.* 2020; Becker 2021; Chen *et al.* 2014; Echebiri 2021; Gil *et al.* 2018; Jia *et al.* 2018; Khan *et al.* 2020; Lues 2021:178):

- **Laissez-faire:** hands-off, distant, tolerant of individual styles and preferences, delegative.
- **Autocratic:** top-down, rigid, compliance-oriented, intimidating.
- **Bureaucratic:** rule-bound compliance, rigid.
- **Transactional:** tactical, managerial, short-term quid-pro-quo bargaining-oriented focus
- **Authoritative:** charismatic, strategic, rational, calculated
- **Democratic:** bottom-up, participative, consultative, responsive, transparent, accountable, serving needs.
- **Transformational:** visionary, inspirational, long-term, big picture-oriented

A combination of authoritative, democratic, transformational leadership and governance approaches seems to be the most optimal to initiate, guide and manage fundamental democratic organisational and societal transformation for sustainable development, especially in the VUCA conditions caused by the current 4IR stage of the information era (Baran & Woznyj 2021:3; Lues 2021:175; Sarkar 2016).

Odumeru and Ogbonna (2013:356) summarised the main attributes of such a leader as someone who “stimulates and inspires (transform) followers to achieve extraordinary outcomes...He/she pays attention to the concern and developmental needs of individual followers; they change followers’ awareness of issues by helping them to look at old problems in a new way; and they are able to arouse, excite and inspire followers to put out extra effort to achieve group goals”.

Needle (2021) regards the main attributes of successful transformational leaders as anticipating change proactively; understanding how to emphatically respond to

transformation stress to build or maintain support for necessary changes; resilience to adapt quickly and effectively to fast-changing circumstances; the ability to inspire subordinates and other team members to enthusiastically accept and promote the envisaged changes and openness, and flexibility and pragmatism to consider and respond positively to alternative strategies that might facilitate and speed up acceptance of the new way of doing things.

Needle (2021) also summarised the pros and cons of transformational leadership as follows:

- **Main advantages:** Following design principles by testing new ideas in practical contexts; allowing team members to experiment with alternative approaches and strategies that refocus them on organisation interests rather than their own uncertainties; leading to team members taking direct ownership of the change process, and uniting staff around a new organisational purpose.
- **Main disadvantages:** Too idealistic visions can discourage employees and disrupt the status quo too much, especially if the drastic changes are seen as change just for the sake of it, instead of benefitting the organisation more, while time-consuming efforts to keep motivation and enthusiasm high might be too much to maintain.

To be successful, transformational leadership interventions must be designed and implemented in such a manner that the advantages of transformational change exceed the disadvantages of such change. Building on these analyses by Needle, Lues (2021) contextualised the requirements of the most appropriate 21st century public leadership styles to achieve the SDG goals by 2030. Bwalya (2018e) summarised and assessed the policy and leadership dimensions of digital government, while Breaugh *et al.* (2023) emphasised the need for collaborative strategies in the design of digital transformation initiatives. Against this background, resilient leadership is also required to overcome strong tendencies of resisting the transition from analogue comfort zones to new digital modes of work, especially by older individuals who might not always be fully digitally literate and would need to upgrade their current skills significantly for this purpose (Lombardi, Cunha & Giustiniano 2021; Oosthuizen 2017).

Atkinson *et al.* (2021:132–134) also summarised the main elements of the transformative leadership vision needed for the development of sustainable global living systems as an understanding of and ability to develop effective strategies to deal with complexity, emergence, sensemaking, network building, context and environment. Pasanen (2022) proposed a detailed systematic service design process approach to nudge employees to facilitate and consolidate digital culture transformation in an organisation, while Nel-Sanders and Thomas (2022) explain how governments can facilitate the establishment of ‘innovation-led entrepreneurial ecosystems’ in society.

Ochara (2013:1) concluded that: “effective organisational leadership in Africa and other developing countries should be anchored in local values; encourage entrepreneurship, take into account opportunities afforded by mobile computing platforms and high diffusion of mobile applications; focus on ethical leadership engagement to spur e-participation and e-democracy; and develop national and regional innovation systems to enable Africa and other developing regions to participate in global knowledge flows”.

Ochara (2013:8) also identified the main digital transformation issues that African leaders have to address in order to achieve sustainable development outcomes on the continent, as ethical and moral, entrepreneurial, institutionalised and innovative leadership, and assessed the implications of digital transformation in Africa. Bwalya (2018a) emphasised the importance of a decolonised approach to digital transformation in Africa. Ndulu *et al.* (2023) compared digital transformation processes in seven African countries and identified what strategies work best in what conditions, while Viik (2017) and Wilson (2004) provide strategic tips to lead digital transformation globally and in Africa. Chaka (2020) and Manana and Mawela (2022) identify comprehensive ranges of skills that employees must have to achieve successful digital transformation in the public sector.

CONCLUSIONS

E-government in Africa is still generally very weak, mainly as a result of the cumulative negative impact of different resource, technological, cultural, social, financial and political policy governance constraints. Most African countries crowd the bottom of the latest UN e-government index.

Most African governments realise the strategic importance of moving as soon as possible to fully-fledged e-government public services delivery platforms. Serious hardware, software, digital literacy and other constraints, however, delay full adoption of crucial 4IR technologies to fast-track the achievement of sustainable development on the continent. The global e-government paradigm shift is still future music in most African nations, and the successful implementation of 4IR in this regard in Africa faces an uphill battle.

South Africa was an early adopter of e-government at the beginning of the 21st century, and was regarded as one of the leaders in this regard in developing countries. It has built up one of the best ICT backbones and local content on the road to e-government in Africa. However, little progress towards more mature digital government transitions has been made over the last two decades in South Africa. Despite these constraints, the current e-government system in South Africa is regarded by the UN as internationally competitive with many more developed nations in Europe and Asia.

NOTE

- * 'Analogue' is used in this article to refer to a manual or mechanical way of doing things, in contrast to an electronic/digital approach. Manual, paper-based files and records are for example regarded as analogue systems, while electronic filing systems and databases are regarded as digital systems.

REFERENCES

- Alblooshi, M., Shamsuzzaman, M. and Haridy, S. 2020. The relationship between leadership styles and organisational innovation: A systematic literature review and narrative synthesis. *European Journal of Innovation Management*. 24:338–370.
- Atkinson, J., Lasbennes, F. and Nabarro, D. 2021. Reflecting on our times: Valuing transformative leadership in real-world 'living systems'. *American Journal of Evaluation*. 42:130–138.
- AU. 2020. *The digital transformation strategy for Africa (2020–2030)*. Available at: <https://au.int/en/RSA-DoCuments/20200518/digital-transformation-strategy-africa-2020-2030>. (Accessed on 28 May 2023).
- Baran, B. and Woznyj, H. 2021. Managing VUCA: The human dynamics of agility. *Organizational Dynamics*. 50:1–8.
- Becker, B. 2021. *The 8 Most Common Leadership Styles & How to Find Your Own*. Available at: <https://blog.hubspot.com/marketing/leadership-styles>. (Accessed on 28 May 2023).
- Braams, R., Wesseling, J., Meijer, A., and Hekkert, M. 2023. Civil servant tactics for realizing transition tasks understanding the microdynamics of transformative government. *Public Administration*. Available at: https://www.researchgate.net/publication/370690513_Civil_servant_tactics_for_realizing_transition_tasks_understanding_the_microdynamics_of_transformative_government. (Accessed on 28 May 2023).
- Breaugh, J., Rackwitz, M. and Hammerschmid, G. 2023. Leadership and institutional design in collaborative government digitalisation: Evidence from Belgium, Denmark, Estonia, Germany, and the UK. *Government Information Quarterly*. 40(2):10178840.
- Bumann, J., Peter, M. 2019. Action fields of digital transformation-a review and comparative analysis of digital transformation maturity models and frameworks. *Open Journal of Business and Management*. 9(4):202140.
- Bwalya, K. 2018a. *Decolonisation of e-government research and practice: exploring contextual issues and opportunities in Africa*. Cape Town: AOSIS. 3–69.
- Bwalya, K. 2018b. Effectiveness of technology adoption models in measuring e-government adoption. In Bwalya, K. 2018. *Decolonisation of e-government research and practice: exploring contextual issues and opportunities in Africa*. Cape Town: AOSIS. 109–136.
- Bwalya, K. 2018c. E-Government maturity models. In Bwalya, K. 2018. *Decolonisation of e-government research and practice: exploring contextual issues and opportunities in Africa*. Cape Town: AOSIS. 137–159.
- Bwalya, K. 2018d. Measuring Individual Adoption and Usage of e-Government Solutions using Principal Component Analysis. In Bwalya, K. 2018. *Decolonisation of e-government research*

- and practice: exploring contextual issues and opportunities in Africa. Cape Town: AOSIS. 161–187.
- Bwalya, K. 2018e, Leadership and Policy Dimensions of e-Government. In Bwalya, K. 2018. *Decolonisation of e-government research and practice: exploring contextual issues and opportunities in Africa*. Cape Town: AOSIS. 73–105.
- Carrijo, P., Alturas, B. and Pedrosa, I. 2023. Similarities and differences between digital transformation maturity models: a literature review. In Kahraman, C., Haktanir, E. (Eds.). 2023. *Intelligent systems in digital transformation. Lecture notes in networks and systems*. 549:33–52. Springer, Cham.
- Chaka, C. 2020. Skills, competencies and literacies attributed to the 4IR Industry 4.0. *Scoping Review*. 46(4):369–399.
- Chen, Y., Tang, G., Jin, J., Xie, Q. and Li, J. 2014. CEOs' transformational leadership and product innovation performance: The roles of corporate entrepreneurship and technology orientation. *Journal of Product Innovation Management*. 31:2–17.
- Cloete, F. 2012. e-Government lessons from South Africa, 2001 – 2011. *Institutions, State of Progress and Measurement, African Journal for Information and Communication (AJIC)*. 12:28–42.
- Cloete, F. 2018a. Moving beyond empty decolonisation mantras to real sustainable empowerment in Africa. *Administratio Publica*. 26(1):58–85.
- Cloete, F. 2018b. Policy Change. Chapter 10 in Cloete, F., De Coning, C., Wissink, H. and Rabie, B. (Eds.). *Improving Public Policy for Good Governance*. (4th ed.). Pretoria: JL van Schaik. Pp 303–344.
- Cloete, F. and Mmakola, D. 2018. Development and Public Policy. Revised Chapter 4 in Cloete, F., De Coning, C., Wissink, H. and Rabie, B. (Eds.). *Improving Public Policy for Good Governance*. (4th ed.). Pretoria: JL van Schaik. Pp 105–133.
- Davenport, T. and Westerman, G. 2018. Why so many high-profile digital transformations fail. *Harvard Business Review*. 9:1–5.
- Delioğlu, N. and Uysal, B. 2022. A review on agile leadership and digital transformation. *Yıldız Social Science Review*. 8(2):121–128.
- Ducass, A. 2017. E-Gov Development in Africa. *Electronic Journal of e-Government*. 15(2):59–62. Available at: <https://academic-publishing.org/index.php/ejeg/article/view/640/603>. (Accessed on 29 May 2023).
- Dunleavy, P. and Margetts, H. 2015. *Design principles for essentially digital governance*. In The 111th annual meeting of the American Political Science Association, 3–6 September 2015. Available at: <http://eprints.lse.ac.uk/64125/>. (Accessed 5 February 2021).
- Echebiri, C. and Amundsen, S. 2021. The relationship between leadership styles and employee-driven innovation: The mediating role of leader–member exchange. *Evidence-based HRM*. 9:63–77.
- Erida, A. and Lotfi, B. 2021. The determinants of organizational change management success: Literature review and case study. *International Journal of Engineering Business Management*. 13:1–15.
- EU. 2019a. Digital transformation. European Parliamentary Research Service. Available at: [https://www.europarl.europa.eu/RegData/etudes/BRIE/2019/633171/EPRS_BRI\(2019\)633171_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2019/633171/EPRS_BRI(2019)633171_EN.pdf). (Accessed on 29 May 2023).
- EU. 2019b. Guidelines and Roadmap for full deployment of e-governance systems in Africa. European Commission: E-Governance Academy. Available at: <https://ega.ee/wp-content/>

uploads/2019/04/eGA_Final-Report-Research-analysis-guidelines-and-roadmap-for-full-deployment-of-e-governance-systems-in-Af.pdf. (Accessed 27 July 2019).

- EU. 2020. eGovernment benchmark 2020–eGovernment that works for the people. Available at: <https://www.capgemini.com/wp-content/uploads/2020/09/eGovernment-Benchmark-2020-Insight-Report.pdf>. (Accessed on 29 May 2023).
- Feola, G. 2015. Societal transformation in response to global environmental change: a review of emerging concepts. *AMBIO: A Journal of the Human Environment*. 44:376–390.
- Fields, Z. and Huesig, S. (Eds.). 2019. *Responsible, sustainable, and globally aware management in the fourth industrial revolution*. IGI Global. Available at: <https://www.igi-global.com/gateway/chapter/full-text-pdf/227812>. (Accessed on 29 May 2023).
- Gil, A. Rodrigo-Moya, B. and Morcillo-Bellido, J. 2018. The effect of leadership in the development of innovation capacity: A learning organization perspective. *Leadership & Organization Development Journal*. 39:694–711.
- Jegede, O. and Ncube, C. 2021. Science, technology, innovation management for industrial development in South Africa: Implications for the fourth industrial revolution. *International Journal of Innovation, Creativity and Change*. 15(9):673–695.
- Jia, X., Chen, J., Mei, L. and Wu, Q. 2018. How leadership matters in organizational innovation: A perspective of openness. *Management Decision*. 56:6–25.
- Kaisara, G. and Pather, S. 2011. The e-Government evaluation challenge: A South Africa Batho Pele-aligned service quality approach. *Government Information Quarterly*. 28:211–221.
- Khan, M., Ismail, F., Hussain, A. and Alghazali, B. 2020. The interplay of leadership styles, innovative work behavior, organizational culture, and organizational citizenship behavior. *Sage Open*. 10:1–16.
- Kirvan, P. 2021. *14 need-to-know tips for digital transformation success*. Available at: <https://searchcio.techtarget.com/tip/14-need-to-know-tips-for-digital-transformation-success?vgnextfmt=print>. (Accessed on 29 May 2023).
- Korachi, Z. and Bounabat, B. 2020. General approach for formulating a digital transformation strategy. *Journal of Computer Science*. 16(4):493–507.
- Kvint, V. and Bodrunov, S. 2022. *Knowledge, technologies, and noonomy*. Cambridge, Mass: Apple Academic Press.
- Lang, K., Hasler, D., Hackl, T. and Ehmig-Klassen, K. 2022. A systems theory-based conceptual framework for holistic digital transformation. *ISPIM Innovation Conference – Event proceedings: Innovating our common future*. Berlin, Germany on 20–23 June 2021. LUT Scientific and Expertise Publications. Pp 1–24.
- Linnér, B-O. and Wibecka, V. 2020. Conceptualising variations in societal transformations towards sustainability. *Environmental Science & Policy*. 106:221–227.
- Lombardi, S., e Cunha, M. and Giustiniano, L. 2021. Improvising resilience: The unfolding of resilient leadership in COVID-19 times. *International Journal of Hospitality Management*. 95:102904. Available at: <https://doi.org/10.1016/j.ijhm.2021.102904>. (Accessed on 29 May 2023).
- Lues, L. 2021. Has public leadership as we know it reached the end of its shelf life? Exploring leadership styles in the 21st century. *Teaching Public Administration*. 39(2):175–191.
- MacLean, D. and Titah. R. 2021. A systematic literature review of empirical research on the impacts of e-government: a public value perspective. *Public Administration Review*. 82(1):23–38.

- Manana, T. and Mawela, T. 2022. Digital skills of public sector employees for digital transformation. Paper at the International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT). Pp 144–150
- Marwala, T. 2021. *Leading in the 21st century – the call for a new type of African leader*. Bryanston, South Africa: Tracey McDonald Publishers. Available at: <https://www.amazon.com/Leading-21st-Century-African-Leader-ebook/dp/B08QCFZQNP> (accessed 18 August 2021).
- Melhem, S., Lee, Y., Dener, C., Yamamichi, M., Priftis, M., Pahlavooni, S., Petrov, O., Chrzanowski, P., Anselme, G. and Sfaxi, M. 2020. *Digital government readiness assessment (DGRA) toolkit V.31*. Washington, D.C.: World Bank Group. Available at: <http://RSA-DoCuments.worldbank.org/curated/en/999901588145595011/Digital-Government-Readiness-Assessment-DGRA-Toolkit-V-31-Guidelines-for-Task-Teams>. (Accessed on 29 May 2023).
- Misuraca, G., Barcevičius, E. and Codagnone, C. (Eds.). 2020. *Exploring digital government transformation in the EU – understanding public sector innovation in a data-driven society*. Luxembourg: European Union. Available at: <https://publications.jrc.ec.europa.eu/repository/handle/JRC11885>. (Accessed on 29 May 2023).
- Morze, N. and Strutyńska, O. 2021. Digital transformation in society: key aspects for model development. *J. Phys.: Conf. Ser.* 1946 012021. Available at: <https://doi.org/10.1088/1742-6596/1946/1/012021>. (Accessed on 29 May 2023).
- Mutula, S.M. 2013. e-Government divide: implications for Sub-Saharan Africa, in information ethics for Africa: Cross cutting themes. In Ocholla, D., Britz, J., Capurro, R. and Bester, C. (Eds.). *Information ethics in Africa: Cross-cutting themes*. Pretoria: ACEIE. 59–66. Available at: https://www.researchgate.net/publication/342707988_Information_Ethics_in_Africa_Cross-cutting_Themes/link/5f2d3a0a92851cd302e566cb/download. (Accessed 25 June 2021).
- Ndlovu, N., Ochara, N.M. and Martin, R. 2022. Influence of digital government innovation on transformational government in resource-constrained contexts. *Journal of Science and Technology Policy Management*. Available at: <https://doi.org/10.1108/JSTPM-11-2021-0173>. (Accessed on 29 May 2023).
- Ndulu, B., Stuart, E., Dercon, D. and Knaack, P. 2023. *Driving digital transformation lessons from seven developing countries*. Oxford: OUP.
- Needle, D. 2021. *Transformational leadership*. Available at: <https://searchcio.techtarget.com/definition/transformational-leadership?vgnextfmt=print>. (Accessed on 19 August 2021).
- Nel-Sanders, D. and Thomas, P. 2022. The role of government in promoting innovation-led entrepreneurial ecosystems. *Africa's Public Service Delivery and Performance Review*. 10(1):a640. Available at: <https://doi.org/10.4102/apsdpr.v10i1.640>. (Accessed on 29 May 2023).
- Ochara, N. 2013. Leadership for the electronic age: towards a development-oriented, socio-technical ontology of leadership. *African Journal of Information and Communication*. Available at: https://repository.up.ac.za/bitstream/handle/2263/39964/Ochara_Leadership_2013.PDF?isAllowed=y&sequence=1. (Accessed on 29 May 2023).
- Odumeru, J. and Ogbonna, I. 2013. Transformational vs. transactional leadership theories: Evidence in literature. *International Review of Management and Business Research*. 2(2):355–361. Available at: <https://www.scinapse.io/papers/77189300>. (Accessed on 29 May 2023).

- OECD. 2020. *Digital Government Index: 2019 results*. OECD Public Governance Policy Papers No. 03. Available at: <https://www.oecd-ilibrary.org/RSA-DoCserver/4de9f5bb-en.pdf?expires=1625580010&id=id&accname=guest&checksum=4DBE79DEF59A3565E5391BAFE6C7D08A>. (Accessed on 29 May 2023).
- OECD. 2021a. *Digital Government Toolkit*. Available at: <https://www.oecd.org/governance/digital-government/toolkit/12principles/>. (Accessed on 29 May 2023).
- OECD. 2021b. *The E-Leaders handbook on the governance of digital government*. OECD Digital Government Studies. OECD Publishing, Paris, Available at: <https://doi.org/10.1787/ac7f2531-en>. (Accessed on 29 May 2023).
- Oosthuizen, J. 2017. *The determinants of fourth industrial revolution leadership dexterity: A proposed framework for 4IR intelligence and subsequent 4IR leadership development*. Paper presented at the 4th International Conference on Responsible Leadership, at GIBS – Johannesburg, South Africa. Available at: https://www.researchgate.net/publication/315114030_the_determinants_of_fourth_industrial_revolution_leadership_dexterity_a_proposed_framework_for_4ir_intelligence_and_subsequent_4ir_leadership_development#fullTextFileContent. (Accessed on 29 May 2023).
- Pasanen, R. 2022. *Leading change and nudging employees towards digital transformation by utilising the service design process*. Available at: https://www.theseus.fi/bitstream/handle/10024/786125/Pasanen_Riikka.pdf?sequence=2. (Accessed on 29 May 2023).
- Peixoto, E., Paulo, H., França, C. and Ramalho, G. 2022. A mapping study about digital transformation of organizational culture and business models. In *Proceedings of the 24th International Conference on Enterprise Information Systems*—Volume 2: ICEIS. 408–417. Available at: <https://www.scitepress.org/Papers/2022/109916/109916.pdf>. (Accessed on 29 May 2023).
- Phillips, J. and Klein, J.D. 2023. Change Management: From Theory to Practice. *TechTrends*. 67:189–197.
- Pratt, M. 2021. *10 digital transformation benefits for business*. Available at: <https://www.techtarget.com/searchcio/tip/Top-10-digital-transformation-benefits-for-business?vgnextfmt=print>. (Accessed on 29 May 2023).
- PWC. 2022. *Part 1: The evolution & characteristics of mature digital government*. Available at: <https://www.pwc.com/m1/en/publications/the-journey-to-digital-government.html>. (Accessed on 29 May 2023).
- Rabie, M. 2013. *Global Economic and Cultural Transformation*. New York: Palgrave Macmillan. Available at: https://doi.org/10.1057/9781137365330_4. (Accessed on 29 May 2023).
- Rahman, S., Teicher, J., Cox, J. and Alam, Q. 2023. Slipstreaming for Public Sector Reform: How Enterprising Public Sector Leaders Navigate Institutional Inertia. *Journal of Public Administration Research and Theory*. 33:4–18. Available at: <https://doi.org/10.1093/jopart/muac006>. (Accessed on 29 May 2023).
- Research ICT Africa. 2020. *Digital futures: South Africa's digital readiness for the 'fourth industrial revolution*. Available at: <https://researchictafrica.net/publication/digital-futures-south-africas-readiness-for-the-4ir/>. (Accessed on 29 May 2023).
- Reyes, J., Morocho, V. and Cedillo, P. 2021 Applying maturity models in organizations for digital transformation: a comparative study. In Rocha, A., López-López, P. and Salgado-Guerrero, J.

- (Eds.). *Proceedings of Communication, Smart Technologies and Innovation for Society (CITIS)*. New York City: Springer. Pp 721–731.
- Rousseau, D. and Ten Have, S. 2022. Evidence-based change management. *Organizational Dynamics*. 51(3):100899.
- RSA-DoC. 2011. *E-Barometer: Measuring the Impact of ICTs on Development in South Africa*. Department of Communications. Pretoria: Government Printer,
- RSA-DTPS. 2016. *National Integrated ICT Policy White Paper*. Department of Tele-Communications and Postal Service. Pretoria: Government Printer,
- RSA-DTPS. 2017. *National e-Government strategy and roadmap*. Department of Tele-Communications and Postal Service. Pretoria: Government Printer,
- Sarkar, A. 2016. We live in a VUCA world: the importance of responsible leadership. *Development and Learning Organizations*. 30(3):9–12.
- Savoldelli, A. and Codagnone, C. 2013. Measuring the public value of e-government: The eGEP2.0 model. *Electronic Journal of e-Government*. 11(1):373–388.
- Stanimirovic, D. and Vintar, M. 2013. Conceptualization of an integrated indicator model for the evaluation of e-government policies. *Electronic Journal of e-Government*. 11(2):293–307.
- UNDESA. 2022. E-government survey 2022. The future of digital government. Available at: <https://publicadministration.un.org/egovkb/en-us/Reports/UN-E-Government-Survey-2022>. (Accessed on 29 May 2023).
- Viik, L. 2017. *Tips to lead digital transformation in Africa*. e-Governance Academy. 3.50 min video. Available at: <https://www.youtube.com/watch?v=d25V3JN1fgk>. (Accessed on 209 May 2023).
- WEF. 2021. *Digital culture: The driving force of digital transformation*. Available at: <https://www.weforum.org/reports/digital-culture-the-driving-force-of-digital-transformation>. (Accessed on 12 Sept 2021).
- Wilson, E. 2004. Leadership in the digital age. In Goethals, G., Sorenson, G. and MacGregor, B. (Eds.). *Encyclopedia of Leadership*. 858–861. Thousand Oaks, London. Available at: <https://citeseerx.ist.psu.edu/viewRSA-DoC/download?doi=10.1.1.454.1745&rep=rep1&type=pdf>. (Accessed on 29 May 2023).

AUTHOR'S CONTACT DETAILS

Prof Fanie Cloete

Research Fellow,
Department of Public Administration and Management,
University of the Free State,
Bloemfontein.
Email: cloetegs@gmail.com
Cell: 083 535 5180

Preparing for Digital Governance

Mapping Competency Domains for Postgraduate Programmes in Public Administration and Management

G van der Waldt

North-West University
Potchefstroom

ABSTRACT

Rapid technological advancement and a highly dynamic world of governance place new demands on public managers to prepare for the digitalisation of the public sector. Public managers need to ensure that they can harness technology's potential to improve governance, enhance government efficiency, engage stakeholders and better serve citizens. This requires investing in skills, competencies and knowledge to successfully navigate public sector institutions towards digital governance. The purpose of this article is to propose a baseline curriculum framework for postgraduate programmes in Public Administration and Management with a view to adequately prepare current and prospective public managers in the field of digital governance. Desktop thematic analysis methodology was used to extract seven core competency domains from an extensive literature survey. Curriculum efforts should enhance public managers' comprehension of these seven domains to support digital transformation and modernisation endeavours in the public sector.

INTRODUCTION

Digital governance is a crucial element of global competitiveness since it enables countries to leverage digital technologies to enhance their economic competitiveness and productivity. Domestically, digital governance promotes good governance with greater transparency, improved service delivery, increased public participation, greater efficiency, and enhanced data-driven decision-making. The scope and extent of technological applications in government has increased exponentially, and

the pace of change is set to accelerate even further. Public managers play a critical role in ensuring that government departments, agencies and entities are effectively utilising digital technologies to improve service delivery, enhance the country's competitiveness, and increase transparency in governance.

Transitioning to digital governance, also known as Gov 3.0, entails significant changes to the traditional systems and processes of public administration and management (PAM). However, the international experience, notably Australia (Edelmann, Mergel and Lampoltshammer 2023), Canada (Jones 2017), India (Shubha 2017), Germany (Pereira, Ronzhyn and Wimmer 2022), Spain (Criado and de Zarate-Alcarazo 2022), Ukraine (Karpenko and Glubochenko 2018), and the USA (Fishenden and Thompson 2013) reveals that digital governance maturity is seriously hampered by the fact that public managers generally lack the necessary competencies to bring the technological advantages of digitalisation to fruition. Information and communication technologies (ICT) and digital transformation in government make it essential to revisit training needs in PAM. This ultimately requires an adjustment of the curriculum of especially postgraduate programmes in the field.

Following desktop thematic analysis methodology, the purpose of this article is to propose a baseline curriculum framework for postgraduate programmes in PAM to adequately prepare current and prospective public managers in the field of digital governance. Such additions and/or adjustments to the existing curriculum intend to deepen fundamental understanding of digitalisation contexts, inclusive of automatisisation, the use of artificial intelligence (AI) and related modernisation endeavours in the public sector.

CONCEPTUALISING AND CONTEXTUALISING DIGITAL GOVERNANCE

Globally, governments are accelerating the utilisation of ICTs to enhance service delivery, transparency, public participation and the general prosperity of society (Charalabidis and Lachana 2020:383). Digital governance is generally regarded to be broader in scope than digital government in the sense that it includes network dynamics with actors from civil society, businesses, citizens and international agencies. In contrast, digital government concerns internal organisational and managerial dimensions associated with the digitalisation of government processes for service delivery (Lachana *et al.* 2018; Pereira, Ronzhyn and Wimmer 2022).

Digitalisation of governance has had a significant impact on the world of government, transforming the way governments operate and interact with key stakeholders. Scholars such as Engvall and Flak (2022), Milakovich (2021) and Misuraca and Viscusi (2014), highlight the following core arenas in which digitalisation has changed the world of governance:

- Improved service delivery: Digitalisation improves the delivery of government services to citizens through applications such as online government portals and mobile applications. These applications have made it easier for citizens to access government services, such as applying for permits or licences, paying taxes, or accessing government and policy-related information.
- Increased efficiency: Digitalisation enables governments to streamline their operations and improve efficiency through the automation of some routine administrative tasks, such as record-keeping and data entry, freeing up staff to focus on more complex tasks.
- Enhanced stakeholder collaboration: Digitalisation enhances stakeholder collaboration and citizen engagement with government. Social media platforms, online forums, and mobile applications have made it easier for citizens to interact with government officials, provide feedback on policies and programmes, and participate in government decision-making.
- Increased openness and transparency: Digitalisation increases openness and transparency in government operations since government institutions can now publish public data and information online, providing citizens with access to government data and facilitating greater political oversight and public scrutiny.
- Data-driven decision-making: Digitalisation enables public managers to make more evidence-based, objective and data-driven decisions. Digital technologies have made it easier to collect and analyse data, providing decision-makers with more accurate and timely information.

Despite its significant potential for greater efficiency in governance, digitalisation poses some challenges for governments, including concerns around data privacy and security, cyberthreats, and the digital divide. It is evident that public managers should navigate these challenges to ensure that digitalisation is managed in a responsible and equitable manner, thereby harnessing the potential benefits of Fourth Industrial Revolution (4IR) technologies in government.

Digital governance is an umbrella term for scholarly inquiry into ICT applications, inclusive of 4IR technologies, such as AI, the internet of things (IoT), and blockchain. McBride, Misnikov and Draheim (2022:121) reason that the term “digital governance” (inclusive of “government”) largely replaced the term “e-government” since e-government is more commonly associated with rather redundant technologies such as e-mail and the use of the internet. Digital governance, on the other hand, is “more all-encompassing and future-oriented, able to include new and rapid technological developments under its umbrella”. Charalabidis, Lachana and Alexopoulos (2022:3) argue that digital governance has accumulated a sufficient “science base” to direct systematic investigations into the “what”, “how”, and “why” dimensions thereof. Also, McBride, Misnikov and Draheim (2022) maintain that digital governance is well on its way to becoming an established discipline since accruing core

theoretical and philosophical bases, relevant theories and methods. However, the domain should be supported by the managerial competence of public officials. The value creation component of digital governance should be sustained by appropriate curricula of qualifications in PAM (cf. Edelmann, Mergel and Lampoltshammer 2023; Kausch-Zongo and Schenk 2022).

DIGITAL GOVERNANCE MANAGERIAL COMPETENCIES

Managerial competencies refer to the knowledge, skills, behaviour patterns, and abilities required by individuals to perform managerial roles and responsibilities effectively (Fanelli *et al.* 2020; Lara, Mogorrón-Guerrero and Ribeiro-Navarrete 2020). In the South African Government context, Chapter 5 of the *Senior Management Service (SMS) Handbook* (DPSA 2003) outlines 11 generic competencies that comprise the SMS Competency Framework. This Framework aims to establish a shared understanding of the critical success factors for performance in the Senior Management Service (senior public managers in post levels 13–16). The Competency Framework has four proficiency levels (basic, competent, advanced and expert) in the following domains:

- a. Strategic capability and leadership
- b. Programme and project management
- c. Financial management
- d. Change management
- e. Knowledge management
- f. Service delivery innovation
- g. Problem solving and analysis
- h. People management and empowerment
- i. Client orientation and customer focus
- j. Communication
- k. Honesty and integrity (DPSA 2003:5)

According to Parrado (2005) and Settles (2005), digital governance competencies are cross-cutting in nature, implying that they support and enhance several managerial competency domains. Based on the SMS Competency Framework, it may be argued that digital governance competencies should be embedded especially in the domains “strategic capability and leadership”, “change management”, “knowledge management”, “service delivery innovation” and “communication”.

The Cabinet of South Africa has approved that the competency assessment for SMS members be made compulsory from 1 April 2008 (DPSA 2009). With effect from 1 April 2020 all senior public managers may only enter the SMS service if they have successfully completed the Public Service Senior Management

Leadership Programme offered by the National School of Government (NSG) (DPSA 2019). For this the NSG awards a tender to one of the higher education institutions to collaborate in the accreditation and registration of a Postgraduate Diploma in Public Affairs and African Governance at NQF level 8. It is envisaged that this Diploma incorporates the 11 competency domains as outlined in the SMS Competency Framework. It is expected that SMS members on post levels 15 and 16 possess at least a postgraduate qualification on National Qualification Framework (NQF) level 8 (NSG 2020). In addition, it is expected that SMS candidates should successfully complete a pre-entry certificate, known as the Nyukela Programme, consisting of three modules, namely “Citizenship”, “Public Administration and African Leadership”, and “Government and I”.

In a digital governance context, Parrado (2005), Settles (2005) and Sithole and Van der Walddt (2016) contend that training programmes should incorporate theoretical models, approaches and methodologies of adjacent or neighbouring disciplines such as economics, political sciences, sociology, business management and information systems. Given the multidisciplinary nature and broad scope of the corpus of knowledge of digital governance, it is, however, a challenge to map the scholarly contours of digital governance, ensuring that qualifications in PAM are all-inclusive and not reductionist in nature. This is especially true for postgraduate qualifications in PAM such as postgraduate diplomas, honours programmes and structured master’s qualifications (e.g. Master in Public Administration). These programmes are typically focused on preparing prospective public managers for the world of governance through specialised knowledge and scholarly competencies to navigate bureaucratic practices, effectively manage public resources, and network with diverse stakeholders (*cf.* Jones 2017; Settles 2005). This article aims to contribute by mapping the technological domain applications that should be incorporated in the curriculum of postgraduate programmes in PAM.

MAPPING DIGITAL GOVERNANCE CURRICULUM DOMAINS

As highlighted above, the nature and scope of digital governance encompass broad multidisciplinary perspectives. A desktop survey of international databases (EBSCOhost, EDS Search, JSTOR, Web of Science, and WorldCat) and a Google Scholar literature review in the fields of digital governance, digital readiness/preparedness, public managerial competencies, ICT applications, and e-governance were used to identify core themes associated with digital governance research. Table 1 reflects on the thematic analysis of literature regarding digital governance; in other words, the methodology followed to identify potential themes proposed for absorption in the curriculum of postgraduate qualifications in PAM. Note that

“domains” and “themes” are used interchangeably. However, “theme” is used in line with thematic analysis methodology and “domain” is used to refer to core competency fields or areas in curriculum design.

Table 1: Collapsing of keywords extracted from literature into core themes

Keywords (“codes”) in literature review	Proposed themes for PAM curriculum
Information dissemination such as government websites and portals, e-newsletters, mobile apps, smartphone technology, budgeting, online reporting	E-government
Citizen interaction, such as tax filing, e-voting, online bill payments, electronic permitting and licencing, online forums, tender systems	
ICT infrastructure, such as broadband networks, data centres, cloud computing, Wi-Fi-networks	
Institutional digital frameworks and administrative systems, such as e-filing, management information systems, geographical information systems, human resource and financial systems, electronic record-keeping	
ICT legal, regulatory and policy framework	
Interoperability of government ICT systems and alignment of networks, standardisation	
Transparency, openness and accountability, such as e-disclosure systems, e-auditing, blockchain technology	
Vision and mission statements and strategic planning	Digital transformation
Political commitment and leadership	
Digital infrastructure development and maintenance	
Change management, fostering innovation and collaboration, culture of adaptability	
Organisational processes and structures to support transformation	
Human resource competencies and capacities (digital readiness, skills, agility to adapt)	
Scalability and flexibility of government transactions and operations through the leveraging of digital technologies	
Data privacy and security	
Societal digital divide and accessibility	
Funding and resource allocation, such as new revenue streams and “business” models	
Performance monitoring and evaluation, such as targets, indicators, and standards	

Keywords (“codes”) in literature review	Proposed themes for PAM curriculum
Data breaches	Cybersecurity
Protection of personal information and identity theft	
Data security (data theft, unauthorised access, destruction)	
National security and cyberthreats (phishing attacks, malware, ransomware)	
Surveillance systems	
Drone technology/unmanned aerial vehicles	
Government case studies (disruption of government operations, damage to reputation)	
Financial losses	
Intelligence operations	
Defence industry: development of advanced weapons systems	
Satellite-based communications and navigation	
Reconnaissance	
Biometric technology	
International law and human rights principles	
Data governance, such as data ownership and sharing	Data analysis
Data management, such as accessibility, collection, storage, dissemination	
Data quality, reliability, bias, accuracy, timeliness	
Data descriptive, predictive and prescriptive analytics (such as use of big data to inform policymaking and gauge public opinion)	
Use of big data and AI in government	
Data applications such as fraud detection, public health systems and emergency response systems	
Data visualisation and decision support systems, such as dashboards	
Data security and privacy concerns	
Data bias and ethics	

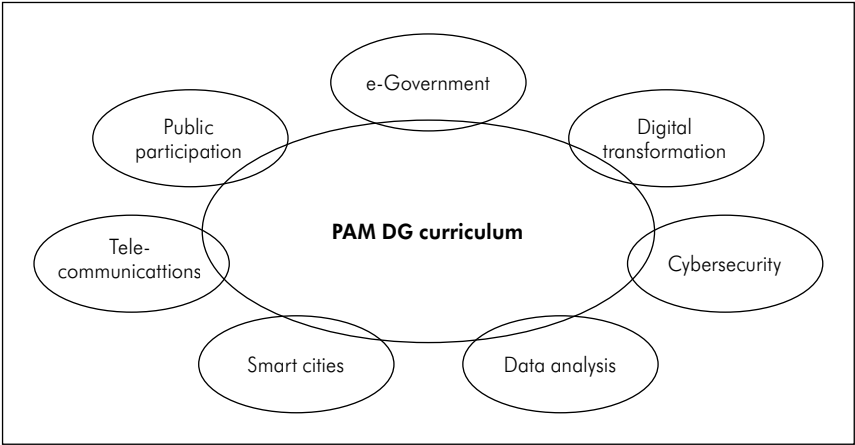
Keywords (“codes”) in literature review	Proposed themes for PAM curriculum
ICT applications/automation in local government/cities	Smart cities
Big data and analytics for service delivery improvement	
Transportation/traffic systems	
Use of sensors, smart metering and monitoring	
Local health systems and monitoring	
Sustainable and liveable cities, such as through the use of sensors, data analysis, and automation	
Emergency response technology	
Surveillance, public safety and security	
City resilience, adaptability and change	
Green energy (such as public lighting, transport, telecommunications, cooling and heating)	
Infrastructure development and maintenance	Telecommunications
Policies and regulations governing telecommunications	
Voice and data network, such as telephone, internet, messaging, video conferencing, and collaboration tools	
Mobile devices, such as supporting remote work and mobile workforce, and to provide mobile access to government services	
Software applications, 5G, and AI to improve the delivery of government services and enhance citizen engagement	
Cloud services and cloud-based collaboration tools and platforms to enable government employees to work together more effectively and efficiently	
Use of VoIP (Voice over Internet Protocol) and unified communications to enable voice and video communications between government agencies and stakeholders	
Implementation of broadband networks and Wi-Fi hotspots to increase internet connectivity and access to government services in rural and underserved areas.	

Keywords ("codes") in literature review	Proposed themes for PAM curriculum
Legislation, regulations and policies governing public participation	Public participation
Use of online platforms and social media to engage citizens	
Participation mechanisms	
Participation levels, quality and expected outcomes	
Citizen data collection methodology and tools	
Government feedback mechanisms	
Transparency, accountability, openness, and responsiveness of government	
Digital divide and access to government services	
Gauging public opinion and satisfaction levels on policy matters	
Trust in government and legitimacy of government	
Establishing citizen-centric digital services	

Source: (Author’s own construction)

Based on this thematic analysis, seven core themes were identified. These themes are illustrated in Figure 1.

Figure 1: PAM digital governance (DG) curriculum themes



Source: (Author’s own construction)

These seven themes should be incorporated into the curriculum of postgraduate programmes in PAM to prepare prospective and current public managers in the why, what, and how application arenas of digital governance. In the subsections below, each of these seven themes is highlighted briefly to expand the corpus of knowledge that should serve as study material content of learning programmes. It should be noted that these themes are highly interrelated; a neat demarcation of each is thus not possible.

Theme 1: E-government

E-government, or electronic government, refers to the use of digital technologies to enhance the delivery of government services and information to citizens, businesses, and other government entities (Malodia *et al.* 2021). It includes the use of technology to provide government services to citizens, such as online portals for accessing information or conducting transactions. E-government can encompass a wide range of activities, including online voting, online tax filing, digital archiving of government records, online public services, and digital communication between government agencies (West 2004).

At its core, e-government is about making government services more accessible, efficient and effective (Twizeyimana and Anderson 2019). By using digital technologies to streamline processes and improve communication, e-government can help to reduce costs, increase transparency, and improve citizen participation and satisfaction. Additionally, e-government can promote better decision-making by providing government officials with access to more accurate and timely data. It is an important tool for modernising government and improving the delivery of public services. It can help to create more responsive, citizen-centric governments that are better equipped to meet the needs and expectations of their constituents.

As far as the curriculum or study content of e-government is concerned, Malodia *et al.* (2021:8) propose a highly useful integrated framework which makes provision for the following subthemes:

- “Citizen orientation (citizen readiness, cultural context, co-creating value, addressing the digital divide)
- Channel orientation (building cooperative norms, building a sustainable economic model, transforming intermediaries)
- Technology orientation (building technological and managerial capabilities)
- Tangible outcomes (cost and time savings, efficiency advantages)
- Intangible outcomes (citizen satisfaction, trust in government)”.

It is proposed that these subthemes be absorbed in the study content of this theme.

It is evident that preparing prospective and current public managers for e-government applications requires a combination of technical training and

leadership development. Public managers need to be proficient in the use of digital technologies to effectively manage e-government applications. This may involve training on specific software applications, hardware, and networking technologies. Successful e-government applications require strong leadership to ensure that they are implemented effectively (West 2004). Public managers should be trained to fulfil roles such as project managers, change agents, and strategic and transformative leaders. As transformative leaders, public managers should foster a culture of innovation. They need to encourage public officials to think creatively and to embrace new ideas and technologies. A further aspect that deserves attention is networking and collaboration between different stakeholders (Twizeyimana and Anderson 2019). Public managers should be trained in how to foster teamwork and stakeholder engagement. Public managers should also be trained in how to engage citizens and to use citizen feedback to improve the quality of e-services.

Theme 2: Digital transformation

According to Eoma and Leeb (2022:2), the term “digital transformation” refers to the “broader, more profound transformations ICT enables in industry, governance structures, and ecosystems”. What Bostrom, Yudkowsky and Frankish (2014:316) label the “wicked” nature of digital transformation, it entails rather complex governance processes and organisational arrangements for utilising digital ICT to enhance government functions and operations, inclusive of the automatisisation of administrative tasks and the digitalisation of public records. It should be seen in the context of broader public-sector transformation, inclusive of policy changes, administrative reforms and rationalisation, modernisation, and continuous process improvement initiatives (Di Giulio and Vecchi 2021).

According to Eoma and Leeb (2022:3), digital transformation has the potential to bring about a “paradigmatic shift” in government by fundamentally altering the way that government operates, interacts with citizens, and delivers services. Digital transformation can involve a wide range of activities, including the adoption of new technologies, the development of new processes and systems, and the implementation of new policies and strategies (Furtado *et al.* 2023). At its core, digital transformation is about leveraging technology to drive innovation and efficiency in government. By using digital technologies to automate processes, improve communication, and enhance data analysis, governments can become more agile, responsive, and effective (Kitsios, Kamariotou and Mavromatis 2023:3). Additionally, digital transformation can help to improve the overall citizen experience by providing more convenient and accessible services. It can also help to drive economic growth and competitiveness by promoting the development of new technologies and fostering innovation.

As far as the inclusion of digital transformation as a curriculum theme in PAM qualifications is concerned, Kitsios, Kamariotou and Mavromatis (2023:6) propose the following dimensions to direct scholarly engagement:

- Organisational arrangements and improved processes
- Services and products (value creation, access, citizen-focused)
- Relationships (establishing and maintaining a network of actors in a digital society)
- Technological applications (enhance transparency, efficiency and effectiveness)
- Business model (new management models, approaches, organisational culture, skills, policies).

It is evident that digital transformation should be embedded in the competency domains of public managers since it is an important tool for modernising government and ensuring that it remains relevant and effective in the digital age.

Theme 3: Cybersecurity

Recent data breaches in governments globally, and also in the South African Government, firmly established this theme in PAM curriculum. Three prominent cases are the Department of Home Affairs, the Department of Justice and Constitutional Development and the Information Regulator.

In 2017, the Department of Home Affairs suffered a data breach where a contractor was found to be selling private information to third parties. The data breach impacted 30 million South African citizens and resulted in the arrest of several individuals involved in the sale of the data. This breach exposed personal information such as ID numbers, addresses, and passport details (IOL Business Report Online 2017). The breach was particularly concerning as the Department of Home Affairs is responsible for issuing identity documents and passports to citizens and visitors. As such, it can be regarded as the custodian of confidential and sensitive personal information. The breach also raised questions about the security of citizen-related information held by government agencies and the ability of these agencies to protect this information from breaches and unauthorised access. The incident highlights the importance of robust cybersecurity measures to protect sensitive personal information from data breaches and other security threats.

The Department of Justice and Constitutional Development (DOJ&CD) reported a data breach in 2021 that resulted in the interruption of its services. The data breach was caused through ransomware, probably spread through phishing emails containing malicious attachments (DOJ&CD 2021). According to Malinga (2021), the ransomware attack also impacted the functionality of the Information Regulator (IR). The IR is overseen by the DOJ&CD and is responsible for monitoring and enforcing compliance with the provisions of the Promotion of Access to

Information Act and the Protection of Personal Information Act (POPIA). The IR reported that in August 2021 38 similar security breaches occurred. It also reported 139 similar data breaches since the promulgation of the POPIA at South African organisations (Moyo 2021).

In addition to these cases, personal details of as many as 24 million South Africans, and nearly 800 000 businesses were compromised by one of the country's largest credit bureaus, Experian (News24 2020). These cases accentuate the need for public managers to gain competencies in the use of technology to protect government systems and data from unauthorised access, theft, or destruction. Data breaches can have serious consequences for public institutions, including financial risks and damage to reputation. Cybersecurity technology is used to protect government networks and data from cyberthreats, such as hacking and cyberespionage.

The literature review further revealed that aspects such as military and intelligence operations, the development of advanced weapons systems, satellite-based communications, reconnaissance, navigation and national security in general are key components of this theme. Defence and national security technology encompasses a wide range of applications, inclusive of drone technology, advanced surveillance systems, and biometric technology, such as facial recognition and fingerprint scanning to identify and track potential threats.

Effective implementation of cybersecurity initiatives requires a high level of expertise, appropriate technological infrastructure, and a supportive regulatory framework (Cullen 2009). Public managers should ensure that cybersecurity concerns are addressed in a manner that is consistent with international law and human rights principles. They should also have the necessary ethical sensitivity and awareness regarding ways of dealing with citizens' privacy, security and confidentiality concerns (Nissenbaum 2004).

Theme 4: Data analytics

Technology is used to collect, analyse, and interpret data to inform government decision-making and policy development (Shabbir and Gardezi 2020). Data analysis comprises a broad field of study geared towards the ways in which data is gathered, analysed and applied. As such it incorporates the use of big data and AI.

Big data typically refers to large and complex data sets that are too voluminous, unstructured, or varied to be processed and analysed using traditional data-processing methods (Erickson and Rothberg 2014:110). These data sets may include a wide variety of information, such as text, images, audio, video, and social media data. The processing and analysis of big data generally involves the use of advanced technologies and techniques, such as machine learning, natural language processing, and data mining. These tools allow public institutions to extract meaningful insights

from complex data sets, enabling them to make more informed decisions, optimise government operations, and improve overall performance.

According to Löfgren and Webster (2020), typical ways big data is used in government include the following:

- Predictive analytics: Governments can use big data to predict future trends and outcomes in areas such as healthcare, public safety, and transportation. For example, by analysing data on traffic patterns and accidents, governments can predict where and when accidents are likely to occur and take steps to prevent them.
- Fraud detection: Governments can use big data to detect fraud in areas such as tax collection, social security benefits, and public procurement. By analysing large amounts of data, governments can identify patterns of fraudulent activity and take steps to prevent it.
- Public health systems: Governments can use big data to monitor public health and identify outbreaks of diseases. By analysing data on symptoms, travel history, and other factors, governments can track the spread of the disease and take steps to contain it.
- Emergency response: Governments can use big data to improve emergency response times and reduce the impact of disasters. By analysing data on weather patterns, traffic patterns, and other factors, governments can predict and prepare for disasters and respond more quickly to emergencies.

Apart from big data, AI has the potential to transform the way governments operate and deliver services to citizens. AI chatbots such as ChatGPT, Microsoft Bing, and Google Bard are geared to fundamentally alter the ways in which governments obtain, analyse, and utilise policy implementation and service-delivery data. According to Chatterjee, Khorana and Kizgin (2022) and Zuidervijk, Chen and Salem (2021), some application possibilities of AI in governance include the following:

- automation of repetitive tasks, such as data entry, record-keeping, and customer service, freeing up government staff to focus on more complex tasks;
- analysis of data and identification of patterns to inform decision-making in areas such as resource allocation, policymaking, and programme evaluation;
- development of chatbots or virtual assistants that can provide citizens with information and assistance on government services;
- enhance citizen engagement through social media analysis, sentiment analysis, and targeted messaging; and
- improve decision-making by providing decision-makers with more accurate and timely information.

The application of AI in governance has the potential to deliver significant benefits for increased efficiency, effectiveness, and citizen engagement. However,

it is important that the adoption of AI is accompanied by strong regulatory frameworks that address concerns around data privacy and security, bias in AI algorithms, and transparency and accountability in decision-making (Charles, Rana and Carter 2022). Additionally, there is a need for a skilled workforce with expertise in AI to successfully leverage its potential in government operations and service-delivery projects.

Theme 5: Smart cities

Current and prospective public managers will either find themselves in a smart city environment or may engage smart city councils on national or provincial government matters. As such, smart cities can be regarded as an essential theme and curriculum dimension of postgraduate programmes.

Smart cities can be regarded as urban areas (cities) that use technologies to create more efficient, sustainable, and liveable cities, such as through the use of sensors, data analysis, and automation (Joshi, Saxena and Godbole 2016:902; Trindade *et al.* 2017:2). Smart cities leverage digital technologies and data analytics to improve the quality of life for citizens, enhance sustainability, and make city operations more efficient (Kitchin 2022; Stübinger and Schneider 2020). Smart cities integrate various types of data and information, including traffic data, energy consumption data, and citizen feedback, to optimise urban services and make them more efficient and effective (Trindade *et al.* 2017). According to Attaran, Kheibari and Bahrepour (2022), examples of smart city applications include intelligent transportation systems that optimise traffic flow, smart lighting systems that reduce energy consumption, and digital platforms that provide citizens with real-time information and services. Kitchin (2022:156) and Van der Walldt (2018:212) add that smart cities rely on advanced technologies such as the IoT, AI, and big data analytics to capture and analyse data, and to automate processes. By using these technologies, smart cities can become more responsive to citizen needs, more efficient in the use of resources, and more resilient to disruptions and emergencies (Toli and Murtagh 2020).

Smart cities represent a vision for the future of urban living, where technology is harnessed to create more liveable, sustainable, and equitable communities. As such, they offer many opportunities to promote innovation, economic growth, and citizen participation (Trindade *et al.* 2017). By leveraging smart city initiatives effectively, governments can enhance their ability to deliver value to citizens, businesses, and other stakeholders, and create more liveable, sustainable, and resilient urban environments. However, they also require careful consideration of issues related to privacy, security, data management, interoperability, standardisation, citizen participation, and ethical concerns. These are typical considerations that should become part of the curriculum of PAM programmes.

Theme 6: Telecommunications

Digitalisation has had a significant impact on telecommunications in government and will continue to influence the competency domains of prospective public managers. Government telecommunication systems are increasingly using digital technologies such as VoIP and digital messaging services to communicate with citizens, businesses, and other stakeholders. This has replaced traditional landline and paper-based communication methods. Government institutions can now respond to citizen inquiries more quickly and effectively, reducing wait times and thereby improve customer satisfaction (Hyland-Wood *et al.* 2021). Mahmood, Weerakkody and Chen (2019) reason that suitable government telecommunication supports digital transformation and increases overall citizen trust and legitimacy of government. Neely and Collins (2018:3) add that telecommunication systems such as the use of social media platforms have a “democratising” effect on government communications since they generally enhance transparency and accountability.

Roztock, Soja and Weistroffer (2019) accentuate the role of ICT on socio-economic development. They argue that four particular dimensions deserve consideration as far as government telecommunication systems are concerned, namely policy issues, stakeholder engagement, suitable technological applications, and the readiness of society to adapt to a digital environment and be able to adopt new technologies. These four dimensions should be embedded in PAM qualifications.

Digitalisation has also enabled government institutions to automate their telecommunications processes, such as call routing and voicemail. This has significantly improved efficiency and reduced costs. Digitalisation of government telecommunication systems furthermore made it possible for government employees to work remotely, which has become increasingly important in the wake of the Covid-19 pandemic. By using digital tools such as video conferencing and cloud-based software, government employees can collaborate and communicate effectively while working from home. Government employees typically utilise mobile devices such as smartphones and tablets to access government telecommunications systems. By analysing data on call volumes, wait times, and other metrics, government institutions can identify areas for improvement and make data-driven decisions to improve service delivery. It is evident that digitalisation of government’s telecommunication systems generally improves internal and external communication and enables new working arrangements such as remote work, the use of mobile devices, and data analytics.

Theme 7: Public participation

A significant focal point of digital governance is the role that ICT plays in enhancing collaboration, participation and engagement with citizens. A people- or

citizen-centred approach to governance characterises socio-economic development efforts in countries such as South Africa and features prominently in national legislation. As such, public participation should be embedded in PAM qualifications. A people-centred approach generally implies that public managers should consider the needs and voices of the public in the design, delivery, implementation, and evaluation of service-delivery programmes and projects. Public managers in all spheres of government should involve the public in decision-making processes. Performance data should also be analysed to monitor and evaluate the overall successes and failures of these programmes and projects.

Tejedo-Romero *et al.* (2022) argue that especially the local sphere of government should utilise digital platforms and e-government mechanisms to enhance community participation. By collecting and analysing data on community feedback and preferences, municipalities can better understand citizen needs and preferences, and tailor their services accordingly. Hovik and Giannoumis (2022) reason that digital technology enables public participation that eventually could lead to broader socio-economic development. According to Legard (2022:50) government institutions, entities, and agencies should develop suitable “e-participation platforms” and share best practices and lessons learnt.

It is evident that digital governance can improve public participation in several ways. Some additional examples include the following:

- Digitalisation of government services can provide citizens with easy access to information about government programmes, policies, and activities. This can help citizens stay informed and engaged in decision-making processes.
- Digital governance can enable online consultations and feedback mechanisms that allow citizens to provide input on government policies and programmes. This can help ensure that government decisions are informed by the views and perspectives of the public.
- Social media platforms to engage with citizens can provide them with updates on government initiatives and programmes. Social media can also be used to gather feedback and input from citizens.
- Digital government can use crowdsourcing to engage citizens in problem-solving and decision-making processes. This can involve asking citizens to provide ideas or feedback on particular matters.
- Digitalisation can make data more accessible and understandable to the public in a way that allows citizens to better analyse and interpret it. This can help citizens better understand government activities and help them to better influence decisions.

By using digital tools and technologies, governments can help ensure that citizens are informed, engaged, and empowered to participate in government decision-making processes.

CONCLUSION

Digital governance is rapidly transforming the public sector and it brings new demands for public managers. To meet these demands, public managers must be prepared to adapt to the new technologies and understand the impact they have on public institutions and the public they serve. They must also be able to effectively use data analytics, digital communication tools, and cybersecurity measures to improve the efficiency and effectiveness of their operations. Public managers should be able to prioritise the ethical use of technology and ensure that the benefits of digital governance are equitably distributed. This means investing in digital literacy and training programmes, collaborating with government stakeholders to identify and address potential issues, and implementing transparent policies and procedures that protect privacy and data security.

Universities have a moral and legal obligation to prepare current and prospective public managers for the challenges that digital governance presents. As digital technologies continue to transform the public sector, it is essential that public managers have the necessary skills and knowledge to effectively leverage these tools. Postgraduate programmes in PAM should play a critical role in this process by providing education and training content to equip public managers with the cognitive and analytical skills they need to navigate the digital landscape. These programmes should cover a wide range of themes, including e-government, digital transformation, cyber security, data analytics, smart cities, telecommunications, and public participation. Such curriculum domains will help to ensure that public managers are well prepared to address the complex and ever-evolving challenges presented by digital governance.

REFERENCES

- Attaran, H., Kheibari, N. and Bahrepour, D. 2022. Toward integrated smart city: a new model for implementation and design challenges. *GeoJournal*. 87(4):511–526. <https://doi.org/10.1007/s10708-021-10560-w>.
- Bostrom, N., Yudkowsky, E. and Frankish, K. 2014. The ethics of artificial intelligence. In Frankish, K. and Ramsey, W.M. (eds) *The Cambridge handbook of artificial intelligence*. Cambridge: Cambridge University Press. Pp. 316–334.
- Charalabidis, Y. and Lachana, Z. 2020. Towards a science base for digital governance. Paper presented at the 21st Annual International Conference on Digital Government Research. June 2020. Pp. 383–389. <https://doi.org/10.1145/3396956.3400062>.
- Charles, V., Rana, N.P. and Carter, L. 2022. Artificial Intelligence for data-driven decision-making and governance in public affairs. *Government Information Quarterly*. 39(4):a101742. <https://doi.org/10.1016/j.giq.2022.101742>.

- Chatterjee, S., Khorana, S. and Kizgin, H. 2022. Harnessing the potential of artificial intelligence to foster citizens' satisfaction: an empirical study on India. *Government Information Quarterly*. 39(4):a101621. <https://doi.org/10.1016/j.giq.2021.101621>.
- Criado, J.I. and de Zarate-Alcarazo, L.O. 2022. Technological frames, CIOs, and Artificial Intelligence in public administration: A socio-cognitive exploratory study in Spanish local governments. *Government Information Quarterly*. 39(3):1–13. <https://doi.org/10.1016/j.giq.2022.101688>.
- Cullen, R. 2009. Culture, identity and information privacy in the age of digital government. *Online Information Review*. 33(3):405–421. <https://doi.org/10.1108/14684520910969871>.
- Department of Justice and Constitutional Development (DOJ&CD). 2021. Justice and constitutional development establishes security breach on IT systems. Official Statement. Pretoria: DJCD. Available at: <https://www.gov.za/speeches/justice-and-constitutional-development-establishes-security-breach-it-systems-20-sep-2021>. (Accessed 4 May 2023).
- Department of Public Service and Administration (DPSA). 2003. SMS Handbook: Chapter 5, Competency Framework (pp. 2–26). Pretoria: DPSA. Available at: https://www.dpsa.gov.za/dpsa2g/documents/sms/publications/sms_hb_ch_5.pdf. (Accessed on 4 May 2023).
- Department of Public Service and Administration (DPSA). 2018. Guideline for the management of competency assessments for the SMS in the public service. Pretoria: DPSA. Available at: <https://www.dpsa.gov.za/dpsa2g/documents/sms/2018/Leadership%20Management%20Competency%20Framework.pdf>. (Accessed on 5 May 2023).
- Department of Public Service and Administration (DPSA). 2019. Practice note on the implementation of the pre-entry certificate for the Senior Management Service (SMS). Pretoria: DPSA. Available at: https://www.dpsa.gov.za/dpsa2g/documents/sms/2019/plsm_19_12_2019.pdf. (Accessed on 5 May 2023).
- Di Giulio, M. and Vecchi, G. 2021. Implementing digitalization in the public sector. Technologies, agency, and governance. *Public Policy and Administration*. 38(2):133–158. <https://doi.org/10.1177/09520767211023283>.
- Edelmann, N., Mergel, I. and Lampoltshammer, T. 2023. Competences that foster digital transformation of public administrations: an Austrian case study. *Administrative Sciences*. 13(2):44–61. <https://doi.org/10.3390/admsci13020044>.
- Engvall, T. and Flak, L.S. 2022. Digital Governance as a Scientific Concept. In: Charalabidis, Y., Flak, L.S. and Pereira, G.V. (Eds.) *Scientific Foundations of Digital Governance and Transformation: Concepts, Approaches and Challenges*. Cham: Springer. Pp. 25–50.
- Eoma, S-J. and Leeb, J. 2022. Digital government transformation in turbulent times: responses, challenges, and future direction. *Government Information Quarterly*. 39(2):1–9 (a101690). <https://doi.org/10.1016/j.giq.2022.101690>.
- Erickson, S. and Rothberg, H. 2014. Big data and knowledge management: establishing a conceptual foundation. *Electronic Journal of Knowledge Management*. 12(2):108–116.
- Fanelli, S., Lanza, G., Enna, C. and Zangrandi, A. 2020. Managerial competences in public organisations: the healthcare professionals' perspective. *BMC Health Services Research*. 20. a303. <https://doi.org/10.1186/s12913-020-05179-5>.
- Fishenden, J. and Thompson, M. 2013. Digital Government, Open Architecture and Innovation: Why Public Sector IT Will Never Be the Same Again. *Journal of Public Administration Research and Theory*. 23(4):977–1004.

- Furtado, L.S., da Silva, T.L.C., Ferreira, M.C.F. and de Macedo, J.A.F. 2023. A framework for digital transformation towards smart governance: using big data tools to target SDGs in Ceará, Brazil. *Journal of Urban Management*, 12(1):74–87. <https://doi.org/10.1016/j.jum.2023.01.003>.
- Hovik, S. and Giannoumis, G.A. 2022. linkages between citizen participation, digital technology, and urban development. In Hovik, S., Giannoumis, G.A., Reichborn-Kjennerud, K., Ruano, J.M., McShane, I. and Legard, S. (eds.) *Citizen participation in the information society: comparing participatory channels*. Cham: Palgrave Macmillan. Ppp. 1–24.
- Hyland-Wood, B., Gardner, J., Leask, J. and Ecker, U.K.H. 2021. Toward effective government communication strategies in the era of COVID-19. *Humanities & Social Sciences Communications*. 8(30):1–11. <https://doi.org/10.1057/s41599-020-00701-w>.
- IOL Business Report Online. 2017. Home Affairs set to investigate massive data breach. 19 October. Available at: <https://www.iol.co.za/business-report/companies/home-affairs-set-to-investigate-massive-data-breach-11622847>. (Accessed on 6 May 2023).
- Jones, P. 2017. The futures of Canadian governance: foresight competencies for public administration in the digital era. *Canadian Public Administration*. 60(4):657–681. <https://doi.org/10.1111/capa.12241>.
- Joshi, S., Saxena, S. and Godbole, T. 2016. developing smart cities: an integrated framework. *Procedia Computer Science*. 93:902–909. <https://doi.org/10.1016/j.procs.2016.07.258>.
- Karpenko, O. and Glubochenko, K. 2018. The Concept of Cultural Competence of the Digital Governance Specialists within Public Administration. *National Academy of Managerial Staff of Culture and Arts Herald*. 14–18. <https://doi.org/10.32461/2226-3209.3.2018.147106>.
- Kausch-Zongo, J. and Schenk, B. 2022. General technological competency and usage in public administration education: an evaluation study considering on-the-job trainings and home studies. *Smart Cities and Regional Development (SCRD) Journal*. 6(1):55–65. <https://doi.org/10.25019/scrd.v6i1.120>.
- Kitchin, R. 2022. Conceptualising smart cities. *Urban Research & Practice*. 15(1):155–159. <https://doi.org/10.1080/17535069.2022.2031143>.
- Kitsios, F., Kamariotou, M. and Mavromatis, A. 2023. Drivers and outcomes of digital transformation: the case of public sector services. *Information*. 14(1):1–14 (a43). <https://doi.org/10.3390/info14010043>.
- Lara, F.J., Mogorrón-Guerrero, H. and Ribeiro-Navarrete, S. 2020. Knowledge of managerial competencies: cross-cultural analysis between American and European students. *Economic Research – Ekonomska Istraživanja*. 33(1): 2059–2074. <https://doi.org/10.1080/1331677X.2019.1585271>.
- Legard, S. 2022. Displacement and citizen participation: a comparison of the enactment of e-participation platforms in Oslo and Madrid. In Hovik, S., Giannoumis, G.A., Reichborn-Kjennerud, K., Ruano, J.M., McShane, I. and Legard, S. (Eds.). *Citizen participation in the information society: comparing participatory channels*. Cham: Palgrave Macmillan. Pp. 49–70.
- Löfgren, K. and Webster, C.W.R. 2020. The value of big data in government: the case of “smart cities”. *Big Data & Society*. 7(1). <https://doi.org/10.1177/2053951720912775>.
- Mahmood, M., Weerakkody, V. and Chen, W. 2019. The role of information and communications technology in the transformation of government and citizen trust. *International Review of Administrative Sciences*. 86(4):708–728. <https://doi.org/10.1177/002085231881679>.

- Malinga, S. 2021. Justice Dept cyber attack spills over to Info Regulator. Johannesburg: ITWeb. <https://www.itweb.co.za/content/xA9PO7NZOBl7o4J8>.
- Malodia, S., Dhir, A., Mishra, M. and Bhatti, Z.A. 2021. Future of e-government: an integrated conceptual framework. *Technological Forecasting and Social Change*. 173. <https://doi.org/10.1016/j.techfore.2021.121102>.
- Milakovich, M.E. 2021. *Digital Governance: Applying Advanced Technologies to Improve Public Service*. (2nd ed.). New York, NY: Routledge.
- Misuraca, G. and Viscusi, G. 2014. Digital Governance in the Public Sector: challenging the Policy-Maker's innovation dilemma. Conference: 8th International Conference on Theory and Practice of Electronic Governance (ICEGOV2014). <https://doi.org/10.1145/2691195.2691286>.
- Moyo, A. 2021. Info watchdog reveals 139 data breaches since POPIA. Johannesburg: ITWeb. Available at: <https://www.itweb.co.za/content/dgp45Ma6WE2qX9l8>. (Accessed 8 May 2023).
- National School of Government (NSG). 2020. Certificate for entry into the SMS (Senior Management Service). Pretoria: NSG. Available at: <https://www.thensg.gov.za/training-course/sms-pre-entry-programme/>. (Accessed on 2 May 2023).
- Neely, S.R. and Collins, M. 2018. Social media and crisis communications: a survey of local governments in Florida. *Journal of Homeland Security and Emergency Management*. 15(1):1–13 (a20160067). <https://doi.org/10.1515/jhsem-2016-0067>.
- News24. 2020 (19 August). Personal details of 24 million South Africans may have been exposed after attack on credit bureau. Available at: <https://www.news24.com/news24/bi-archive/the-personal-details-of-millions-of-south-africans-have-just-been-hacked-2020-8#:~:text=Experian%20says%20the%20personal%20information,law%20enforcement%20and%20regulatory%20authorities>. (Accessed on 5 May 2023).
- Nissenbaum, H. 2004. Privacy as Contextual Integrity. *Washington Law Review*, 79(1):119–158.
- Pereira, G.V., Ronzhyn, A. and Wimmer, M.A. 2022. Building Digital Governance Competencies: Baseline for a Curriculum and Master Programme. In Charalabidis, Y., Flak, L.S. and Pereira, G.V. (Eds.) *Scientific Foundations of Digital Governance and Transformation: Concepts, Approaches and Challenges*. Pp. 361–383. Cham: Springer. https://doi.org/10.1007/978-3-030-92945-9_14.
- Roztocki, N., Soja, P. and Weistroffer, H.R. 2019. The role of information and communication technologies in socioeconomic development: towards a multi-dimensional framework. *Information Technology for Development*. 25(2):171–183. <https://doi.org/10.1080/02681102.2019.1596654>.
- Shabbir, M.Q. and Gardezi, S.B.W. 2020. Application of big data analytics and organizational performance: the mediating role of knowledge management practices. *Journal of Big Data*. 7:1–17, a47. <https://doi.org/10.1186/s40537-020-00317-6>.
- Shubha, V. 2017. Leading Digital Transformation with e-Governance Competency Framework. ICEGOV '17: Proceedings of the Special Collection on eGovernment Innovations in India. Pp.11–17. <https://doi.org/10.1145/3055219.3055223>.
- Sithole, V.E. and Van der Waltd, G. 2016. An e-governance training model for public managers: the case of selected Free State provincial departments. *Administratio Publica*, 24(1):137–162.
- Stübinger, J. and Schneider, L. 2020. Understanding smart city: a data-driven literature review. *Sustainability*. 12(20):a8460. <https://doi.org/10.3390/su12208460>.

- Tejedo-Romero, F., Araujo, J.F.F.E., Tejada, A. and Ramírez, Y. 2022. E-government mechanisms to enhance the participation of citizens and society: exploratory analysis through the dimension of municipalities. *Technology in Society*. 70:a101978. <https://doi.org/10.1016/j.techsoc.2022.101978>.
- Toli, A.M. and Murtagh, N. 2020. The concept of sustainability in smart city definitions. *Frontiers in Built Environment*. 6:1–10 (a77). <https://doi.org/10.3389/fbuil.2020.00077>.
- Trindade, E.P., Hinnig, M.P.F., da Costa, E.M., Marques, J.S., Bastos, R.C. and Yigitcanlaret, T. 2017. Sustainable development of smart cities: a systematic review of the literature. *Journal of Open Innovation: Technology, Market, and Complexity*. 3(11):1–14. <https://doi.org/10.1186/s40852-017-0063-2>.
- Twizeyimana, J.D. and Anderson, A. 2019. The public value of e-government: A literature review. *Government Information Quarterly*. 36(2):167–178. <https://doi.org/10.1016/j.giq.2019.01.001>.
- Van der Waltd, G. 2018. City governance resilience, SMART Cities, and big data: towards a meta-interface framework. In Van der Steen, M., Burger, M. and Burger, W. (eds.) *The governance of transitions – the transitions of governance: cases from southern Africa*. Cape Town: SUN Press. Pp. 209–232.
- West, D.M. 2004. E-government and the transformation of service delivery and citizen attitudes. *Public Administration Review*. 64(1):15–27.
- Zuiderwijk, A., Chen, Y-C. and Salem, F. 2021. Implications of the use of artificial intelligence in public governance: a systematic literature review and a research agenda. *Government Information Quarterly*. 38(3):a101577. <https://doi.org/10.1016/j.giq.2021.101577>.

AUTHOR'S CONTACT DETAILS

Prof Gerrit van der Waltd

Research Professor: Public Governance

Focus Area: Social Transformation

North-West University

Private Bag X6001

Potchefstroom

Tel. 018 2991633

E-mail: Gerrit.vanderwaltd@nwu.ac.za

Fiscal Decentralisation

Lesotho Subnational Government Financing

T J Mohapi*

Department of Public Administration and Management
University of South Africa

V T Sambo

Department of Public Administration and Management
University of South Africa

G M Ferreira

Department of Public Administration and Management
University of South Africa

ABSTRACT

The Lesotho government implemented decentralisation as a policy for promoting popular participation and better local public service, for securing socio-economic wellness for the Basotho. Decentralisation was operationalised in 1997 through the Local Government Act 6 of 1997 (Lesotho Local Government Act 6 of 1997). In 2014 the government of Lesotho adopted the national decentralisation policy as a comprehensive reform framework for building strong and effective local authorities for ensuring economic and social benefits to the people. In the public financial sector, the focus is on the development of fiscal decentralisation legislation to ensure functional fiscal relations and public financial management to support the decentralisation process. In this regard, the study on which this article is based, focused on domesticating a fiscal decentralisation methodology for Lesotho. The article considers among other things the conceptual nature of fiscal decentralisation and subnational government financing in Lesotho. The study was conducted using the mixed methods research methodology. Data was collected through document analysis, closed-ended questionnaires and interviews within local councils in the four districts namely: Maseru, Berea, Mophale's Hoek and Thaba-Tseka, to ensure representation of all the regions of the country. The respondents who were purposively selected, included elected councillors and technical staff. The article recommends the development of a comprehensive subnational government framework to improve

resource mobilisation for subnational government financing. The findings of the research highlight the importance of financial independence and self-reliance of subnational government. The findings also show that subnational governments need to have effective financial management systems to guarantee the availability and management of financial resources.

INTRODUCTION

Decentralisation in Lesotho is a constitutional matter as Section 106 of the Constitution Act 5 of 1993 stipulates that parliament shall create necessary authorities for enabling urban and rural communities to determine their affairs and to develop themselves (Lesotho Constitution Act 5 of 1993). Lesotho implemented decentralisation as a developmental policy to promote equitable economic development in all the districts. According to the National Decentralisation Policy of 2014, the objectives of decentralisation include: promoting access to quality public services, promoting equitable economic development across the country and securing local autonomy through capacitated decentralised institutions, among others (Lesotho Ministry of Local Government and Chieftainship 2014:xi). The Lesotho decentralisation process is an economic development strategy aimed at ensuring equitable development and access to quality public services through democratic and accountable local institutions that promote citizens' participation in governance.

In decentralised systems, all government spheres have different responsibilities and functions that require financial resources to be delivered/executed. Allain-Dupre (2018:3) asserts that the process of allocating authority and responsibilities between government spheres is an important aspect of the decentralisation process as it has a significant impact on the success of the process. The inter-governmental relations between government spheres are important for the success of decentralisation processes. This is applicable to intergovernmental fiscal relations, whereby division of fiscal authority and responsibilities is considered to have a critical impact on the performance of decentralisation processes (Kim & Dougherty 2018:14). The intention of intergovernmental fiscal relations is to ensure that all spheres of government have adequate resources to finance their mandate. This article is about fiscal decentralisation with specific focus on Lesotho's subnational government finance. It will explore the nature of fiscal decentralisation and highlight the importance of subnational government financing in enabling subnational governments to finance and deliver on their mandate.

The assignment of expenditure responsibilities to subnational government requires equivalent allocation of financial resources for performing them.

Subnational governments experience a mismatch between expenditure responsibilities and revenue sources restraining them from performing their functions (OECD 2016:21). The absence of adequate financial resources at subnational spheres is more likely to be counter-productive, hence the importance of subnational finance. In acknowledging the importance of subnational finance, the African Development Bank Group (2019:1) argues that the inability of subnational governments to assume their responsibilities due to financial challenges would result in deterioration of quality and quantity of public services and increase budget deficits. According to Smoke (2019:3) subnational government finance comprises all financial resources for financing subnational developmental (capital) and recurrent activities. Subnational development finance is accessed through development transfers, borrowing and public-private partnership while recurrent finance is mobilised through own-source revenue and intergovernmental transfers (Smoke 2019:4–8). Central to subnational finance is the need to strengthen the subnational resource base, to facilitate mobilisation of adequate financial resources and ensure their effective utilisation. According to the African Development Bank Group (2019:3), the success of subnational governments in performing their functional assignments depends on the structure of intergovernmental relations, structure of governance, institutional capacity, mechanism for financing subnational governments and legislation. The objective of subnational finance is then to enhance local development by considering the structure of intergovernmental fiscal relations, the financial management system, the mechanism for financing subnational governments and the legal frameworks governing operations (ADBG 2019:3).

BACKGROUND AND RATIONALE

The method of allocating fiscal responsibilities between different government spheres has consequences for economic growth, government efficiency, participation and ultimately the performance of the decentralisation process (Reingewertz 2014:2). This is because the fiscal capacity of government spheres has a notable impact on their ability to deliver on their mandate. According to Rahim (2013:3), the primary objective of fiscal decentralisation is to strengthen a subnational government's finances and its capacity to deliver services, by giving it revenue authority and expenditure responsibilities. Subnational governments have the challenge of the widening gap between revenue supply and spending needs. Adamtey, Obeng and Sarpong (2020 in Mohamp 2023), argue that many subnational governments in sub-Saharan African countries are unable to provide quality public services to communities mainly because of their inability to mobilise adequate resources. According to the authors there is no consensus among

scholars on the best strategies for promoting greater revenue mobilisation by subnational governments for them to make adequate fiscal resources available to deliver services (Adamtey *et al.* 2020 in Mohapi 2023). Lesotho is not an exception to this challenge as subnational governments remain under-funded with low local revenue generation and a weak public financial management and accountability system to support quality service delivery (Lesotho Ministry of Local Government and Chieftainship 2014:5).

The Lesotho fiscal decentralisation is explained through intergovernmental transfers, as transfers are the only fiscal decentralisation element that is practised at Lesotho subnational governments. Lesotho is a unitary state with two spheres of government, national and local government; the local government consists of four types of local authorities namely: district councils, municipal councils, urban councils and community councils (Lesotho Local Government Amendment Act 5 of 2004). The national government is the primary financier of local authorities, as subnational governments depend on national transfers to finance their entire budget. Nyane (2016:67), observes that in most cases, including Lesotho, fiscal decentralisation is implied as transfers from national to subnational governments due to weak local revenue. In acknowledging the national government as the principal financier for subnational government, the government of Lesotho caters for subnational governments' operational and capital budget through intergovernmental transfers (Daemane 2012:203). Kali (2020:15), argues that the challenge is not limited to dependency on finances but also on the allocations that are inadequate, unpredictable and often erratic in timing.

Lesotho's subnational governments do not have revenue and expenditure powers as they remain beholden to the national government for budget management (Lesotho Ministry of Local Government and Chieftainship 2014:8). Although the National Decentralisation Policy of 2014 announced devolution as a mode of decentralisation for the country, the fiscal devolution is not yet in place, as Lesotho's subnational governments do not have sources of revenue and cannot expend local revenue. Consequently, Lesotho's local authorities become non-autonomous institutions with unfunded mandates. In analysing the decentralisation process, Nyane (2016:69) points out that fiscal decentralisation in Lesotho is still in its infancy as subnational governments depend on the national government for sources and rates of revenue, expenditure and budget management. Sefeane (2020:2) maintains the assessment and identification of potential sources of revenue remains a purview of national government.

According to Kali (2020:15), Lesotho's subnational governments do not have authority to independently develop budget and mobilise resources and constantly face the challenge of insufficient financial resources. The National Decentralisation Policy of 2014, acknowledges that local councils are not sufficient structures for achieving development because they are under-funded

(Lesotho Ministry of Local Government and Chieftainship 2014:2). Lesotho adopted the National Decentralisation Policy of 2014 as a platform for reforming decentralisation. The decentralisation reforms mandate government to develop various legislation to support the decentralisation process towards achieving its mandate. In the public financial sector, the government is to develop comprehensive fiscal decentralisation frameworks focusing on structure and functioning of fiscal relations, and public financial management, to support decentralisation processes. In this regard, the article intends to propose to the government, effective policy tools to assist subnational governments to raise adequate revenue to finance their mandate. Consequently, the objective of the article is to recommend strategies for improving resource mobilisation for financing Lesotho's subnational governments.

The article will contextualise fiscal decentralisation with a specific focus on subnational government financing. This will be followed by a brief description of the research design and methodology of the study and a discussion of the findings. The conclusion of the study will summarise the article as well as present the recommendations.

In summary, fiscal decentralisation is a critical component of the decentralisation process as it has an outcome determining impact on the process. The above discussion indicated the primary objective of fiscal decentralisation and the growing gap between revenue supply and expenditure demands at subnational government as the major challenge towards delivery of quality services. The multiple fiscal decentralisation challenges of Lesotho's subnational governments were highlighted. In addressing the challenges, the National Decentralisation Policy of 2014 was adopted as a bedrock for reforms proposing development of comprehensive legislative frameworks to support devolution. The article would contribute by recommending effective policy tools to strengthen subnational finance.

CONTEXTUALISING FISCAL DECENTRALISATION

The literature defines fiscal decentralisation within budget parameters. Fiscal decentralisation refers to the authority of subnational governments to independently manage their budget in terms of revenue and spending (Digdowiseiso, Murshed & Bergh 2022:1). According to Otto and Danquah (2021:2), fiscal decentralisation is about sharing taxing and spending responsibilities between spheres of government, whereby subnational governments are accorded revenue and expenditure autonomy. In this regard, fiscal decentralisation is assessed through the extent to which subnational governments have sufficient resources, generate their own revenue, and the extent to which transfers are stable and predictable (Nyane 2019:8).

Osei (2017:8) explains fiscal decentralisation through assignment of expenditure and revenue functions, intergovernmental transfers and subnational borrowing as its core elements.

According to Hendriks (2014:16), fiscal decentralisation can be regarded as part of intergovernmental fiscal relations as it covers principles of fiscal relations between national and subnational governments. Fiscal decentralisation entails the capacity of subnational governments to independently generate and manage their financial resources and have authority over expenditure (Rodriguez-Acosta 2016:26). In this regard, fiscal decentralisation is charged with generation and management of fiscal resources at subnational governments to ensure service delivery. Tkachenko (2020:74), reasons that fiscal decentralisation should incorporate effective financial management systems to ensure an effective and efficient use of public funds and delivery of quality services to the local population. The public financial management system thus refers to a set of strategies for collecting, allocating, spending and accounting for public resources (Morgner & Chene 2014:2; Cheruiyot, Oketch, Namusonge & Sakwa 2017:212).

THEORETICAL PERSPECTIVES

The theoretical base of fiscal decentralisation assists in understanding factors influencing the success of fiscal decentralisation, the principles guiding the allocation of expenditure sources and preferred intergovernmental transfers design system. The literature identifies the first-generation and second-generation theories in explaining fiscal decentralisation.

First-generation theory

The first-generation theory of fiscal decentralisation explains it from an economic perspective and identifies three main functions of government as: allocation of resources, income distribution and macroeconomic stability (Yushkov 2015:405). Ejibowah (2018:221), explains that in line with the subsidiary principle, the first-generation theory assigns distribution and stabilisation functions to national government, and the allocation function to subnational government. The subsidiary principle, points to the need to organise responsibilities in a way that functions are performed by the smallest, least centralised and capable authority (Martinez-Vazquez, Lago-Penas & Sacchi 2016:3). The main objective of the subsidiary principle is to enhance efficiency and accountability by linking service benefits to the costs of their provision through local taxes or service fees (Sharma 2015:20).

In assigning expenditure responsibilities, the first-generation theory considers correspondence between the sphere of government and beneficiaries (Hart & Welham 2016:9; Boadway & Eyraud 2018:15). Thus, national government is expected to provide services that are intended to benefit the entire nation and that their provision goes beyond subnational government territories. The theory allocates benefit tax to subnational government and reserves all other tax bases for national government (Jha 2012:9). Ozkok and Cutcu (2021:3), explain that this benefit principle results in financial challenges including vertical imbalances, horizontal inequalities and spill over/externalities that the first-generation theory proponents suggest should be addressed through intergovernmental transfers. These financial challenges are the existence of fiscal gaps resulting from the discrepancy between subnational governments' financial resources and spending requirements (Hendriks 2014:20). In acknowledging the fiscal gaps, the first-generation theory supports the engagement of intergovernmental transfers as a source of subnational governments' revenue. Ankamah (2012:33) asserts that intergovernmental transfers contribute to fiscal decentralisation by promoting strategic planning through financial projections on the expected allocation for subsequent years thereby allowing autonomous decisions on how the resources can be utilised.

Design of intergovernmental transfers

Intergovernmental transfers refer to a multidimensional component of fiscal decentralisation whereby finances are transferred from one government sphere to another (Hart & Welham 2016:12; Leon 2019:14). Central to intergovernmental transfers is the asymmetry between subnational expenditure needs and revenues supply, hence their primary objective is to complement the subnational governments' revenue and ensure availability of resources to address the inevitable fiscal gaps between and among spheres of government (Lual 2018:7; Shotton & Gankhuyag 2019:31). In relation to various objectives, the literature identifies the following intergovernmental transfers:

- Transfer for addressing vertical fiscal gaps: Hendriks (2014:21) explains a vertical fiscal gap as a financial structure in which national government has excess funds relative to its spending needs, while subnational government has revenue shortfall relative to its spending responsibilities. In such a situation, the intergovernmental transfer is designed to address the gap between subnational government's spending responsibility and revenue supply (Hobdari, Nguyen, Salvatore & Ruggiero 2018:14).
- Transfer for addressing horizontal fiscal gaps/equalisation: A horizontal fiscal gap refers to the disparity between spending needs and financial resources among subnational governments at the same level (Al-Samarri & Lewis 2021:12). Boadway and Eyraud (2018:20) explain that the transfer for

addressing a horizontal fiscal gap intends to promote national equity and efficiency by reducing disparity in the ability of subnational governments to offer similar services at a similar tax rate.

- Transfer for correcting externalities: Externalities occur when the services provided by one subnational government benefit citizens beyond its borders, hence the transfer is designed as a compensation strategy to mitigate the impact (Jha 2012:10; Boadway & Eyraud 2018:21).
- Transfer for supporting national standards: Boadway and Eyraud (2018:21), explain that intergovernmental transfers may also be designed to encourage subnational governments to implement expenditure programmes that satisfy national standards and priorities that, without the transfer, would not be given priority at local level.

In general, the first-generation theory prescribes the minimal assignment of tax authority to subnational government resulting in fiscal gaps. The fiscal gaps are addressed through coherent intergovernmental transfers' design.

Second-generation theory

The second-generation theory complements the first-generation theory as it enriches the scope of fiscal decentralisation by considering incentives generated by institutions (Valdesalici 2018:3). Ejobowah (2018:224) explains that the second-generation theory subscribes to the basis of functional assignment of the first-generation theory, but puts more emphasis on the revenue assignment. The second-generation theory highlights the need for subnational governments to generate their own revenues for financing their functional responsibilities (Ejobowah 2018:224). In this regard, the second-generation theory asserts that subnational governments should have their own revenue to finance their functions and prefer minimal intervention by national government (Martinez-Vazquez *et al.* 2016:4). Similarly, Jha (2012:2) points out that the important issue becomes the incentives or motivation yielded by the tax collection authority at the subnational sphere. The theory capacitates subnational government with bargaining power and promotes autonomy (Leon 2019:13). The second-generation theory emphasises self-reliance and financial independence of subnational governments.

In promoting subnational governments' fiscal autonomy and limiting national government intervention, the literature considers the concept of hard budget constraints and soft budget constraints. According to Ejobowah (2018:225), hard budget constraints refer to a situation where institutions take full responsibility for their financial activities by funding their expenditure with their own revenue, and internalise any deficits without considering external intervention. Contrarily, in soft budget constraints institutions dispatch losses and depend on external bail

out (Ejobowah 2018:225). The practicality of hard budget constraints is joined to the existence of market-based and rule-based institutions, whereby government structures would be designed in a way that they motivate and encourage public officials to strive for economic prosperity (Leon 2019:13). Ejobowah (2018:225) explains that in a market-based setting, subnational governments become independent actors with autonomous taxing and spending powers, and they can borrow money and manage their financial systems. The rule-based institutional design refers to the legislative frameworks that prevent subnational governments from taking part in fiscal malpractices that can cause excessive losses (Ejobowah 2018:225). On the one hand, these legal instruments set the parameters within which subnational governments can exercise their full fiscal autonomy without disturbing the macroeconomic stability. On the other hand, the legal instruments protect subnational governments from national government intervention in their spending and taxation decisions.

The first-generation and second-generation theories agree on the engagement of a subsidiary principle in assigning functional responsibilities, while they differ in revenue assignments. In maintaining macroeconomic stability, the first-generation theory resorts to minimal assignment of tax authority to subnational government, and this is supplemented by national government transfers to address fiscal imbalances. Complementary, second-generation theory advocates for the autonomy of subnational governments by matching expenditure and revenue assignments to ensure availability of local revenue. The second-generation theory is recommended for developing comprehensive subnational government financing mechanisms. The adoption of a market-based and rule-based system would create an enabling environment for developing robust intergovernmental fiscal frameworks and policies to finance vibrant subnational government. In a market-based system, subnational governments would be financially independent and disciplined institutions that can independently mobilise resources to finance their mandate. The rule-based system would promote subnational fiscal responsibility and prevent financial malpractices, resulting in autonomous and creditworthy subnational governments capable of managing their financial system.

RESEARCH METHODOLOGY

The research design and methodology for the study was mixed-method research, explained within the pragmatism philosophy. According to Okesina (2020:16) the pragmatism paradigm advocates for a research approach that is realistic and allows use of multiple perspectives by mixing paradigms, approaches and methods of data collection and analysis. Shannon-Baker (2016:325) explains that the combination of methods, approaches and paradigms in one study, is

for complementing the strength and weaknesses of each perspective. In this study, convergent mixed-method research design was used for complementarity and triangulation reasons. Walker and Baxter (2019:3) define the convergent design as a parallel design that engages both qualitative and quantitative methods simultaneously. The study engaged closed-ended questionnaires, open-ended interviews and documentary review to collect data. The study respondents (councillors and technical employees) were selected through purposive sampling, to secure inclusion of information-rich participants in all areas of interest (Baltes & Ralph 2020:4).

The study population consisted of council management teams, councillors and finance officers of District Councils, Urban Councils, Community Councils and Municipal Councils from three districts (Berea, Thaba-Tseka and Mohale's Hoek), treated as a homogenous group of local authorities' employees. The questionnaires were distributed among council management teams (except their chief executive officers who were to be interviewed), council finance officers and members of the finance committee (except chairpersons who were to be interviewed). Of 74 distributed questionnaires 44 were completed and returned. Interviews were held with chief executive officers of councils and chairpersons of finance committees of participating councils. Of 70 employees 26 were interviewed in their respective offices as the exercise reached saturation.

A literature review and relevant empirical documents were used to formulate the interview schedule and questionnaire questions which had similar content for triangulation purposes. The research used a closed-ended four scale Likert type questionnaire, coded categorically with valueless numbers in which "1 is assigned Agree, 2=Partly Agree, 3=Disagree and 4=Totally Disagree". Newberry, O'Leary and Israel (2021:2) explain that in categorical items, the numbers assigned represent the referred category without an attached value. Univariate statistical analysis was used to analyse data collected through the questionnaires. Additionally, content analysis was used to analyse data collected through interviews. The data was converted into text and then coded for establishing a connection between the data and identified themes. This enabled the extraction and examination of all segments of data to summarise and categorise it for demonstrating observed themes, patterns and views about the subject under study.

FINDINGS AND DISCUSSIONS

This section presents reporting and analysis of data. Table 1 presents data collected through questionnaires and Table 2 the data collected through interviews. The reporting and analysis of findings of the research is on the identified fiscal decentralisation elements and the literature views on such elements.

Table 1: Reporting and analysis of findings from data collected through questionnaires

Questions	Agree	Partly agree	Disagree	Totally disagree	Interpretation
CONCEPTUALISING FISCAL DECENTRALISATION Literature review: Fiscal decentralisation is the allocation of budget powers in terms of revenue and expenditure as well as intergovernmental transfers between government spheres, whereby subnational governments are given the authority through a legal framework to manage their fiscal system and ensure the availability of resources to finance their mandate (Hendriks 2014:17; 'Nyane, 2019:8).					
Q1.1 Allocation of budget powers as revenue management to subnational government.	0%	11%	21%	68%	89% of the respondents disagree with the view that national government allocates budget powers as revenue management to subnational governments.
Q1.2 Allocation of budget powers as expenditure management to subnational management.	0%	18%	32%	50%	82% of the respondents disagree with the view that national government allocates budget powers such as expenditure management to subnational government.
Q 1.3 Support of local service delivery through intergovernmental transfers.	54%	39%	7%	0%	93% of the respondents agree that national government supports local service delivery through intergovernmental transfers.
Q1.4 Subnational government financial management is guided by multiple financial regulations.	59%	27%	9%	5%	86% of the respondents agree that subnational government financial management is governed through the use of multiple financial regulations.
REVENUE MANAGEMENT Literature review: As local governance bodies charged with the development of their communities, subnational governments have a responsibility to manage local revenue for financing their functions (Ejobowah 2018:224).					
Q2.1 Authority to determine revenue sources.	41%	27%	9%	23%	68% of the respondents agree that subnational governments have authority to determine sources of revenue, while 32% argue against it.

Questions	Agree	Partly agree	Disagree	Totally disagree	Interpretation
Q2.2 Authority to set taxes and fee rates within subnational territory.	29%	9%	23%	39%	62% of finance officers and councillors that completed the questionnaire are of the view that subnational governments do not have the authority to determine taxes and fee rates within their respective territories.
Q 2.3 The capacity of subnational government to collect revenue.	48%	20%	30%	2%	68% of the respondents agree that subnational governments have administrative capacity to collect data.
Q 2.4 Ownership of bank accounts by subnational governments.	0%	0%	89%	11%	All the respondents disagree with the statement that all councils have bank accounts according to the Local Government Act of 1997.
Q2.5 Availability of tax administration regulations at councils.	25%	9%	30%	36%	66% of the respondents argue that subnational governments do not have tax administration regulations.
EXPENDITURE MANAGEMENT Literature review: For subnational governments to perform their responsibility of serving local communities, they need to allocate and use resources responsively, efficiently and effectively (Tkachenko 2020:74).					
Q3.1 Development of developmental needs and agenda through District Development Coordination Committee.	48%	27%	20%	5%	75% of the respondents agree that districts identify local development needs and establish their development agenda through the District Development Coordination Committee.
Q3.2 Reflection of local economic and social needs in the district budget.	54%	32%	14%	0%	86% of the respondents agree that district budget policy reflects economic and social preferences of local citizens.
Q3.3 The use of local revenue for local development.	9%	11%	25%	55%	The majority of the respondents (80%) disagree that the locally collected revenue is used to finance local development projects.

Questions	Agree	Partly agree	Disagree	Totally disagree	Interpretation
Q3.4 Availability of budget allocation strategies among councils.	7%	20%	30%	43%	73% of the respondents are against the view that there are predetermined budget allocation strategies among councils in the district.
INTERGOVERNMENTAL TRANSFER Literature review: Intergovernmental transfer refers to a multidimensional component of fiscal decentralisation whereby finances are transferred from one sphere of government to another for addressing fiscal gaps (Leon 2019:14; Shotton & Gankhuyag 2019:31).					
Q4.1 Transfers for addressing asymmetry between revenue and expenditure.	39%	16%	25%	20%	55% of the respondents agree that subnational governments receive transfers to address the asymmetry between expenditure and revenue, while 45% of the respondents disagreed.
Q4.2 Grand transfers for local development.	75%	25%	0%	0%	All the respondents agree that subnational governments receive grant transfers from national government.
Q4.3 Sharing locally collected revenue.	0%	39%	16%	45%	61% of the respondents disagree with the view that councils receive a share of revenue collected in their territory.
Q4.4 Authority to determine the use of transfer funds.	52%	36%	7%	5%	88% of the respondents agree that local authorities have authority to determine how the transferred funds are utilised.
Q4.5 Availability of transfer system regulations.	50%	27%	0%	23%	77% of the respondents agree that there is a regulatory framework for governing the transfers system.
Q4.6 Suitability of transfer system as a funding strategy.	41%	43%	11%	5%	84% of the respondents are of the view that the transfer system is the best funding strategy for councils.

Questions	Agree	Partly agree	Disagree	Totally disagree	Interpretation
SUBNATIONAL GOVERNMENT FINANCIAL RESOURCES Literature review: Subnational governments are independent financial actors with the responsibility to finance their expenditure with their own revenue (Ejobowah 2018:224).					
Q5.1 The need for a broad revenue base for subnational government.	86%	14%	0%	0%	All the respondents support the view that subnational governments need a broad revenue base for generation of revenue to finance their mandate.
Q5.2 Availability of multiple revenue collectors at subnational government.	57%	16%	23%	4%	73% of respondents agree that there are multiple revenue collectors at the districts.
Q5.3 The share of national budget as subnational government income.	100%	0%	0%	0%	All respondents agree that subnational governments receive a share of national budget as income.
Q5.4 Availability of budget allocation formula for subnational government.	18%	7%	32%	43%	75% of the respondents disagree with the view that there is a budget allocation formula that determines the national budget allocation for subnational government.
Q5.5 Sustainable and predictable income for subnational governments.	0%	0%	86%	14%	All respondents disagree with the view that subnational governments have sustainable and predictable income.
Q5.6 Capacity of subnational governments to borrow funds to finance their activities.	0%	34%	48%	18%	66% of the respondents disagree with the view that subnational governments can borrow funds to finance their activities.
Q5.7 Availability of funding model for local service delivery.	52%	27%	21%	0%	79% of the respondents agree that the country has a funding model to support local service delivery.
Q5.8 The failure to deliver services due to limited funds.	81%	19%	0%	0%	All the respondents are of the view that subnational governments fail to deliver services due to budget constraints.

The data collected through questionnaires shows that Lesotho's decentralisation lacks the concept of fiscal decentralisation as the financial practices at subnational government are incompatible with the definition of the concept as described in the literature. The critical issues that are highlighted from the collected data include lack of authority for revenue and expenditure management, absence of subnational government regulatory frameworks and the dependence on intergovernmental transfers. This would negatively affect subnational governments' efforts to mobilise resources.

Table 2: Reporting and analysis of findings from data collected through interviews

Elements of Fiscal Decentralisation	Literature review	Interview responses (from Council chief executive officers and chairpersons of finance committees)
Contextualising fiscal decentralisation	Fiscal decentralisation is the allocation of budget powers in terms of revenue and expenditure as well as intergovernmental transfers between government spheres, whereby subnational governments are given the authority through a legal framework to manage their fiscal system and ensure the availability of resources to finance their mandate (Hendriks 2014:17; 'Nyane 2019:8).	The majority of respondents explained that local authorities receive transfers as their primary income to finance their activities. They said local revenue rates and collection are administered through national government frameworks and subnational governments are not allowed to spend the collected revenue for local development. The fiscal relations between government spheres were explained through intergovernmental transfers. One councillor argued that there is no fiscal decentralisation framework because parliamentarians are not ready to implement it.
Revenue management	As local governance bodies charged with the development of their communities, subnational governments have a responsibility to manage local revenue for financing their functions (Ejobowah 2018:224).	The majority of respondents explained that local authorities identify sources of revenue through bylaws, but the bylaws have to be approved by national government and to date none have been approved. The common view was that subnational governments do not have tax administration policies; they use national policies that also determine the tax and fee rates. They stated absence of subnational revenue management legislation, limited revenue sources and inability to spend collected revenue as major challenges at local level.

Elements of Fiscal Decentralisation	Literature review	Interview responses (from Council chief executive officers and chairpersons of finance committees)
Expenditure management	In order for subnational governments to perform their responsibility of serving local communities, they need to allocate and use resources responsively, efficiently and effectively (Tkachenko 2020:74).	The interviewees explained that budget policy reflects social and economic needs of local citizens. They also indicated that councils do not have a predetermined budget allocation formula. Councils resort to internal arrangements that change annually, on the best ways to use the limited budget to finance all the councils in the district in an effort to secure the operations. The interviewees suggested the development of financial management legislation to strengthen local expenditure management.
Intergovernmental transfer	Intergovernmental transfer refers to a multidimensional component of fiscal decentralisation whereby finances are transferred from one sphere of government to another for addressing fiscal gaps (Leon 2019:14; Shotton & Gankhuyag 2019:31).	The common themes that emerged among the interviewees was that local authorities receive transfers to finance their entire budget and have authority to determine their development agenda as well as how the transferred funds are used. The transfer funds were explained as a primary source of income; therefore the vertical fiscal gap is not applicable because councils do not have revenue supply. One Finance Manager asserted that transfers are the best funding strategy for local authorities although they are currently regulated through national government policies.
Subnational government financial resources	Subnational governments are independent financial actors with the responsibility to finance their expenditure with their own revenue (Ejobowah 2018:224).	The common view of the interviewees was that broadening the revenue base would improve revenue generation and service delivery at local spheres. They explained that there is no budget allocation formula, the budget estimates/ceilings are set by the national government. Subnational governments continually experience budget shortages that negatively affect their ability to deliver services.

DISCUSSION OF FINDINGS

This discussion considers the themes that emerged from the interviews and questionnaire responses and the literature in relation to conceptualisation of fiscal decentralisation, revenue and expenditure management, intergovernmental transfers and financial resources as critical factors in financing subnational government.

Contextualising fiscal decentralisation

The findings on the conceptual nature of fiscal decentralisation show that the Lesotho decentralisation process lacks the fiscal decentralisation component. The country's fiscal decentralisation is explained through intergovernmental transfers. This is because transfers are the only element of fiscal decentralisation that is practised in Lesotho's subnational government. This is incompatible with the definition of the concept of fiscal decentralisation as described in the literature. In addition to transfers, the literature considers devolution of budget powers as revenue and expenditure management, and the subnational financial management system as critical elements of fiscal decentralisation (Tkachenko 2020:74; Digdowiseiso *et al.* 2022:1). Fiscal decentralisation is a prerequisite for the success of the decentralisation process and enables subnational governments to achieve good governance, accountability and improve efficiency in service delivery (Adamtey *et al.* in Mohapi 2023). It is important for the decentralisation process to have a comprehensive fiscal decentralisation design to guide the devolution of financial resources and ensure financial independence and accountability of subnational governments. The fiscal decentralisation design that incorporates assignment of revenue and expenditure management, intergovernmental transfers and an effective local financial management system would enable subnational governments to be autonomous and fiscally disciplined entities. Lesotho needs to develop an inclusive fiscal decentralisation legislation with all elements of fiscal decentralisation to support the decentralisation process. This will enable subnational governments to mobilise adequate financial resources to finance their mandate.

Revenue and expenditure management

Local revenue management is a critical part of fiscal decentralisation as fiscal decentralisation is described through the delegation of budget powers such as revenue management. The second-generation theory advocates for subnational governments to generate their own revenue through local revenue sources for financing their functional responsibilities (Ejobowah 2018:224). In addressing the questions about local revenue management, the respondents explained that

local authorities do not identify revenue sources nor set the tax and fee rates. Although the Local Government Act 6 of 1997 permits revenue management at subnational governments, respondents explained that the execution of the Act depends on national government as subnational governments are tasked with developing bylaws for revenue management (Lesotho Local Government Act 6 of 1997). Such bylaws still need to be authorised by national government and to date, none of the proposed bylaws have been approved. Consequently, councils do not have revenue management policies as they cannot identify and manage revenue sources without approved bylaws. In this regard, national government has the power to identify sources of revenue and set tax rates resulting in councils relying on transfers as their sole source of income.

In the absence of subnational revenue management policies, the interviewed respondents declared that local authorities use national government policies to collect revenue. The Treasury Regulations of 2014, require all the collected revenue to be deposited into the consolidated fund and become part of national revenue (Lesotho Ministry of Finance 2014). Irrespective of their ability to develop a budget policy with economic and social preferences of local communities as reflected in the findings, subnational governments do not have expenditure management powers as they cannot use locally collected revenue to finance their budget. The inability to spend local revenue and absence of a budget allocation formula result in unpredictable income that continually affects subnational government expenditure. The literature highlights the importance for subnational governments to have budget in terms of revenue and expenditure management, for the following reasons, as it would:

- improve public revenue mobilisation and secure subnational financing,
- promote subnational financial independence,
- improve financial planning by matching expenditure needs to revenue supply to avoid unfunded mandate,
- improve accountability and responsiveness at subnational government, and
- enable subnational governments to decide on the level and structure of their expenditure (Martinez-Vazquez *et al.* 2016:4; Ejobowah 2018:225; Leon 2019:13).

Intergovernmental transfers

Intergovernmental transfers are a source of subnational government income. The reviewed literature and feedback from respondents reveal that subnational governments have transfers as their primary source of income as the country does not have an established funding model. This finding contradicts the basis for the need for transfers as presented in the fiscal decentralisation literature. The primary objective of intergovernmental transfers is to complement the subnational revenue for addressing the inevitable fiscal gaps between and among spheres of

government (Lual 2018:7). However, based on the findings, the respondents view transfers as a suitable source of funding for Lesotho's subnational governments. It is recommended that the country develops a domesticated intergovernmental transfer system that regulates fiscal relations between, and among government spheres. Ankamah (2012:33), argues that an intergovernmental transfer system designed to support subnational financial management, should be transparent with allocations, present predictable amounts for institutions, and promote local autonomy in utilising the resources. The nature and design of an intergovernmental transfer system is important for country context.

Subnational government financial resources

Financing subnational government stems from the view that finance follows functions. This view is explained in two ways: First, subnational governments should have enough resources to finance their mandate, and second, the source of subnational government financing is important (Hart & Welham 2016:11). In this regard, the composition of subnational government revenue is highlighted by considering the balance between the fraction of locally raised revenue and the fraction of national transfers in the subnational government revenue. The findings indicate that subnational governments need a broad revenue base to generate enough revenue to finance their mandate. In responding to questions 5.3 and 5.4, the majority of respondents who completed the questionnaire agreed that local authorities receive transfers as their basic income from national government without a formula (*cf.* Table 1). The common theme among interviewed respondents was that there is no formula, but national government determines the budget ceiling for local authorities. It was also discovered that local authorities do not have predictable and sustainable income as the budget estimates continually decline, resulting in failure to deliver services. Without predictable and sustainable income, local authorities are not able to explore borrowing as a source of income.

The primary objective of fiscal decentralisation is to strengthen subnational governments' finances and their capacity to effectively deliver on their mandate. Fiscal decentralisation directs the decision on the generation and utilisation of financial resources at subnational governments. The literature refers to a subnational government financial management system as a critical part of fiscal decentralisation (Hendriks 2014:17). Public financial management is a combination of regulations, processes, procedures and systems utilised by government spheres to generate and allocate resources, expend the resources, report on the expenditure, and undertake audits for accountability purposes (Cheruiyot *et al.* 2017:212). Public financial management concentrates on the prioritisation and effective utilisation of scarce resources.

CONCLUSION

Fiscal decentralisation is a core component of the decentralisation process. The financing of subnational governments is central to fiscal decentralisation reforms. The Lesotho decentralisation reforms intend to address the fiscal aspect of the decentralisation process. In this regard, the study makes a scientific contribution to policy mandate as it assists the country in its endeavour to develop comprehensive fiscal decentralisation legislation. The author contends there is a need to develop a constructive fiscal intergovernmental framework and policies to strengthen, finance and support vibrant subnational governments. In designing the subnational government framework to improve resource mobilisation, Lesotho has to consider the following elements:

- Diversification of subnational government income sources, for instance explore all possible income sources such as local revenue, intergovernmental transfers, donations and loans.
- Devolution of revenue management authority to allow subnational government to manage local revenue sources and promote local autonomy.
- Designing of a sustainable intergovernmental transfer system with clear allocation formula and predictable allocations.
- Establishing a subnational financial management system to direct and manage local financial practices in an effort to ensure efficient generation and safeguarding of subnational financial resources.

The study found that the Lesotho decentralisation process lacks the fiscal component, because fiscal decentralisation is practised as intergovernmental transfers. The decentralisation process does not have fiscal decentralisation legislation to guide and manage a subnational government financial management system. It was also discovered that local authorities do not have revenue management authority and cannot expend the locally collected revenue. In the absence of revenue sources, the study's findings indicate that local authorities depend on national government transfers to finance their functions and this results in inefficiencies in service delivery. The author believes that the inclusion of all the above identified elements would improve resource mobilisation for subnational government financing and enable the delivery of efficient services in the Lesotho public service.

NOTE

- * The article is part of the primary author's Doctor of Philosophy thesis: Mohapi, T.J. 2023. *Fiscal Decentralisation: Domesticating a methodology for Lesotho*. Pretoria: University of South Africa, under the supervision of Prof V.T. Sambo and co-supervision of Prof G.M. Ferreira.

REFERENCES

- Allain-Dupre, D. 2018. *Assigning Responsibilities across levels of Government: Trends, Challenges and Guiding Principles for Policy-makers*. Paris: OCED.
- Al-Samarrai, S. and Lewis, B. 2021. *The Role of Intergovernmental Fiscal Transfers in Improving Education Outcomes*. Available at: <http://openknowledge.worldbank.org>. (Accessed on 6 April 2022).
- Ankamah, S. 2012. The Politics of Fiscal Decentralisation in Ghana: An Overview of the Fundamentals. *Journal of Public Administration Research*. 1(1):33–41.
- Baltes, S. and Ralph, P. 2020. *Sampling in Software Engineering Research: A Critical Review and Guidelines*. Available at: <https://arxiv.org/pdf/2002.07764.pdf>. (Accessed on 12 December 2020).
- Boadway, R. and Eyraud, L. 2018. *Designing Sound Fiscal Relations Across Government Levels in Decentralized Countries*. Available at: <http://dx.doi.org/10.5089/9781484387788.001>. (Accessed on 26 February 2020).
- Cheruiyot, M.P., Oketch, J.R., Namusonge, G.S. and Sakwa, M. 2017. Effect of Public Financial Management Practices on Performance in Kericho Country Government, Kenya: A Critical Review. *International Journal of Education and Research*. 5(12):211–223.
- Daemane, M.M.M. 2012. *Decentralisation in Lesotho: Prospects and Challenges*. Thesis submitted in fulfilment of the requirement of the degree of Doctor of Philosophy in Development Studies. Johannesburg.
- Digdowniseiso, K., Murshed, S.M. and Bergh, S.I. 2022. *How effective is Fiscal Decentralisation for Inequality Reduction in Developing Countries?* Available at: <https://doi.org/103390/su14010505>. (Accessed on 7 July 2022).
- Ejobowah, J.B. 2018. The Second-Generation Theory of Fiscal Federalism: A Critical Evaluation. *Perspectives on Federalism Journal*. 10(1):219–242.
- Hart, T. and Welham, B. 2016. *Fiscal Decentralisation: A Public Financial Management Introductory Guide*. Available at: <https://www.oid.org/publications/10623-pfm>. (Accessed on 14 August 2019).
- Hendriks, C.J. 2014. *Intergovernmental Fiscal Relations: The Paradox between the Vertical Fiscal Imbalance and Responsiveness and Accountability in Northern Cape Provincial Government*. Unpublished doctoral thesis in Public Administration and Management. Bloemfontein: University of the Free State.
- Hobdari, N., Nguyen, V., Salvatore, D. and Ruggiero, E. 2018. *Lessons for Effective Fiscal Decentralisation in Sub-Saharan Africa*. Available at: www.researchgate.net/publication/326886728. (Accessed on 2 March 2020).
- Jha, P.C. 2012. *Theory of Fiscal Federalism: An Analysis*. Available at: <https://mp.ra.ub.uni-muenchen.de/41769/>. (Accessed on 17 April 2020).
- Kali, M. 2020. Challenges and Prospects of Decentralisation in Lesotho. *International Journal of Humanities, Arts and Social Studies*. 5(2):11–20.
- Kim, J. and Dougherty, S. 2018. *Fiscal Decentralisation and Inclusive Growth*. Seoul: OECD Publishing.

- Leon, V. 2019. *Decentralisation and Development in the World's First Black Republic*. Thesis submitted in fulfilment of the requirements for the degree of Doctor of Philosophy. New York: Milano School for International Management, Urban Affairs and Policy.
- Lesotho. 1993. *The Constitution of Lesotho Act 5 of 1993*. Maseru: Government Printers.
- Lesotho. 1997. *Local Government Act 6 of 1997*. Maseru: Government Printers.
- Lesotho. 2004. *Local Government Amendment Act 5 of 2004*. Maseru: Government Printers.
- Lesotho. 2014. *National Decentralisation Policy*. Maseru: Government Printers.
- Lesotho. 2014. *Treasury Regulations*. Maseru: Government Printers.
- Lual, T.N. 2018. *Intergovernmental Fiscal Transfers in Germany, Nigeria and Ethiopia*. Available at: www.grin.com/document/464935. (Accessed on 1 April 2020).
- Martinez-Vazquez, J., Lago-Penas, S. and Sacchi, A. 2016. *The Impact of Fiscal Decentralisation: A Survey*. Available at: <https://www.researchgate.net>. (Accessed on 24 February 2020).
- Morgner, M. and Chene, M. 2014. *Public Financial Management: Topic Guide*. Available at: <https://knowledgehub.transparency.org>. (Accessed on 27 July 2020).
- Newberry, M.G., O'Leary, J.L. and Israel, G.D. 2021. *The Savvy Survey #15: Survey Responses and Data Entry*. Available at: <http://edis.ifas.ufl.edu/publication/PD079>. (Accessed on 17 December 2021).
- 'Nyane, H. 2016. Assignment of Functions to Local Authorities in Lesotho. *Commonwealth Journal of Local Governance*. (19):58–74. Available at: <https://doi.org/10.5130/cjlg.v0i19.5449>. (Accessed on 15 August 2018).
- 'Nyane, H. 2019. A critique of the newly proposed model of decentralisation in Lesotho. *Commonwealth Journal of Local Government*. (22):1–21. Available at: <https://doi.org/10.5130/cjlg.vi22.7074>. (Accessed on 20 June 2020).
- OECD (Organisation for Economic Cooperation and Development). 2016. *Subnational Governments around the world: Structure and Finance*. Available at: <https://www.oecd.org-regional-regiona-policy>. (Accessed on 11 November 2022).
- Osei, E. 2017. *Fiscal Decentralisation in Ghana: Progress and Challenges*. Institute of Fiscal Studies Paper No. 10. Available at: <https://www.ifsgghana.org>. (Accessed on 6 April 2022).
- Otoo, I. and Danquah, M. 2021. *Fiscal Decentralisation and Efficiency of Public Service delivery by Local Government in Ghana*. Available at: <https://doi.org/10.35188/UNU-WINDER/2021/028-3>. (Accessed on 6 April 2022).
- Ozkok, Y. and Cutcu, I. 2021. *Does Fiscal Federalism matter for economic growth? Evidence from the United States*. Available at: <https://doi.org/10.1080/00036846.2021.1998337>. (Accessed on 4 May 2022).
- Rahim, A.A. 2013. *Fiscal Decentralisation in Ghana: Challenges and Way forward*. A Term Paper Presented as Part of the Assessment for Development Finance – PI445. Accra, Ghana: Kwame Nkrumah University of Science and Technology.
- Reingewertz, Y. 2014. *Fiscal Decentralisation-A Survey of the Empirical Literature*. Available at: <http://dx.doi.org/10.2139/ssrn.2523335>. (Accessed on 22 May 2020).
- Rodriguez-Acosta, C.A. 2016. *The Impact of Decentralisation and New Intergovernmental Relations on Public Service Delivery: A Comparative Analysis of Colombia and Paraguay*. A dissertation

- submitted in partial fulfilment of the requirements for the degree of Doctor of Philosophy in Public Affairs. Florida: International University, Miami.
- Sefeane, K. 2020. *Integration of Lesotho Local Government Authorities in the European Cooperation Programming Process 2021–2027*. Available at: <http://www.knowledge-uclga.org>. (Accessed on 10 May 2021).
- Shannon-Baker, P. 2016. Making Paradigms Meaningful in Mixed Methods Research. *Journal of Mixed Methods Research*. 10(4):319–334.
- Sharma, P.S. 2015. *Functional Analysis and Assignment of Government*. Available at: <https://www.undp.org/content/dam/nepal/dcs/generic>. (Accessed on 5 May 2022).
- Shotton, R. and Gankhuyag, U. 2019. *Fiscal Transfers in Asia: Challenges and Opportunities for Financing Sustainable Development at the Local Level*. Available at: [www.undp.org/dam.RBAP-2019-Fiscal-Transfer-in-Asia](http://www.undp.org/dam/RBAP-2019-Fiscal-Transfer-in-Asia). (Accessed on 1 April 2020).
- Smoke, P. 2019. *Improving Subnational Government Development Finance in Emerging and Developing Economies: Towards a Strategic Approach*. ADBI Working Paper 921. Available at: <https://www.adb.org/sites/default/publication/485851/adbi-wp921.pdf>. (Accessed on 12 November 2022).
- Tkachenko, L. 2020. Public Finance Management: Challenges and Opportunities. *Athens Journal of Business and Economics*. 6(1):73–98.
- Valdesalici, A. 2018. *Defining Fiscal Federalism*. Available at: www.researchget.net. (Accessed on 19 February 2020).
- Walker, C. and Baxter, J. 2019. Method Sequence and Dominance in Mixed Methods Research: A Case Study of the Social Acceptance of Wind Energy Literature. *International Journal of Qualitative Methods*. 18(1):1–14.
- Yushkov, A. 2015. Fiscal Decentralisation and Regional Economic Growth: Theory, Empirics and the Russian Experience. *Russian Journal of Economics*. 1(4):404–418.

AUTHORS' CONTACT DETAILS

T J Mohapi

University of South Africa
Tel.: +266 589 49 523
E-mail: m.mohapi@ymail.com

G M Ferreira

University of South Africa
Cell.: +278 346 84 067
E-mail: ferregm56@gmail.com

V T Sambo

University of South Africa
Tel.: 012 429 4355
E-mail: Esambovt@unisa.ac.za

Service Delivery Challenges within Rural Communities

The Case of the Nkomazi Local Municipal Area

B R Shongwe*

School of Public Management, Governance and Public Policy
University of Johannesburg

D F Meyer

School of Public Management, Governance and Public Policy
University of Johannesburg

ABSTRACT

Municipalities across the Republic of South Africa continuously experience challenges in providing quality basic services to their communities. Communities that experience poor service delivery often engage in violent and disruptive service delivery protests. These service delivery protests are typically triggered by the failure of municipalities to provide communities with basic quality services such as electricity, water, tarred roads, sanitation and waste removal. In general, corruption, financial mismanagement and unproductive operations are among the challenges that contribute to ineffective service delivery by municipalities. This article will comprehensively outline that service delivery by South African municipalities is widely perceived to be poor in various areas. The focus of this study was to identify key challenges that affect service delivery in the Nkomazi Local Municipality area. This article adopted a qualitative research approach for data collection. The data was collected using semi-structured interviews with municipal administrators within various departments, six ward councillors and four public relations (PR) councillors. The research findings of the qualitative interviews are rigorously interpreted and analysed. Some of the key challenges that hinder the municipality from effectively delivering quality basic services to its communities are caused by poor planning, financial mismanagement, corruption and lack of community participation.

INTRODUCTION

The Republic of South Africa has three spheres of government: the national, provincial and local governments (Booyesen 2012:5). The local government sphere comprises 257 municipalities that are tasked with the provision of basic services such as water, sanitation services, recreational parks and gardens, cemeteries, community halls, libraries, tarred roads, electricity and many other services that enhance the lives and dignity of community members (Kanye 2014 in Shongwe 2023). South Africa consists of eight Category A Metropolitan Municipalities, 205 Category B Local Municipalities and 44 Category C District Municipalities (Kgobe 2020:57). Municipalities are obliged to provide services in a free and impartial manner (Beyers 2015 in Shongwe 2023) and should be accountable to the electorate (Masuku 2019:122). Local government authorities need to ensure that the promotion of social and economic development is adequately addressed (Beyers 2015 in Shongwe 2023). According to Kanye (2010 in Shongwe 2023), it is the mandate of government institutions such as municipalities to ensure efficient service delivery that enhances the welfare of the community members under its watch.

Local municipalities must enhance and promote community participation (Madumo 2014:132). Community members ought to be involved in all planning and decision-making processes within the municipality (Kanyane 2014:6). This will contribute positively to effective service delivery. Community members must be allowed to participate freely in all municipality systems and be encouraged to pay for services. The local municipality will receive recognition and accolades for performing well (Masuku 2019:124). According to Madzhivandila and Asha (2012:371), the Integrated Development Plan (IDP) plays a crucial role in service delivery as it is a tool that promotes and enhances public participation; this is where community members voice their demands in a formal setting during public meetings. Community members are vital in formulating the IDP (RSA 2000). Effective consultation, formulation and implementation of the IDP will contribute immensely to effectively delivering various community services (Brynard 2012:6).

According to Naidoo and Ramphal (2018:82), local municipalities are the types of local government close to the needs and demands of the citizens. A municipality has the right to govern, on its initiative, the local government affairs of its community, subject to national and provincial legislation (RSA 1996). Municipalities must be a driving force for providing basic services to people across various communities (Masuku 2019:121). The municipality is the relevant institution of the sphere of government that directly interacts with communities as it has direct access to the local community's everyday affairs (Kanyane 2010: 85). However, most local municipalities within the Republic of South Africa are tainted and perceived to be ineffective to enhance the quality of service delivery (RSA 2009:6).

This study sought to reveal the key challenges that prevent effective service delivery to communities under the Nkomazi Local Municipality and provides mechanisms and strategies to address these challenges thoroughly. This study is among some of the studies that specifically focus on service delivery at a local municipality level since the establishment of municipalities across the country in 2000. This study is distinctive as it identifies and analyses specific challenges that hinder local government authorities in providing basic services. For that reason, this study is unique because it contributes to theoretical knowledge concerning service delivery. The research provides a better understanding of the link between service delivery and the policies and legislations of the country, thereby enhancing the sense of responsibility and ownership of the mandate to both the government and recipients of basic services. The significance of this study is that it provides knowledge and theories of effective strategies that address hindrances to service delivery within local municipalities. This study contributes substantially to issues concerning the provision of theoretical knowledge on the branch of public management and government institutions responsible for capacity building within local government.

THEORETICAL FRAMEWORK

This study utilised the Public Administration and the New Public Management (NPM) theories. Public administration refers to the actions and processes taken by the government of the day to deliver essential basic services to the people it serves. The basic quality services being delivered to people by the government are meant to enhance the lives of community members. Public administration also refers to all government-related activities, institutions and statutes (Bvuma 2014:8). Stroh (2014:7) outlines that public administration refers to the delivery of basic quality services to the society by the government and is guided by the variety of needs and demands of the society.

According to the *Constitution of the Republic of South Africa* (1996), “public administration must be governed by the democratic values and principles enshrined in the *Constitution of the Republic of South Africa*, which include the following principles extracted from Section 195(1) of the Constitution: (a) A high standard of professional ethics must be promoted and maintained; (b) efficient, economical and effective use of resources must be promoted; (c) public administration must be development-oriented; (d) services must be provided impartially, fairly, equitably and without bias; (e) people’s needs must be responded to, and the public must be encouraged to participate in policy-making and (f) public administration must be accountable” (RSA 1996).

This theory links to this study since it is interrelated with governmental processes, public policymaking and the provision of public services by public servants to

the public. It is suited for this article because public institutions and administrators are supposed to be at the forefront of enhancing social, economic and political development.

Denhardt and Catlaw (2015:127-131) mention that the NPM theory originated as a mechanism for practical developments in the operations of public institutions around the 1970s. It attempted to resolve the fiscal and political crises confronting governments worldwide during that time. During the 1970s, the NPM approach diagnosed the problems in public institutions, such as inefficient use of organisational resources and ineffectiveness in implementing public programmes and projects. This approach, as a remedy, encouraged restrictions on governmental growth, privatisation of government assets and the use of contracts instead of direct service provision (Maserumule 2011:28).

Maserumule (2011:29-30) states that the NPM is part of the contemporary features of theoretical and current worldviews. The NPM has thus far contributed to world discourse on how government should enhance development and performance management. This theory has many theoretical features and is invaluable for having contributed a great deal to how governments can enhance production and performance management and ensure proper and effective financial management and accountability (Mbandlwa & Mishi 2020:3).

This study associates itself with the NPM approach because the South African government enacted several legislative frameworks to enhance its performance and ensure accountability. That legislation is The Public Finance Management Act 1 of 1999, Local Government: Municipal Finance Management Act 56 of 2003 and Local Government: Municipal Systems Act 32 of 2000 (Maserumule 2011:31). These legislative frameworks are entrenched in the foundations and features of the NPM which incorporates aspects such as decentralisation, economic and social development, performance management and outsourcing (Matsiliza 2012:69). For example, the Municipal Systems Act 32 of 2000 and the Municipal Finance Management Act 56 of 2003) state that it is a prerequisite for municipalities to ensure the continued involvement of community members and other stakeholders during processes of integrated development planning as well as overall budgeting processes (Maserumule 2011:31).

LITERATURE REVIEW

Local government is among the three governmental spheres, including the provincial and national governments, and is responsible for the local administration of public services. A local government sphere is autonomous and entitled to regulate matters relating to local affairs. Municipalities have the power to make and enforce by-laws (Madue 2016:8). Local government, specifically municipalities, have roles

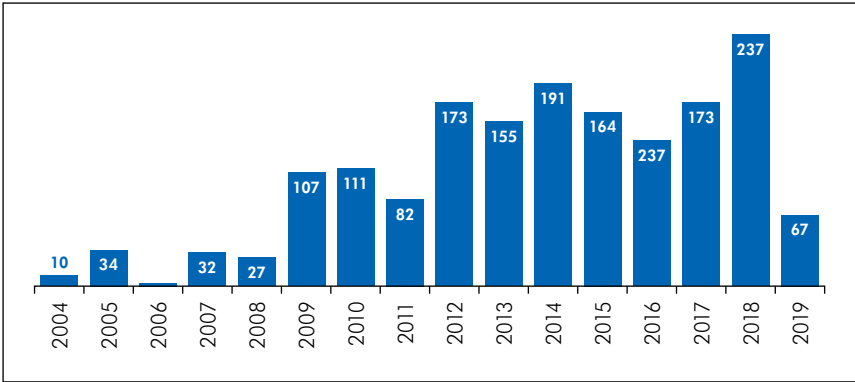
and functions which differ entirely from those in the provincial and national spheres. However, the similarity between the three spheres of government is that of ensuring that services are delivered to the people without discrimination (Kanyane 2010:16).

According to Mbandlwa (2018:17-18) providing quality basic services to the people should be at the forefront of local government to uplift the dignity of the poor and previously disadvantaged groups affected by the apartheid system. Many municipalities in the country are not doing well concerning service delivery in their respective communities. Several challenges within local municipalities continue to hinder providing basic services to communities such as that some municipalities are struggling to draw up an effective budget to address community service delivery issues. The failure of municipalities to draft budgets ensures that they find it hard to sustain themselves financially and operate effectively, inevitably rendering such municipalities failures and incompetent (Beyers 2016 in Shongwe 2023).

Service delivery refers to the provision of basic quality services and goods that are classified as tangible public goods and intangible services (Beyers 2016 in Shongwe 2023). The effective delivery of basic services means setting up strategies and plans that will be implemented by a government institution like a municipality (Sefala 2009:31). Service delivery can be effectively addressed if accountability and ethical leadership are the driving forces for public officials (Mbandlwa 2018:21). Beyers (2015 in Shongwe 2023) asserts that most South Africans have lost confidence in the government's ability to govern efficiently, particularly in local government. Service delivery challenges all over South Africa have become severe, and this is epitomised by the rise of service delivery protests in various parts of the country, year after year (Ngcqoyiya 2018:31).

Figure 1 illustrates the trend for service delivery since 2004. It shows that service protests are rising yearly (Asha and Makalela 2019:18).

Figure 1: Service delivery protests every year (2004–2019)



Source: (Municipal IQ in Shongwe 2023)

According to the consolidated general report on local government audit outcomes (MFMA 2019/2020), more than 70% of municipalities in the country are finding it difficult to manage their finances; hence, they also struggle to enhance the provision of basic services in the communities they serve (RSA 2020:36). The 2020 Auditor-General's report outlines that one in five municipalities is on the verge of collapsing. The financial state of many municipalities is dire, to the extent that most of them would soon find it hard to operate properly, which is a significant concern (RSA 2021:84-86). More than 70% of municipalities are struggling to operate satisfactorily, maintain the provision of basic services, and ensure that they consistently spend their allocated annual budget sustainably. When a local municipality cannot provide basic services to communities, residents' trust in the municipality diminishes, creating the conditions for violent service delivery protests (Kgobe 2020:5).

Mabizela and Matsiliza (2020:1-5) state that the responsibility of municipalities is to ensure effective and efficient delivery of basic and essential services to communities, such as water, electricity, housing and proper sanitation. There is an uneven distribution of basic services between rural and urban areas, whereby urban areas enjoy better service than rural areas. Scholars have alluded to the capacity of municipalities, which is a prerequisite for service provision in local government. Many municipalities across the country lack adequate capacity to provide essential services to communities. The current state of local government in rural areas shows that most rural municipalities still have backlogs for basic service delivery.

The White Paper on Local Government of 1998 (RSA 1998:27) outlines the mandate for municipalities, which is to ensure that they effectively address service delivery concerns such as infrastructure development, the housing backlog and many other services that citizens require. Disturbingly, most municipalities struggle to provide adequate services to ensure that there is a level of effectiveness and efficiency. Municipalities need help to conduct effective oversight, contributing to the growing concern about the lack of services in communities. Monitoring and evaluation systems are not adequately administered, which contributes to a lack of accountability (Mabizela & Matsiliza 2020:2).

Municipalities in rural areas and townships could more effectively deliver basic services to citizens. This is seen by widespread demonstrations and service delivery protests across rural and township municipalities as citizens no longer have confidence and trust in municipalities because they fail to ensure that basic services are delivered to communities (Siddle & Koelble 2016:16-17). Mabizela and Matsiliza (2020:4-7) state that service delivery challenges result from several issues, including corruption, lack of capacity within municipalities, lack of effective oversight and inadequate budget from the national and provincial government provided to municipalities.

Mabizela and Matsiliza (2020:6-7) conducted a study examining service delivery challenges experienced by communities in the Okhahlamba Municipality in KwaZulu- Natal. They discovered that the Okhahlamba Local Municipality is failing to provide essential services to communities within its jurisdiction, resulting in a backlog for service delivery. It was found that the municipality is experiencing several service delivery setbacks, and this was mostly triggered by its human resource and capacity constraints. It was also discovered that in the area of Maswazini, one of the communities where the study was conducted, community members stated that there was minimal service delivery. They said they did not have portable running water, and sanitation services were of poor quality. The findings also reveal that there is inadequate housing for citizens.

According to COGTA (2015 in Mabizela and Matsiliza 2020:6), municipalities are experiencing various challenges that hinder service delivery, such as inadequate technical skills, knowledge, leadership and management skills. Accordingly, public administrators and ward councillors need to gain the necessary skills and knowledge to perform their duties in delivering services to communities (IDASA 2010:9). Municipal administrators and ward councillors are closely linked to communities; they interact with community members to continually enhance. Local government authorities must devise strategies for acquiring human and financial resources to improve the lives of community members (Mabizela & Matsiliza 2020:5-6).

A report by The Institute for Democratic Alternative in South Africa (IDASA) (2010:6) (in Mabizela and Matsiliza 2020:6-9), concurs that municipalities are facing considerable challenges in financial mismanagement, corruption and non-compliance with financial legislation for local government. The failure of most municipalities in the country to control their finances determines those municipalities' success and failure in sustaining themselves and providing services to communities. Primarily, the lack of financial management contributes to a great degree of failure in local economic development, infrastructure development, strategic growth, and service delivery.

Mabizela and Matsiliza (2020:5-7) assert that political interference is another major contributor to service delivery constraints in Okhahlamba Local Municipality. Political parties involve themselves in issues of the municipal council and the administration of the municipality which creates problems for the smooth implementation of programmes for local and economic development policies and integrated development planning. There is no proper planned platform for public participation to enhance public service delivery in Okhahlamba. The main factors that hinder service delivery processes are mostly internal, strategic and operational. The findings indicated that the Okhahlamba Local Municipality does not provide sufficient services to the entire community under its jurisdiction. This is because of a lack of financial and other limited resources, political influence and

decision-making, a lack of community involvement unless there is a crisis and too many parties and stakeholders with interests that get involved in the formulation and implementation of service delivery projects and programmes (Mabizela & Matsiliza 2020:5-7).

Matsiliza (2012:67) asserts that the *Constitution of the Republic of South Africa* Act 108 of 1996, Section 40 (1) and (2) outlines duties and statutes for governing in the sphere of local government. This section of the Constitution assigns local government and municipalities with objectives to embark on and ensure that these objectives are attained. Municipalities are given the assignment of ensuring that there is effective infrastructure development in communities and that there is local economic and social development. Accordingly, provincial departments of cooperative governance monitor and supervise the operations of municipalities across the country. These governments provide oversight as they supervise, monitor and support municipalities countrywide. Politically, the sphere of local government is largely controlled by the African National Congress (ANC) because it controls the majority of municipalities in the country, except in some cases like the Western Cape Province, where the City of Cape Town and several municipalities are controlled by the Democratic Alliance (DA) and some minority political parties (Mngomezulu 2020:39).

Mngomezulu (2020:38-40) conducted a study to investigate the state of political meddling in municipal administration at uMlalazi Local Municipality in KwaZulu-Natal, South Africa. For the past several years, there has been an increase in service delivery protests in the country because many people have grown frustrated with the lack of essential and effective service delivery. Corruption has become a common and systemic practice in most municipalities. South African municipalities continue to experience challenges with addressing political meddling in the affairs of the municipal administration, creating tension and squabbling between politicians and municipal administrators. Mashamaite (2014:12) identifies what appears to be avoidance of the principle of the separation of powers between politicians and municipal administrators in many municipalities across the country. According to Mngomezulu (2020:39-40), political interference in most municipalities creates conflicts about how the municipalities are governed. Such clashes are prevalent in municipal projects and tenders where politicians typically show their interests. Politicians want to force municipalities to employ their cadres despite them not having the relevant qualifications, which means that the people who deserve to be employed are overlooked.

Munzhedzi (2020:94) argues that political interference affects service delivery to a large extent because the employment of underqualified people creates problems for the functioning of the municipality, leading to poor service delivery. Mngomezulu (2020:42-43) further states that political interference affects service delivery precisely because when an incompetent person is employed, they may fail

to execute their duties satisfactorily and the deployment of unqualified officials will impact service delivery. The ANC's cadre deployment policy supports the view that people must be employed in government positions because they are loyal cadres of the organisation, which impacts service delivery. Disturbingly, during IDP processes in uMlalazi Local Municipality, municipal development projects are routinely allocated to members of the ANC, despite some of them not undergoing financial verification and scrutiny to check if they are indeed capable of undertaking such a project. Some municipalities, including uMlalazi Local Municipality, do not comply with the country's financial legislation, such as the MFMA.

Munzhedzi (2020:95-97) further claims that municipalities are characterised by their engagement in corrupt activities and maladministration. Surprisingly, those who violate legislative frameworks in municipalities are not held accountable and continue their malpractice, unscathed. Municipal managers, administrators, and politicians closely linked to municipalities practise misconduct regularly, despite several legislative frameworks and laws established to guard against it and ensure proper accountability and the rule of law. The institutions tasked with monitoring public finances and tracking the performance of municipalities are the municipal councils and Municipal Public Accounts Committees (MPACs), the Auditor-General and the Public Protector, among others.

Masiya, Davids and Mangai (2019:20-23) conducted a study assessing service delivery and public perception of a municipality's ability to deliver essential services in various communities in South Africa. Many developing countries face service delivery challenges, which have exacerbated a lot of violent service delivery protests that result in the loss of life and damage to property. Several municipalities in South Africa have also experienced, and some are still experiencing, service delivery protests accompanied by extreme violence and the destruction of property. Scholars have argued that such protests are mainly a result of the failure of municipalities to own up to their promise of providing citizens with basic quality services. The countrywide protests and demonstrations about poor service delivery result from poor housing quality, lack of water and sanitation services, failure by municipalities to address unemployment and the lack of electricity in many households.

Asha and Makalela (2019:2-3) mentioned in their study that service delivery challenges remain unresolved. Accordingly, most communities in the country have expressed unhappiness with the government and municipalities' failure to provide adequate basic services. Municipalities should promote their developmental role to overcome the centralised and fragmented planning practices of the apartheid era. The IDP is the municipality's blueprint for encouraging community participation whereby community members raise their needs, demands and concerns so that a municipality can address them; however, municipalities are struggling to implement what the IDP requires them to do.

RESEARCH METHODOLOGY

This study adopted a qualitative research approach. Creswell (2013:50) describes the qualitative research approach as a unique technique often used to collect data describing a particular social phenomenon. This approach is often used to get a deeper understanding and experience of how a specific individual or group of people within a specific environment views and understands the world around them. Bekwa, Grobler, Olivier, Sadiki and Van Niekerk (2019:41) indicate that the advantage of using the qualitative research approach is that it gives valuable and rich data about the participants' perspectives on their views and experiences about the world around them. Since the research approach was qualitative, a multiple case study design was used as the interpretivist research paradigm.

A multiple case study design was utilised because this research involved several people, the officials (elected ward councillors and appointed municipal officials), who are the main research subjects. Bekwa *et al.* (2019:142) define a case study design as a type of qualitative research design which researchers utilise to investigate a single entity or phenomenon in a specific environment. The case study design was selected because the sample population was small, comprising 16 participants from the municipality. In this study, the researcher used semi-structured interviews and document analysis for data collection.

According to Maree (2010:122), an interview is a two-way conversation with the researcher asking the participants questions to collect data and learn more about the participants' ideas, beliefs, views, opinions and behaviours. In this study, open-ended questions were used in the interview guide because open-ended questions allow the researcher to extract data on the participants' points of view. The interviews were conducted at the offices of Nkomazi Local Municipality sites with municipal administrators from various departments in the municipality and ward and PR councillors.

The interview guide questions had the following sub-headings that were open-ended questions with reservations for probing questions:

- Understanding of service delivery
- The role of local government
- Public participation
- Integrated Development Planning
- The roles and responsibilities of councillors and administrators within the municipality
- Financial management and auditing within the municipality
- Monitoring and evaluation within the municipality

These sub-headings provided detailed, rich data concerning the status quo of service delivery and its challenges within the municipality.

Document analysis was chosen as a form of secondary data collection to gain insight into the service delivery challenges in the Nkomazi Local Municipality by addressing the everyday needs of the communities under its jurisdiction. Document analysis involved thoroughly analysing the municipality's IDP from 2017 to 2022. The Auditor-General's report on municipalities for 2019/2020 was also analysed. Additionally, the 2020 Auditor-General's Covid-19 Municipal Relief Funding Report (Covid-19 MRFR) provided a detailed analysis of the current state of affairs within the municipality. During this research, numerous reports from SALGA and COGTA were analysed to gain insight into the affairs of the Nkomazi Local Municipality. Newspaper articles, independent newspaper columns, media publications and press statements, and relevant and valuable official reports were critically scrutinised.

The data was analysed through thematic analysis. Braun and Clarke (2006:78) explain that the thematic data analysis approach involves discovering, interpreting and reporting patterns and clusters of meaning within the dataset. There are specific steps taken during thematic data analysis, namely collecting, organising, transcribing, coding, describing and categorising interview notes, which were then transcribed and repeatedly read to identify emerging patterns.

This study investigated obstacles to municipal administrators and councillors executing their day-to-day activities of service delivery provision. Interviews were conducted to find out how the municipality addressed and managed these difficulties. In addition, document analysis was performed, and theoretical concepts were derived from the literature and municipal documents to obtain rich data.

RESULTS AND DISCUSSIONS

The following issues were identified as the biggest service delivery challenges within the Nkomazi Local Municipality. They are distinguished as institutional and external factors contributing to service delivery challenges.

Institutional factors contributing to service delivery challenges

Internal service delivery challenges within the municipality refer to all the internal issues found within the rural institution, and include:

Corruption

Nkomazi Local Municipality has been embroiled in corruption scandals over the past few years (The South African 2020). During the Covid-19 pandemic in 2020, the municipality was reportedly involved in a PPE procurement scandal. The Hawks raided and seized critical documents relating to possible R27 million

PPE irregularities (City Press 2020). The Hawks' spokesperson Brigadier Hangwani Mulaudzi confirmed that on 27 August 2020

"...the hawks raided the Nkomazi Local Municipality following investigations relating to alleged Covid-19 procurement fraud wherein the appointment of contractors and service providers were inconsistent with the requisite procurement regulations" (The South African 2020).

It is believed that the allegations of corruption were made three weeks before the Hawks raided the municipality's offices. It was reported in some newspapers that the municipal manager of Nkomazi Local Municipality,

"...had left officials confused when he allegedly hogged the Covid-19 budget and appointed companies without a bidding process, violating the Municipal Finance Management Act and Treasury regulations for tender above R200 000" (City Press 2020).

Fraud

Fraud is also among the challenges that were discovered within the municipality. It is believed that the municipality was embroiled in alleged fraudulent activities during the Covid-19 pandemic. According to the Auditor-General's Covid-19 Municipal Relief Funding Report (CMRF), the municipality was among the top 12 municipalities with a heightened risk of susceptibility to fraud. The CMRF outlined that the Auditor General sampled all the metropolitan municipalities and several district and local municipalities to audit their expenditure trend during the Covid-19 pandemic. It was believed that Nkomazi Local Municipality assigned a service provider to do fumigation in March 2020, which cost the municipality R2 620 000. The Auditor-General raised concerns about the overspending for a single service for one month and voiced unhappiness over the lack of compliance to appointment and payment of service providers without following the proper processes and procedures for procurement policies within the municipality (RSA 2020:36-38).

According to the Auditor-General's CMRF report, Nkomazi Local Municipality was among the top four municipalities in the country with overpayments in service delivery related to the supply of water tankers during the Covid-19 pandemic in 2020. The Auditor-General's findings revealed that the municipality had assigned service providers to deliver water (with water tankers) to villages that experienced water shortages, and the service providers overcharged the municipality for the water tankers and transportation of water to and from villages. However, what was disturbing from the findings was that all the water tankers, irrespective of size or capacity, were paid the same flat rate for water transportation to villages, regardless of

Table 1: Municipalities with overpayments during Covid-19 municipal relief funding

Province	Municipality	Project	Possible Overpayment
Eastern Cape	OR Tambo District Municipality	Water supply using tankers	R 43 140
Mpumalanga	Bushbuckridge Local Municipality	Water supply using tankers	R 228 800
Mpumalanga	Mbombela Local Municipality	Water supply using tankers	R 84 837
Mpumalanga	Nkomazi Local Municipality	Water supply using tankers	R 461 380
Total			R 818 157

Source: (Auditor-General 2020)

the distance that these water tankers had travelled. Another alarming revelation was that documentation detailing this contractual agreement between the municipality and the service providers was hard to find. According to the Auditor-General's report, the municipality incurred a loss of R461 380 due to the municipality's ignorance and noncompliance in determining the appropriate rates for travelling different kilometres to pay all the service providers (RSA 2020:66-67).

Human resource issues

A human resource issue was also identified as one of the challenges contributing to the lack of effective service delivery within the municipality (*City Press* 2020). The municipality was believed to be involved in nepotism and the 'jobs-for-cadre' scandal (*The South African* 2020). Additionally, there needed to be better recruitment processes and procedures within the municipality. In 2019, the municipality's human resource and finance departments overpaid staff members, incurring ZAR18 million with the overpayment of salaries. According to the municipality spokesperson, Cyril Ripinga, *"An employee made a blunder by paying salaries instead of travel allowances on the 15th of November 2019"*. Thereafter, the municipal council did not pay staff members their salaries 10 days later, on 25 November, and workers decided to embark on a strike. They forcefully closed the Driekoppies water treatment stations and prevented the municipality from rendering other essential services, which interrupted the provision of services to the communities within the municipality.

This put pressure on the council, and they had no other option but to accept that they had paid out a 'bonus' and thereby incurred an extra R18 million for salary payments in 10 days. Money that was supposed to be channeled to service

delivery was now lost due to the incompetence of staff members in the finance department. The council tried to persuade workers to collaborate to find strategies for recovering the money, but workers kept their position. The council implored workers to deduct the money from their salaries voluntarily. However, workers refused as they said they could not pay back the money because they were generally in debt. They threatened to shut down the municipality if it continued to attempt to recoup the money (*The South African* 2020).

Political interference

The Nkomazi Local Municipality has become more politicised than ever. It is sometimes difficult to distinguish between politicians and municipal administrators. There is a prevalence of cadre deployment and nepotism within the municipality. The issue of political interference in the municipality undermines the autonomy of its human resources department. The municipality is governed by the ruling ANC party, which has an outright majority in the municipal council (NKLM IDP 2020/2021). The ANC's influential political leaders have seized the duties and responsibilities for recruiting and selecting workers within the municipality without following the proper procedures and protocols. The political deployment of underserving and unqualified persons ensured that even individuals who did not qualify were selected for managerial and strategic positions simply because they were loyal members of the political party (*City Press* 2020).

Political allegiance and loyalty over service delivery were the order of the day at the municipality. Around October 2020, community members from the Nkomazi sub-region staged a protest at the municipality's head office in Malelane to voice their dissatisfaction about the employment of more than 200 workers without following proper procedure. However, these claims were refuted by the spokesperson of the municipality, Cyril Ripinga, who said

"...these posts were advertised [in 2019] and were meant to be filled early in 2020 but the procedure was delayed by the outbreak of Covid-19 and the lockdown" (Mpumalanga News 2020).

The fact that the municipality was embroiled in a battle for the election of the executive mayor also contributed to the lack of service delivery in communities under the municipality (*City Press* 2020).

External service delivery challenges

The following are external factors contributing to delivery challenges within the municipality: Illegal immigration, vandalism, and illegal dumping. These external factors are problems beyond the municipality's operations or reach; the

municipality might be unable to control or prevent them. The following external factors were identified as the most problematic affecting service delivery within Nkomazi Local Municipality.

Illegal immigration

Communities across the municipalities have expanded considerably over the past 10 years. New informal settlements have ‘mushroomed’ all over the municipality’s jurisdiction, burdening the municipality’s budget. The increase in the population due to immigration means that the municipality’s revenue would decrease because some funds had to be channeled towards providing basic services like water to these newly established villages. The influx of illegal immigrants from neighbouring countries like Eswatini and Mozambique contributes to population growth, putting a strain on the municipality’s finances. The immigration of foreigners contributes to the high rise of illegal land invasion, causing the expansion of new villages within the Nkomazi sub-region. This created problems for the municipality regarding effective planning (NKLM IDP 2020/2021).

Vandalism

Another issue that continues to derail the provision of basic services to communities is the vandalism of public and private properties during service delivery protests. Community members destroy basic infrastructures like roads, schools, community halls, cemeteries and clinics during service delivery protests or any other protest they embark on. This, in turn, delays the progress of service delivery by the municipality. There is also significant vandalism of water infrastructure like pipes which usually occurs through illegal water connections by community members (NKLM IDP 2020/2021).

The Auditor-General asserts that nationwide municipalities lose between 15 to 30% of water annually. Approximately 84% of municipalities with the mandate to provide communities with water services suffered a loss of R9.82 billion of wasted water that was supposed to be distributed effectively to needy communities. It was noted that some municipalities explained their water losses in rand value and how they lost such water; others opted to take no accountability by not disclosing how they had lost their water (RSA 2022:43-45).

Illegal dumping

The issue of litterage continues to be an obstacle to effective service delivery by the municipality. Residents from various communities, dump rubbish near or inside dams and streams the municipality should use to provide clean, quality water to community members. The garbage also blocks and destroys water infrastructure like pipes and water treatment plants. This creates a burden for the municipality in

the provision of clean drinking water. It also contributes to illnesses that originate from drinking polluted water (NKLM IDP 2020/2021).

According to the Auditor-General, municipalities have a duty and an obligation to provide communities with clean drinking water and sanitation services. The provision of water is a key service delivery area for all municipalities in the country. However, in some municipalities, water is not adequately conserved by effectively managing and controlling its use (RSA 2022:34).

Results from qualitative interviews

The study sought to investigate service delivery challenges in Nkomazi Local Municipality. Data collection was done by conducting face-to-face semi-structured interviews with 16 participants. There were six municipal administrators within various departments, six ward councillors and four PR councillors. The study undertook to understand service delivery challenges within the Nkomazi Local Municipality. The data collected was meant to reflect the strategic importance of Nkomazi Local Municipality in service delivery and, more specifically, in advancing the lives of community members under its jurisdiction. This research focused on the challenges that Nkomazi Local Municipality faces in terms of service delivery. From the findings of this study, when analysing responses from the participants as well as through document analysis, Nkomazi Local Municipality is confronted with the following challenges that hinder effective service delivery:

Poor planning

The municipality needed to plan properly how it wanted to achieve its set objectives of delivering essential services to communities. It needed to have strategic plans for effectively and efficiently utilising its resources. Inadequate planning contributed to mishaps with service delivery to communities. The municipality needs to be more responsive to the needs and demands of communities, as seen by lengthy delays in addressing critical matters such as water shortages and other service delivery issues. If community members are not involved by the municipality, the municipality cannot focus on delivering services that are essential to communities.

Financial mismanagement

The study found an element of noncompliance with financial legislation by the municipality. Municipal officials do not comply with procurement policies. This, in turn, creates a burden on the municipality's finances as procurement processes and procedures are routinely disregarded by the municipality. This contributes to the inadequate delivery of services as funds for the provision of such essential services are usually exhausted due to illicit and irregular expenditure.

Human resource issues

Nkomazi Local Municipality seems to need to conduct proper recruitment procedures for its personnel. The municipality does not uphold the principle of transparency during its recruitment, shortlisting and interview processes. Proper procedures are not adequately followed, and political influence is allowed during these processes.

Corruption

Corruption was rated among the most severe factors contributing to the decline in municipal service delivery in the entire country. Corruption is a major challenge for many local municipalities in the country, and Nkomazi Local Municipality is no exception.

Political interference

This issue continues to be of great concern in many municipalities all over the country. Municipalities have been 'captured' by top politicians everywhere, and Nkomazi Local Municipality is no exception. A municipality must note that it is responsible for ensuring service delivery and not for honouring the interests of politicians. Political interference affects how a municipality is governed; communities often suffer because of lack of service delivery.

Illegal immigration

The influx of legal and illegal immigrants from neighbouring Eswatini and Mozambique negatively impacted the Nkomazi Local Municipality's budget. The immigrants have to be catered for by the municipality once they are in the country as they engage in land invasion and the illegal occupation of land. This places a burden on the municipality's finances.

Vandalism

Community members engage in vandalism whenever they want to show their unhappiness over the lack of service delivery to their communities. Community members damage public and private property during service delivery protests and should own up to that. Community members seem unaware that the facilities they are destroying need to be preserved for future generations and that they must take responsibility for and guard against vandalism as these properties are there to enhance their lives and that of the future generation.

Illegal dumping

There was gross illegal dumping in many communities within the Nkomazi Local Municipality. This poses a hazard to communities and the environment as a

whole. Poisonous substances may result in children's deaths due to excessive and illicit garbage dumping.

CONCLUSION AND RECOMMENDATIONS

The study focused on attempts to understand the main challenges the Nkomazi Local Municipality faces regarding service delivery within its communities. Service delivery is guaranteed by the *Constitution of the Republic of South Africa* of 1996. It is a constitutional right of the members of the various communities under the Nkomazi Local Municipality. The study sought to understand the challenges the municipality faces in providing effective service delivery by interviewing municipal administrators and ward and PR councillors. The 16 participants in this study were representative of the views of a specific group of respondents attached to the Nkomazi Local Municipality. Their views may not represent the entire local municipality.

The study further explored the problems and challenges facing the municipality with service delivery in depth by analysing important documents. The data obtained from document analysis was linked with the data gathered through interviews with the participants of the study. Finally, possible mechanisms for addressing service delivery problems experienced by Nkomazi Local Municipality were suggested.

The following recommendations are made for the municipality to address the challenges that hinder effective service delivery to communities under its jurisdiction.

Improved planning

Proper planning must be implemented within the municipality. In addition, the municipality must monitor and implement IDP forums. Municipalities should effectively and continuously involve community members in the affairs of the municipality by ensuring regular and constant communication with members of the communities. For example, there should be effective formulation and implementation of the IDP, which requires community members' involvement. The municipality needs to develop community development initiatives (outreach programmes) that will inform community members on basic IDP skills and the importance of public participation.

In addition, the municipality should involve ward committees because they play a crucial role in ensuring the effective dissemination of information from the municipality to the communities. Public participation is beneficial and should be promoted as it enables communities to provide input about their needs and demands and what should be done by the municipality to enhance their lives. The

ward committees are essential to local government as they promote transparent, effective, and good governance.

Improved financial management

There should be proper monitoring and evaluation of financial usage within the municipality and accountability needs to be enforced. Community members must also pay for the services they receive from the municipality to supplement the institution's annual budget. The municipality must devise strategies to ensure that it can manage its annual allocated budget by doing proper and effective planning to sustain itself and remain operational. Most importantly, there should be consequences for municipal officials who neglect financial legislation and engage in financial mismanagement. Law enforcement agencies must be allowed to investigate and prosecute anyone who violates the law.

Effective human resource management

The municipality must recruit capable and qualified personnel. The municipality must conduct workshops and training for newly recruited staff members so that they can perform better. The municipality must be transparent with its recruitment, shortlisting and interview processes. Proper procedures must be followed, and political influence should not be allowed during these processes.

Prospective job applicants should undergo rigorous checks and verifications for the municipality to employ only qualified and appropriate personnel. A municipal administrator employed in the finance department should possess a financial qualification because a municipality requires a person with relevant experience and expertise to deal with fiscal matters. The municipality should refrain from employing people based on their political affiliations. In short, political interference should be disregarded at all costs during the municipality's recruitment drive. Performance agreement contracts should be designed and implemented.

Corruption control

The Prevention and Combating of Corrupt Activities Act 12 of 2004 and the Local Government: Municipal Finance Management Act 56 of 2003 must be strenuously enforced to address issues around corruption and maladministration within the municipality. Transgressors must be prosecuted by ensuring that when an employee is found to have been engaged in corrupt activities, they must be reported to law enforcement agencies to ensure it does not occur in future. The municipality must deal with corruption decisively and municipal officials must be held accountable at all times.

Political interference control measures

The municipality must ensure that it does not allow itself to engage in political issues but instead focus on service delivery. Recruitment of personnel must not be politically motivated. Instead, it must be on merit. Deployment should be aligned with qualifications. The majority political party within a municipality should lead the way by ensuring that it commits to effectively implementing municipal policies, programmes and activities that will assist the municipality in sustaining itself. This means that community members should hold the municipal council accountable. Members of the majority political party should not engage in numerical arrogance. They must regularly be held accountable. Transparency, democracy and good governance should be a municipality's key pillars and cultures.

Illegal immigration control

The Department of Home Affairs needs to investigate this issue as it is beyond the municipality's capacity. Border management should be reinforced, and there should be strict rules for those entering the country either illegally or legally.

Dealing with vandalism

The municipality must work closely with law enforcement agencies to report cases of vandalism, and the perpetrators of such vandalism must be prosecuted. Community members need to take ownership of their community facilities and report any vandalism of public property.

Dealing with illegal dumping

The Nkomazi Local Municipality, in collaboration with community members and other stakeholders, must develop awareness campaigns and educate community members about the dangers of excessive and illicit garbage dumping. The community must know that illicit dumping poses significant potential environmental hazards. Communities need to take care of their environment and keep it clean and safe at all times. One way of teaching is by the so-called three 'Rs': reduce, reuse and recycle, whereby people are taught that waste material can be recycled to maintain a sustainable environment. Community members must be taught that waste material can be reused and recycled into valuable items and that they can even open businesses and sell these. Plastic, glass and cardboard boxes can reproduce items such as caps, rubber mats, drinking cups and many other kitchen utensils.

NOTE

- * This article is based on an unpublished Master's dissertation (M.A. Public Management and Governance) under the supervision of Prof D Meyer: Shongwe, B. 2023. *Service delivery challenges within rural communities: The case of the Nkomazi Local Municipal area*. Johannesburg: University of Johannesburg.

REFERENCES

- Asha, A.A. and Makalela, K.I. 2019. Challenges in the Implementation of Integrated Development Plan and Service Delivery in Lepelle-Nkumphi Municipality, Limpopo Province. *International Journal of Economics and Finance Studies*. 12(1):1-15.
- Bekwa, N. Grobler, S., Olivier, B., Sadiki, M. and Van Niekerk, A. 2019. *Economic and management research for HMEMS80* (Unisa custom edition). London: Pearson Education Limited.
- Beyers, L.J.E. 2016. Service delivery challenges facing municipalities: A case of Fetakgomo Local Municipality in Sekhukhune District Municipality, Limpopo Province. *Bangladesh e-Journal of Sociology*. 13(2):167-178.
- Booyesen, S. 2012a. Conclusion. In Booyesen, S. (Ed.). *Local elections in South Africa: Parties, people and politics*. Bloemfontein: Sun Press. Pp 351-359.
- Braun, V. and Clarke, V. 2006. Using thematic analysis in psychology. *Qualitative Research. Psychology*. 3(2):77-101.
- Brynard, D.J. 2012. *Ethics in public administration and administrative justice*: Only Study Guide 2 for PUB3707. Pretoria: University of South Africa.
- Bvuma, J. 2014. *Public Administration: Foundations of Public Administration: Rights and Obligations in Public Administration*: Only Study Guide 1 for PUB2601. Pretoria: University of South Africa.
- City Press. 2020. Available at: <https://www.news24.com/citypress/news/municipality-pays-r18m-in-salaries-twice-but-workers-refuse-to-give-it-back-20191125>. (Accessed on 21 October 2021).
- Creswell, J.W. 2013. *Research design: Qualitative, quantitative, and mixed-methods approaches*. Boston: Sage Publications.
- Denhardt, R.B. and Catlaw, T.J. 2015. *Theories of Public Organization*. Stamford: Cengage Learning.
- Institute for Democratic Alternatives in South Africa. 2010. *The state of local governance and service delivery in South Africa: Issues, challenges, and solutions*. Pretoria: IDASA.
- Kanyane, M.H. 2010. Service delivery in question. In Kondlo, K. and Maserumule, M.H. (eds.). *The Zuma administration: Critical challenges*. HSRC Press: Cape Town.
- Kanyane, M.H. 2013. Exploring challenges of municipal service delivery in South Africa (1994-2013): *Africa's public service delivery and performance review*. 21(3):90-110.
- Kanye, M.H. 2014. Exploring challenges of municipal service delivery issues in South Africa. *Africa's public service delivery and performance review*. 2(1):90-110.

- Mabizela, H. and Matsiliza, N.S. 2020. Uncovering the gaps in the provision of services in the rural Okhahlamba Municipality of KwaZulu-Natal province. *Africa's Public Service Delivery and Performance Review*. 8(1):1-9.
- Madue, S.M. 2016. In the name of ignorance: Trampling on the powers of institutions supporting constitutional democracy in South Africa. *Journal of Public Administration and Development Alternatives*. 1(2):1-13.
- Madzivhandila, T.S. and Asha, A.A. 2012. Integrated development process and service delivery challenges for South Africa's local municipalities. *Journal of Public Administration*. 4(2):1-18.
- Maree, K. (Ed.). 2010. *First steps in research*. (6th ed.). Pretoria: Van Schaik Publishers.
- Maserumule, M.H. 2011. *Good governance in the new partnership for Africa's development (NEPAD): A public administration perspective*. Unpublished PhD Thesis. Pretoria: University of South Africa.
- Masiya, T., Davids, Y.D. and Mangai, M.S. 2019. Assessing service delivery: Public perception of municipal service delivery in South Africa. *Theoretical and Empirical Researches in Urban Management*. 14(2):20-40.
- Masuku, M.M. 2019. Effective governance in South Africa: Reality or myth? *African Journal of Democracy and Governance*. 6(3):1-19.
- Mashamaite, K. 2014. Public service delivery protests in a democratic South Africa: A dilemma for local municipalities. *Mediterranean Journal of Social Sciences*. 25(5):41-59.
- Masiya, T., Davids, Y.D. and Mangai, M.S. 2019. Assessing service delivery: Public perception of municipal service delivery in South Africa. *Theoretical and Empirical Researches in Urban Management*. 14(2):20-40.
- Masuku, M.M. 2019. Effective governance in South Africa: Reality or myth? *African Journal of Democracy & Governance*. 6(2&3):117-134.
- Matsiliza, N. 2012. Participatory monitoring and evaluation: Reviewing an inclusive approach in South Africa's government-wide monitoring and evaluation: Africa's public service delivery and performance review. *Africa's Development Watch*. 8(2):67-83.
- Mbandlwa, Z. 2018. *Assessing the ward councillors' leadership characteristics and their impact on service delivery in eThekweni Metropolitan Municipality*. Unpublished MA Public Management Dissertation. Durban: Durban University of Technology.
- Mbandlwa, Z. and Mishi, S. 2020. Ward councillors' leadership characteristics and their impact on service delivery: Case of eThekweni Metropolitan Municipality, South Africa. *Journal of Local Government Research and Innovation*. 1(6):1-13.
- Mngomezulu, S. 2020. Political Interference in the administration of service delivery in Umlalazi Local Municipality of KwaZulu-Natal, South Africa. *Journal of Economics and Behavioral Studies*. 12(1):38-45.
- Munzhedzi, P.H. 2020. Evaluating the efficacy of municipal policy implementation in South Africa: Challenges and prospects. *African Journal of Governance and Development*. 9(1):89-105.
- Mpumalanga News. 2020. Nkomazi Local Municipality mum on service delivery issues. Available at: <https://mpumalanganews.co.za/402609/nkomazi-local-municipality-mum-on-service-delivery-issues/>. (Accessed on 30 October 2021).

- Naidoo, C. and Ramphal, R.R. 2018. The factors that affect public participation for effective municipal service delivery: A case of ward committees. *South African Journal of Industrial Engineering*. 29(4):82-93.
- Ngcqoyiya, C. 2018. Perception issues and challenges of service delivery: A case study of Umzimkhulu Municipality, in KwaZulu-Natal, South Africa. *IOSR Journal of Humanities and Social Science*. 23(03):71-78.
- Nkomazi Local Municipality. 2019. *Integrated Development Plan: 2020/2021. Malelane: Nkomazi Municipality Office of the Municipality Manager*. Available at: <http://cgta.mpg.gov.za/IDP/Ehlanzeni2018-19/NkomaziPart-1.pdf>.
- News 24. 2020. <https://www.news24.com/news24/southafrica/news/hawks-raids-nkomazi-municipality-over-alleged-r27m-covid-19-tender-fraud-20200731>. (Accessed on 26 June 2021).
- Republic of South Africa. 1996. *The Constitution of South Africa*, 1996. Cape Town: Government Printer.
- Republic of South Africa. 1998. *Local Government: Municipal Structures Act*. (Act No. 117 of 1998). Pretoria: Government Printer.
- Republic of South Africa. 1998. *White Paper on Local Government*, 1998. Pretoria: Government Printer.
- Republic of South Africa. 2000. *Municipal Systems Act*. (Act No. 32 of 2000). Pretoria: Government Printer.
- Republic of South Africa. 2009. *The state of local government in South Africa. (Department of Cooperative Governance and Traditional Affairs). Report*. Pretoria: Government Printer.
- Republic of South Africa. 2017. *The state of basic service delivery in South Africa: In-depth analysis of community survey 2016 data (Statistics South Africa)*. Pretoria: Government Printer.
- Republic of South Africa. 2020. *The Covid-19 Municipal Relief Funding Report (CMRF)*. Pretoria: Government Printer.
- Republic of South Africa. 2021. *The Third Special Report on the Financial Management of Government's Covid-19 Initiatives*. Pretoria: Government Printer.
- Republic of South Africa. 2021. *Consolidated general report on local government audit outcomes (MFMA 2019/2020)*. Pretoria: Government Printer.
- Republic of South Africa. 2022. *Consolidated general report on local government audit outcomes (MFMA 2020/2021)*. Pretoria: Government Printer.
- Sefala, M.J. 2009. Nature and purpose of developmental local government: Case of the Limpopo Province. *Journal of Public Administration*. 44(4):1-18.
- Siddle, A. and Koelble, T.A. 2016. Local government in South Africa: Can the objectives of the developmental state be achieved through the current model of decentralised governance? *Swedish International Centre for Local Democracy*. Research report no. 7. Exakta Print, Malmö.
- Stroh, E.C. 2014. *Public management skills: Only study guide for PUB3701*. Pretoria: University of South Africa.
- The South African*. 2020. Available at: <https://www.thesouthafrican.com/news/ppe-corruption-scandal/>. (Accessed on 6 August 2020).

AUTHORS' CONTACT DETAILS

Mr Bonginkosi R. Shongwe

University of Johannesburg
Kingsway Avenue Campus,
Auckland Park
Johannesburg
Cell: 0820929262
Email: bshongwe111@gmail.com

Prof Daniel F. Meyer

School of Public Management,
Governance and Public Policy
University of Johannesburg
Kingsway Avenue Campus,
Auckland Park
Johannesburg.
Cell: 082 8505656
Email: dfmeyer@uj.ac.za

Organisational Variables Influencing Strategic Management in Public Organisations

Critical Considerations for Strategy Execution

C B Sibiya*

School of Management, IT & Governance
University of KwaZulu-Natal

M Subban

School of Management, IT & Governance
University of KwaZulu-Natal

ABSTRACT

The article focuses on the organisational variables influencing the execution of strategy in public sector organisations. After comprehensive strategies have been developed, significant challenges are often encountered during the strategy execution process. The growing needs of communities coincide with the shrinking resources in public administration where cost-effective means and efficiency are the main concerns to contribute to more effective service delivery with less resources. The article conceptualises the public strategic management process, provides several perspectives for strategic management in the public sector and highlights the public service planning framework and other plans. It then discusses the organisational variables influencing strategy execution in public organisations. The study found that the ability to develop and execute new strategies means the difference between effective service delivery and failure on the part of public sector organisations and that the majority of strategies in public organisations fail in the strategy execution phase.

INTRODUCTION

Strategic management mechanisms have been introduced in public administration globally to enhance capacities and organisational performance caused by

increasing economic and political challenges (Bryson and Alston 2005). Pearce and Robinson (2007: 3) maintain that strategic management is regarded as the senior management role and responsibilities that results in strategy development, application and monitoring with a set of strategic objectives within a sector in which organisations function. Strategy execution is regarded as a practice that transforms the strategic plan into a sequence of activities that ensure that the predetermined strategic objectives of the organisation, are effectively realised (Thompson and Strickland 2003:365). For many years, strategy development was widely viewed as a more significant part of the strategic management process than strategy execution. However, contemporary studies find that strategy execution is a prerequisite for effective organisational performance and service delivery (Wauters 2017 in Sibiya 2018).

Bryson (2010 in Sibiya 2018) concurs that strategic management is a process that integrates strategic development and strategy implementation on an ongoing basis to realise legal mandates. Strategy execution, rather than strategy development alone, is a prerequisite for effective organisational performance.

Scarcity of resources affects effective organisational performance and service delivery, and most organisations have processes for translating strategy into desired outcomes. However, the strategies do not produce the desired outcomes. Ehlers and Lazenby (2007:213) view execution as a means of focusing on the operational plans, and to address any unanticipated deviations.

Strategy execution is essential, but demanding because execution activities have a longer time frame and involve more individuals, which becomes a challenge as the public service is a complex environment (Brynard 2005:651,653). The incapacity of public organisations to efficiently accomplish their set strategic goals is one of the factors that limits their success, which requires constant investment in managing resources (Madue 2008:200).

Research studies revealed that there are many organisational detracting factors to the effective execution of strategy (Kaplan and Norton 2004:277). Rahimnia, Polychronakis, and Sharp (2009:246-260), concurred on the “blemishes” in strategic planning as the contributor to the inhibitors to strategy execution. The authors highlighted key inhibitors of strategy execution within the organisation, which were: planning results, organisational arrangements, individuals (people), as well as leadership. When key people in the development of a strategy are not participating in its execution, the ownership is often lost.

The challenges and the key organisational drivers and inhibitors of effective strategy execution are discussed in this article as important organisational variables influencing strategy execution in public sector organisations. These include: linking budgets to strategy execution; audit report perspectives; leadership and management; organisational structure and culture and Information Communication and Technology (ICT).

CONCEPTUALISING THE PUBLIC STRATEGIC MANAGEMENT PROCESS

When examining features of the public strategic management process, it is important to delineate the concept of public strategic management. There is no clarification of what strategic management processes precisely involve (Stoney 2001:27-33). However, Walker, Andrews, Boyne, Meier, and O’Toole (2010:524), argued that strategic management processes are made up of two significant organisational processes namely, the strategy development, which requires crafting strategies, objectives and actions to have a strategic plan that determines the future ideal of the organisation; and strategy execution complemented by resource allocation and strategy control and evaluation. The strategic management process is the central integrative process that gives the organisation a sense of direction and ensures a concerted effort to realise strategic goals and objectives (Poister and Streib 1999 in Sibiya 2018).

Bryson (1988:118) describes strategic management largely as a well-organised effort to produce significant results and activities that define ‘what’ an organisation is, ‘what’ it does, ‘why’ such services, programmes and projects exist; and ‘how it should be done’. In contrast, Abdulwahid, Muhamad Sehar Arshad and Iftekhar (2013), view strategy planning as an important integrated process. However, in terms of this article, the application of strategy is seen as the key process that has a longer time frame than planning, involves many people and has more task complexities. The Public Service Commission (2008:13-15) states that Head of Departments (HoDs) have unplanned transitions from one leader to another, taking over predetermined strategy objectives that cause instability and inconsistencies in public sector organisations. Poister, Pitts and Edwards (2010 in Sibiya 2018) concede that strategic planning approaches in public sector organisations are not always designed at improving organisational performance of ongoing programmes.

Thompson, Strickland and Gamble (2008:256) have introduced a strategic management model that has five phases of planning, as reflected in Table 1:

Table 1: Strategy management model

One	Two	Three	Four	Five
Define the organisation and the vision and mission	Setting strategic goals and objectives	Developing of performance targets	Executing of the strategic plan	Monitoring and evaluation of performance and reformulating the strategy
↑	↕	↕	↕	↑
Feedback				

Source: (Thompson, Strickland and Gamble 2008:256)

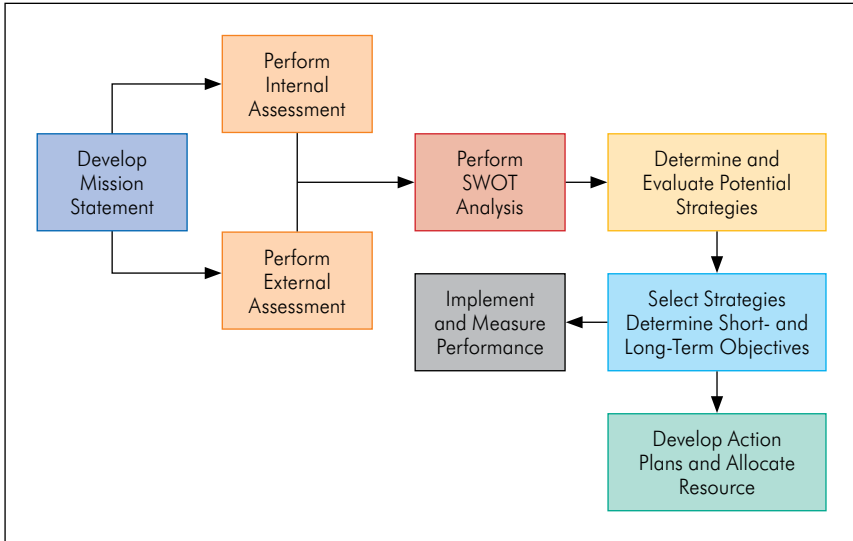
Inferring from Table 1, leadership does not necessarily go through the phases as indicated. The *first three* phases identify the philosophy that underpins strategic planning processes in the organisation. *First*, the boundaries between phases are sought to change the basic method of managing the strategic planning processes. Leadership and senior managers describe the vision and mission of the organisation. *Second*, is the link between the vision and mission that leads to the strategic goals and objectives. *Third*, it involves the development of performance targets that link to strategic objectives. The *fourth* phase consists of a series of steps performed by various resources within the organisational levels. The execution of strategies needs planning and managing processes, which is a large part of management, since customers expect better and customised services. Management ought to learn to respond to customers' needs and render effective services and products. This also involves not only managing, but also the administrative burden to turn the strategic plans into desired results, and the processes take time to produce the products. The *fifth* phase, focuses on the effects and not the causes. It focuses on the monitoring and evaluation and is both the end and the beginning of the strategic management cycle. Developing and executing strategy must be regarded as something that is constant and progressive.

In contextualising strategy execution, there is no generally accepted definition of strategy execution because the separation of strategy development and execution causes fragmentation in a top-down strategy approach *versus* a bottom-up strategy development that are noticeable factors in public sector organisations, asserts Hupe (2014 in Sibiya 2018). It is important to understand the bigger picture before starting with the execution of the strategy. The strategy management model in Table 1 lacks detail as there is no clear transition from Phases One, Two and Three to Phase Four, and it can be viewed as a top-down approach that does not provide an understanding of the strategic plan, philosophy or changes, argues Hupe (2014 in Sibiya 2018). On the other hand, in a bottom-up approach, employees at all levels participate in strategic processes and need to better understand the role of their activities so that there are relationships between operations, senior management and middle managers and officials at all levels in directing their actions within set rules and objectives, (Hill 2003 in Sibiya 2018).

According to Pearce and Robinson (2007:3), management should provide direction and certainty in organisations by taking strategic, administrative and operational decisions so that the employees at all the levels will understand the organisational goals within the organisation. In addition, these decisions are contributors towards the improvement of service delivery, not only on the strategic management level, but on all employees who implement these strategies.

Colgan, Rochford and Burke (2016: 27) assert that effective leadership and management are crucial for the effectiveness of public sector strategies. However,

Figure 1: Strategic planning model



Source: (Stavros, Cooperrider and Kelly 2014:4)

Smets, Jarzabkowski, Burke and Spee (2015:942-943) view leadership effectiveness as most visible when an organisation develops its strategy, but the visibility of leaders when executing the strategy is minimal. Thus, information such as setting objectives, utilisation of resources and monitoring the process are additions to the original strategic models. Stavros, Cooperrider and Kelly (2014:4) indicate a characteristic strategic management process in contemporary management literature, as reflected in Figure 1.

Drawn from Figure 1, the transition *from* strategy development *to* strategy execution remains a gap, which is complex, since it involves many people who operate at different levels of the organisation (Harrington and Williams 2004:297-306). Neuman (2006:111) states that the primary strategy is to view an organisation as a system of improving the performance of each organisational component. According to Stavros *et al.* (2014:5), a major emphasis of the strategy planning is to set the stage of defining and coordinating the interdependent factors or operations by improving service delivery or productivity. Hupe (2014 in Sibiya 2018) contends that it is difficult, if not impossible, for public managers to state the overall mission and objectives of their organisations, as they are not all encompassing and measurable as revenue results. In contrast, Mukherjee and Mukherjee (2017 in Sibiya 2018) view the desirable outcomes as likely to rely on embracing behavioural changes on the planning processes of programmes in which ownership by the communities ought to be the key factor.

The lessons learned from Figure 1 concerning the strategic planning model have implications in the day-to-day functioning of any organisation. The application of the strategies provides an advantage to the leadership and senior management during a transition to focus on organisational performance that will propel public sector organisations to effective service delivery. However, Bovaird, Stoker and Loeffler (2016 in Sibiya 2018) urge that citizens more importantly, should be involved in the planning process.

PERSPECTIVES FOR STRATEGIC MANAGEMENT IN THE PUBLIC SECTOR

Strategic management processes which embrace public policies and planning as part of a broader strategy process, gained significant ground in theory and in practice. Strategic management is now prominent because of the introduction of the National Treasury Strategic Framework where strategic management processes become the core of the public sector organisations for generating insights into the complex direct and indirect links among public strategic management processes (Wolf and Floyd 2013 in Sibiya 2018). Theoretically, the context of public strategic management is linked to the level of inclusiveness and convention of the process. The strategic planning processes have received more attention because of the South African Excellence Model, New Public Management (NPM), New Public Governance (NPG), government outcomes and the National Development Plan, 2030 (NDP), and the idea is to create an effective service delivery model. There are possibilities of applying strategic theories in public organisations, especially by the National Treasury Strategic Framework, but in the public sector, the emphasis is on cost which is a more prevalent feature, especially when defining delivery systems (Sibiya 2018).

The hypothetical assumption that public strategic plans lead to positive outcomes, is highlighted by Walker, Andrews, Boyne, Meier and O'Toole (2010) who indicate that strategies vary across organisations, and organisational variables to be taken into account are diverse. For them, strategic plans shape the impact of constraints on performance, and is in theory, both directly and indirectly linked to performance (Walker *et al.* 2010).

Public senior managers are the “guardians” to ensure the continuity of government service delivery and influence the collective processes (Cook 1998 in Sibiya 2018). Within government, planning of and budgeting for programmes and projects cannot occur properly without policy considerations that form the basis for tracking and assessment of programmes of action (Valadez and Bamberger 1994:59-61).

Public policy development is based on the Election Manifesto and the Medium-Term Strategic Framework, including the 12 national strategic outcomes,

which were established through comprehensive consultations at both ministerial and administrative levels (RSA Guide to the Outcome Approach in Sibiya 2018). The strategy links to planning frameworks and other programmes of action, budget processes to promote accountability, as well as the need for monitoring and reporting (National Treasury 2010:1-2). The National Treasury's policy defines the scope within which the organisation can operate and empowers senior management to deal with the strategy decision. According to Jones and George (2003:277), organisational planning must contain five phases for the allocation of roles and responsibilities for strategy execution, develop operational plans with schedules, allocate the necessary resources, and ensure accountability of responsible managers.

For the scope and parameters within which public sector organisations function and the decisions to be taken by the executive authority or senior management, organisational planning is viewed as a bigger picture approach that blends objective analysis of values, goals, and priorities to map a future direction (Poister 2010:246). Government departments formulate strategy in accordance with Chapter 11 of the White Paper on the Transformation of the Public Service of 1995.

Lorange (1998 in Sibiya 2018), indicates that successful execution begins with the development phase and a failure to find that link between strategy development and strategy execution is a step towards strategy failure. Strategy development is basically entrepreneurial in nature and requires thorough analysis, decision-making and innovation. However, Chapman (2002:18), concurs with Lorange that strategy execution requires administrative and managerial skills and an ability to analytically envisage impediments that might arise.

Naidoo (2005:109) states that the slow pace of execution by some government departments can be ascribed to a shortage in management competencies. Kaplan and Norton (1996:10), agreed that leadership can strategically drive policy processes by the effective utilisation of public servants. According to Bana and Shitindi (2009:61) an organisation without a strategy is viewed as purposeless and unsure of its service delivery and production processes.

Thompson and Strickland (2003:365), view strategy execution as an important part of organisational performance for realising predetermined objectives to enhance service delivery. There has to be a move from bureaucratic rule-driven systems in public service management to an outcomes-based approach to government's effective service delivery. Public service organisations are supposed to function within the NPM model, where the focus is on achievement of measurable results and effective service delivery. Despite the quest for efficiency through NPM and disapproval of unethical conduct by the public service, there are still complaints of unreliable behaviour by public servants, including senior management (Lues and Bester 2007:93). Welman, Kruger and Mitchell (2005:12, 20), emphasised the lack of understanding of the complexity within which public

organisations are operating. This can be strengthened by organisational policies, organisational culture, organisational structure and systems, which are key for effective public management practices. Kreitner and Kinicki (2010: 66-67) support Welman, Kruger and Mitchell that the values are concepts that pertain to desirable behaviours and are important for the individual and organisation to succeed in pursuing organisational strategy.

The government's strategic approach is based on an outcomes-based approach, as advanced by the National Treasury (2010:2) in South Africa. Public organisations are to adopt strategic development to support leadership and senior managers to contain their budgets (fiscal "harshness") and maintain funding for the government's priority areas. Although the strategic development process and strategic choices are made and executed, there are organisational factors that affect its effectiveness in the realisation of predetermined objectives (Elbanna 2006 in Sibiya 2018).

PUBLIC SERVICE PLANNING FRAMEWORK AND OTHER PLANS

Efficiency is achieved by linking the policymaking process with the resources needed to make the policy a reality. In South Africa, the public demands for effective service continue to grow as people's expectations about service delivery continually rise. The significant number of participants involved in the execution includes middle management, production staff, and the users of the public service, who are crucial for effective organisational performance (Guth and MacMillan 1986:313-327). Cook (1998 in Sibiya 2018) finds substantial merit in successful strategy execution being through people who embrace difficult challenges and continue to build an organisational culture that encourages work commitment on a daily basis. With public policy as the focal point of public administration, the strategic execution process is intended to get the organisation to where it wants to be (Canfield 2005 in Sibiya 2018).

Funds are always insufficient and have to be carefully managed to get the most out of resource constraints, taking cognisance of efficiency and effectiveness. Pencheon, Guest, Melzer and Muir Gray (2008:3787) consider the relationships between plans, programmes, the organisational principles and transformation as the important factors of effective service delivery. The relationships between policy, strategy, and implementation with models, are illustrated in Table 2:

Drawn from Table 2, integration between policy, strategy and implementation is stimulated by governments' needs to regulate and link organisations' performance targets with budget allocations. Bryson (1995 in Sibiya 2018) supported this model as it contains the purpose, policies, programmes, activities and results that outline what an organisation wants to achieve. Henry (2008:8) concurs that

Table 2: Relationships among policy, strategy and implementation

Policy	Strategy	Implementation
Particular philosophies accepted by management are frequently built on policy as evidence. Influenced by policy mandate of the services, politics, public opinion, and the media. Instance: It is the strategy of the government to convert policies into programmes and strategies.	Important steps that need to be taken are critical to deliver the services over a 5-year period. Needs to be supported by strategy goals and objectives. Example: Pre-determined strategic objectives set (5) years, guide the operations and resources allocation of the public sector organisation to improve service delivery and sustained projects.	Ensure that agreed set of objectives and targets provide common platform to better leverage budget and other resources alignment and translate strategy into action. Example: The application of strategy enabling a public sector organisation to measure its progress in realising the agreed initiatives to advance strategic objectives for growth and project implementation.

Source: (Pencheon, Guest, Melzer and Muir Gray 2008:3787)

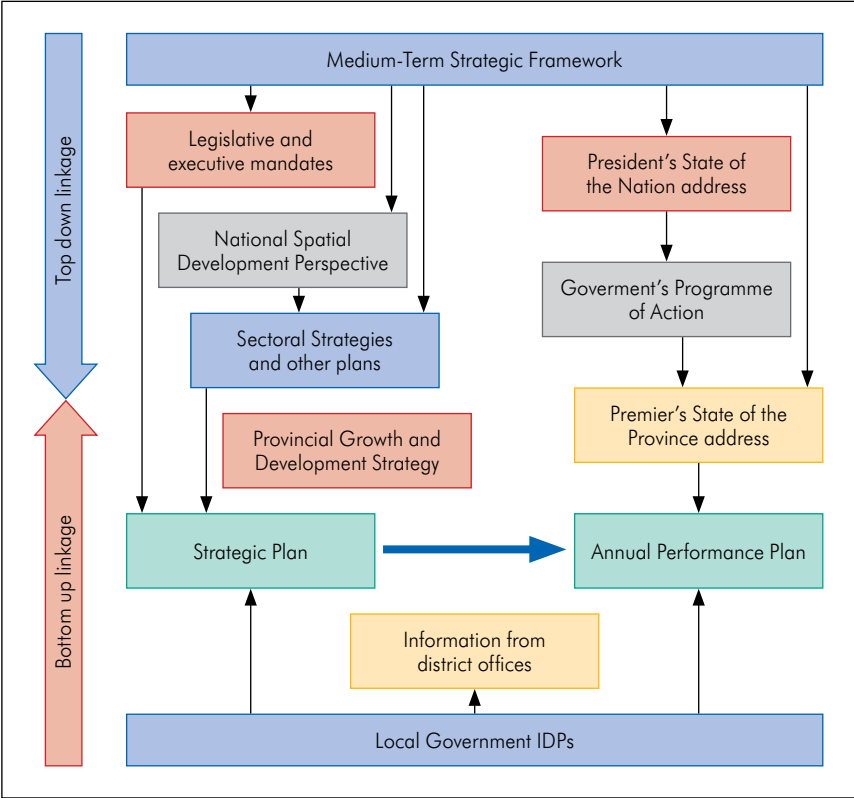
functional strategy is basically responsible for the provision of support to business strategy and it is the responsibility of managers in the different functional units to ensure that their activities are geared to assist and complement the efforts towards achieving the business strategy.

The National Treasury (2010:4) gives guidelines for the Public Service Planning Framework that serves as a strategic development plan in public sector organisations. The framework is preceded by a number of policy decisions from national to provincial, as well as local municipality levels and the spread of resources to give effect to the processes. The Public Service Planning Framework and other plans start from the Medium-Term Strategic Framework (MTSF) backed by the legislative mandates through the State of the Nation address, the Government's Programme of Action and what resources would be required via the State of the Province address. Together, they lead to the development of the Provincial Growth and Development Strategy and the formation of strategy for the various departments at provincial and local spheres. The process is about coherent activities backed by set policy frameworks that eventually concentrates on resources for realising pre-determined strategic objectives. The challenge of the Public Service Planning Framework is the environment from national to provincial and specifically public sector organisations within the different targets and spreading of resources through various agendas.

Figure 2 illustrates the frameworks in terms of the strategic intent that is translated to service delivery.

Each public sector organisation must develop a five-year strategic plan, which aligns with the Government's Programme of Action, as expressed in the MTSF. However, it is expected that each government department or public organisation ought to have strategic plans. Hughes (2003 in Sibiya 2018), identifies challenges

Figure 2: Public service planning



Source: (National Treasury 2010:4)

in the Public Service Planning Framework and other plans as compared to private sector organisations. Hughes (2003 in Sibiyi 2018), is of the view that appropriate benefits of the Public Service Planning Framework and other plans requires capacity for the execution of strategies and set plans that should be identified and developed during the planning processes. This implies that the leadership and management role is also to provide strategic direction and contribute towards organisational developments that could be viewed as coherent sets of activities executed at strategic management positions. However, Naidoo (2004 in Sibiyi 2018) asserts that inadequate alignment of strategy planning and strategy implementation could lead to ineffective service delivery and the realisation of predetermined strategic objectives.

As reflected in Figure 2, public sector organisations' planning consists of a number of legislative mandates, sectoral strategies and the Provincial Growth and Development Strategy with role-players outside the organisations that make

central contributions to the strategic planning and other plans. These different plans should be integrated into a single strategic plan for a public sector organisation. Guth and MacMillan (1986:313-327) state that the challenge of strategy application is to link strategic controls, since imbalance of strategic controls fails to outline the boundaries of the leadership and management role that can impact negatively on the public sector organisation. Kaplan and Norton (1996:10) concur that strategy-focused leadership can strategically drive policy processes by the effective utilisation of public servants.

ORGANISATIONAL FACTORS AFFECTING STRATEGY EXECUTION

The strategies are carried out by employees who are performing activities within their job descriptions that contribute towards converting strategies into tangible and intangible service delivery (Dye 2005 in Sibiya 2018). The planning process in Figure 2 impacts all employees in different ways at different component levels in the current rapidly changing environment that necessitates integrated management approaches. Despite the quest for efficiency and disapproval of unethical conduct by public managers, there are still accusations of disreputable behaviour (Lues and Bester 2007:93). Kreitner and Kinicki (2010: 66-67) concur with Lues and Bester that the values that pertain to desirable behaviours in pursuing the organisational strategy have been lacking in public sector organisations. Government also noticed that the policy development phase had not produced the desired strategic outcomes, hence the focus changed to service delivery during 1997 to 2003 (Sibiya 2018).

As stated in Matters, Leeds and Carroll (2015 in Sibiya 2018), strategic changes happen effectively in the organisation which have a real effect on all aspects of the organisational factors such as leadership, management, the people involved, its culture, structures and processes. Without structured performance, management information strategy becomes pointless when it is executed, and within this aspect, the study contributed to alignment of organisational factors into effective strategy execution.

Naidoo (2005:109) argues that the slow pace of policy implementation by some public service departments can be attributed to a deficiency in leadership competencies. Horwitz (2006 in Sibiya 2018) views this as a lack of an integrated approach to improve strategy execution in public sector organisations. This implies that public service managers need to regularly rethink and realign service delivery plans and budgets to the strategic programmes determined in terms of the government department's purpose (Dickey1992:124-126).

Successful strategy execution is key for any organisation and is a crucial service delivery driver in public sector organisations. Many studies indicate that more time is

allocated to the strategic planning processes than strategy implementation, and the problem of strategies often failing, not because of insufficient strategy planning, but because of the lack of execution. According to Rajasekar (2014 in Sibiya 2018), higher rates of strategy failure (between 50 and 80%) are recorded in most organisations.

Strategy execution interacts with organisational variables that affect the overall realisation of predetermined strategic objectives (Okumus 2003:871-882). Hunter (2007:20) agreed with Okumus that the execution of strategy needs people with capacity and other resources to be allocated and managed at all component levels. Cloete (1995 in Sibiya 2018) urges that public administration is the collection of moral principles, norms, attitudes, values and obligations to ensure that the Government's Programme of Action remains a central focus on the promotion of the general welfare of all members of the population.

Kaplan and Norton (1996:195) are of the opinion that failure to link action programmes and resources allocation to long-term strategy priorities becomes an obstacle in implementing the strategy. The causes of failure in strategy execution relate to the abilities, process and activities that are needed to bring strategy to success (Cocks 2010 in Sibiya 2018). Sibiya (2018) further contends that strategy execution is exercising authority over resources, employees and their work. Accordingly, successful organisations adapt by improving their internal processes, systems, competencies and services.

A study was conducted by Nelson, Martin and Power (2008:2) where they invited thousands of staff members, about 25% of whom came from executive ranks, to complete an online assessment of their organisations' capabilities. The staff members represented more than 1000 organisations, public sector organisations and non-profit organisations from over 50 countries (Nelson, Martin and Power 2008:2). Their finding was that three out of every five staff members rated their organisation as weak at strategy execution. It seems to be losing strategic focus and has to be redirected to ensure continued focus on a desirable future (Nelson, Martin and Power 2008:2).

The importance of organisational factors enabling the effective functioning of public sector organisations and the effectiveness of strategy execution, are highlighted and discussed below.

Linking budgets to strategy execution

Once the strategy is developed, government departments consider a variety of public policies to address both the financial management capacity and improvement of service delivery. The linking of budgeting to strategy execution emphasises the involvement of the broader role-players in the allocation of resources. According to Chitiga-Mabugu (2016:168), government uses policies to guide the application of different programmes and actions to realise anticipated objectives.

The budget programme structures provide the relationship between the predetermined strategic objectives and operational budgets to ensure value-for-money. Consequently, the public leaders and managers are accountable for public funds allocated to the public sector organisations, as voted by Parliament and provincial legislatures. From the perspective of budgeting linked to strategy execution, one may consider financial management to be an important systematic variable influencing service delivery (De Coning 2006 in Sibiya).

The Public Finance Management Act 1 of 1999 (PFMA) in the context of budgeting and Treasury Regulations is key in the government budgeting frameworks and highlights good governance. Integrated budgeting planning is vital in the allocation of funds and strategies that are developed to ensure compliance with the PFMA statutory requirements concerning allocated funds for the implementation of government programmes. According to Radomska (2014:77), linking the budget to a strategy becomes a major challenge for senior managers, hence, it is critical for budget plans to link to strategy to ensure that strategic objectives and priorities are budgeted for and achieved as deliverables. As a result of social complexity, it implies that public sector organisations should do more with limited resources and consider the inputs, activities, outputs, outcomes and impacts of the organisational performance. The challenge is that the budget focus is on the short-term perspective, and planning generally takes three to five years over the MTEF. Given South Africa's development challenges, a strategy needs to inform the allocation of resources so that historical inequities can be progressively eliminated. However, the operational plans have to be developed within the context of available resources informed by longer-term planning and priorities.

Audit report perspectives

The Auditor-General of South Africa's (AGSA) responsibility is as required by section 188 of the *Constitution of the Republic of South Africa*, 1996, section 4 of the Public Audit Act 25 of 2004 and the PFMA. Each public sector organisation should have developed at least five-year strategic plans, which are linked to the MTSF and the Government's Programme of Action. Within this context, the realisation of predetermined strategic objectives has become a key responsibility to the public leaders and managers (Bryson 2004:50). It is important for public sector organisations to develop strategic plans, and have the ability to execute them, although it is often more difficult than their formulating.

AGSA audits public sector organisations' strategies based on the predetermined strategic objectives, whether the organisations have been able to realise their set objectives, and how the public resources were utilised during the financial year, from 1 April to 31 March of the following year. AGSA expressed an opinion based on the typologies of three criteria: unqualified report, meaning

that the organisation has delivered on its predetermined objectives, which is a clean report. The qualified report, means the organisation has made significant strides relating from performance to implement its predetermined objectives with certain specific recommendations to be implemented by the organisation. The disclaimer report, is when it is difficult for the AGSA to express opinions, which then means that the organisation has not progressed according to predetermined objectives.

The AGSA (in Sibiyi 2018) revealed that government departments and public entities have challenges in the realisation of their predetermined strategic objectives. The AGSA reports were based on actual service delivery performance and the realisation of predetermined strategic objectives against the set performance indicators and targets in the strategic plan. The focus was also on the annual performance plans and report on the findings that were presented before the relevant legislature and Parliament. The AGSA has given many accounts of the organisational factors behind the failure to deliver effective services and the remedial actions that should be taken for many other implementation failures.

Sibiyi (2018) highlights that the following are prevalent deficiencies reported in almost all Auditor-General reports found in the public sector organisations:

- Unclear reporting as a result of prolonged processes of filling leadership or senior management positions;
- Implementation of projects not aligned to predetermined strategic objectives as a result of lack of control;
- Failure to follow monitoring processes; and
- Imbalance of budgeting planning for the two core functions (agriculture and environmental affairs) resulting in non-result-oriented actions.

An AGSA (in Sibiyi 2018) snapshot of lessons from past failures included requirements for success such as:

- know the organisational responsibilities; and
- the role of management to own the process and be an active leader in management.

One of these challenges, is the high number of vacancies which exists in senior management positions within departments and public entities in the South African public sector which has an adverse effect on effective strategy execution. Findings of the Auditor-General (in Sibiyi 2018) particularly concerning leadership and governance, are significant and ought to be addressed in the public sector organisation. Departments and public sector organisations are facing problems in their employment processes, including delays in the verification of qualifications and vetting of applicants, and having to advertise posts numerous times because applicants are not suitable for the positions.

Leadership and management

Mothae and Sindane (2007:151) observe that the execution of policies in public organisations is the duty of public officials who need to understand them. The leadership could be the solution to many challenges experienced by public servants during strategy execution, stresses Mfene (2008:212). Aligned with this point of view, Naidoo (2005:109) argues that deficiency in strategy execution has been identified by leaders who are asking themselves why things do not work as planned. Public managers face significant resistance to change at all levels in the public service.

Kaplan and Norton (1996:10, 199) indicate that there should be a link between the activities of leadership in providing strategic direction and policy implementation. Leadership is one of the key contributors towards effective strategy execution (Pienaar *et al.* 2008:13). Leadership has a strong impact through their day-to-day tasks, which gives them direct contact with the customers (Brynard 2005 in Sibiya 2018). To ensure employees are acting ethically when performing their work and making decisions, management can take several measures by appointing people who are behaving ethically in the position, and use the labour relations process against people who are not performing ethically; also, to employ strong measures to promote ethical behaviour (Lues and Bester 2007:94).

Leinonen and Juntunen (2007 in Sibiya 2018) contend that rational ways of public administration, call for a new type of leadership and management. Adopting a new type of public sector leadership and management will enable decisions on how to design or reconfigure public sector organisations to develop strategic plans and build organisational capabilities (Nag, Hambrick and Chen 2007 in Sibiya 2018). Leadership should translate strategy into manageable actions for the middle managers' adaptation, since they are considered as generally responsible for the success or failure of an organisation, which needs communication, resources and autonomy.

The responsibility of leadership to provide strategic direction and ensure functionality to strategy execution is of concern (O'Reilly 2010:12). The importance of leadership and management has to start at the first-line supervisor up to the senior management and Chief Executive Office on organisational performance (Hoffman and Jones 2005). The significance of leadership is defined as one of the main strategic contributors in the realisation of pre-determined objectives (Pienaar, Muell, and Cronje 2008:13).

Stock (2007 in Sibiya 2018) indicates that it is essential that the leaders accept accountability for organisational performance, comprising financial management and human resources. Leadership and management are multi-functional which encompasses direction and attainment of results through others (Huey 1994:42-50). Kaplan and Norton (1996:199) indicate that there should be a link between

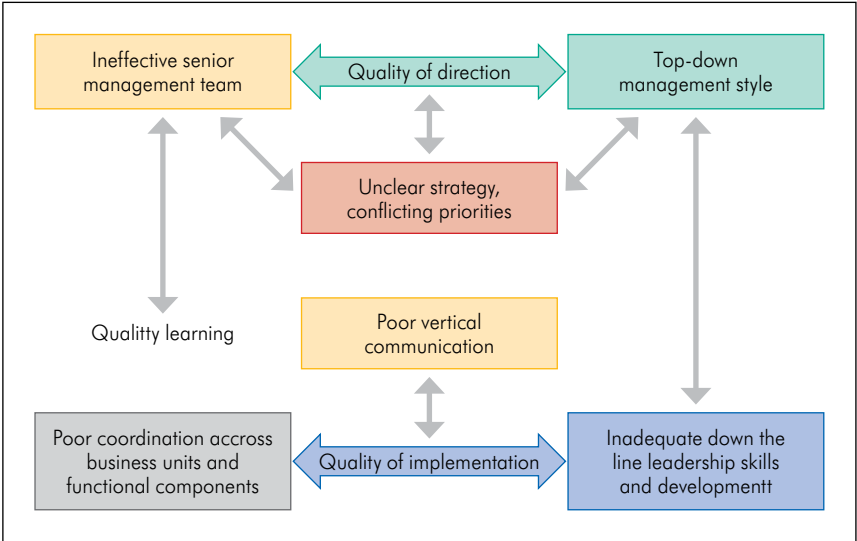
the activities for the realisation of the strategy. It is evident that leaders could prove to be an essential element of the change from improvement to institutionalisation. The question can then be asked, why it is so difficult to improve strategy execution in most government departments.

The study conducted by the Public Service Commission (2008:viii-ix) on the turnover rate of HoDs reflects that one of the major challenges of the public sector organisations is to keep skilful leaders and managers in the public service. It also showed that it takes about six months to a year for the new leaders to settle in. It was found that a new HoD establishes a new strategy, and is therefore unable to account for progress that took place during the erstwhile HoD's term. A lack of continuity to fulfil the strategic mandates impacts policy implementation.

Beer and Eisenstat (2000:37) identified 'six strategic killers' of strategy execution and learning. Their focus is on the leadership from a different perspective. Beer and Eisenstat (2000:37) proposed that the lack of effective leadership and changing priorities will result in poor coordination because employees will reflect uncertainty about senior management. They (2000:29) indicate that the 'six strategic killers' are risks for effective execution of the strategy. Their views are reflected in the graphic illustration in Figure 3 on how the 'strategic killers' interact.

Figure 3, provides elements of 'strategic killers' that need to be tackled during the strategic management processes. Poor vertical communication is at the core as it presents integrated strategic management approaches. The top-down

Figure 3: Interaction of 'six strategic killers'



Source: (Beer and Eisenstat 2000:29)

approach has a tendency to bypass middle managers and go directly to the lower-level employees with orders to get things done. This approach is viewed as causing uncertainty and inconsistency. It results in lack of leadership where senior managers do not provide strategic direction to their subordinates about the common strategic objectives. This also affects the quality of strategic direction. The issues of poor coordination across business units and functional components affect effective reporting and monitoring of organisational performance. This is viewed as influenced by an ineffective senior management team that skipped the middle management levels to give instructions to the lower-level employees. Lack of strategy management associated with lack of effective management as a top-down approach, is where only leadership and senior managers know and understand the bigger picture of the organisational strategy. Organisations should be looking for people who will be able to work across the organisations' components.

Pacheco, Caulliraux and Spiegel (2014 in Sibiya 2018) contend that ineffective senior management teams and leadership approaches are viewed as major organisational factors affecting the quality of learning that impacts the coordination of activities and developing skills and competencies in the lower-level management to keep employees committed to strategy execution. Interaction of the 'six strategic killers' model shows how these factors can influence strategy execution. Dransfield (2001 in Sibiya 2018) urges that effective sharing of information is needed in developing a plan. According to Hitt, Ireland and Hoskisson (2007:500), the importance of leadership and senior management arose from providing strategic direction and developing management controls for developing and sustaining operational attitude and moral principles while improving organisational performance.

Ehlers and Lazenby (2007:213) say management must share information and ensure that employees understand the strategic change and realise their roles and contributions. Kaplan and Norton's findings (1996:10) concur that strategy-focused leadership could be driven by policy process as a central point for the management of organisational factors and deployment of resources. In support of this, Dransfield (2001:85) maintains that a number of approaches can be applied to deal with strategic change and staff empowerment.

Organisational structure and culture

According to Huey (1994 in Sibiya 2018), organisational structure and culture requires strategic leadership and management to integrate and align them to strategy execution. Leadership is also considered the key element to decide on how to structure staff and lay out plans in building organisational capabilities (Nag, Hambrick and Chen in Sibiya 2018). Ethics refers to the values and principles which are manifested in leadership's behaviour that impacts organisational performance.

More than ever, it seems that the success of organisations and the well-being of society are affected by ethical choices made by the leadership in the organisations.

While strategy relates to all facets of management, strategy execution interacts with organisational variables that affect the overall outcome. Consequently, though, the demands and expectations on government to deliver public services are unlimited, government resources to meet these demands are limited. The test of a strategy is in its execution where employees and leadership have been recognised as key organisational factors to ensure effective strategy execution (Kaplan and Norton 2004:277). It stands to reason that strategy execution is a thoughtful process for the management of any organisation.

Kaplan and Norton (1996:10) view strategy-focused leadership as a strategically-driven organisational structure and culture, since strategy execution requires both tangible resources and processes. O'Reilly (2010 in Sibiya 2018) views the allocation of resources in two phases, phase one is focused mainly on strategic planning and articulation of the pre-determined objectives; and phase two involves allocation of tangible resources such as senior managers involved in the planning processes.

Slater, Olson and Hult (2010 in Sibiya 2018) state that there is a need for the leadership and management to consider the alignment of organisations towards new strategies. Bhimani and Langfield-Smith (2007), maintain that the alignment of the organisational structure and culture to the strategy needs to be monitored, since organisational structure typically segregates the functions and duties into different levels of the organisation. Bushardt, Glascoff, and Doty (2011:60-65) concur that the link between organisational structure and strategy affects organisational culture. Progression of the public service transformation does not seem to have produced the envisaged result according to Nengwekhulu (2009:342,345). Hrebiniak (2006) in a study conducted of obstacles to effective strategy implementation concurs and found that insufficient information sharing, unclear responsibilities and accountabilities work against predetermined strategic objectives. However, the impact of organisational culture and implementing strategies does exist. Rajasekar (2014 in Sibiya 2018) emphasises alignment of organisational structures and interaction between the employees and management, as well as beneficiaries of organisational services concerning the 'rendering of services or doing business'. The organisational structure plays a critical role to allocate human resources to execute the strategy. The key challenges are employees' perceptions towards strategy and behaviours towards clients or beneficiaries of the services.

Rapert and Wren (1998:289-294), have accepted the double nature of the organisational structure, covering both the decentralisation and centralisation models, as well as the communication channels within the set structure. According to Hillman and Siam (2014 in Sibiya 2018), the influence of organisational structures and cultures impacts organisational performance. Alton and Ikavako (2002 in Sibiya 2018) concur that organisational structures and values plays a key role in

sharing information and communication concerning the strategy execution and organisational performance.

Organisational culture is an important key factor of ethical behaviour in an organisation. Unethical behaviour can take many forms, including the misuse of property and public resources, public officials taking a percentage on government contracts, contracting government business to themselves, friends and relatives, and receiving excessive performance bonuses or offers from government contractors, suggests Clapper, De Jager and Fourie (2002:33). Finally, organisations can train and develop their employees on ethics, and how to deal with ethical challenges.

Information, Communication and Technology

Leaders and managers should effectively share strategic information with all stakeholders within and outside the organisation. Communication within the public sector organisation has a vital role in strategy development, but also in the effective strategy execution. O'Reilly (2010 in Sibiya 2018), indicated that leaders and managers at different levels of the organisation should create strategy awareness among the employees that would enhance service delivery. The ICT systems should be utilised and effectively managed as the responsibility of leadership and management.

ICT is regarded as an important factor of organisational effectiveness through understanding the roles, and to create a sense of ownership to all role-players (Noble 1999 in Sibiya 2018). Long and Franklin (2004:311) concur that communication is key to implementation of policies and strategies, while Hrebiniak (2005:15, 23) urges that the absence of effective ICT will contribute towards conflicting priorities.

Forman and Argenti (2005:255-256) maintain that organisations where employees have access to information through open and supportive ICT networks are inclined to exceed those with more limiting communication situations. Long and Franklin (2004:311) view ICT as a key factor to share strategic information with middle management who are at the interface of strategy and operations. These authors also view ICT as a significant organisational factor towards effective strategy execution.

CONCLUSION

Strategic planning is a tool to ensure the benefits of realising predetermined strategic objectives, however, literature indicates there are organisational factors that need to be effectively managed to ensure that strategic objectives are effectively achieved.

The article discussed the influence of organisational variables on the execution of pre-determined objectives and strategic goals that impact the beneficiaries. In so doing, the quality of life of communities would be improved. The article was based

on the public service strategy execution, whereby leaders are involved in the formulation of strategy and for taking ownership of the strategy execution in accordance with their mandate. The study contributed towards research in the field of public management in particular relating to the synergistic linkage of leadership, management and governance and strategy execution to counter the challenges experienced in operationalising strategic management towards efficacy in service delivery.

NOTE

- * This article is partly based on a Doctor of Administration thesis completed under the supervision of Prof Mogie Subban, entitled: Sibiya, C.B. 2018. *Examining Strategic Management Processes in Public Organisations: A Case Study of KwaZulu-Natal Department of Agriculture and Environmental Affairs*. Durban: University of KwaZulu-Natal.

REFERENCES

- Abdulwahid, S., Muhamad K.U., Sehar, Z., Arshad, M.S. and Iftekhar, K. 2013. Why do public Sector organizations fail in Implementation of Strategic Plan in Pakistan. *Interdisciplinary Journal of Contemporary Research Business*. 2(9):108-120.
- Bana, A. and Shitindi, E. 2009. Performance management in the Tanzania public service: A paper presented at conference on Governance Excellence Managing Human Potential held at Arusha International conference centre. Tanzania. Available at: <http://www.capam.org/documents/bana.benson.paper.pdf>. (Accessed on 20 July 2016).
- Beer, M. and Eisenstat, R.A. 2000. The silent killers of strategy implementation and learning. *Sloan Management Review*. 41:29-40.
- Bhimani, A. and Langfield-Smith, K. 2007. Structure, formality and the importance of financial and non-financial information in strategy development and implementation. *Management Accounting Research*. 18:3-31.
- Bryson, J. 1995. *Strategic Planning for Public and Non-profit Organization: A guide to strengthening and sustaining organisational achievement*. San Francisco: Jossey-Bass.
- Bryson, J.M. 2004. *Strategic planning for public and nonprofit organizations: a guide to strengthening and sustaining organizational achievement*. (3rd ed.). San Francisco: Jossey-Bass.
- Bryson, J.M. and Alston, F.K. 2005. *Creating and Implementing Your Strategic Plan—A Workbook for Public and Non-profit Organizations*. (2nd ed.). San Francisco: Jossey-Bass.
- Bushardt, S.C., Glascoff, D.W. and Doty, D.H. 2011. Organizational Culture, Formal Reward Structure, and Effective Strategy Implementation: A Conceptual Model. *Journal of Organizational Culture, Communication and Conflict*. 15(2):57-71.
- Chapman, J.A. 2002. A framework for transformational change in organisations. *Leadership and Organisation Development Journal*. 23(1):16-25.

- Chitiga-Mabugu, M. 2016. The application of economy-wide techniques as a tool to understand policy impacts. *African Journal of Public Affairs*. 9(2):166-179.
- Clapper, V., De Jager, J. and Fourie, L. 2002. The state of public service ethics: an exploratory study. *Politeia*. 21(1):25-36.
- Colgan, A., Rochford, S. and Burke, K. 2016. *Implementing Public Service Reform – Messages from the literature*. Dublin:Centre for Effective Services.
- Dickey, T. 1992. *Budgeting – a practical guide for Better Business Planning*. London: Kogan Page.
- Ehlers, T. and Lazenby, K. 2007. *Strategic Management: Southern Africa concepts and cases*. Pretoria: van Schaik.
- Ferlie, E. and Ongaro, E. 2015. Strategic Management in Public Services Organizations: Concepts, Schools, and Contemporary Issues. *International Public Management Journal*. 19(4):596-599.
- Forman, J. and Argenti, P.A. 2005. How Corporate Communication Influences Strategy Implementation, Reputation and the Corporate Brand: An Exploratory Qualitative Study. *Corporate Reputation Review*. 8(3):245-264.
- Guth, W.D. and MacMillan, I.C. 1986. Strategy Implementation versus Middle Management Self Interest. *Strategic Management Journal*. 7(4):313-327.
- Harrington, D. and Williams, B. 2004. Moving the quality effort forward – the emerging role of the middle manager. *Managing Service Quality*. 14(4):297-306.
- Hitt, M.A., Ireland, R.D. and Hoskisson, R.E. 2007. *Strategic Management: Competitiveness and Globalization*. (7th ed.). Ohio: Thomson/South Western.
- Hofmann, D.A. and Jones, L.M. 2005. Leadership, collective personality, and performance. *Journal of Applied Psychology*. 90:509-522.
- Hrebiniak, L.G. 2005. *Making Strategy Work: Leading Effective Execution and Change*. Upper Saddle River, NJ: Wharton School Publishing.
- Hrebiniak, L. 2006. *Obstacles to effective Strategy Implementation. Organizational Implementation and Control*. Boston: McGraw-Hill.
- Huey, J. 1994. The new post-heroic leadership. *Fortune Magazine*. 21 February 1994. Pp. 42-45.
- Hughes, O.E. 2003. *Public Management and Administration. An Introduction*. (3rd ed.). New York: Palgrave-Macmillan.
- Hunter, D. 2007. *Managing for Health*. London: Routledge.
- Jones, G.R. and George, J.M. 2003. *Contemporary Management*. (3rd ed.). Boston, MA: Irwin McGraw-Hill.
- Kaplan, R.S. and Norton, D.P. 1996. *The Balanced Scorecard: Translating strategy into action*. Boston: Harvard Business School Press.
- Kaplan, R.S. and Norton, D.P. 2004. *Strategy Maps: Turning Intangible Assets into Tangible Results*. Boston, MA: Harvard Business School Press. P.31.
- Kreitner, R. and Kinicki, A. 2010. *Organizational Behavior*. (9th ed.). New York: Irwin McGraw-Hill.
- Long, E. and Franklin, A.L. 2004. The paradox of implementing the Government Performance and Results Act: Top-down direction for bottom-up implementation. *Public Administration Review*. 64:309-319.

- Lues, L. and Bester, C.L. 2007. The impact of legislative directives and human behaviour on the ethical conduct of public Service Managers. *Journal of Public Administration*. 42(5):93-108.
- Madue, S. 2008. Policy implementation in a turbulent environment. *Journal of Public Administration*. 43(3):197-206.
- Mfene, P.N. 2008. Administrative leadership and public policy implementation in the public service: Nelson Mandela Metropolitan University. *Journal of Public Administration*. 43(1):207-218.
- Mothae, L. and Sindane, M. 2007. A strategy-focused leadership for improving public policy implementation and promoting good governance. *Journal of Public Administration*. 42(5):1-2.
- Naidoo, G. 2005. A transformative African leadership and governance framework to improve public service delivery in South Africa. *Journal for Public Administration*. 42(8):795-798.
- National Treasury. 2010. *The Framework for Strategic Plans and Annual Performance Plans*. Pretoria: Government Printer.
- Nelson, G.L., Martin, K.L. and Power, E. 2008. The Secrets to Successful Strategy Execution. *Harvard Business Review*. 86(6):60-70.
- Nengwekhulu, R.H. 2009. Public service delivery challenges facing the South African public service. *Journal of Public Administration*. 44(2): 341-363.
- Neuman, W.L. 2006. *Social research methods: Qualitative and quantitative approaches*. (6th ed.). Boston: Allyn and Bacon.
- Okumus, F. 2003. A framework to implement strategies in organizations. *Management Decision*. 41(9):871-882.
- Pearce, J.A. and Robinson, R.B. 2007. *Formulation, Implementation and Control of Competitive Strategy*. (9th ed.). Boston, MA: McGraw-Hill Irwin.
- Pencheon, D., Guest, C., Melzer, D. and Muir Gray, J.A. 2008. *Understanding the Theory and Process of Strategy Development: Strategy communication and strategy implementation in relation to health care—Oxford Handbook of Public Health Practice*. Oxford: Oxford University Press.
- Pienaar, R., Muell, D.C. and Cronje, S. 2008. Strategy implementation: overcoming the challenge. *Management Today*. 24(2):2-15.
- Poister H.T. 2010. The Future of Strategic Planning in the Public Sector: Linking Strategic Management and Performance. *Public Administration Review*. 70:246–254.
- Public Audit Act 25 of 2004. Pretoria: Government Printer.
- Public Finance Management Act 1 of 1999. Pretoria: Government Printer.
- Public Service Commission. 2008. *The Turn-over Rate of Heads of Department and its Implications for the Public Service*. Pretoria: Government Printer.
- Radomska, J. 2014. The role of managers in effective strategy implementation. *International Journal of Contemporary Management*. 13(3):77-85.
- Rahimnia, F., Polychronakis, Y. and Sharp, J.M. 2009. A Conceptual Framework of Impeders to Strategy Implementation from an Exploratory Case Study in an Iranian University. *Education, Business and Society: Contemporary Middle Eastern Issues*. 2(4):246-261.
- Rapert, M. and Wren, B. 1998. Reconsidering organizational structure: A dual perspective of frameworks and processes. *Journal of Managerial Issues*. 10(3):287-302.

- Sial, A., Usman, M.K., Zufiqar, S., Satti, A.M. and Khursheed, I. 2013. Why Do Public Sector Organizations Fail in Implementation of Strategic Plan in Pakistan? *Public Policy and Administration Research*. 3(1):232-240.
- Sibiya, C.B. 2018. *Examining Strategic Management Processes in Public Organisations: A Case Study of KwaZulu-Natal Department of Agriculture and Environmental Affairs*. Durban: University of KwaZulu-Natal.
- Smets, M., Jarzabkowski, P., Burke, G.T. and Spee, P. 2015. Reinsurance trading in Lloyd's of London: Balancing conflicting-yet complementary logics in practice. *Academy of Management Journal*. 58: 32-970.
- Stavros, J.M., Cooperrider, D. and Kelley, L. 2014. Strategic Inquiry! Appreciative Intent: Inspiration to SOAR A New Framework for Strategic Planning. *International Journal of Appreciative Inquiry*. 1(1):205-302. Available at: <https://www.researchgate.net/publication>. (Accessed on 25 January 2016).
- Stoney, C. 2001. Strategic Management or Strategic Taylorism? A Case Study into Change within a UK Local Authority. *International Journal of Public Sector Management*. 1(1):27-42.
- Thompson, A.A. and Strickland, A.J. 2003. *Strategic Management: Concepts and Cases*. (13th ed.). New York: McGraw-Hill.
- Thompson, A.A. Jr, Strickland III, A.J. and Gamble, J.E. 2008. *Crafting and executing strategy: The quest for competitive advantage-Concepts and cases*. (15th ed.). New York: McGraw-Hill Irwin.
- Valadez, J. and Bamberger, M. 1994. *Monitoring and evaluating social programs in developing countries: a handbook for policymakers, managers and researchers*. Washington DC: EDI Development Studies, World Bank.
- Walker, R.M., Andrews, R., Boyne, G.A., Meier, K.J. and O'Toole, L.J. 2010. Wakeup Call: Strategic Management, Network Alarms, and Performance. *Public Administration Review*. 70(5):731-741.
- Welman, C., Kruger, F. and Mitchell, B. 2005. *Research Methodology*. Cape Town, South Africa: Oxford University Press.
- White Paper on the Transformation of the Public Service. 1995. *Government Gazette*. 388(18340). Pretoria: Government Printer.

AUTHORS' CONTACT DETAILS

Dr Collen Bongani Sibiya

School of Management, IT & Governance
College of Law & Management Studies
University of KwaZulu-Natal
Private Bag X54001
Durban
4000
Cell: 0823019471
Email address: collensibiya82@gmail.com

Prof Mogie Subban

School of Management, IT & Governance
College of Law & Management Studies
University of KwaZulu-Natal
Private Bag X54001
Durban
4000
Cell: 0823734303
Email address: subbanm@ukzn.ac.za

Data and Information Security Governance

A Prerequisite for Improved Protection of Public Sector Data and Information Practices

L Masilela

School of Public Management, Governance and Public Policy
University of Johannesburg

D Nel-Sanders

School of Public Management, Governance and Public Policy
University of Johannesburg

ABSTRACT

The government, as one of the biggest collectors and distributors of Public Sector Data and Information (PSDI) assets, must be constantly aware of the risks associated with handling such large volumes of data and information. To stay ahead and counteract sensitive data and information theft, the government must understand the importance of data and information security governance (DISG) and how it can be utilised as an effective approach to improve data and Information Security (IS) systems. The goal of DISG is to ensure a comprehensive approach towards the overall protection of data and information, its confidentiality, as well as the integrity of any organisation, in the public or private sector. The aim of this article is to focus on understanding the role and importance of DISG practices and principles at the national sphere of government in the South African public sector, namely, the Department of Energy (DoE), former Department of Science and Technology (DST) and former Department of Environmental Affairs (DEA). A qualitative approach is followed using a case study strategy. Primary data was collected through semi-structured interviews. Findings are presented through an analytical framework consisting of the people, systems, processes, risks and controls. The study concludes that there is a heightened need for research and development initiatives in the study and institutionalisation of DISG management practices in the public sector.

INTRODUCTION

The government and its departments have had to ensure that they remain active and responsive to global trends that have made their way into the public sector. The dynamic and complex environment, due to the technological changes and global revolutions, has had a direct impact on the process of receiving, capturing, storing, retrieving, and exchanging of data and information assets in the public sector. With government being a centre point of sharing information, this initially sparked an interest in the government to analyse and determine the value of its PSDI. This required an in-depth analysis of the risks associated with the misuse, theft, and distortion of PSDI. The study focuses on understanding the role and importance of DISG at the national sphere of government in the South African public sector and determining the nature of governance of PSDI assets. A preliminary literature review of relevant concepts and themes is conducted. First, is a conceptual analysis of data governance, information governance and information security governance (ISG). Second, a discussion of the relationship between information technology (IT) governance and corporate governance. Third, a discussion of the scientific and methodological approach to the study. Fourth, the research findings and results will be presented. Lastly, the article will provide recommendations for future research proposals for DISG practices and principles for improved public sector management functions at the national sphere of government.

CONCEPTUALISING DATA GOVERNANCE

According to Webber, Otto and Österle (2009:22), the term “data governance” can be defined as “the allocation of decision-making and related duties in the management of data within an enterprise”. This definition means that senior management in an organisation must implement and execute specific structural frameworks that will assist in sound decision-making rights and responsibilities to protect and manage an organisation’s data assets (Khatri and Brown 2010:148). Organisations in the private and public sectors, utilise data governance to manage, protect, and create value to all the data they collect, sort, distribute, and destroy (Webber *et al.* 2009:15; Khatri and Brown 2010:148). The value of data in an organisation is often determined by its usability and accessibility to employers, supply chain partners, citizens, stakeholder partners, investors, and other external institutions. Organisations source, collect, refine, and transfer data assets into meaningful, useful, and reliable formats for generating valuable information and knowledge assets in an organisation. Data must be collected in a specific organisational context for:

- determining the strengths, weakness, threats, and opportunities triggered by its internal and external environments;

- specifying organisational roles, responsibilities, structures, and functionalities of senior management and subordinate personnel;
- identifying and determining the needs and wants of its consumer base and adopting and transitioning to the latest trends;
- analysing its financial growth and stability to determine increases and decreases in profits and losses;
- designing and implementing strategic plans, programmes, and objectives to increase overall productivity and efficiency in an organisation;
- identifying, analysing, and mitigating internal and external risks that may hamper operational efficiency; and
- implementing tools and techniques to ensure effective monitoring and evaluation of all organisational operations (Khatri and Brown 2010:149).

CONCEPTUAL CLARIFICATION

The following section clarifies important concepts related to this research, including information governance, ISG, IT governance and corporate governance.

Conceptualising information governance

According to the Ministry of Economic Affairs and Communications (MEAC 2014:20), information governance can be defined as “a set of processes, roles, policies, standards and indicators which together ensure the efficient and effective use of information and thus the achievement of an organisation’s goals and objectives”. The information that an organisation possesses is critical for its operational functionalities, because it stipulates and provides guidelines for the decision-making structures, functions, strategies, processes, and systems that are drawn from the information assets compiled and given to senior management. For an organisation to plan strategically and achieve its organisational goals and objectives, it needs accurate, reliable, factual, and unbiased data and information assets to make sound business decisions (MEAC 2014:14). Information governance exists purely because data governance exists. Data is collected, refined, stored, and transferred internally for various organisational purposes before being transformed into valuable information.

Data must therefore be handled and managed in a sensitive manner because without data, an organisation cannot have information to assist it in making feasible and sound management decisions (MEAC 2014:14). Information governance in an organisation maximises the use of information to increase productivity and to ensure the efficiency of top-, middle-, and lower-level functionalities. The implementation of information governance in an organisation allows senior management to:

- gain a holistic perspective of and insight into customer tastes and trends;
- establish, design, and implement innovative solutions to tackle challenges associated with information governance; and
- be legally compliant with sectoral legislative rules, laws, and regulations in its industry and markets (Alles and Piechocki 2012:20; Webber *et al.* 2009:17).

Information governance is interrelated and connected to knowledge management because information governance assists in answering the following questions for senior management:

- how does information travel and rotate in an organisation?
- how accurate, factual, and reliable is the information that is transmitted in an organisation?
- who is responsible for the collection, storage, and authorisation of particular data and information assets?
- how is that information protected against unauthorised access, misuse, fraud, and loss? (MEAC 2014:14).

Finding the answers to these questions will assist senior management in: (1) ensuring effective communication structures and channels between management and its personnel; and (2) delegating responsibility to competent and knowledgeable staff to ensure that information is utilised and distributed correctly to the right person, at the right time, and for the right reasons (MEAC 2014:14).

Conceptualising information security governance

Technological advancements globally have resulted in organisations in the public and private sectors collecting, storing, using, and distributing information assets at a significantly increasing pace. Organisations must utilise their information assets strategically to perform their daily functional activities and, most importantly, to secure their assets through effective security systems. To achieve this, organisations in the public and private sectors must protect their data and information because this is one of their most valuable assets (IT Governance Institute 2006:7-8). To understand the term “information security governance”, one must first understand what “information security” is. IS can be defined as those efforts undertaken by an organisation to protect its critical information assets through computerised systems and hardware that collect, store, and transmit information internally and externally to and by users. ISG is “defining information security objectives for the organisation based on corporate strategic goals, thus ensuring alignment between information security and organisational objectives” (IT Governance Institute 2006:15).

Organisations are constantly using information assets to run their daily functional operations and it is critical to keep the systems, as well as their information, safe,

secure, and adequately protected at all times. To achieve ISG, senior management cannot only focus on the security systems alone, there must also be emphasis on:

- knowing how the information is collected;
- who collects and stores the information;
- where and how it is processed and stored; and
- who has been granted access and authority to that information and how that information is being protected (Westby and Allen 2007:21).

Implementing ISG in an organisation can assist in achieving the following:

- formulating and implementing conducive policies, frameworks, and models to ensure ISG;
- improving operational, managerial, and technical functions and structures;
- assigning specific roles and responsibilities to achieve effective ISG throughout the organisation;
- ensuring accountability and transparency for all information assets that are collected, stored, and distributed to internal and external users;
- having in place sound management principles, practices, strategies, and programmes geared towards the achievement of satisfactory ISG; and
- having in place systems, applications, and key performance indicators to conduct monitoring and evaluation processes on how the organisation is progressing in its ISG goals and objectives (Westby and Allen 2007:21).

The link between IT governance and corporate governance

According to the King IV Report (Institute of Directors in Southern Africa [IoDSA] 2016:52), IT governance can be defined as “a framework that supports effective and efficient management of IT resources to facilitate the achievement of an organisation’s strategic objectives”. Due to the way in which technology has advanced over the years, organisations in the public and private sectors have been exposed to the global community. Organisations have moved away from the centred and traditional approach of face-to-face business activities, to dealing with clients, consumers, suppliers, and various other stakeholders via the dissemination of information across the Web (Iyamu 2011:3). The use of IT has allowed organisations to have more direct interaction with their internal and external stakeholders and as a result, organisations must ensure that they handle and manage their data and information assets in such a way that it corresponds with the values, mission, and objectives that they portray. The use of IT has resulted in organisations going the extra mile to ensure and foster a “culture of care” in how they govern the use of IT in their daily activities (Hüner, Ofner and Otto 2009:231).

IT governance has become interrelated and connected to corporate governance, because organisations use technology to run their daily activities and it

has become essential and necessary for senior management to add a governance approach to the use of IT to ensure that it meets the organisation's business goals, vision, mission, and objectives (Hüner *et al.* 2009:231). This indicates that IT governance and corporate governance work hand in hand, because IT governance entails processes implemented by senior management to strategically assist an organisation to meet its business goals through the use of IT. Corporate governance, on the other hand, is about how boards and executive management ensure the protection of IT assets through sound management practices that:

- protect an organisation's IT;
- correctly handle and store crucial data and information assets through IT;
- mitigate and minimise the risks associated with theft, hacking, fraud, and loss of valuable and highly classified data and information assets;
- find ways to implement strategies, programmes, and mechanisms for using resources in a sustainable manner; and
- create and foster a culture of inclusive senior management efforts in all aspects related to data and information governance in an organisation (Eckerson 2014:7; Hardy 2006:55).

The value of data and information to organisations has resulted in a global shift towards establishing and implementing guidelines and compliance legislation to protect data and information assets in organisations (Rossouw and Van Vuuren 2013:20; IT Governance Institute 2006:23).

These guidelines and compliance legislation were established for the purpose of ensuring that management is knowledgeable of:

- how data and information is collected;
- the channels in which it is distributed; and
- how the data and information is handled and managed in an organisation (Rossouw and Van Vuuren 2013:20; IT Governance Institute 2005:23).

The use of technology to run daily activities has resulted in public and private sectors prioritising the implementation of corporate governance. The accelerating pace of technological innovations has resulted in a global village, where investors, stakeholders, and customers have easy access to various types of data and information assets and are able to analyse, investigate, question, and constructively critique the manner in which an organisation is being governed by its senior management (Bahrman 2011:10). As a result, organisations have become more visible to and accessible by the public, which motivates the directors and gatekeepers of organisations to ensure that they have the necessary skills, knowledge, and capabilities to safeguard and protect all of their critical data- and information-related assets. Incorporating corporate governance assists senior organisational management (SOM) to create and implement structures and functions that will allow:

- setting goals, objectives, and targets for the organisation;
- establishing and implementing strategic plans and programmes to achieve goals and objectives;
- creating key performance indicators to measure success and challenges; and
- finding effective and efficient tools and techniques to monitor and evaluate the overall progress of corporate governance in the organisation (Bahrman 2011:10-11).

In the South African context, corporate governance is guided by the King Code. The King Code of Corporate Governance specifically lays the foundation of how organisations in South Africa, in the public or private sphere, should govern themselves. The King Code explicitly defines and details the primary expectations of business leaders across South Africa on how they should conduct and govern their organisations to achieve sound corporate governance (IoDSA 2016:6). According to IoDSA (2016:10), corporate governance in South Africa entails:

- creating and implementing a framework that specifies the rights, responsibilities, procedures and relationships among stakeholders;
- instilling good governance that leads to effective leadership from management;
- ensuring that leaders define an organisation's strategy, create direction, and encourage ethical values and behaviours that guide an organisation's business strategy; and
- finding economically viable and sustainable methods to utilise an organisation's resources (IoDSA 2016:10).

RESEARCH METHODOLOGY

The chosen methodological approach for this research study was a qualitative research approach. The qualitative research approach was selected for this particular research study because it is primarily concerned with how the social world is interpreted, perceived, understood, and experienced by others. The choice of using qualitative research for a research project is aimed at allowing researchers to collect data by approaching individual(s) or group(s) with the goal and aim of finding out more about these individual(s) or group(s). This is done to collect rich and in-depth raw data that can be used by the researcher as a basis for theorising in the research project. The relationships and interactions that occur between the researcher and the participants is a key element in the research project because it leads to the development and generation of new concepts (Bryman 2012:8; Denzin and Lincoln 2000:10; Flick 2013:14). Experienced, knowledgeable, and skilled SOM from the DoE, DST and DEA were interviewed. These included SOM who primarily deal with DISG in their respective government departments. The

sample size included 14 research participants; four from the DoE, six from the DST and six from the DEA. A semi-structured interview schedule was used. The interviews were aimed at identifying, analysing, investigating, and understanding the measures that have been applied by SOM to ensure the protection of PSDI. The research results were analysed and discussed using a self-constructed analytical framework that consisted of five elements, namely people, systems, processes, risks and controls of each department.

INTEGRATED APPROACH OF DISG PRACTICES

The strategic combination of data governance and ISG practices and principles has been identified as one of the various approaches that could be implemented and incorporated in private and public sector institutions to improve the management of an organisation's data and information systems and processes (IT Governance Institute 2006:7). According to Webber *et al.* (2009:22), data governance involves efforts made by an organisation to ensure the "availability, usability, integrity and security" of data assets in an organisation. ISG, on the other hand, consists of the tools, mechanisms, frameworks, and regulations developed and implemented by an organisation to direct and control IT security (IT Governance Institute 2003:15). The strategic alignment of the two abovementioned management functions is an integrated governance approach that could significantly improve an organisation's accountability framework by overseeing and ensuring that SOM and its stakeholder partners in an organisation implement effective, efficient, and feasible management controls geared towards the mitigation of data and IS-related risks and challenges. Utilising this integrated approach of DISG as a strategic tool provides a holistic view and approach to how organisations in private and public sector institutions can implement strategic approaches to improve the processes and systems of data and information collection, handling, processing, storing, archiving, and destroying (Brotby 2008:8; Moulton and Coles 2003:45-52).

Assessing the DISG practices within the Departments of Energy, Science and Innovation and Environmental Affairs

The data and information that an organisation collects and stores plays a critical role in its daily activities and in its decision-making functions. The amount, complexity, and range of the different types of data and information assets that an organisation holds have made SOM more concerned about the manner in which data and information assets are collected, handled, managed, distributed, and utilised in an organisation. An organisation's DISG initiatives should not be counter-productive, rigid, and centralised because it could result in partially overlapping

functions, which could lead to other functions being performed poorly or left uncompleted (British Academy and The Royal Society 2017:59).

According to Chalker (2014:7), “data is being collected at an accelerated rate, beyond humans’ capability to process it with the traditional manual ways”. As a result, organisations around the globe are experiencing the complications associated with transforming data into information and therefore making SOM realise the need for improved and advanced approaches, strategies, and programmes to organisations’ current DISG roles and responsibilities. “Governance is not about doing things right; it is about doing the right things” (Chalker 2014:2). An organisation’s DISG must be flexible so that it has enough resources and capabilities to be adaptable and to implement changes internally to survive in its external environment, but must also be rigid enough to steer the organisation in the right direction and keep the organisation, its employees, and resources grounded and sustainable in the long term (British Academy and The Royal Society 2017:60; Chalker 2014:7-8).

The challenges of ensuring that an organisation’s data and information is adequately protected have made SOM realise that DISG plays a crucial role in ensuring that an organisation reaches its goals and objectives. The trend of IS in organisations has made SOM realise the necessity of IS to stay up to date and relevant in the global community. DISG should be a higher-order management function, not only an IT function, and should be embedded throughout the organisation. SOM is responsible for ensuring strategic guidance to the management personnel, who carry out and oversee their daily activities. To achieve IS, both governance and management commitment and actions must be implemented. For DISG practices and principles to be fully effective and efficient within the public sector, there must be a link and synergy between people, systems and processes throughout the organisation as one variable cannot exist without the other.

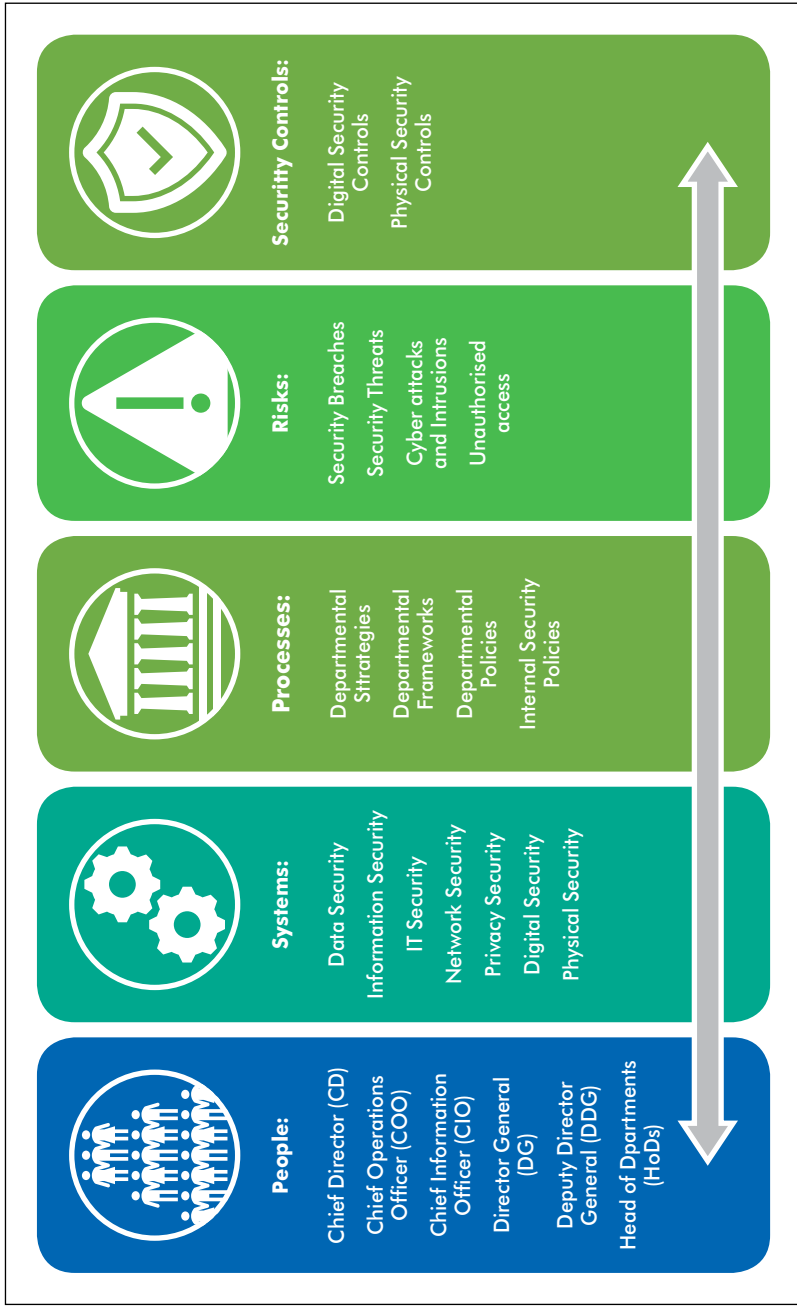
RESEARCH FINDINGS

The following section will analyse and assess the integrated DISG practices and principles of three national government departments, the DoE, former DST and the DEA. The data presented in this section is the results and findings of a research study undertaken for a master’s dissertation. The interview results and interpretations discussed are a generalisation of the interview data derived from the semi-structured interview questionnaire.

Figure 1 provides a simple illustration of the people, systems and process variables that have been analysed within the DoE, DST and DEA. The following briefly provides an explanation of the three variables:

- **People:** particularly for this article focuses on SOM who are primarily responsible for developing, implementing, and overseeing of departmental DISG

Figure 1: Analytical Framework: Departmental People, Systems and Institutions



Source: (Author's own illustration)

- practices and principles at the national sphere of government within the three respective government departments;
- **Systems:** this includes all digital and physical systems that have been developed and implemented to operate the departments' data and information systems, the IT infrastructure and security systems;
 - **Processes:** this includes all the internal legislative policies and frameworks that each department has institutionalised to aid and enhance DISG practices and principles throughout the organisation;
 - **Risks:** these include all digital and physical risks that could negatively impact the departments' abilities and capabilities to protect, recover and restore their PSDI to its original state after a digital or physical intrusion has emerged; and
 - **Security controls:** this includes all tangible and intangible activities, processes and behaviours that have been incorporated towards protecting PSDI to achieve effective and efficient DISG practices and principles.

Tables 1 to 3 provide tabulated results of the research study undertaken in the DoE, DST and DEA. The left-hand side of the Table provides the headings of the self-constructed analytical framework which includes people, systems, processes, risks and security controls. These are identified as the main themes (factors) under the scope of DISG practices. The right-hand side of the Table consists of all the subthemes (variables) of DISG practices institutionalised by SOM in order to ensure effective and efficient security measures of its PSDI. The primary goal and purpose of Tables 1 to 3 were to provide a systematic analysis of the self-constructed analytical framework for each main theme to be identified and analysed in each department to ensure repetitiveness, reliability, and rigour of an institutional analysis of each department. In addition, the subthemes in Tables 1 to 3 will also allow for the cross-comparison of data findings of identified DISG practices of each department to analyse, assess and determine their similarities and differences.

Table 1: People, systems, processes, risks and security controls of the DoE

Department of Energy	
Factor(s)	Variables
Departmental People	■ Director: Information Technology Programmes ■ Director: Information Security Officer ■ Director: Knowledge and Records Management ■ Deputy Director General: IT Specialist ■ Knowledge and Information Management Forums and Committees ■ Risk Committees ■ Facilities Management

Department of Energy	
Factor(s)	Variables
Departmental Systems	■ Electronic Document Management System ■ Top 20 Critical Security Controls ■ Internal Servers ■ Biometrics ■ Firewalls ■ Intrusion detection systems ■ Anti-virus programs and updates ■ Patch management systems ■ Password encryption ■ Access control
Departmental Processes	■ Information Security Policy ■ Firewall Policy ■ Backup Policy ■ Logical Access Control Policy ■ Acceptable Use Policy ■ IT Security Policy ■ Minimum Information Security Standards of 1996 (MISS) ■ Control Objective for Information and Related Technologies (COBIT) Framework ■ International Organisation for Standardisation (ISO) and the International Electrotechnical Commission (IEC) 27001 ■ Security Clearance Certificate
Departmental Risks	■ Bribery ■ Theft of classified information ■ Fraudulent activities ■ Looting and theft of money ■ Trading government industry secrets ■ Lack of steel cabinets for paper-based data and information ■ Unintentional disclosure of sensitive data and information ■ Lack of vigilance ■ Missing files from the departmental registry ■ Digital devices to capture, share and copy sensitive data and information
Departmental Security Controls	■ Audit processes by the Auditor General ■ Departmental Continuity / Disaster Recovery Plan ■ Training sessions and seminars with the Human Resource and Development Unit in conjunction with the Records Management ■ Password encryption ■ Biometrics ■ CCTV cameras in real time mode ■ Access control and security boom gates ■ Guest registry

(Author's own construction)

Table 2: People, systems, processes, risks and security controls of the DST

Department of Science and Technology	
Factor(s)	Variables
Departmental People	▪ Chief Information Officer ▪ Director of Information Systems ▪ Senior Business Analyst ▪ Senior Manager: Knowledge, Information and Records Management ▪ Systems Development Manager ▪ IT Security and Infrastructure Management
Departmental Systems	▪ Document Management System ▪ Intrusion Detection System ▪ Enterprise Content Management System
Departmental Processes	▪ Email, Intranet, and Internet Policy ▪ Computing Asset Replacement Policy ▪ Infrastructure Protection Policy ▪ Security Policy ▪ Clean Desk Policy ▪ Control Objective for Information and Related Technologies (COBIT) Framework ▪ International Organisation for Standardisation (ISO) and the International Electrotechnical Commission (IEC) 27001
Departmental Risks	▪ Loss of critical data and information ▪ Financial losses ▪ Unauthorised access ▪ Fraudulent activities ▪ Loss of tacit information and knowledge ▪ Denial of access virus ▪ Missing Physical Files and Documents ▪ Unethical conduct by departmental staff ▪ Malicious virus attacks
Departmental Security Controls	▪ Training and awareness sessions and seminars ▪ Installation of firewalls ▪ Intrusion detection systems ▪ Anti-virus programs and updates, ▪ Patch management systems ▪ Password encryption ▪ Biometrics ▪ CCTV cameras in real time mode ▪ Access control and security boom gates ▪ Electric fences ▪ Two system registry at reception

(Author's own construction)

Table 3: People, systems, processes, risks and security controls of the DEA

Department of Environmental Affairs	
Factor(s)	Variables
Departmental People	■ Chief Directorate – Government Information Technology Officer ■ Chief Director: Knowledge and Information Management ■ Director: IT Security and Infrastructure Management ■ Deputy Director General: IT Business Systems Manager
Departmental Systems	■ Email Archiving Solution ■ Data Linkage Prevention solution ■ Access Control Systems ■ Electronic Document Management System
Departmental Processes	■ The Governance of IT Policy ■ IT Security Policy ■ IT Governance Framework ■ IT Security Protection Strategies ■ Department of Public Service and Administration: Framework for IT Security ■ Control Objective for Information and Related Technologies (COBIT) Framework ■ International Organisation for Standardisation (ISO) and the International Electrotechnical Commission (IEC) 27001
Departmental Risks	■ Large scale power outages ■ Physical security of the premises would be disabled because the biometric functions / processes / systems would be inactive ■ Intrusions by malware, phishing and hacking ■ IT Breaches: unfamiliar / unknown flash drives ■ Fraud and corruption ■ Reputational damage ■ Loss of institutional memory ■ Records Management Units not fully incorporated or covered in each department's overall business continuity plan
Departmental Security Controls	■ Mobile Device Management Programme ■ Flash drives with encrypted keypad ■ Training and awareness sessions and seminars ■ Installation of firewalls ■ Intrusion detection systems ■ Anti-virus programs and updates, ■ Patch management systems ■ Password encryption on the IT systems and critical PSDI assets ■ Biometrics ■ CCTV cameras in real time mode ■ Access control and security boom gates ■ Electric fences ■ Registry at reception

Source: (Author's own construction)

Integrated approach of the processes of the DoE, DST and DEA

SOM is responsible for ensuring strategic guidance to departmental personnel, who carry out and foresee all activities, roles, functions, and responsibilities in the area and field of DISG. To achieve effective and efficient DISG goals and objectives towards the protection of PSDI assets, government departments must prioritise their legal, regulatory, and compliance policies to adequately protect and safeguard their critical PSDI assets. To adequately manage, mitigate, and minimise the potential risks associated with the latest global revolutions, SOM must formulate and implement strategic policies, frameworks, and models geared towards effective DISG management practices. Each government department has a unique set of goals and objectives that must be achieved on a monthly, quarterly, and annual basis. As a result, it is imperative that SOM identifies, establishes, and implements a DISG policy, strategy, and framework that will ensure that it benefits from improved DISG management practices and principles in the long term.

Government departments could consider forming internal review committees, which will review and provide constructive feedback about which internal policies, strategies, and frameworks lack DISG management practices and principles. The proposed internal review committee should also consider intensive research and development initiatives, programmes, and projects towards the formulation and implementation of DISG policies, frameworks, strategies, and models. Depending on the nature and the requirements of the individual public sector institutions, this may include the development of single or multiple policies, frameworks, models, and strategies towards achieving effective DISG management practices. The policies, frameworks, models, and strategies that SOM incorporates and integrates into current organisational policies, strategies, and frameworks must ensure synergy between IT governance, data governance, and ISG to achieve and maintain effective DISG management practices and principles.

The key to having sound DISG policies, frameworks, models, and strategies in the public sector is to ensure that the chosen policies, frameworks, models, and strategies are not too complex and difficult to understand and execute by employees, investors, stakeholders, suppliers, service consultants and providers, regulatory agencies, and other various stakeholders that the organisation is associated with. For effective DISG management practices, SOM must seek policies, frameworks, models, and strategies that cater for the unique organisational needs and objectives of each department; a “one-size-fits-all” approach will not help the departments to achieve their desired DISG goals and objectives. SOM should consider the formulation and implementation of newly constructed DISG management policies, frameworks, and models that will effectively guide SOM in making decisions that will strategically align current governance activities with improved DISG management practices without disturbing current operations.

Integrated approach of the people of the DoE, DST and DEA

The use of PSDI assets plays a crucial role in assisting public sector institutions to perform their daily operational functions to effectively and efficiently deliver public sector goods and services to the country's citizenry. There must be coordinated efforts between legal compliance and DISG management practices and functions. Public sector institutions must identify and employ personnel who will assist them in achieving their strategic goals and objectives of incorporating and integrating DISG management practices and functions into their current organisational functions. The three departments already have IT and IS departmental directorates and officers in place; however, SOM must re-evaluate which standards, procedures, baselines, and processes will be required to implement and execute DISG management practices and functions throughout the organisation.

The global revolutions have made the accessibility and usability of ICT systems and processes a significant tool that can assist in achieving desired DISG in each respective government department. The government must identify and explore approaches that will allow the integration, synergy, and standardisation of DISG management practices and functions. This will assist the government in finding effective measures and tools to minimise the duplication of roles, functions, and responsibilities in the management of IT systems and infrastructure. To achieve this, there must be standardisation of DISG control measures in the public service. The government must establish and implement a Security Standard Committee that deals specifically with DISG in the public service. There must also be operational efficiency by improving the alignment of SITA security with the DISG needs of each government department. More specifically, this Security Standard Committee would need to examine the specific security needs of each department and customise DISG policies, strategies, frameworks, and models in such a manner that they reflect and integrate strategic operational needs with the desired DISG management practices and functions of each department.

Executive management must also be involved and more participative in Records Management Units. The Records Management Units must be taken more seriously in the departments. This can begin with SOM making provisions to incorporate the Records Management Units into the departments' business continuity plans before it can explore and construct generic or even specific security systems, processes, and corporate cultural plans. The three departments could develop and implement legal and regulatory compliance indicators and monitoring and evaluating procedures of all regulatory policies that have been stipulated by the state, as well as those that have been implemented by supporting institutional structures such as the DPSA, SITA, and the SSA. Enforced compliance with internal rules, laws, and regulations must be improved as there are some instances of non-compliance.

The need for ensuring that internal policies are adhered to could be significantly improved by introducing frequent and regular assessments and auditing of legal and regulatory compliance in a non-hostile manner, not on the same level as conducted by the Auditor-General of South Africa. This could provide consistent communication and feedback reports, advice, and guidelines on problem areas that must be improved in each respective government department. The three departments could also consider establishing and implementing benchmarking tools or processes against best practices to revise, develop, and implement new internal policies, strategies, frameworks, and models that could improve the compliance in terms of protecting PSDI through advanced DISG management practices and functions.

Integrated approach of the systems of the DoE, DST and DEA

The internal IT security systems that have been implemented thus far have yielded the necessary security measures and controls towards the departments' strategic IT and IT security goals and objectives as there have been minimal security intrusions or incidents in the departments. The departments' physical information protection must, however, be reviewed, particularly in terms of identifying and implementing feasible measures to counteract the challenges of exiting departmental floors/rooms that have locked doors and biometrics, as they are ineffective if they are left open, unattended, or disabled due to power failures. To counteract this challenge, the departments must accelerate the internal digitisation processes of paper-based documents that are currently situated in departmental basements and physical registries into an electronic system in the Records Management Units to minimise these types of risks and challenges. In the Records Management Units, all departmental systems and processes that will be developed and implemented for the digitisation of paper-based documents should not be diagnosed as IT systems but rather be developed, approved, implemented, and standardised based on the stipulations and guidelines as set out by the NARS of South Africa Act of 1996. These government departments should also consider the amendment of the NARS of South Africa Act of 1996 to make provision for rules, laws, and regulations for the digitisation of paper-based records to be compliant and legally protected.

There should be synergy among all processes and functions in the departments, particularly between the Records Management Units and the IT departments. The roles, responsibilities, and functions in the processes related to the collection, classification, handling, storage, and protection of critical PSDI assets are currently not synchronised. The departments' security policies and records management policies must be aligned and integrated to create a centralised perspective for the protection of PSDI assets in the departments, as in most instances

the processes and systems are individualised to cater for some but not all needs in the area of security in the Records Management Units. Integrating the departments' processes of the collection, classification, handling, storage, and protection of PSDI as a whole could minimise the duplication of functions. Second, scarce resources should be used to establish an informative knowledge centre for accessible and reliable PSDI assets to the general public. Finally, the departments as a whole should ensure that they are equipped with the integrated and aligned skills, knowledge, and best practices for handling and protecting PSDI assets.

The departments have adequate measures, processes, and controls for the storage of their electronic PSDI assets; however, this is not true for their paper-based records. The government needs an intensive digital centralisation process of its paper-based records. There are procedures in place that consist of the manual scanning of paper-based records and uploading these records on the EDMS; however, this process is a time-consuming and costly exercise that the departments are currently using. The DPSA is in the process of developing and implementing an initiative towards the use of cloud storage for PSDI. This initiative has been identified as the "Cloud First" programme. The DPSA has, since May 2018, been working on formulating the "Cloud First Draft Paper" as its first cloud-based policy in the public sector. This policy, when enacted, intends to provide effective and efficient guidelines for the improvement of PSDI storage and security measures.

The introduction of artificial intelligence (AI), cognitive systems, and BCT into the public service has indicated that there is much room for the government to grow, develop, and improve in terms of its public service delivery objectives. This is an important indicator because it illustrates that government departments have the opportunities and can create capabilities for their offices to upgrade and enhance IT infrastructure to accommodate AI, cognitive systems, and BCT into the public service. Upgrading the government's technological equipment, skills, and expertise in AI and cognitive systems can accelerate public sector institutions to achieve more modern DISG management standards and practices.

Standardisation of security control measures in the public service is needed. To keep up with global trends and advanced industrial revolutions, the initial financial investments and resources allocated to national government departments must be reviewed. This is because government departments need to purchase new IT systems, infrastructure, and programmes that will produce effective and improved IT security tools and mechanisms that are internationally recognised and standardised. The government is currently undergoing intensive and lengthy restructuring in the public sector. One of the key elements that must urgently be addressed is the need for a larger pool of financial resources and initial investments from the DPSA, SITA, and the SSA towards DISG investments that will meet the strategic goals and objectives of departments' PSDI security needs. The

principle of value for money is crucial as government departments should take the initiative and present cost-effective strategies and budgets for the identified security systems and processes that will gear the departments towards their individualised needs for effective and efficient DISG management practices, systems, and processes. The DPSA could then benchmark these proposed DISG strategies and budgets against international best practices to evaluate and determine whether they will achieve the government's DISG.

The SSA has been experiencing major backlogs, which have resulted in some departmental staff not yet receiving security clearance certificates. This often causes internal organisational and structural challenges because employees who apply for or are promoted to senior management positions must comply with the requirements of having a valid security clearance certificate from the SSA for particular senior management positions. This also impacts employees who work in conjunction with senior management in the collection, classification, and storage of classified/confidential/critical/sensitive data and information assets as those employees may not be able to participate in certain meetings and reviewing of documents due to the absence of the security clearance certificate. The SSA should consider the possibility of outsourcing or privatising the functions of security vetting to obtain and grant departmental employees clearance certificates as it is one of the major challenges and security risks that must be addressed due to increased criminal activities in the public sector.

Integrated approach of the risks and controls of the DoE, DST and DEA

The technological environment is constantly evolving which requires consistent efforts to keep up with best practices in terms of PSDI. More financial and human resources must be directed towards integrated risk and data and information management practices for DISG. One of the challenges identified in the public sector is the lack of understanding risks and relationships between IT, IT security, and IS in general. Departmental training and awareness strategies and approaches must be changed because SOM has stated that it does not achieve its desired training results of having effective and efficient IT security for its internal systems and infrastructure. More awareness programmes and sessions are needed; particularly one-on-one sessions would be more preferable with targeted audience groups. It would give departmental staff more informed sessions, in a relaxed environment where they can freely ask questions without feeling they lack understanding, and improve listening and comprehension skills about the nature and complexity of IT. Every employee must also be trained regularly in the areas of IT security, and make the effort and time to familiarise themselves with general IT security risks, challenges, and best practices. Another significant element would be to ensure

that all employees are knowledgeable and familiar with all the necessary policies, frameworks, and guidelines as this would assist in improving working relationships and trust between the different levels of organisational management, subordinates, support, and executive staff. This could significantly improve the integration of communication channels and structures where the gap between SOM and subordinates can be narrowed and all employees can feel that their inputs and insights are significant, useful, and meaningful towards improved DISC management practices and principles.

Determining the entire organisation's training for skills and knowledge development in the area of DISC will be heavily dependent on the various needs of each directorate in the DoE, DEA, and DST. All employees in the DoE, DEA, and DST undergo induction processes and programmes on a monthly basis, as well as security awareness sessions every month. Employees are also required to sign declaration of secrecy forms when they commence work. The declaration of secrecy form provides a general overview to departmental employees about which information assets can be made accessible to the general public and which information assets should be declared as classified/confidential/critical/sensitive. The declaration of secrecy form is guided by the provisions of the PAIA of 2000. The Legal Services department conducts PAIA sessions with all employees on a regular basis.

SOM and all departmental staff have put certain measures in place, which, if used correctly, will be able to identify which types of data and information assets are subject to the violation of policy, legal, and regulatory requirements in the public service. This data and information would first have to be classified and considered confidential/sensitive/secret based on the content of the data and information assets. If the data and information is considered to be confidential/sensitive/secret/classified, then it could have very harmful and damaging ramifications for the national departments. The leakage of confidential/sensitive/secret/classified data and information assets would compromise the confidentiality, integrity, and access of the departments, and financial losses could be incurred due to security systems, processes, and software being compromised.

In addition, the reputation of the three departments would be tarnished, because it will also impact the level of trust, accountability, and security between the national departments, their private investors, supply chain partners, citizens, local and international bodies, and various other stakeholders. The leakage of confidential/sensitive/secret/classified data and information assets would cause immense problems and serious consequences for state security, such as possible wars between countries if top-secret military data and information assets were leaked. This could cause national working relationships to be strained, which presents gaps and opportunities for external intruders to capitalise on the mistrust of the government to infiltrate further attacks on critical IT infrastructure and systems. Certain departments will be disabled, delayed in their operations,

or irresponsible due to the lack of the availability of lost or stolen documents, or documents that have been tampered with. Lastly, these financial risks would cripple the efficiency of the departments if certain resources were restricted due to the leakage of particular data and information.

CONCLUSIONS

For the government to improve its efforts towards the institutionalisation of DISG management practices, its public sector institutions must have clearly defined roles, responsibilities, and functions. Government departments are often challenged by the bureaucratic systems and processes embedded in public sector administration functions. This unfortunately results in the duplication of functions and unstandardised public sector outputs and the mismanagement of scarce resources. Reinventing the roles, responsibilities, and functions of public sector institutions in the protection of their PSDI assets through the institutionalisation of DISG management principles and practices could improve the processes and synergy of data governance, information governance, and ISG. This will improve the processes related to the collection, classification, storage, and transmission of PSDI between the government and its citizenry in creating and maintaining a reliable, factual, and unbiased information society.

The policy and institutional contexts of the protection of PSDI assets plays a critical role in the establishment and achievement of effective and sound DISG management policies, practices, systems, and standards in South Africa. The government has made tremendous strides by implementing policies, strategies, and frameworks that are aligned to the protection of its PSDI assets. The government has proved its willingness to abide by international best practices; a primary example is the PoPI Act of 2013, which was influenced by the Council of Europe Convention, the EU's Data Protection Directive, and the OECD's guidelines. This indicates that the government in general has the right types of policies in place for efforts geared towards DISG management practices; however, it is often challenged in the areas of implementation, compliance, monitoring, and evaluating of its policies.

The security landscape of IT on a local and international scale is constantly evolving and changes daily. This requires consistent efforts to keep up with best practices to adequately protect PSDI assets and to minimise the risks associated with the theft, misuse, unauthorised access, and fraudulent activities associated with cybercrime. The government must implement proactive measures and approaches for the protection of PSDI through effective DISG management practices that have a high focus on DISG risk management policies, strategies, models, and frameworks for the adequate identification and mitigation of internal and external risks that could significantly hamper the protection of PSDI.

NOTE

*This article is partly based on a master's dissertation: Masilela, L. 2019. *Data and Information Security Governance in the Departments of Energy, Science and Technology, and Environmental Affairs*. Johannesburg: University of Johannesburg, completed under the supervision of Prof Danielle Nel-Sanders.

REFERENCES

- Alles, M. and Piechocki, M. 2012. Will XBRL improve corporate governance? A framework for enhancing governance decision making using interactive data. *International Journal of Accounting Information Systems*. 13(1):91-108.
- Bahrman, D.P. 2011. *Evaluating and Improving Organisational Governance*. Florida: Institute of Internal Auditors Research Foundation.
- British Academy and The Royal Society. 2017. *Data Governance Landscape Review*. London: The Royal Society.
- Broby, W.K. 2008. *Information Security Governance: Guidance for Information Security Managers*. Rolling Meadows: T-Governance Institute.
- Bryman, A. 2012. *Social Research Methods*. New York: Oxford University Press.
- Chalker, A. 2014. *Data Governance Overview*. Atlanta: Protiviti Inc. Available at: <https://pdfs.semanticscholar.org/162f/177ad31d5a342977bcc8112cb42666bad57b.pdf>. (Accessed on 25 July 2019).
- Denzin, N.K. and Lincoln, Y.S. 2000. *Handbook of Qualitative Research*. (2nd ed.). London: Sage Publications.
- Eckerson, W. 2014. *Data Governance for the Enterprise: Trends in the Use of Data Quality, Master Data Management and Metadata Management*. Available at: <http://www.techtarget.com>. (Accessed on 22 May 2018).
- Flick, U. 2013. *The Sage Handbook of Qualitative Data Analysis*. London: Sage Publications.
- Hardy, G. 2006. Using IT governance and COBIT to deliver value with IT and respond to legal, regulatory and compliance challenges. *Information Security Technical Report*. 11(1):55-61.
- Hüner, K.M., Ofner, M. and Otto, B. 2009. Towards a maturity model for corporate data quality management. *Association of Computing Machinery (ACM)*. 1(1):231-238.
- Institute of Directors in Southern Africa (IoDSA). 2016. *Draft of King IV Code for Governance Principles*. Johannesburg. South Africa: IoDSA.
- IT Governance Institute. 2003. *Board Briefing on IT Governance*. (2nd ed.). Rolling Meadows: IT Governance Institute.
- IT Governance Institute. 2006. *Information Security Governance: Guide for Boards of Directors and Executive Management*. (2nd ed.). Rolling Meadows: IT Governance Institute.
- Iyamu, T. 2011. The architecture of information in organisations. *South African Journal of Information Management*. 13(1):1-9.

- Khatri, V. and Brown, C.V. 2010. Designing data governance. *Communications of the ACM*. 53(1):148-152.
- Ministry of Economic Affairs and Communications (MEAC). 2014. *Information Governance: Current Situation Analysis and Implementation Strategy*. Republic of Estonia: Government Printer.
- Moulton, R. and Coles, R. 2003. *Applying Information Security Governance: Computers and Security*. Oxford: Elsevier, Ltd.
- Rossouw, D. and Van Vuuren, L. 2013. *Business Ethics*. (5th ed.). Cape Town: Oxford University Press.
- Webber, K., Otto, B. and Österle, H. 2009. One size does not fit all: A contingency approach to data governance. *ACM Journal of Data and Information Quality*. 1(1):1-27.
- Westby, J. and Allen, J. 2007. *Governing for Enterprise Security (GES) Implementation Guide*. Carnegie Mellon University: Software Engineering Institute.

AUTHOR'S CONTACT DETAILS

Ms L Masilela

School of Public Management, Governance
and Public Policy
College of Business and Economics
University of Johannesburg
Auckland Park
South Africa
Cell: 0782086006
Email: luciamasilela@gmail.com

Prof D Nel-Sanders

School of Public Management, Governance
and Public Policy
College of Business and Economics
University of Johannesburg
Auckland Park
South Africa
Cell: 0824631776
Email: daniellen@uj.ac.za

Editorial policy

1. *Administratio Publica* is a quarterly journal. The aim of the journal is to promote academic scholarship in Public Administration and Management and related fields.
2. *Administratio Publica* publishes original theoretical and empirical manuscripts in Public Administration, Public Management and Public governance related fields (teaching, research and learning). This includes articles, debates, research notes, review articles and book reviews in English.
3. All articles, research notes and review articles are submitted to the Chief Editor. The Chief Editor decides whether a proposed article falls within the ambit of scholarly material and research that supports the objectives of ASSADPAM. The article is then submitted to subject specialist referees for evaluation. Normally, these referees are members of ASSADPAM. In the case of any one of two reviewers rejecting the article completely, the editorial committee considers the reports, and may appoint another independent reviewer.
4. Anonymity is maintained by omitting the name of the author and all identifying footnotes or other references in the text. Referee reports may be made available to authors but the names of the referees will be withheld. The particulars of referees will only be made known to the Department of Higher Education, for purposes of auditing the requirements as set out by the Policy for the Measurement of Research Output of Public Higher Education Institutions (Higher Education Act 101 of 1997).
5. Copyright of published material vests with ASSADPAM (Association of Southern African Schools and Departments of Public Administration and Management). Authors, however, bear full responsibility for the contents of their own contributions. Opinions expressed in the journal are those of the authors, and are not necessarily subscribed to by the Chief Editor, the editorial committee, the Editorial Board, or the Executive Committee of ASSADPAM.
6. Accepted papers will, as far as possible, be published in the sequence in which they were received. A page fee will be charged for all articles approved for publication. This fee is subject to annual adjustment.

Presentation of manuscripts before publication

Formatting requirements

- Typewritten on one side of the paper and language edited. A4 format; one and a half spacing with generous margins; statistics are to be presented in table format, and graphic images in boxed diagrams.
- Contributions should preferably be between 6 000–8 000 words. Contributions less than 4 500 words will be considered as viewpoints and not as full articles for subsidy purposes. Each article must be preceded by an abstract (maximum length 200 words). Abstracts should represent the contents of the article concisely.
- Title of Paper (Centre, Bold, 12pt Times New Roman/Arial, Caps), Name and Surname, Affiliation (name of university/institution), (Align left, Bold 12pt Times New Roman).
- Biographical details: Please provide full details of title, surname and initials, and first name, address, fax, cell, work number etc at the end of the references/bibliography and not under the title on the first page.
- **ABSTRACT:** (12pt Times New Roman, Title in Caps), 200 words. Do not format, shade or place in textbox or table.
- **HEADINGS:** Do not number headings in text
 - **HEADING 1** (Bold, 12pt Times New Roman, Caps)

- **Sub-heading 1.1** (Bold, 12pt Times New Roman)
- *Sub-heading 1.1.1* (Bold, Italic, 12pt Times New Roman)
- Sub-heading 1.1.1.1 (Underline, 12pt Times New Roman)
- When listing, please use bullet form.
- Please do not format or use any special style in your article.
- All tables and figures should be numbered and given a heading e.g. Table 1: Population Growth before the table is inserted. The source must be provided at the bottom of the table.
- Remember to give credit to the original authors from which the table or figure was adapted.
- References in the text: Thornhill (2012:128), (Nhlabathi and Mubangizi 2012:55), (Cloete et al. 2008:4). First time three or more authors are used, type out all the authors, thereafter et al.
- **REFERENCES:** Bibliographical particulars arranged alphabetically, no numbering and according to the simple Harvard method e.g.:
 - **Book:** Gildenhuis, J.S.H. and Knipe, A. 2000. *The Organisation of Government: An Introduction*. Pretoria: Van Schaik.
 - **Article:** Auriacombe, C.J. 2007. Deconstructing the public interest for application in South Africa. *Administratio Publica*. November 2007. 15(1):28–52.
 - **Official publication:** Public Service Commission. 2008c. Report on the Audit of Reporting Requirements and Departmental Monitoring and Evaluation Systems within National and Provincial Government. Public Service Commission. Pretoria: Government Printers.
 - **Internet reference:** Patton, M.Q. 1999. Utilization-Focused Evaluation in Africa, training lectures presented at the African Evaluation Association, 13–17 Sept. 1999. Nairobi. Kenya. Available at: www.afrea.org/documents/document.cfm?docID=64 (Accessed on 25 March 2009).
 - **Title of acts in italics** e.g. *Public Service Act, 1994* (Act 103 of 1994) or Public Service Act 103 of 1994. Be consistent with the referencing.
 - **Full reference:** South Africa (Republic). 2000. *Local Government: Municipal Systems Act, 2000* (Act 32 of 2000). Pretoria: Government Printers.
- **AUTHOR'S CONTACT DETAILS:** The title, the name(s), e-mail addresses, telephone and fax numbers and postal address(es) of the author(s), to whom correspondence should be addressed.
- A separate application first page(s) should contain the proof of language editing by a qualified language practitioner, title, the name(s) e-mail addresses, telephone and fax numbers and postal address(es) of the author(s) names and to whom correspondence and the invoice for page fees should be addressed. The article is to commence on a new page in a separate file. Language editing fees will be added to the page fees if the language editing is not satisfactory.
- Two kinds of references are distinguished; notes and bibliographical particulars. Notes should be numbered consecutively and be inserted at the end of the text and not in footnotes.
- A statement should accompany all manuscripts (at the end of the document when submitting the paper) in which the author(s) state(s) that the contribution has not been published, is not being published, or is not being considered for publication elsewhere, and will not be submitted for publication elsewhere unless rejected by the Editorial Committee. The author(s) need to declare that they are willing to pay the page fee, if the article is approved for publication that the article has not been submitted to any conference proceedings, is not plagiarised and is checked on Turnitin before submission.
- Confirmation of provisional acceptance of articles will be forwarded to authors on receipt by the Chief Editor. Authors will be given notice of the outcome of peer reviews within twelve weeks of submission. It is important to know that an article submitted in the second half of the year may only be published in planned editions of the following year.

Administratio Publica

ISSN 1015-4833